BONUS QUESTION BANK

600 ABA-Style Board Review Questions

With full answers and explanations

A free companion to the Anesthesia Crisis and Emergency Handbook, 2027 Edition

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How to Use This Question Bank

This bank holds 600 additional questions, 20 for each of the 30 chapters of the handbook, in the same order as the chapters. Every question has five answer choices, lettered A to E, and one best answer. None uses an image.

The questions come first and the answers follow. Work a group of 20 without looking ahead, commit to an answer for every question, and only then turn to the Answers and Explanations section, which begins on a new page. Read every explanation, including for the questions you answered correctly.

The questions are original. They are not drawn from any examination and do not predict or reproduce examination content. "ABA-style" describes only the format and content focus. This bank is independent and is not affiliated with, sponsored by, or endorsed by the American Board of Anesthesiology.

The contents page and the links at the top of each group are clickable, and the page numbers beside them always work. If the links do not respond in your viewer, open the file in a PDF reader app or in your phone browser's PDF viewer and use the page numbers.

Clinical practice changes as evidence changes. Confirm every decision and every dose against current guidelines, product information, and your own institution's protocols before applying it to patient care.

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QUESTION GROUP 1

The Anatomy of an Intraoperative Crisis

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Questions 1 to 20. Choose the single best answer for each.

1. A hospital safety committee reviews an event in which a wrong drug was given to a patient and a delayed alarm response followed. The review finds a look-alike ampule stored in the wrong tray, an incomplete machine check, and an alarm limit set too wide, all in the same room on the same morning. Which of the following best explains how this event reached the patient?

A. A single clinician lacking knowledge was the sole cause, and the other findings were incidental

B. The patient had unusually high physiologic reserve, which masked the earlier warnings

C. Holes in several successive defenses lined up and allowed a small trigger through

D. Fast pattern recognition was used in place of slow analysis at one step of the case

E. The event arose from an unpredictable fault that no defense could have caught

2. Four patients are disconnected from the ventilator at the same moment when a circuit comes apart at the elbow. Which of the following patients is likely to become hypoxemic first?

A. A healthy young man who was preoxygenated for an elective knee arthroscopy

B. A lean older woman who is euthermic and breathing room air before induction

C. A healthy adult who was preoxygenated and is lying supine and afebrile

D. A fit middle-aged man in the semirecumbent position after full preoxygenation

E. A pregnant woman with fever at induction for an urgent operation

3. A 63-year-old man is undergoing an open colectomy under general anesthesia. The saturation has fallen from 99 to 92 percent over two minutes. The anesthesiologist suspects a loose probe, wiggles it, and then replaces it. Over the next two minutes the saturation reads 89 percent, the heart rate has risen to 110, and the chest appears to move less than before. A colleague asks whether the probe is truly the problem. Which of the following best describes the error in the anesthesiologist's approach?

A. Using slow deliberate analysis before a rapid pattern was available to guide care

B. Failing to treat a physiologic state before a firm diagnosis had been made

C. Remaining in fast pattern recognition after the pattern stopped fitting

D. Overestimating the patient's reserve before the case began

E. Hindsight bias in judging the decision after the outcome was known

4. A 40-year-old healthy man is undergoing a laparoscopic hernia repair. Within one minute of insufflation, the saturation drifts from 100 to 97 percent. The capnogram, airway pressure, blood pressure, and heart rate are unchanged, and the surgeon is working smoothly. The anesthesiologist is deciding how seriously to take a change that appears small. Which of the following is the best approach?

A. Wait until the saturation reaches a threshold such as 90 percent before acting

B. Check ventilation and the airway now, since an early change costs little to check

C. Treat the reading as artifact because most alarms are false in the operating room

D. Ignore the change until the surgeon has completed the current step

E. Repeat the measurement every ten minutes until a clear downward trend is established

5. Two operating rooms are managing different problems. In room A, a loose connection allows a slow circuit leak that has reduced the delivered tidal volume by a small amount every few minutes for half an hour. In room B, a patient who was stable one minute ago has collapsed abruptly after receiving a drug. Which of the following correctly matches the skill that each situation rewards?

   A. Room A rewards reading the trend over time, and room B rewards speed of action

   B. Room A rewards speed of action, and room B rewards reading the trend over time

   C. Both rooms reward waiting to see whether a clear pattern develops over time

   D. Both rooms reward obtaining laboratory confirmation before any action is taken

   E. Room A rewards reading the trend, and room B rewards a search for the cause before treatment

6. A department audits errors in its operating rooms over a year. Which of the following times in a case are errors most likely to cluster, and where should extra attention be placed?

   A. During the stable maintenance phase of routine cases that last under an hour

   B. Around transitions such as induction, positioning, emergence, and handover

   C. During the first routine machine check performed each morning

   D. Only in cases that involve a trainee or a first case of the day

   E. Only at the moment of surgical incision, when stimulation is greatest

7. A new anesthesiologist tells her mentor that she silences roughly forty alarms each morning because nearly all of them turn out to be false, and she is worried she will one day fail to respond to a real one. She asks how to protect herself against slowing down over a long career. Which of the following is the best advice from her mentor?

   A. Build a small number of habits that make the correct check automatic

   B. Maintain constant maximal vigilance for every alarm throughout the shift

   C. Turn alarm volume down so that fewer false alarms interrupt concentration

   D. Widen all alarm limits so the monitor sounds less often during the case

   E. Rely on the surgeon to notify the team of any change on the monitor

8. A 50-year-old woman is under general anesthesia for a cholecystectomy when the saturation alarm sounds and the number falls from 98 to 91 percent in under a minute. The anesthesiologist has no information about the cause and is working alone with the surgeon already dissecting. She has time for only a couple of actions before more is known. Which two actions should come first?

   A. Replace the oximeter probe, then repeat the blood pressure measurement

   B. Draw an arterial blood gas, then review the anesthetic record with the nurse

   C. Disconnect the surgeon's monopolar cautery, then recalibrate the capnograph

   D. Call for help, then give 100 percent oxygen at a high flow

   E. Stop the surgery, then deepen the anesthetic and reassess the circulation

9. A 65-year-old man is undergoing a hip arthroplasty under general anesthesia. The capnography trace suddenly goes flat. The chest rise appears questionable from the head of the bed, and the pulse oximeter still reads 99 percent. The anesthesiologist is unsure which monitor to believe and asks the room which is more trustworthy at this moment. Which of the following is the best interpretation?

   A. No ventilation or no circulation until proven otherwise; trust the waveform

   B. The patient is adequately ventilated, since the saturation lags by many seconds and is normal

   C. The saturation is wrong and the flat trace is artifact until a blood gas is received

   D. The waveform is unreliable and chest rise alone is sufficient to confirm ventilation

   E. The flat trace reflects a single shallow breath and will recover without intervention

10. A 58-year-old woman is in the operating room for a laparoscopic hysterectomy when the electrocardiogram suddenly shows a flat line and the alarm sounds. The oximeter waveform and the capnogram look entirely normal, and the patient appears well perfused. The anesthesiologist is deciding whether to begin chest compressions. Which of the following is the best next action?

A. Begin compressions immediately because the rhythm display shows no activity

B. Wait for the next noninvasive blood pressure cycle before checking the leads

C. Feel for a carotid or femoral pulse and check whether a lead has come off

D. Silence the alarm and proceed because the patient is unconscious and cannot be disturbed

E. Check the oximeter probe site first, since it is the monitor most likely to fail

11. A 45-year-old man develops sudden hypotension and bronchospasm five minutes after induction for a knee arthroscopy. An antibiotic and a neuromuscular blocker were given in quick succession, and a second syringe was drawn up by the nurse. The anesthesiologist cannot recall which was given first or the order of the doses. The team is waiting for direction. Which of the following is the best next step?

A. Assume drugs are not responsible because the induction passed without incident

B. Review aloud what was given, in what dose, and when, and name the likely trigger

C. Look first at the monitor for a technical fault before reviewing any drug

D. Wait for the surgical incision to learn whether the response continues

E. Accept that all recent drugs were given at standard doses and so cannot be responsible

12. A 35-year-old man under general anesthesia has a falling blood pressure with a slow heart rate. Which of the following is the best reading of this pattern?

A. It points toward loss of volume or vasodilation with compensation

B. It points toward a hypermetabolic state with rising carbon dioxide production

C. It points toward a leak in the circuit between the machine and the patient

D. It points toward drugs, a vagal event, or a primary cardiac problem

E. It points toward a pneumothorax with a rising airway pressure

13. A 70-year-old woman is hypotensive during a bowel resection, and the anesthesiologist has called for help and given oxygen. One minute later, the blood pressure is lower than it was at the start and the saturation is also falling. The anesthesiologist asks whether to keep exploring the cause broadly or to commit to a specific pathway for the leading diagnosis. Which of the following is the best next step?

A. Continue a broad diagnostic search until the cause has been confirmed

B. Repeat the first minute's assessment without changing the approach

C. Wait for the surgeon to propose a diagnosis before choosing a pathway

D. Move to the algorithm for the leading diagnosis and stop exploring

E. Perform a full examination of every organ system before giving any treatment

14. A 68-year-old man is hypotensive and tachycardic during a knee operation performed under a thigh tourniquet, and bleeding appears greater than expected. Despite fluid and a vasopressor, the pressure cannot be stabilized while the surgeon continues, and the surgeon is reluctant to stop. Who is best placed to say how long the patient can wait, and what should be said?

A. The surgeon, who should decide alone whether the patient can tolerate more surgery

B. The circulating nurse, who should negotiate with the surgeon on behalf of the team

C. The anesthesiologist, who should wait until the patient is stable before raising the matter

D. The circulating nurse, who should relay the anesthesiologist's request later in the case

E. The anesthesiologist, who should state plainly that the surgery must pause

15. A 56-year-old woman remains hypoxemic after fifteen minutes of treatment for presumed aspiration. A resident suggests that the team list the most dangerous explanations that could still be missed. Which of the following is the best reason to start with the lethal diagnoses?

A. The most common diagnosis is the one most likely to explain the findings

B. The lethal diagnoses are always the easiest to confirm with a simple test

C. The team will be less likely to disagree with a serious diagnosis

D. The lethal diagnoses are the ones that the monitors identify first

E. The most dangerous diagnosis is the one that cannot be afforded to miss

16. In a simulated crisis, an experienced anesthesiologist forgets to stop the volatile agent and does not repeat a rescue dose that is needed. In the debrief he says that he knew both steps and cannot explain why he did not do them, and he asks whether the emergency manual would really have helped him. Which of the following best explains the benefit of a cognitive aid in this situation?

A. It turns recall into reading and protects steps that judgment tends to skip

B. It replaces clinical judgment, so the team need not decide among options

C. It shortens a crisis by removing the need for communication among team members

D. It is useful mainly for trainees and offers little to experienced clinicians

E. It shortens the case by removing disagreement among the team members

17. A residency program wants its trainees to be able to carry out the first line actions for rare crises in the correct order without prompting, both in the examination and in a real event. The program director notes that under stress clinicians attend to fewer cues and favor actions that are familiar over actions that are correct. Which of the following is the most effective preparation?

A. Read each emergency chapter several times so the knowledge is thoroughly understood

B. Rehearse the response in simulation so that it is available under stress

C. Ask trainees to commit the algorithm to memory and recite it silently

D. Rely on the supervising anesthesiologist to remember the steps during any event

E. Attend a lecture that summarizes the most recent guideline changes

18. A department reviews a case in which the patient died after a rare complication, although the anesthesiologist's decisions were reasonable given the information available at the time. A committee member argues that the decisions must have been poor because the outcome was so bad, and wants to change practice on that basis. Which of the following best describes the flaw in this argument?

A. Anchoring caused the committee to fix on the first explanation offered

B. Outcome bias makes decisions look worse than they were when the result is bad

C. Premature closure caused the committee to stop the search at one cause

D. Diagnostic momentum carried a label from the start of the case into the review

E. Fixation on a plan that had stopped working during the event itself

19. During an emergency, the lead anesthesiologist says loudly to the room that someone should bring the blood, and no one moves. Two minutes later, when a colleague asks when epinephrine was last given, no one can remember. The team's reply to each problem is a shrug. Which of the following changes would best prevent both problems from recurring?

A. Assign tasks to people by name and name one person as recorder

B. Speak louder to the whole room and ask each person to keep a private mental note

C. Give all tasks to the most senior helper and have the leader record the events

D. Ask the surgeon to remain silent and the nurse to leave to avoid distraction

E. Ask the circulating nurse to watch the monitors and call out any change

20. A 62-year-old man is hypotensive during a laparotomy. A junior nurse has noticed that the suction canister has filled quickly and that the blood pressure has been falling for several minutes. She does not speak because the senior team appears busy. The anesthesiologist, afterward, wants to reduce the chance of this happening in the next case. Which of the following is the most effective step?

A. Ask staff to raise concerns only after the case has ended and the patient has left

B. Assign all monitoring duties to the most senior person in the room alone

C. State before the case that anyone who sees something should speak up

D. Remind the team that senior clinicians are not to be interrupted during an event

E. Give the nurses a written protocol and ask them to review it before the case

QUESTION GROUP 2

Monitoring That Warns Early

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Questions 21 to 40. Choose the single best answer for each.

21. A hospital committee is writing a monitoring policy for its operating rooms and procedure suites. The chair asks when the American Society of Anesthesiologists standards require continual end-tidal carbon dioxide analysis, and how that applies to patients who are sedated but not intubated. Which of the following statements best reflects the standard? The committee also wants the policy to cover settings outside general anesthesia.

A. Required only during general anesthesia that lasts more than one hour with controlled ventilation

B. Required only when a neuromuscular blocker has been given and ventilation is controlled mechanically throughout

C. Required only after auscultation of the chest has confirmed that the tube lies in the trachea and not the esophagus

D. Required only for patients who have underlying lung disease, obesity, or a predicted difficult airway

E. Required with a tracheal tube or supraglottic airway, and sought in moderate and deep sedation

22. A 70-year-old man with chronic obstructive lung disease is anesthetized in the lateral position for a thoracic procedure. His end-tidal carbon dioxide reads 28 mm Hg, but the clinical picture suggests he may be hypoventilated. The anesthesiologist wants to use the monitor reliably for the rest of the case. Which of the following is the best approach?

A. Assume the end-tidal value equals the arterial value and adjust ventilation to it alone

B. Discard the capnograph because the lateral position makes the value uninterpretable

C. Draw an arterial sample every ten minutes and ignore the capnograph between samples

D. Draw one arterial sample, learn his gradient, and read the monitor with that offset

E. Increase minute ventilation until the end-tidal value reaches the normal range for a healthy adult

23. During a robotic prostatectomy, a 60-year-old man's end-tidal carbon dioxide gradually falls from 36 to 22 mm Hg over five minutes. The waveform keeps its shape but looks smaller. The heart rate, blood pressure, saturation, and airway pressure have not changed from baseline. The anesthesiologist wants to determine what has happened. Which of the following is the best next step?

A. Look for a partial disconnection or a leak around the cuff

B. Treat for massive hemorrhage and request blood products from the blood bank

C. Treat for malignant hyperthermia and prepare to give dantrolene

D. Replace the exhausted carbon dioxide absorbent and recheck the waveform

E. Assume pulmonary embolism and begin anticoagulation immediately

24. A 24-year-old man is under general anesthesia for repair of a fractured forearm. Over fifteen minutes he becomes tachycardic and his jaw feels rigid. His end-tidal carbon dioxide has climbed from 38 to 58 mm Hg despite a higher minute ventilation, the baseline returns to zero on every breath, and the temperature has not yet changed. Which of the following is the most likely explanation?

A. Rebreathing from exhausted carbon dioxide absorbent

B. A hypermetabolic state such as malignant hyperthermia

C. Hypoventilation from an inadequate tidal volume

D. Absorption of insufflated carbon dioxide from the surgical field

E. A displaced tube advanced into one main bronchus after repositioning

25. A 55-year-old man is turned prone for a spine operation. Shortly afterward, the capnogram shows a sloping upstroke with no alveolar plateau, and the peak airway pressure has increased. The saturation has not yet changed. The anesthesiologist asks what the waveform shape is telling her. Which of the following is the most likely explanation? The surgeon asks whether the case can continue, and the nurse notes that the tube was repositioned during turning.

A. Rebreathing from a failed one-way valve in the breathing circuit

B. Return of spontaneous breathing during a partial neuromuscular block

C. Cardiogenic oscillations as the heart beats against the lungs

D. Obstruction to airflow from a kinked tube, secretions, or bronchospasm

E. Esophageal placement of the tube with gas moving in and out of the stomach

26. A 38-year-old woman is anesthetized and paralyzed with a neuromuscular blocker that was given 40 minutes ago. The end-tidal carbon dioxide trace shows a normal plateau with a distinct notch in its midportion on each breath. The anesthesiologist is asked what the notch indicates. Which of the following is the best answer? The surgeon has just asked whether any further relaxant is needed, and the train of four has not yet been checked.

A. The patient is rebreathing because the absorbent is exhausted

B. The heart is beating against the lungs in an apneic patient

C. The patient is making an inspiratory effort against a partial block

D. The sampling line has become kinked or has been disconnected

E. Gas entering the sample from a leaking mask with a high fresh gas flow

27. A 66-year-old man is in cardiac arrest during surgery and compressions are under way. The anesthesiologist is watching the capnogram, which has read about 14 mm Hg for several minutes. Suddenly the end-tidal value jumps to 44 mm Hg with a normal waveform, though no one has yet felt a pulse. Which of the following is the most appropriate interpretation and action?

A. The tube has moved into the esophagus, so remove it and start mask ventilation

B. The value is an artifact of hyperventilation and should be ignored until a pulse is felt

C. The rescuers are now compressing too deeply, so reduce the depth of compression

D. The compressions are too fast, and this falsely raises the end-tidal value

E. This may be return of spontaneous circulation, so pause to check the rhythm and pulse

28. A 72-year-old woman has been in cardiac arrest for eight minutes during a laparotomy complicated by bleeding. Compressions are being given, and the capnogram shows a waveform with a value of about 8 mm Hg. A resident asks whether this means resuscitation should be stopped. Which of the following is the best response? The team is following the resuscitation protocol, and a second clinician has arrived to help.

A. The low value is conclusive proof that the patient cannot recover, so stop resuscitation now

B. The low value shows that the tube is misplaced and should be removed and replaced immediately

C. Correct compressions, rescuer fatigue, or volume loss, and continue resuscitation

D. The low value should be ignored because it does not depend on cardiac output or perfusion

E. Increase the ventilation rate sharply to bring the value up toward the normal range

29. An intubated 48-year-old woman is under anesthesia when the capnogram suddenly goes flat. The chest rises with each breath, breath sounds are present on both sides, the blood pressure is normal, and the saturation is 99 percent. The sampling line has beads of moisture in it. Which of the following is the best next action? The surgeon is waiting, and the nurse offers to bring another circuit while the anesthesiologist decides what to do first.

A. Remove the tube immediately and ventilate with a mask because the tube must be misplaced

B. Stop ventilation until the trace returns to avoid barotrauma

C. Begin chest compressions because a flat trace signals cardiac arrest in this patient

D. Check the sampling line and water trap within seconds while continuing to ventilate

E. Increase the fresh gas flow, which will clear the line of moisture and restore the trace

30. During a difficult laryngoscopy in a 40-year-old man, the pulse oximeter reads 99 percent for the entire attempt, which lasts about a minute and a half. Within seconds of the tube being passed and the first breath, the reading is 81 percent and then falls further. The resident asks why the monitor gave so little warning. Which of the following best explains this?

A. Lung reserve was consumed before the reading fell, then the curve steepened

B. The oximeter overestimates the saturation by a fixed amount in all patients

C. The oximeter measures only venous blood and so reports a delayed, lagging value

D. The reading is averaged over minutes, so it cannot register a change at all

E. The probe must have slipped from the finger, since the display changed suddenly

31. A 52-year-old man in the intensive care unit is breathing 100 percent oxygen through a ventilator after an aspiration event. The pulse oximeter reads 94 percent with a good plethysmographic waveform. A resident says that a reading of 94 percent is acceptable. Which of the following is the best interpretation? She asks whether this reading means the patient has plenty of reserve if ventilation is interrupted.

A. The patient has a large oxygen reserve because the saturation is above 90 percent

B. A large shunt or a poor signal, so the reading says little about reserve

C. The oximeter is reading carboxyhemoglobin as oxyhemoglobin and is falsely high

D. The tissues are well oxygenated because the reading is above the usual threshold

E. The probe is on a limb with a tight tourniquet and is reading venous blood

32. A 36-year-old woman has a pulse oximeter reading that has settled at about 85 percent for an hour. Her blood looks chocolate brown on sampling, and giving 100 percent oxygen does not change the number. She is comfortable and has no lung disease. The anesthesiologist asks which test will settle the diagnosis. Which of the following is the best answer?

A. A repeated measurement with a different pulse oximeter on the opposite hand

B. A chest radiograph to look for pulmonary edema or consolidation

C. An arterial blood gas with co-oximetry to measure methemoglobin

D. A capnogram to confirm that ventilation of the lungs is adequate

E. Carboxyhemoglobin is raised and is being read as oxyhemoglobin

33. A 58-year-old Black man is recovering from a long operation. The pulse oximeter shows 93 percent and the capnogram is normal. He is mildly tachypneic, and the team is considering a transfer to the ward. The nurse points out the studies on oximeter accuracy. Which of the following is the best next step? The surgical team is waiting for a decision on whether he can leave the recovery unit.

A. Accept the reading as accurate because oximeters are calibrated equally for all patients

B. Assume the true value is lower by exactly three points and adjust oxygen accordingly

C. Disregard all oximeter data and rely only on the respiratory rate and the chest examination

D. Use the trend, the capnogram, and the clinical picture, and send an arterial blood gas

E. Treat the reading as falsely low, since darker skin makes oximeters underestimate

34. An anesthesiologist is giving a short talk to residents about a case of an elderly woman who was undergoing a long hip operation. The saturation was 98 percent, the blood pressure was normal, and the capnogram was normal, but the patient was noted to be gray, sweating, and tachycardic under regional anesthesia with sedation. Which of the following is the most important lesson?

A. The numbers are reassuring, so the appearance of the patient can be disregarded

B. The oximeter should be replaced because it must be reading falsely high

C. The patient should be reassured and the monitors reviewed again in thirty minutes

D. The monitors show stable values, so continue the current plan and recheck in an hour

E. The patient is right, and the numbers may hide compensation that is about to fail

35. An elderly woman is at high risk for awareness during a long operation under volatile anesthesia with a neuromuscular blocker. The department has no processed electroencephalographic monitor available on that day. The chief asks whether a protocol using an alarm on the end-tidal anesthetic concentration would be an acceptable alternative. Which of the following is the best answer? The department wants a clear policy and cannot postpone the case.

A. Yes, in high-risk patients the alarm performed as well as a processed index

B. No, a processed index is an absolute requirement for any patient who has been given a blocker

C. No, end-tidal concentration cannot be measured during volatile anesthesia

D. Yes, but only because the processed index is unreliable in every elderly patient

E. Yes, but only when the patient is given a total intravenous technique

36. A resident repeatedly silences a recurrent low blood pressure alarm during a stable case, saying that the alarm is always false. The attending asks the resident to explain the safe approach to such alarms. Which of the following is the best answer? The attending, called to the room after the third alarm, finds the resident muting it without looking at the patient or the transducer.

A. Silence each alarm quickly so that it does not distract the surgeon or the team

B. Never silence an alarm before its cause is found, and fix the artifact or limit aloud

C. Set all limits as wide as possible at the start so the monitor sounds as seldom as possible

D. Disable the alarm for the case and rely on the clinical impression of the circulation

E. Reduce the alarm volume so it does not distract during stable periods

37. A 64-year-old man has an arterial line zeroed at the heart level. During the case the table is raised 20 centimeters for the surgeon, and the transducer is not releveled. The anesthesiologist wants to estimate how much the reading will be in error because of the height difference. Which of the following is the best estimate? The surgeon is concerned that the pressure on the monitor now looks unusually low.

A. About 2 mm Hg

B. About 7 mm Hg

C. About 30 mm Hg

D. About 45 mm Hg

E. About 15 mm Hg

38. A 70-year-old woman is hypotensive after induction for a major abdominal operation. A central venous catheter reads 4 mm Hg. A resident states that the low value shows she needs a fluid bolus. The attending disagrees and explains how to use the central line. Which of the following is the most accurate statement? She has already received fluid, and the surgeon is asking for a decision about a further large bolus.

A. A pressure below 5 mm Hg proves fluid responsiveness in a ventilated patient

B. It poorly predicts fluid responsiveness, so it should not decide on a bolus

C. The main purpose of the catheter is to measure pressure, not to deliver drugs

D. The pressure value is more reliable than its waveform for assessing volume status

E. The central venous waveform is uninformative in all circumstances and should be disregarded

39. A 46-year-old man is positioned sitting for a posterior fossa craniotomy with a central venous catheter in place. Suddenly the end-tidal carbon dioxide falls sharply and the blood pressure drops. The anesthesiologist suspects an embolism through an open venous system. Which of the following is the most appropriate immediate action? The surgeon has just warned that a venous sinus may have been opened.

A. Give a bronchodilator for presumed bronchospasm and raise the fresh gas flow

B. Reduce the volatile agent because the anesthetic is deeper than needed for surgery

C. Replace the carbon dioxide absorbent and run a leak test of the circuit

D. Aspirate through the central catheter to confirm and treat air embolism

E. Treat for tension pneumothorax with decompression of the affected side

40. A 68-year-old woman has been hypotensive for twenty minutes in the operating room despite fluid and a vasopressor. The cause remains unclear and could be hypovolemia, ventricular failure, tamponade, embolism, or a valvular problem. Which of the following is the most useful next step to settle the question quickly? The arterial pressure trace and the electrocardiogram show no clear cause, and the capnogram is stable.

A. Perform echocardiography to distinguish among these causes within minutes

B. Insert a pulmonary artery catheter and wait for the pressures to guide therapy

C. Interpret the central venous pressure value to decide the cause of instability

D. Send the patient to radiology for a formal study before making any decision

E. Place a central venous catheter and use the pressure to guide fluid over the next hour

QUESTION GROUP 3

Teams, Communication, and Cognitive Aids

Contents | Answers for this group | Previous group | Next group

Questions 41 to 60. Choose the single best answer for each.

41. A 50-year-old woman becomes hypotensive and tachycardic during a hysterectomy. The anesthesiologist is told by a resident that bleeding may be occurring, but dismisses the idea and instead pursues a series of rare endocrine explanations for twenty minutes, giving no attention to the obvious possibility. In the classification of fixation errors used in crisis resource management, which of the following best describes this error?

A. This and only this, in which one diagnosis is clung to despite contrary evidence

B. Persistent failure to revise a plan that has stopped working despite new findings

C. Everything but this, in which the obvious diagnosis is dropped for exotic ones

D. Failure to project how the situation will look in five minutes if nothing changes

E. Anchoring on a diagnosis supplied by the previous clinician at handover

42. During a long case, a surgical nurse notices that the suction canister is filling quickly, the surgeon has just said the field is wet, and the anesthesiologist, who has been entering the record, has not looked at the monitor since the blood pressure fell. Each person knew one piece of the picture. In the three levels of situational awareness, at which level did this team most likely fail?

A. Projecting the course of events, because the pressure trend was not extrapolated by anyone

B. Understanding what the elements mean, because the findings were noticed but misread

C. Declaring the emergency, because no named leader had been designated in advance

D. Perceiving the relevant elements, because nobody noticed the information

E. Closed-loop communication, because the orders were not repeated back

43. A 28-year-old woman is undergoing an emergency cesarean delivery under general anesthesia. The attending anesthesiologist is busy with the airway when the patient begins to bleed heavily. A third-year resident is better placed to see the whole room, and a more senior colleague is standing in the doorway. Which of the following best describes who should lead in the first minutes?

A. The most senior person present, because leadership follows rank and experience

B. The surgeon, because the operation is the source of the current bleeding

C. The person who has known the patient the longest and knows the history best

D. The person who is physically strongest and can perform the most hands-on tasks

E. Whoever can see the whole room, because leadership is a function, not a rank

44. A 57-year-old man is hypotensive during a bowel resection. The anesthesiologist, the only clinician at the head of the bed, is placing an arterial line, drawing up a vasopressor, calling the blood bank, and trying to record the times. The anesthesiologist is falling behind on all tasks. Which of the following is the best way to address the workload?

A. Assign tasks early, match each to the helper's skill, and state what is expected

B. Complete each task personally, one after another, so that none is done carelessly

C. Ask the helpers to wait until the blood pressure is stable before assigning tasks

D. Allow the helpers to choose tasks themselves to promote independent action

E. Divide the tasks equally among all helpers regardless of their skills or training

45. In one operating room, all team members debate each decision during an emergency, and no one finally decides on a course of action; treatments drift and are repeated. In another room, junior staff observe an error but say nothing because the attending is known to be intimidating. Which statement best describes the authority gradients in these rooms?

A. The first room is too steep and the second room is too flat

B. Both rooms are too flat, since neither has produced a good decision

C. Both rooms are too steep, since neither team has a working leader

D. Both rooms are balanced, since both debate and silence are acceptable

E. The first room is too flat and the second room is too steep

46. A circulating nurse sees the anesthesiologist about to hang a bag of blood that she believes belongs to another patient. She wants to speak up clearly. In the structure taught for assertive communication, which of the following is the best order for her statement? The anesthesiologist is focused on the monitor and has not noticed the label, and the nurse wants to be heard without delaying care.

A. Propose an action, state the observation, ask for agreement, and then state the concern

B. State the concern, wait until the case is over, and then state the observation

C. Ask for agreement, state the concern, and then propose an action if there is time

D. State the concern as a question and wait for the leader to notice the problem

E. State the observation, state the concern, propose an action, and ask for agreement

47. A 72-year-old woman has had a falling saturation for ninety seconds during a laparotomy. The resident is unsure of the cause and has given 100 percent oxygen. The resident hesitates to call the attending because the problem might resolve and calling might seem to show incompetence. The attending has a rule about when to call. Which of the following is the best rule?

A. Call only after the patient has deteriorated further and the cause has been identified

B. Call only when two separate treatments have failed and the patient is in extremis

C. Call at the end of the case, when the full details can be reviewed together

D. Call if the cause is unclear in a minute or two, or the first treatment fails

E. Call only after the cause has been identified so that help can be directed

48. A nurse is asked to confirm the identity of a blood product against the patient before it is hung during a hemorrhage. The anesthesiologist hesitates, thinking a check would waste time and imply distrust of the nurse. The department wants all cases to follow the same practice. Which of the following best explains why cross-checking should be routine?

A. Checks are only needed in emergencies, and routine use leads to alarm fatigue

B. Each check takes seconds, stops a class of error, and carries no accusation

C. Checks transfer responsibility for error from the clinician to the nurse

D. Checks are required by law, so their clinical benefit need not be considered

E. Cross-check only in emergencies, since routine checking slows ordinary cases

49. During a resuscitation, the leader asks a nurse for a dose of fifteen units of a drug. Another nurse hears it as fifty, and the two begin drawing up different amounts. The team uses closed-loop communication in general. Which additional practice would most reliably prevent this misunderstanding? The second nurse asks which amount to draw up, and the leader has already moved on to the next task.

A. Speak the number more loudly so that everyone in the room hears it the first time

B. Use standard abbreviations for drugs so that orders can be given quickly

C. Say each digit of numbers that sound alike, and give drug, amount, and route

D. Give the order in general terms so that the nurse can choose the amount

E. Use a short nickname for each drug so that orders are quick and easy to recall

50. During a prolonged crisis in an operating room, several members are working with different ideas of what is wrong. One is treating bronchospasm, another is preparing for transfusion, and the surgeon thinks the problem is anesthetic depth. The leader wants to prevent duplicated effort and contradictory actions. Which of the following is the best step?

A. Ask each member to keep working on the diagnosis that seems most likely to that person

B. Wait until the crisis is over and then reconcile the different views at the debrief

C. Limit communication to the surgeon and a single helper so that the room stays quiet

D. Announce what is thought to be happening, what has been done, and what is wanted next

E. Allow each member to continue independently until the surgeon calls a halt

51. In a hemorrhage, the anesthesiologist assigned the airway to a junior resident and the drugs to the nurse. After several minutes, the resident's speech slows, repeated attempts at the same task are made, and the airway is not improving. The leader notices these signs. Which of the following is the best response? The leader had spent the first minutes assigning tasks and then returned to watching the monitors.

A. Leave the assignment in place because rigid roles prevent confusion in a crisis

B. Reassign the task plainly, without blame, and brief the new holder on progress

C. Tell the resident that more effort is needed and wait for some improvement first

D. Take no action until the resident asks for help and then replace them silently

E. Ask the resident to continue and send the nurse to stand by at the doorway

52. A 45-year-old woman is having a cesarean delivery under spinal anesthesia when her blood pressure drops and the team begins to speak urgently. One team member says that they are losing her. The patient, who is awake, becomes tearful and asks what is happening. Which of the following is the best approach to communication in this setting? The operation is proceeding, and a student nurse and a family member are also in the room.

A. Use neutral words and a steady tone, and address the patient when appropriate

B. Continue speaking freely because patients cannot understand during an emergency

C. Ask the patient not to speak or ask questions until the emergency is over

D. Whisper all comments so that the patient cannot hear any of the team's remarks

E. Ignore the patient's presence and address only the other members of the team

53. A 68-year-old man with a known risk of massive bleeding is scheduled for a major vascular operation. The anesthesiologist wants to prepare the team in advance. Which of the following best describes the content of an effective two-minute briefing? The surgeon and the nurses will be in the room and have time to meet briefly before the patient arrives.

A. Review the patient's complete history in order and omit the specific hazards of the case

B. Confirm only that the surgeon has completed a satisfactory consent discussion

C. Name the hazard, the plan if it occurs, who does what, and the first sign of failure

D. Describe in general terms how the team would respond to any possible emergency

E. Review the plan for the case only after the incision has been made

54. A department chief is asked to justify the use of emergency checklists for a skeptical senior colleague who believes experienced clinicians should hold everything in their heads. The chief cites the main evidence. Which of the following best summarizes it? He asks what the evidence is in simulation and in real practice.

A. Large trials in real patients have proved that checklists reduce mortality in emergencies of every kind

B. Simulation studies show no difference, but patients clearly prefer the clinicians who use a checklist

C. Simulations show fewer omitted steps with a checklist, but real-world outcome data are limited

D. The evidence applies only to junior trainees and is entirely absent for experienced clinicians

E. Observational studies in real patients show that checklists increase the number of omitted steps

55. A 33-year-old woman develops severe hypotension and bronchospasm immediately after induction, and the anesthesiologist suspects anaphylaxis. A resident reaches for the emergency manual and begins to read the introduction. Which of the following is the best way to sequence the response? The surgeon asks why the team is waiting while the resident reads, and the blood pressure is still falling.

A. State the problem and first actions from memory, then open the manual

B. Read the whole page aloud before giving any treatment to avoid missing a step

C. Wait until the manual has been located and the page found before beginning treatment

D. Avoid the manual entirely because it signals panic to the surgical team

E. Delegate treatment to the surgeon while the anesthesiologist locates the manual

56. After a cardiac arrest during surgery, the anesthesiologist writes the record. A colleague realizes later that a dose time was recorded incorrectly. The anesthesiologist wants to fix the error and also thinks about adding a comment on why the surgical team caused the problem. Which of the following is the best action? The record is to be reviewed later by the department.

A. Overwrite the incorrect time so that the record shows only the correct value

B. Add a comment describing how the surgical team contributed to the event

C. Leave the error uncorrected, since the review will find the correct times later

D. Add a dated correction without altering the original entry, in factual terms

E. Ask a nurse to correct the record, since only nurses may amend the chart

57. Three days after a serious intraoperative cardiac arrest from which the patient recovered, a senior anesthesiologist who led the resuscitation tells a colleague that he is sleeping poorly and doubting his skills. He has not reported the event to the institution's safety system. Which of the following is the best response from the department?

A. Reassure him that these feelings are rare and will resolve without any support

B. Tell him to avoid discussing the event so that it does not affect other staff

C. Remove him from clinical work permanently pending a formal investigation

D. Ask him to discuss the event only with the family and not with colleagues

E. Check on him, offer relief, use peer support, and report the event

58. A 74-year-old woman is being transferred from the operating room to the intensive care unit after a complex operation. She is mechanically ventilated and on a vasopressor infusion. The anesthesiologist wants to make the transfer safe. Which of the following is the best practice at the receiving unit? The intensive care nurses are ready to receive her, and the transport team is waiting at the door.

A. Leave the written record at the nursing station and return to the operating room immediately

B. Give a verbal summary at the bedside and wait until the baseline is confirmed

C. Give the summary in the corridor so that the receiving staff can start preparing the bed

D. Hand over once the patient is in the bed and leave while the nurses connect the monitors

E. Telephone the summary to the unit and send the record with the transport staff alone

59. A resident is taking over from a faculty member in the middle of a long case. The faculty member gives a clear summary and asks whether the resident has any questions. The resident says no, though the plan for the next hour is unclear to the resident. Which of the following is the best way for the faculty member to improve this handover?

A. Invite questions and ask what the resident would do if the patient worsened

B. Ask whether the resident understands the plan, and accept a yes or no answer

C. Hand over the plan again in the same words, only more slowly and clearly

D. Assume the resident understood because the resident did not ask any questions

E. Ask the resident to write down the plan and submit it after the case

60. An anesthesiologist is preparing to meet the family of a patient who had a serious intraoperative adverse event. The cause is not yet fully known, and the surgeon wishes to give a firm explanation in the first conversation. Which of the following is the best approach to the conversation? The surgeon is anxious about liability and asks to speak for the team.

A. Give a definite explanation of the cause right away, even if it later proves incorrect

B. State what is known plainly, avoid blame, express sympathy, and promise to return

C. Postpone the conversation until the formal investigation has been fully completed

D. Share only the outcome and avoid any acknowledgment of the event itself

E. Ask the surgeon to speak first so that the anesthesiologist can follow that account

QUESTION GROUP 4

Emergency Drug Physiology

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Questions 61 to 80. Choose the single best answer for each.

61. A 62-year-old man in pulseless electrical activity is receiving chest compressions when a resident asks why a 1 mg bolus of epinephrine is given in cardiac arrest, since the same drug at low infusion rates mainly increases heart rate and contractility. Compressions are ongoing, the airway is secured, and a second intravenous line is in place. The anesthesiologist explains the principal reason for the large bolus. Which of the following is the best explanation?

A. Beta-2 relaxation of bronchial muscle improves ventilation during chest compressions

B. Beta-1 stimulation directly restores organized contraction in the arrested myocardium

C. Alpha-1 constriction raises aortic diastolic pressure and coronary perfusion pressure

D. V1a receptor stimulation reduces myocardial oxygen demand during the low flow state

E. Inhibition of phosphodiesterase-3 increases the contractility of the arrested ventricle

62. A 49-year-old woman who takes metoprolol develops hypotension at 60 over 30, wheezing, and flushing minutes after rocuronium. Her heart rate before induction was 58, and no other drugs have been given since the rocuronium. She has received two intramuscular doses of epinephrine, and the response has been poor. Anaphylaxis is the working diagnosis and a third dose is being prepared. Which of the following is the best management of the poor response?

A. Stop epinephrine because beta-blockade makes it ineffective and give phenylephrine alone

B. Replace epinephrine with an antihistamine as the main treatment for the hypotension

C. Give dobutamine as a continuous infusion to overcome the receptor blockade in the heart

D. Continue epinephrine and add glucagon or vasopressin because beta-blockade blunts it

E. Give sodium bicarbonate to correct the acidosis that is blunting the epinephrine response

63. A 58-year-old man has been in septic shock for four days on norepinephrine and vasopressin. She has remained on both infusions since admission, and the lactate and creatinine have been stable. His serum sodium has fallen from 138 to 124 mmol per liter, and his fingertips are now dusky and cool. The pressure is adequate. Which of the following drug effects best accounts for both findings?

A. Norepinephrine, through beta-2 shift of sodium into cells and digital vasodilation

B. Norepinephrine, through renal V2 receptor activation and loss of sodium in the urine

C. Vasopressin, through V2 free water retention and constriction of digital vessels

D. Vasopressin, through beta-1 stimulation causing renal sodium loss and cutaneous steal

E. Norepinephrine, through V1a stimulation of the collecting duct and digital vasospasm

64. A 55-year-old woman is under general anesthesia for a seven-hour operation. The surgeon reports minimal blood loss, the depth of anesthesia has not changed, and the end-tidal agent concentration is stable. Ephedrine 10 mg raised the pressure well early in the case, but the third and fourth doses produce progressively smaller rises, and the heart rate is 55. Which of the following best explains the declining response?

A. Ephedrine is rapidly converted to phenylephrine, which has a shorter duration of action

B. Repeated doses down-regulate V1a receptors on vascular smooth muscle within an hour

C. Ephedrine acts only on beta-2 receptors, which become saturated at higher doses

D. Repeated doses are metabolized more rapidly each time by hepatic enzymes in the liver

E. Repeated doses deplete the releasable stores of norepinephrine in nerve terminals

65. A 72-year-old man is bleeding during a laparotomy. The anesthesia record shows that the infusion has been increased three times, and the surgeon is still controlling a bleeding vessel. Over two hours the norepinephrine requirement has doubled, and the pH is 7.12, the temperature is 34.5 degrees Celsius, and several liters of measured blood loss have not been replaced. The surgeon asks the anesthesiologist to increase the infusion again. Which of the following is the best next step?

A. Double the norepinephrine rate again and reassess the pressure after an hour of observation

B. Switch to repeated ephedrine boluses, because the infusion has stopped working in this patient

C. Stop the infusion entirely and allow the pressure to find its own level for a time

D. Restore volume, correct acidosis, and warm the patient, then reassess the need for more drug

E. Add a large dose of ephedrine and continue to raise the infusion at half-hourly intervals

66. A 60-year-old woman is receiving a norepinephrine infusion through a 22-gauge intravenous catheter in the forearm. The catheter was placed in the emergency department several hours earlier and has not been rechecked since. The skin around the site has become blanched, cool, and firm, and the infusion pump is not alarming. Which of the following is the most appropriate immediate management?

A. Stop the infusion at once, and infiltrate the affected area with phentolamine

B. Continue the infusion at a lower rate and apply a cold compress to the site

C. Infiltrate the area with additional epinephrine to limit the spread of fluid

D. Elevate the arm and observe the site, since pale skin resolves without treatment

E. Apply a warm compress and give a fluid bolus through the same vein

67. A 67-year-old man has refractory vasoplegia after cardiopulmonary bypass despite high-dose norepinephrine and vasopressin, and methylene blue is proposed. Echocardiography shows acceptable cardiac function, and the surgeon reports that there is no residual bleeding. His medications include sertraline for depression. Which of the following is the most important concern with this plan?

A. Methylene blue blocks V1a receptors and will cancel the benefit of the vasopressin

B. Methylene blue potentiates neuromuscular blockade and will prolong postoperative paralysis

C. Methylene blue raises serum potassium sharply through a shift out of the cells

D. Methylene blue causes severe bronchospasm through stimulation of muscarinic receptors

E. Methylene blue can precipitate serotonin syndrome in patients taking serotonergic drugs

68. A 63-year-old man with a creatinine of 4.2 mg per deciliter is in a low output state, and an inotrope is started after norepinephrine. The echocardiogram shows a poorly contracting left ventricle and a normal right ventricle, and the central venous pressure is unremarkable. Several hours later he remains hypotensive with a low systemic vascular resistance, and the effect seems to persist after the infusion is reduced. Which of the following drugs and properties best fits this picture?

A. Dobutamine, which causes unopposed alpha-1 constriction and renal accumulation

B. Phenylephrine, which depletes norepinephrine stores and vasodilates over time

C. Vasopressin, which acts at V2 receptors and accumulates in renal failure

D. Norepinephrine, which vasodilates by beta-2 action and accumulates in renal failure

E. Milrinone, which vasodilates the vessels and accumulates in renal failure

69. A 78-year-old man with severe aortic stenosis becomes hypotensive at 70 over 40 after induction of general anesthesia. The electrocardiogram shows no ischemic changes, and the surgeon is ready to begin once the pressure is restored. The rhythm is sinus at 62 beats per minute and the volume status is adequate. Which of the following is the best choice for support of the pressure?

A. Phenylephrine, to restore afterload without raising the heart rate

B. Dobutamine, to raise contractility and shorten the ejection time

C. Glycopyrrolate, to raise the heart rate and the cardiac output

D. Ephedrine in repeated doses, to raise both the rate and contractility

E. Epinephrine, to raise contractility and rate while keeping the pressure up

70. A 68-year-old man with long-standing hypertension, whose usual pressure is 165 over 95, is receiving a spinal anesthetic for a hip repair. The block level is T6, he has received a liter of crystalloid, and there is no bleeding. His mean pressure is 62 mm Hg on a norepinephrine infusion, he has become dull and slow to answer, and the urine output has fallen. Which of the following is the best approach?

A. Accept the pressure, since a mean of 65 is the goal for every patient

B. Lower the target further to reduce the work of the heart during the spinal

C. Stop the norepinephrine and give a diuretic to improve the urine output

D. Raise the target above 65 to suit his baseline and reassess perfusion

E. Keep the pressure at 65 and treat the confusion with a low dose of an anxiolytic

71. A 35-year-old man with known ventricular preexcitation develops an irregular, rapid, wide-complex tachycardia at 220 beats per minute during surgery. The monitor shows a rapid irregular rhythm with a varying QRS width, and defibrillator pads are on the chest. He is now confused and the pressure is 76 over 40. Which of the following is the most appropriate treatment?

A. Adenosine given as a rapid push followed by a flush of saline

B. Synchronized cardioversion with sedation if time allows

C. Diltiazem given as an intravenous bolus over two minutes

D. Esmolol given as an intravenous bolus followed by an infusion

E. Verapamil given as an intravenous bolus over two minutes

72. A 60-year-old man has stable monomorphic ventricular tachycardia at 170 beats per minute with a pressure of 112 over 70 during a cardiac catheter procedure. He is comfortable and awake on the table, and a defibrillator with pads attached is at the bedside. The decision is to give amiodarone 150 mg, and a colleague prepares to push it over 30 seconds. Which of the following is the best advice?

A. Give it as a rapid push followed by a flush, because its half-life is only seconds

B. Withhold it, because amiodarone is contraindicated in a patient who has a pulse

C. Give it over 10 minutes only if the corrected QT interval is already prolonged at baseline

D. Give it as a rapid bolus together with calcium to prevent the bradycardia and hypotension

E. Give it over 10 minutes, because rapid injection causes hypotension and bradycardia

73. A 45-year-old woman receives a large volume of bupivacaine for a plexus block and then develops agitation, a wide-complex bradyarrhythmia, and a seizure. The block was performed with ultrasound guidance, and the injection was given quickly with aspiration reported as negative. A colleague proposes lidocaine for the wide-complex rhythm. Which of the following is the best response?

A. Give lidocaine, since it is the standard drug for any wide-complex rhythm

B. Give adenosine, since the rhythm is most likely a supraventricular tachycardia

C. Suspect local anesthetic toxicity and avoid adding a sodium channel blocker

D. Give amiodarone as a rapid push to terminate the rhythm before the next seizure

E. Give esmolol to slow the ventricular rate and reduce the risk of further seizures

74. A 64-year-old woman who received ondansetron and haloperidol has a serum potassium of 3.0 mmol per liter and develops runs of polymorphic ventricular tachycardia with a lengthening QT interval. Her magnesium has not been measured, and the QT interval was borderline on the preoperative electrocardiogram. She has a pulse and the pressure is adequate between runs. Which of the following is the best management?

A. Give magnesium, replace potassium, and stop drugs that prolong the QT interval

B. Give an amiodarone bolus to suppress the ventricular ectopy and shorten the runs

C. Give calcium chloride to stabilize the membrane and shorten the QT interval

D. Give esmolol to slow the ventricular rate and raise the threshold for torsades

E. Give adenosine as a rapid push to terminate the polymorphic ventricular tachycardia

75. A 52-year-old heart transplant recipient has a heart rate of 38 and hypotension after induction of anesthesia. The surgical team reports no bleeding, and the ventilation and the oxygen saturation are normal. Atropine 1 mg intravenously produces no change in the rate. Which of the following is the most appropriate next step?

A. Repeat atropine up to the maximum cumulative dose before trying other drugs

B. Give glycopyrrolate, which acts at the sinus node by a different mechanism

C. Give neostigmine to prolong the effect of any remaining vagal tone on the sinus node

D. Give sodium bicarbonate to correct the conduction block in the denervated heart

E. Give a direct beta agonist such as epinephrine and ready transcutaneous pacing

76. A 54-year-old man has residual neuromuscular blockade at the end of an operation, and neostigmine is given alone. The train-of-four shows four twitches with fade, and the anticholinergic was not drawn up with the reversal. Within two minutes the heart rate falls from 78 to 40 and the airway fills with secretions. Which of the following best explains the event and its treatment?

A. Nicotinic overstimulation at the neuromuscular junction; give an additional dose of neostigmine

B. Unopposed muscarinic effect of neostigmine; give an anticholinergic such as glycopyrrolate

C. Beta-adrenergic blockade by neostigmine; give isoproterenol to restore the heart rate

D. Residual relaxant acting on the cardiac receptors; give sugammadex at the high dose

E. Vagal reflex from the oropharyngeal suction catheter; give ephedrine and continue the reversal

77. A 40-year-old trauma patient has received 12 units of packed red cells with citrate preservative. A central line has not yet been placed, and warmed fluids are running while the blood bank sends further units. The pressure is falling despite vasopressor use, the contractility looks poor on echocardiography, and the ionized calcium is 0.7 mmol per liter. Peripheral access is a small hand vein. Which of the following is the best management?

A. Give sodium bicarbonate to raise the ionized calcium and support contractility

B. Give calcium through a large vein or central line and remeasure the ionized level

C. Give calcium chloride rapidly into the hand vein to restore the level quickly

D. Increase the vasopressor dose, because ionized calcium does not affect contractility

E. Give magnesium to correct the deficit, since citrate chelates magnesium alone

78. An anesthesiologist is about to give a 10 mcg bolus of epinephrine for hypotension. The induction is an emergency, and two people are drawing up drugs at the same time. Two ampules are on the tray, one labeled 1 in 1000 and one labeled 1 in 10,000, and the nurse notes that the ratio notation is easy to confuse. Which of the following is the best way to prevent a tenfold error?

A. Rely on the ratio notation, since it is the traditional standard that all anesthesiologists know

B. Dilute the drug freehand into a syringe until the volume looks suitable for a small bolus

C. Express the concentration as mass per volume and use a premixed or pharmacy-prepared syringe

D. Draw up the dose from memory and confirm the effect by watching the pressure response

E. Draw the contents of the ampule into an unlabeled syringe to save time at the bedside

79. A resident reaches for a vial of ephedrine from the usual place in the drawer and draws up rocuronium from a look-alike vial placed beside it. Both vials are of similar size with similar labels, and the tray is rearranged by each user between cases. The error is caught before injection. The department reviews the event. Which of the following is the best classification and response?

A. A mistake, best addressed by teaching the resident more about the pharmacology

B. A slip, best addressed by separating look-alike drugs and changing storage

C. A mistake, best addressed by a cognitive aid that selects the correct drug

D. A slip, best addressed by counseling the resident to concentrate harder in the future

E. A slip, best addressed by a second opinion on the choice of drug

80. An anesthesiologist realizes that the syringe just injected into a patient was a neuromuscular blocker and not the intended vasopressor. The anesthesiologist was interrupted twice while preparing the drugs, and no second person checked the labels. The patient is still receiving ventilation by the machine and the pressure is stable. Which of the following is the best approach to the error?

A. Treat the effects quietly and discard the syringe to avoid alarming the team and the patient

B. Complete the operation first and report the error at the next departmental review meeting

C. Wait for signs of harm to appear before telling the surgeon or any other member of the team

D. Stop, treat the effects, call for help, keep the vial and syringe, and disclose afterward

E. Treat the effects, then return the vial to the tray so that it is not wasted

QUESTION GROUP 5

Machine, Gas, and Power Failures

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Questions 81 to 100. Choose the single best answer for each.

81. A new resident is completing the checkout of an anesthesia workstation that has just passed its automated self-test. The case is an elective one starting in ten minutes, the pipeline is connected, and the surgeon is waiting. The attending asks which single item provides a backup for failure of nearly every machine function and is not verified by the self-test. Which of the following is the best answer?

A. A vaporizer filled to the maximum line with the agent planned for the case

B. A fully charged battery indicator on both the ventilator and the monitor

C. A scavenging system that has been connected to the active suction outlet

D. A working self-inflating resuscitation bag within reach of the bed head

E. A spare breathing circuit and filter packed in the sealed original wrapper

82. During the daily checkout, a technician opens the oxygen cylinder on the machine, reads the gauge, and then closes the cylinder valve again while the pipeline supplies the room. The checkout is part of the routine at the start of every day, and the pipeline supplies the room during the case. A student asks why the cylinder is closed. Which of the following is the best reason?

A. An open cylinder leaks gas back into the pipeline and reduces the pressure in the wall supply lines

B. An open cylinder raises the delivered oxygen fraction above the value set on the flowmeter control

C. An open cylinder damages the regulator whenever the pipeline gas is flowing at the same time

D. Open cylinders lose their contents through the regulator even when the machine is switched off

E. An open cylinder silently takes over if pipeline pressure falls, spending the reserve unnoticed

83. After a vaporizer is changed between cases, the low-pressure leak test is performed with all vaporizers turned off and passes. The surgeon is ready to start, and the vaporizer display reads the set value with the fresh gas flow at two liters per minute. During the case the anesthesiologist smells agent near the machine, and the delivered concentration seems lower than set. Which of the following is the best next step?

A. Repeat the test with each vaporizer on, since a cap or mount leak shows only then

B. Accept the earlier result, since a test that passes with vaporizers off is conclusive

C. Increase the dial setting on the vaporizer and monitor the end-tidal agent concentration

D. Replace the carbon dioxide absorbent, because the smell must be coming from the canister

E. Repeat the test with only the nitrous oxide flowmeter open and each vaporizer off

84. During a pre-use test of the breathing circuit, a technician wants to confirm that the unidirectional valves open and close in the correct directions, because an incompetent valve would cause rebreathing. The checklist for the day has items for the high-pressure, intermediate, and low-pressure systems, and the circuit has just been replaced. Which of the following steps tests this?

A. Occluding the Y-piece and confirming that the circuit pressure holds steady

B. Reading the oxygen analyzer while the sensor is exposed to room air

C. Squeezing the reservoir bag and watching the valves open and close

D. Opening the cylinder valve and reading the pressure on the gauge

E. Calibrating the flow control valves against a reference flowmeter

85. A vaporizer containing sevoflurane was knocked over while a machine was being wheeled between rooms and then set upright. The vaporizer interlock is intact, and the machine is being prepared for the first case of the day. The technician returns it to use without any further check. Which of the following is the main hazard?

A. The vaporizer will deliver no agent at all until the filler cap is replaced

B. A hypoxic gas mixture will result because the proportioning system is disabled

C. Carbon monoxide will form in the vaporizer chamber as the agent heats up

D. Liquid agent may pass downstream and deliver a dangerously high concentration

E. Nitrous oxide will be delivered in place of oxygen through the interlock

86. A 55-year-old woman is ventilated by hand and the peak pressure is high. The tracheal tube appeared correctly placed at induction, and the breath sounds were equal at that time. After disconnecting the circuit, the anesthesiologist gives breaths with a self-inflating bag and finds that it is hard to squeeze. Which of the following is the best next step?

A. Pass a suction catheter through the tracheal tube to test its patency

B. Replace the breathing circuit hoses with a new set from the cart

C. Disconnect the scavenging hose from the circuit and watch the pressure

D. Open the adjustable pressure-limiting valve fully on the machine

E. Replace the sampling line and recheck the capnogram trace

87. During a five-hour operation, a 48-year-old man's peak pressure has risen gradually and the tidal volume has fallen, mimicking bronchospasm. He is a smoker, the peak pressure was normal at the start of the case, and the saturation is 97 percent. Auscultation shows no wheeze, a suction catheter passes freely down the tube, and a nebulized drug was given through the circuit two hours ago. Which of the following is the most likely cause?

A. An obstructed oxygen pipeline supplying the anesthesia machine

B. A heat and moisture exchanger clogged with secretions or drug

C. Exhausted carbon dioxide absorbent in the canister of the machine

D. An incompetent inspiratory unidirectional valve in the circuit

E. A kinked sampling line attached to the gas analyzer at the Y-piece

88. The airway pressure rises in a 62-year-old man without explanation, the bellows fail to descend, and the tube is patent. The surgical team has not changed the position of the table, and the end-tidal carbon dioxide trace is normal in shape. The anesthesiologist suspects back pressure from the waste gas system. Which of the following is the best way to test this?

A. Increase the fresh gas flow and watch whether the bellows begin to descend

B. Deflate the cuff of the tracheal tube and listen for an audible leak

C. Disconnect the scavenging hose from the circuit and watch the pressure

D. Close the pressure-limiting valve fully and watch the airway pressure gauge

E. Switch from volume control to pressure control and watch the waveform

89. A patient is being hand ventilated with a bag while the team searches for a leak. The cuff pressure was checked an hour ago, and the connection at the sampling line has been inspected. The anesthesiologist occludes the Y-piece, sets a moderate fresh gas flow, and presses the flush valve. The circuit pressurizes normally. Where is the leak most likely to be?

A. In the breathing hoses or the absorber canister on the machine

B. In the bellows housing or the ventilator drive mechanism itself

C. In the patient interface, such as the cuff or Y-piece fitting

D. In the pipeline connection or the cylinder yoke on the machine

E. In the adjustable pressure-relief valve of the breathing system

90. In a 70-year-old woman, the capnogram baseline has risen above zero and the end-tidal carbon dioxide has climbed from 36 to 50 mm Hg despite a higher minute ventilation. She has been under anesthesia for three hours on a low fresh gas flow with sevoflurane. The absorbent in the canister has changed color throughout. Which of the following is the best next step?

A. Replace the exhausted absorbent and check the valves for rebreathing

B. Deepen the anesthetic, since the rise reflects increased carbon dioxide production

C. Treat for pulmonary embolism, since the end-tidal value is unexpectedly abnormal

D. Withdraw the tracheal tube by two centimeters to improve the ventilation

E. Treat for malignant hyperthermia, since the end-tidal carbon dioxide has risen

91. A hospital pipeline fails during a long case, and the machine is running on a cylinder with a limited supply. The procedure cannot be stopped safely for about an hour, and the biomedical engineers are on their way to the room. The anesthesiologist is asked how to make the oxygen last as long as possible. Which of the following is the best approach?

A. Raise the fresh gas flow to keep the circuit full and use the flush to refill the bag

B. Lower the fresh gas flow, ventilate by hand or avoid oxygen drive gas, and avoid the flush

C. Keep the ventilator on its usual setting because the drive gas is not drawn from the cylinder

D. Switch to a low-flow technique and use the oxygen flush button for each manual breath

E. Run the ventilator at a higher tidal volume so that fewer breaths are needed

92. A 40-year-old patient has been stabilized on cylinder oxygen after the analyzer showed that the wall outlet was delivering a wrong gas. The monitor shows stable hemodynamics, and the engineering department has not yet reported any problem with the gas supply. Two other operating rooms report falling saturations in their patients, and the ward nurses report unexplained hypoxemia. Which of the following is the best additional action?

A. Continue the case on the cylinder and report the event after the patient leaves

B. Reconnect the wall supply after confirming that the pressure gauge reads normal

C. Assume the pipeline has cleared and recalibrate the oxygen analyzer for reassurance

D. Wait for the pipeline pressure alarm to sound before notifying anyone

E. Alert the other rooms, the charge nurse, and the engineering department at once

93. The anesthesiologist has stabilized a patient after an oxygen supply event in which the outlet delivered nitrogen, and the case is finished. The surgeon has left the room, the recovery area is waiting for the patient, and the next case is scheduled in the same room. The charge nurse asks what should be done with the room and the equipment. Which of the following is the best answer?

A. Return the room to service after a checkout, because the machine was not at fault

B. Purge the outlets with oxygen and discard the logs once the patient has recovered

C. Resume scheduled cases using cylinders only and defer the investigation for about a week

D. Send the machine to be cleaned and recalibrated before it or the outlets are examined

E. Remove the room from service, label the machine and outlets, and save samples and logs

94. A 58-year-old man is under general anesthesia when building power fails and the displays go dark. The abdomen is open, the surgical team is in the middle of the laparotomy, and the cause of the failure is not yet known. The machine is plugged into a white wall outlet and not a red one, and the hospital generator is running. A technician asks what the anesthesiologist should expect. Which of the following is correct?

A. The equipment will run on its batteries, which last half an hour to a few hours

B. The equipment will remain on full power because the generator supplies every outlet

C. The equipment will restore itself within ten seconds through the emergency generator

D. The equipment will stop at once because the batteries are used only for the monitors

E. The equipment will switch to compressed air and continue normally for several hours

95. A power failure has left a 50-year-old man ventilated by hand with a self-inflating bag. The surgeon has agreed to pause, and a battery-powered syringe pump with propofol and an opioid is available in the room. The workstation controls gas flow electronically and has no mechanical backup, so no inhaled agent can be delivered, and the outage is expected to last. Which of the following is the best way to maintain anesthesia?

A. Rely on the vaporizer, which is mechanical and delivers agent without gas flow

B. Allow the patient to emerge and stop the operation until the power returns

C. Increase the electronic flow controls to their maximum to force agent delivery

D. Increase the vaporizer dial to deliver agent from the residual gas in the circuit

E. Switch to intravenous anesthesia with propofol, supplemented with an opioid

96. Power returns after a ten-minute outage, and the workstation lights up showing a ventilation mode and settings. The anesthesia record shows that the settings were changed twice before the failure, and the alarms were silenced earlier. The patient is being hand ventilated and is stable. A technician suggests reconnecting the ventilator immediately because the machine has recovered. Which of the following is the best response?

A. Reconnect the ventilator because the displayed settings match the pre-failure values

B. Take the machine out of use for the rest of the day and complete the case using only a bag

C. Restart the machine repeatedly until all the alarms have been cleared and silenced

D. Reset and check the machine and alarms before resuming, not assuming the old settings

E. Reconnect the ventilator after silencing the alarms that remain active from the outage

97. Before a 66-year-old patient's tonsil and neck procedure, the team is performing a fire risk assessment. The operating room nurse has the surgical plan in hand, and the patient will receive oxygen by mask under the drapes. They ask which scenario meets the advisory's definition of a high risk case. Which of the following is correct?

A. Knee arthroscopy under spinal anesthesia with a tourniquet and no supplemental oxygen

B. Pelvic surgery under general anesthesia with a cuffed tube and no ignition source

C. Surgery above the xiphoid with an open oxygen source and an ignition source in use

D. Lower limb surgery under a nerve block with oxygen by mask and no electrosurgery

E. Hip surgery under general anesthesia with electrosurgery and a cuffed tracheal tube

98. During a tracheostomy performed with electrosurgery, flames appear inside a 54-year-old man's tracheal tube. The inspired oxygen concentration in the circuit is high, and the surgeon has stopped and called for saline. The surgeon stops and the anesthesiologist recognizes an airway fire. Which of the following is the most appropriate immediate action?

A. Increase the oxygen flow to blow out the flames and continue ventilation through the tube

B. Leave the tube in place and pour saline over the drapes at the surgical field

C. Deflate the cuff and advance the tube deeper into the trachea to move the flame

D. Remove the tube, disconnect the circuit, and flood the airway with saline

E. Close the pressure-limiting valve and increase the oxygen flow to inflate the lungs

99. During a facial procedure, the anesthesiologist stops the gases, the drapes are removed, and flames on the burning material are doused with saline. The charge nurse has not yet arrived, and the fire extinguisher stands in a wall cabinet near the doorway. The fire flares up again on the surgical towels and is spreading across the table. Which of the following is the best next action?

A. Use a carbon dioxide extinguisher, activate the alarm, and evacuate if necessary

B. Pour more saline onto the table and wait to see whether the flames subside

C. Place a dry drape over the flames to smother them while the surgery continues

D. Increase the oxygen flow to the patient and continue the operation near the fire

E. Leave the fire to burn out, since the drapes have already been removed from the field

100. The oxygen analyzer alarm sounds repeatedly in a 44-year-old patient receiving a low fresh gas flow of oxygen and nitrous oxide, although the saturation is 99 percent. The inspired oxygen sensor was calibrated at the start of the day, and the flowmeter shows oxygen at 0.5 liters per minute. The inspired oxygen reads lower than the set limit. A colleague suggests silencing the alarm because the patient looks well. Which of the following is the best response?

A. Silence the alarm and lower the limit because the oximeter reading is normal

B. Ignore the alarm until the saturation falls, since the oximeter is the better monitor

C. Turn up the nitrous oxide to compensate for the fall in inspired oxygen

D. Increase the oxygen flow to flush any nitrous oxide from the circuit

E. Treat the alarm as real, check the flows and circuit, and use a second analyzer

QUESTION GROUP 6

Predicting and Preparing for the Difficult Airway

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Questions 101 to 120. Choose the single best answer for each.

101. A 45-year-old man has an airway examination that is entirely normal, with a Mallampati class of 1, good mouth opening, and a normal thyromental distance. The resident and the attending found the same normal findings, so no difficult airway plan was made. Laryngoscopy proves unexpectedly difficult. A resident asks how the examination could have missed it. Which of the following is the best explanation?

A. The tests are highly sensitive, so a normal examination must have been recorded incorrectly

B. The Mallampati class detects nearly all difficult laryngoscopies, so the examiner erred

C. Difficulty occurs only in patients who have a history of previous airway surgery or radiation

D. Bedside tests have modest sensitivity, so most unexpected difficulty follows a normal test

E. Difficulty is mainly caused by operator inexperience and is unrelated to any patient predictor

102. A 63-year-old man has a Mallampati class of 3, a short thyromental distance, and inability to bring the lower incisors in front of the upper incisors. The team will adjust the plan according to how these findings are interpreted, and the first attempt would otherwise use a standard blade. His neck extension is limited. He is scheduled for an elective operation under general anesthesia. Which of the following is the best interpretation and plan?

A. Each test alone is reliable, so this combination guarantees difficulty and requires cancellation

B. The findings are individually weak, so proceed with direct laryngoscopy and no special plan

C. Several abnormal findings raise the odds, so plan videolaryngoscopy or an awake approach

D. Only the Mallampati class matters, so the other findings can be disregarded in the plan

E. Two abnormal results are acceptable, so plan an ordinary induction with a larger blade

103. A 52-year-old man with a large neck mass and previous neck radiation is scheduled for surgery. Palpation of the neck with the patient lying still shows no clear landmarks, and the room is booked for the afternoon. The airway team predicts difficulty, and the cricothyroid membrane cannot be felt on palpation. Which of the following is the best step before induction?

A. Defer any attempt to find the membrane until front-of-neck access is actually required

B. Identify the cricothyroid membrane with ultrasound and mark it before induction

C. Rely on the landmarks of the thyroid cartilage and the sternal notch during an emergency

D. Plan to use a supraglottic airway as the only rescue and omit front-of-neck planning

E. Mark the cricoid cartilage instead, because it is easier to palpate in most patients

104. A 30-year-old man in a motor vehicle collision arrives with a cervical collar and needs intubation. Neurologic examination is limited, and radiographs of the cervical spine are not yet available, so the spine is treated as unstable. A first attempt with direct laryngoscopy and manual in-line stabilization gives a poor view. Which of the following is the best approach for the next attempt?

A. Direct laryngoscopy again with firmer in-line stabilization to hold the neck immobile

B. Hyperextension of the head to improve the view while the collar remains tightly in place

C. Videolaryngoscope with a bougie, the collar loosened, and an assistant holding the head

D. Awake nasal intubation despite the patient's agitation and a suspected skull base fracture

E. Perform a blind nasal intubation with the collar tightened and the head kept midline

105. A 40-year-old woman is intubated with a hyperangulated videolaryngoscope blade. The patient is paralyzed and preoxygenated, the saturation is 100 percent, and a bougie and a stylet are both available on the cart. The screen shows an excellent view of the glottis, but the tracheal tube repeatedly fails to advance toward the cords. Which of the following is the best interpretation?

A. The view is probably inaccurate, so the blade should be replaced by a standard Macintosh blade

B. The tube will always pass when the view is good, so the problem must be the cuff

C. Videolaryngoscopy is unsuitable for this patient, so an awake technique should be started

D. The view is correct but the tube is too small, so choose a larger tube size

E. A good view does not ensure passage, so use a bougie or a stylet matched to the blade

106. A 2-year-old child needs a general anesthetic for a dental procedure. The airway is normal on examination, the parents are anxious, and the surgeon wants to start as soon as possible. The anesthesiologist is considering how long the child can tolerate apnea after induction. Compared with a healthy adult, which of the following is the correct consideration?

A. The child tolerates apnea longer than an adult because of a larger functional residual capacity

B. The child tolerates eight minutes of apnea after preoxygenation, as does a healthy adult

C. The child needs no preoxygenation because the oxygen consumption is lower than in adults

D. Desaturation occurs within a minute or two of apnea, so the first attempt must be prepared

E. The child needs less preoxygenation because the airway is shorter than an adult airway

107. A 65-year-old man with sepsis and acute hypoxemic respiratory failure on high-flow oxygen needs urgent intubation. He has been on a high-flow oxygen device at 60 liters per minute, and the intensive care team is at the bedside. His saturation is 88 percent and he is tiring. The team asks which preoxygenation strategy has been shown in a large trial to reduce hypoxemia during intubation compared with a face mask. Which of the following is the best answer?

A. Three breaths of oxygen from an anesthesia mask with the patient supine

B. Noninvasive positive pressure ventilation delivered before induction

C. Preoxygenation by simple nasal cannula only, with the face mask omitted

D. A single minute of room air breathing before the induction drug is given

E. Preoxygenation with oxygen through a nasal cannula for thirty seconds

108. A 58-year-old woman with severe pulmonary hypertension and a failing right ventricle needs intubation for respiratory failure. Her echocardiogram shows a dilated right ventricle with a flattened septum, and an inhaled vasodilator is available. Her saturation is 90 percent and her systemic pressure is marginal. Which of the following is the best plan?

A. Give a standard induction dose, then treat any collapse with a large fluid bolus once it has occurred

B. Accept some hypercarbia during a prolonged laryngoscopy and use high peak airway pressures to ventilate

C. Avoid hypoxia and hypercarbia, support pressure with a vasopressor, and ready an inhaled vasodilator

D. Induce rapidly with a high dose of propofol to blunt the response to the laryngoscopy and the tube

E. Delay the vasopressor until after induction so that the pressure response can be observed first and treated

109. A 60-year-old man has been in shock for days on maximal catecholamine infusions and now needs intubation. He has received norepinephrine at a high rate for two days, and his lactate remains elevated. The team is choosing an induction agent for a hemodynamically fragile patient. Which of the following is the most relevant hazard of ketamine in this setting?

A. It causes profound adrenal suppression lasting several days after a single dose

B. It lowers systemic vascular resistance by blocking alpha-1 receptors directly

C. It can depress the heart when the catecholamine system is exhausted

D. It abolishes the airway reflexes and prevents the use of apneic oxygenation

E. It raises pulmonary vascular resistance and is contraindicated in every shocked patient

110. A 45-year-old man with a bowel obstruction and pneumonia has a saturation of 88 percent despite oxygen by mask and is tiring. His abdomen is distended, he vomited twice before arrival, and the intensive care team is ready to proceed. A full stomach calls for a rapid sequence induction, but his oxygen reserve is very low. Which of the following is the best approach?

A. A strict rapid sequence with no mask ventilation under any circumstances until the tube is placed

B. An inhalational induction with spontaneous breathing, to avoid any risk of aspiration or apnea

C. Delay of the induction until the bowel obstruction has been decompressed by the surgeon or by a tube

D. Intubation under deep sedation with remifentanil and topical anesthesia before any relaxant is given

E. A modified rapid sequence with gentle low-pressure mask ventilation, since hypoxia is worse

111. A 29-year-old woman in early labor weighs 140 kg, has a Mallampati class of 4, and a short neck. She has had no neuraxial analgesia so far, and the obstetric team expects a possible cesarean delivery if labor does not progress. She asks about pain relief. The anesthesiologist is concerned about the airway if an urgent cesarean delivery becomes necessary. Which of the following is the best plan?

A. Wait until a cesarean delivery is needed and then decide on the technique of anesthesia

B. Avoid neuraxial analgesia and rely on intravenous opioids until delivery occurs

C. Plan a general anesthetic with routine direct laryngoscopy for any operative delivery

D. Place an early epidural so that the block can be extended if surgery is needed

E. Neuraxial block is contraindicated by the obesity, so choose general anesthesia

112. A 35-year-old man with a predicted difficult airway has been given an induction drug. The preoxygenation was adequate, the head is elevated, and a second anesthesiologist is present with a videolaryngoscope ready. Mask ventilation is easy and the saturation is 100 percent. The anesthesiologist is deciding whether to give a neuromuscular blocker now. Which of the following is the best approach?

A. Withhold the relaxant for the whole case because blockade always worsens mask ventilation

B. Confirm mask ventilation first, then give the relaxant to improve compliance and view

C. Give the relaxant before checking the mask ventilation to save time at induction

D. Give the relaxant only after the first failed laryngoscopy has proved that it is needed

E. Give a second dose of the induction drug and ventilate by mask before the relaxant

113. A 60-year-old man has an airway tumor with stridor at rest and a critical narrowing seen on computed tomography. He is sitting upright because he cannot lie flat, and the surgeon and anesthesiologist have reviewed the imaging together. He is tiring but cooperative. The team discusses whether an awake flexible bronchoscopic intubation is the safest option. Which of the following is the best choice?

A. Intravenous induction with a short-acting relaxant followed by videolaryngoscopy

B. Awake surgical airway under local anesthesia, planned with the surgeon early

C. Inhalational induction with deep sedation and gentle manipulation of the airway

D. Awake flexible bronchoscopy with heavy sedation to reduce the cough and the anxiety

E. Awake nasal intubation with a flexible scope and topical anesthesia only

114. A 56-year-old woman is scheduled for an awake fiberoptic intubation in 30 minutes. The operating room is being prepared, the nurse has laid out the topical anesthetic and a flexible scope, and the patient has consented. The resident plans to give glycopyrrolate at the moment the scope is passed because the secretions are not yet a problem. Which of the following is the best advice?

A. Give it early, ideally 15 minutes or more before topicalization starts

B. Give it only once the scope has been passed and secretions are seen in the view

C. Give it after the tube is in place, to dry the airway before extubation

D. Omit it, because an antisialagogue has no effect on the topical anesthetic

E. Give it after topicalization is complete, since it is not needed before then

115. A 68-year-old woman has a lean body weight of 60 kg and an actual weight of 110 kg. The equipment has been laid out, including the topical anesthetic solutions and the scope, and the plan has been discussed with her. An awake intubation will use nebulized lidocaine, gargles, sprays through the scope, and a transtracheal injection. The anesthesiologist calculates the maximum total lidocaine dose before starting. Which of the following is correct?

A. 990 mg, calculated from the actual body weight and counted for each route separately

B. 540 mg, counting all routes together and recording what has been given

C. 540 mg for the sprays and a further 540 mg for the injected lidocaine

D. 270 mg, because mucosal absorption makes the sprayed doses count double

E. 630 mg, calculated from lean weight with an extra allowance for the injected dose

116. A 32-year-old man with facial trauma, raccoon eyes, and clear fluid from the nose requires an awake intubation for a predicted difficult airway. The anesthesiologist has explained the procedure to him, and the scope, the oral conduit, and a nasal vasoconstrictor are on the tray. He is cooperative. The anesthesiologist considers the route for the flexible scope. Which of the following is the best choice?

A. The nasal route, because it gives the straightest and shortest path to the glottis

B. The nasal route after a topical vasoconstrictor to reduce the nasal bleeding

C. The oral route with an intubating conduit to prevent biting of the scope

D. The oral route with the scope passed alone, with no protective conduit in place

E. A surgical airway under local anesthesia, since bronchoscopy is contraindicated

117. During an awake bronchoscopic intubation, the tube hangs up at the arytenoids as it is advanced over the scope in a 59-year-old man. The scope tip is above the carina with the tube mounted on it, the patient is calm, and the capnograph is connected. The scope tip is in the trachea above the carina. Which of the following is the best maneuver?

A. Advance the tube with firm steady pressure to push it through the arytenoids

B. Rotate the tube clockwise while pulling the scope back a few centimeters

C. Withdraw the scope completely and induce general anesthesia at once to start again

D. Rotate the tube counterclockwise to reorient the bevel, keeping the scope still

E. Change to a larger tube and push it firmly through the arytenoids without rotating it

118. An awake fiberoptic intubation in a 61-year-old patient has failed after two attempts. A second operator, a videolaryngoscope, and an equipped front-of-neck kit are in the room, and the surgeon is available. The patient is calm, breathing comfortably, and saturating 98 percent. The operator wants to repeat the same technique with deeper sedation. Which of the following is the best next step?

A. Repeat the same technique with deeper sedation until the intubation succeeds

B. Induce general anesthesia and proceed immediately with direct laryngoscopy

C. Give a neuromuscular blocker to improve conditions for another bronchoscopy attempt

D. Repeat the attempt using the same tube and a stiffer scope with more force

E. Stop and reassess, then change the device, operator, route, or plan

119. A team is briefing for a predicted difficult airway. The airway time-out has been completed, the equipment is checked, and the nurse is ready to bring the front-of-neck kit. A resident asks what plan C means in the strategy being stated. The anesthesiologist explains the sequence from plan A to plan D. Which of the following is the correct description of plan C?

A. Return to mask ventilation and decide whether to wake the patient

B. Front-of-neck access with a scalpel when oxygenation fails

C. A supraglottic airway used for oxygenation or as a conduit

D. A repeat attempt at laryngoscopy with a different blade and a bougie

E. A decision to cancel the case and extubate in the recovery area

120. During the first laryngoscopy in a 70-year-old man with a predicted difficult airway, the saturation falls from 98 to 93 percent. The oximeter tone has dropped, the second anesthesiologist is watching the monitor, and a bag and mask are ready at the head of the bed. The team had agreed before induction to stop and reoxygenate at 93 percent. The operator says the tube is almost in and asks for one more try. Which of the following is the best response?

A. Continue laryngoscopy until the saturation reaches 90 percent before changing the plan

B. Stop laryngoscopy and reoxygenate, since the threshold agreed before induction was reached

C. Make the additional attempt because the glottis is almost in view and desaturation is slow

D. Increase the induction drug dose to deepen anesthesia before the next attempt at laryngoscopy

E. Continue until the saturation reaches 85 percent so that the operator can learn from the case

QUESTION GROUP 7

Failed Intubation and the Cannot Intubate, Cannot Oxygenate Pathway

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Questions 121 to 140. Choose the single best answer for each.

121. A resident preparing a teaching session asks which sequence of plans the Difficult Airway Society pathway follows when an airway is failing. Movement is downward only when the step above has failed, and each step has an exit. Which of the following lists the plans in the correct order?

A. Tracheal intubation, supraglottic airway, final face mask attempt, front-of-neck access

B. Tracheal intubation, face mask ventilation, supraglottic airway, front-of-neck access

C. Supraglottic airway, tracheal intubation, final face mask attempt, front-of-neck access

D. Tracheal intubation, supraglottic airway, front-of-neck access, final face mask attempt

E. Face mask ventilation, tracheal intubation, supraglottic airway, front-of-neck access

122. A 44-year-old man has had four attempts at laryngoscopy during a difficult induction. A colleague observes that the view on the fourth attempt was worse than on the first, and that the patient, who could be mask ventilated after the first attempt, is now difficult to ventilate. His saturation has drifted to 94 percent between attempts. Which of the following best explains this progression?

A. Repeated blade passes cause bleeding, edema, and secretions that worsen the view and the mask seal

B. Cumulative neuromuscular blockade stiffens the chest wall further with each successive attempt at laryngoscopy

C. Repeated laryngoscopy gradually shifts the larynx anteriorly and widens the laryngeal inlet over the sequence of attempts

D. Anesthetic depth lightens with each interval between attempts and allows the cords to close against the blade

E. Cumulative laryngoscope pressure compresses the trachea and narrows it below the cricoid during each attempt

123. A 32-year-old woman at 36 weeks of gestation needs urgent general anesthesia for fetal bradycardia. Before induction, the anesthesiologist recognizes that her oxygen reserve is small and her consumption is high. Her saturation is 98 percent on high-flow nasal oxygen, the team is experienced, and a video laryngoscope and a second-generation supraglottic airway are ready. Which of the following is the best action at the bedside briefing?

A. Plan three full attempts before changing anything, since the ceiling of three is a quota that applies to all patients

B. Leave the stopping saturation undecided so that the team can respond to how the first attempt goes in the moment

C. Rely on apneic oxygenation alone and plan to continue attempts until a tube has been placed in the trachea

D. Limit the briefing to the drug doses and leave all airway decisions to the senior colleague who is attending

E. Agree the saturation at which attempts stop and mask ventilation resumes, since the second may be the last safe attempt

124. An anesthesiologist is intubating a 55-year-old man for emergency surgery. Laryngoscopy shows a poor view. The anesthesiologist considers calling for a senior colleague but notes that a call now might prove unnecessary. His saturation is 99 percent, a senior colleague is in the next theater, and the difficult airway cart is in the corridor. Which of the following is the most appropriate timing for calling for help?

A. After the third failed attempt, once the ceiling has been reached

B. Only when the saturation falls below an agreed threshold

C. After a supraglottic airway has failed and mask ventilation is difficult

D. When the patient has been awakened and the operation has been rescheduled

E. After the first failed laryngoscopy, and again aloud at the second

125. A 60-year-old woman is about to receive induction drugs for a laparotomy. The anesthesiologist has a video laryngoscope available, but the attending suggests starting with direct laryngoscopy and moving to video only if the first two attempts fail. The patient has a small mouth opening and a receding mandible. Her saturation is 99 percent and both airways are being prepared by an experienced assistant. Which of the following is the best response?

A. Start with direct laryngoscopy, since two failures are needed to justify changing the device that is in use at the time

B. Use a hyperangulated blade without a stylet so that the tube follows the line of the view toward the glottis on the first attempt

C. Use the video laryngoscope first, since predicted difficulty makes it the better opening device, not a late rescue

D. Delay the video laryngoscope until after a supraglottic airway has been placed and found to ventilate adequately

E. Begin with a bougie through direct laryngoscopy and reserve video laryngoscopy for the fourth attempt if needed

126. A 48-year-old man has had two attempts at direct laryngoscopy with a Macintosh blade. Only the epiglottis is visible. The anesthesiologist passes a bougie beneath the epiglottis and feels a series of clicks, then a firm stop. The saturation is 95 percent and the attending anesthesiologist is at the bedside. Which of the following does this finding indicate?

A. The bougie tip is in the esophagus, where the lack of resistance produces soft clicks

B. The bougie tip is in the trachea, crossing tracheal rings and meeting hold-up at the carina

C. The bougie has entered the vallecula, and the tube should now be advanced blindly

D. The bougie has coiled in the pharynx, and it should be withdrawn and the technique repeated

E. The bougie is at the level of the cords, where resistance is felt but clicks are absent

127. A 70-year-old man has failed intubation after the permitted attempts, and the team places a supraglottic airway to restore oxygenation. The anesthesiologist is choosing between a first-generation and a second-generation device. His saturation is 91 percent and the team has agreed that further laryngoscopy is not wanted. Which of the following is the best reason to prefer the second-generation device?

A. It allows conventional intubation through the device without any additional equipment or special skill

B. It removes the need to confirm ventilation with capnography once the cuff has been inflated

C. It permits unlimited insertion attempts because it seats more reliably than a first-generation device

D. It ventilates at higher pressures and offers some protection against regurgitation through a drain channel

E. It guarantees protection against aspiration, so that cricoid pressure and head-up positioning can be omitted

128. A supraglottic airway has been inserted in a 63-year-old woman after failed intubation. Peak pressures are high, an audible leak is present at the mouth, and the capnography trace is flat. The assistant proposes squeezing the bag harder. Her saturation is 86 percent and the cuff has been inflated to the manufacturer's volume. Which of the following is the best next step?

A. Remove and reinsert the device after relubrication, using a full jaw thrust or a different size

B. Leave the device in place and increase the squeeze of the bag to overcome the leak at the mouth of the device

C. Inflate the cuff to a much higher pressure while leaving the device where it sits at present in the pharynx

D. Advance the device deeper without removing it, in order to improve the seating of the tip behind the larynx

E. Replace the device with a tracheal tube passed blindly beside it into the pharynx without any view

129. A 52-year-old man cannot be ventilated by mask or supraglottic airway after failed intubation, and the saturation is 78 percent. Before declaring an emergency, a colleague notices that the circuit has come apart at the elbow connector and the pressure-limiting valve was left closed on the machine. Which of the following is the best conclusion?

A. The faults are unrelated to the airway failure and should be corrected after front-of-neck access

B. Circuit disconnection is excluded as a cause once the capnography trace has been flat for some time

C. The faults matter only for mask ventilation and do not affect ventilation through a supraglottic airway

D. Equipment faults can mimic a failed airway, so check the circuit and valve as well as the patient

E. Equipment faults are unlikely in a failed airway, so attention should stay on the patient's anatomy

130. A 41-year-old man is being ventilated well through a second-generation supraglottic airway after failed intubation. The department has a flexible bronchoscope in the room and a clinician skilled in using it. A trainee proposes removing the supraglottic airway to attempt direct laryngoscopy again. The bronchoscope, a tube, and an exchange catheter are on the cart, and the anesthesiologist is experienced with the technique. Which of the following is the best alternative?

A. Remove the device so that the bronchoscope can reach the larynx without an obstruction in the way of the view

B. Withdraw the device slightly and attempt to pass a tube blindly beside it into the trachea without a view

C. Leave the device in place and perform a scalpel cricothyroidotomy at once, although oxygenation is adequate

D. Pass a bronchoscope loaded with a tube or exchange catheter through the device, oxygen continuing

E. Leave the device in place and attempt nasal intubation in the paralyzed patient without vision of the glottis

131. A 58-year-old man cannot be ventilated by mask after induction with propofol, fentanyl, and rocuronium, and the chest feels stiff. A nerve stimulator is applied, and no twitches are seen in response to train-of-four stimulation. His saturation is 92 percent and the video laryngoscope has not provided a view. Which of the following is the best interpretation?

A. Laryngospasm or chest wall rigidity is likely, so a further dose of relaxant will help the ventilation

B. The stiffness is not muscle tone, so look for glottic obstruction, a leak, or a stiff chest

C. Incomplete blockade is likely, so a second dose of rocuronium should be given at once to the patient

D. The result is unreliable, so the cause should be assumed to be light anesthesia and more propofol given

E. Complete blockade is the cause of the stiffness, so sugammadex should be given before any other step

132. An anesthesiologist is setting up for a rapid sequence induction in a 36-year-old man, planning rocuronium. Sugammadex is stored in the drug room down the hall. The team believes it can be fetched if needed. Induction is about to begin, the saturation is 99 percent, and the patient has no airway predictors. Which of the following is the best change to the plan?

A. Draw up sugammadex and place it within reach before induction begins

B. Fetch sugammadex only when intubation has failed and the saturation has begun to fall

C. Replace rocuronium with succinylcholine so that sugammadex is not required

D. Keep sugammadex in the drug room and rely on waiting for spontaneous recovery

E. Draw up sugammadex only once the first attempt at intubation has failed

133. After multiple laryngoscopy attempts and fentanyl at induction, a 50-year-old man has been successfully oxygenated through a supraglottic airway. A colleague proposes giving sugammadex and removing the device, saying that reversal will solve any remaining airway problem. He is breathing six times a minute and the saturation is 97 percent on oxygen. Which of the following statements best describes the limitation of reversal?

A. It restores breathing and also rapidly clears laryngeal edema from repeated laryngoscopy within minutes

B. It reverses ventilatory depression from opioids and hypnotics in addition to the neuromuscular block

C. It guarantees a patent airway because the pharyngeal muscles regain full tone after the reversal

D. It reverses rocuronium only if it is given before the first attempt at intubation has been made

E. It restores the patient's own breathing but does not clear laryngeal edema or opioid depression

134. A 67-year-old woman has been rescued with a supraglottic airway after failed intubation, and the team has decided to wake her. Rocuronium was given 25 minutes ago, and sugammadex has been administered. The recovery room is expecting an ordinary emergence. Which of the following should be documented at the stimulator before the device is removed?

A. Four visible twitches on stimulation

B. A sustained tetanic response for five seconds

C. A train-of-four ratio of at least 0.9

D. A negative inspiratory force judged adequate

E. A train-of-four count of two with fade

135. A 38-year-old man requires urgent laparotomy for bleeding after trauma. Intubation has failed, and a second-generation supraglottic airway is ventilating well with a saturation of 97 percent. The surgeon wants to begin immediately. The surgeon is scrubbed, blood loss is continuing, and a senior anesthesiologist is available. Which of the following is the most appropriate approach?

A. Wake the patient as a rule, because waking is the only acceptable option after a failed intubation

B. Proceed using the supraglottic airway without discussing a fallback, since oxygenation is adequate now

C. Make a further attempt at direct laryngoscopy before beginning, because the operation is urgent

D. Wake the patient only if the surgeon agrees, because the surgeon decides the airway plan in this case

E. Proceed with the device after agreeing a spoken fallback with a senior colleague and the surgeon

136. A 29-year-old woman is having a cesarean delivery under general anesthesia for a nonreassuring fetal heart rate. Intubation fails, and a supraglottic airway ventilates her well. Her blood pressure and saturation are stable, and the obstetrician is ready to deliver. The fetal heart rate tracing remains abnormal, and the obstetrician is waiting with the patient prepared. Which of the following is the primary consideration in deciding whether to wake her?

A. Her own safety, and whether waking her is more dangerous than continuing

B. The fetal heart rate tracing, which should override concerns about an unsecured airway

C. The preference of the obstetrician regarding the timing of delivery

D. The wish to avoid any delay, since time to delivery is the only measurable outcome

E. The gestational age of the fetus, which alone determines whether to proceed

137. A 45-year-old man is being awakened after failed intubation rescued by a supraglottic airway. The team has reversed the neuromuscular block. As emergence proceeds, he begins to obstruct, with a falling saturation. He is coughing, moving his arms, and the saturation has fallen to 88 percent. The head of the bed is raised, the pharynx has been suctioned, and the supraglottic device is still available. Which of the following is the best plan?

A. Repeat laryngoscopy at once to secure the airway before he wakes further and the view worsens

B. Give additional rocuronium and attempt intubation with a different blade after clearing the secretions

C. Extubate the supraglottic device and suction while he continues to obstruct against the closed airway

D. Return to the supraglottic device or mask ventilation, not repeat laryngoscopy

E. Increase the dose of an opioid antagonist and wait for spontaneous breathing to improve over time

138. A 63-year-old man who had a failed intubation has been awakened and the operation was postponed. The team plans an awake technique for the next attempt. The team wants to avoid a repeat of the first attempt and has a bronchoscope, an antisialagogue, and a surgical airway kit ready. Which of the following describes a feature of this approach?

A. Deep sedation to ensure that the patient is unaware and still, with topical anesthesia limited to the nose

B. Induction of general anesthesia first, followed by bronchoscopy once the patient is fully asleep and paralyzed

C. Light sedation, thorough topical anesthesia of the airway, and capnography confirmation before induction

D. A brief preparation with topical anesthesia applied immediately before the bronchoscope is passed through the nose

E. Complete topical anesthesia of the nose only, with the pharynx and larynx left untreated before the bronchoscope

139. A 56-year-old man is in a cannot intubate, cannot oxygenate situation with a flat capnography trace after failure of intubation, a supraglottic device, and a two-handed mask attempt. A team member questions why the airway should be declared by evidence of failure rather than by a low number on the pulse oximeter. Which of the following is the best answer?

A. The oximeter is unreliable, so a saturation below a set value should never trigger the decision to cut

B. Declaration requires that the saturation fall below 70 percent before the scalpel is permitted in the room

C. Declaration depends on a surgeon being present to confirm that the neck can be accessed safely

D. Declaration depends on the pulse oximeter reaching the lowest value that it can display on the monitor

E. Failed intubation, supraglottic airway, and mask ventilation, with capnography as the judge

140. During a difficult airway case in a 46-year-old man, an anesthesiologist hesitates to cut because a further attempt from above might work. The anesthesiologist asks what principle should govern the decision. The patient is paralyzed, the saturation is 74 percent, and a two-person mask attempt has just failed. The anesthesiologist says that one more try from above might work. Which of the following is the best answer?

A. Continue attempts from above while the patient remains paralyzed, since a scalpel airway is a last resort

B. Wait for the surgeon to arrive, because the trained surgical hand should always perform the cut

C. Judge it by the saturation in the next minute, since an unneeded early scalpel costs only a scar

D. Defer the decision until the saturation has fallen below a specific threshold set earlier by the team

E. Continue one further attempt only if the capnography trace shows any waveform before cutting

QUESTION GROUP 8

Front-of-Neck Access

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Questions 141 to 160. Choose the single best answer for each.

141. A 40-year-old man is in a cannot intubate, cannot oxygenate emergency, and the cricothyroid membrane is palpable. The anesthesiologist is deciding where on the membrane to place the transverse stab incision. The saturation is 72 percent, the scalpel, bougie, and tube are open on the tray, and the larynx is stabilized. Which of the following is the best placement, and why?

A. The lower half of the membrane, because the cricothyroid arteries run along the upper border

B. The upper half of the membrane, because the vocal folds lie well above and are safely distant

C. The center of the membrane, because the vessels are lateral and the midline has no arteries

D. The thyroid cartilage just above the membrane, because it is firmer and easier to enter

E. The cricoid cartilage just below the membrane, because it is thinner and easier to enter

142. A 47-year-old woman with a short neck and no visible thyroid prominence is in a cannot intubate, cannot oxygenate emergency. The anesthesiologist thinks she has found the cricothyroid membrane by palpation but is concerned about a common error in women. Her saturation is 74 percent and the kit is open on the tray. Which of the following is that error?

A. Mistaking the cricoid for the first tracheal ring and cutting too low

B. Mistaking the thyroid cartilage for the cricoid and cutting too high

C. Mistaking the hyoid bone for the thyroid notch and cutting laterally

D. Mistaking the thyroid isthmus for the membrane and entering the esophagus

E. Mistaking the membrane for the interval between tracheal rings and cutting too far laterally

143. An anesthesiologist practicing the laryngeal handshake places the thumb and middle finger on either side of the hyoid bone and slides them down to the thyroid cartilage. The index finger then travels down the midline over the thyroid notch and the cartilage. The room is quiet, the patient is awake, and the examination is part of preparation for the airway plan. At what point does the finger identify the cricothyroid membrane?

A. When it meets a firm ring directly below the thyroid cartilage, with a soft depression above

B. When it reaches the sternal notch and then rolls back up over the first tracheal ring

C. When it feels a bony prominence in the midline that moves with swallowing

D. When it drops into a soft depression, with the firm cricoid ring directly below

E. When it reaches the hyoid bone, which is the first bony landmark in the midline

144. A linear high-frequency ultrasound probe is placed transversely over the thyroid cartilage in a patient scheduled for an anticipated difficult airway. The operator uses the acronym TACA while sliding the probe toward the feet. The patient is awake and lying flat with the neck extended, and the findings will be marked on the skin. Which finding is expected at the first position over the thyroid cartilage?

A. A dark arch with a bright line at its base

B. A row of dark beads along a bright line

C. A dark triangle with a bright anterior rim

D. A bright horizontal line with no structure anterior to it

E. A bright vertical line with acoustic shadowing behind it

145. A 58-year-old woman with obesity and a predicted difficult airway is scheduled for surgery in 20 minutes. The department has an ultrasound machine in the room. A colleague suggests keeping the machine ready to use during a cannot oxygenate emergency if needed. She has a body mass index of 44, and the team has otherwise planned the airway in detail. Which of the following is the best response?

A. Mark the membrane with ultrasound before induction, while awake, since it is a planning tool

B. Scan during the emergency, since the machine adds certainty when the saturation is falling fast

C. Skip the scan, since palpation alone is reliable in most patients with obesity and a short neck

D. Perform the scan after induction so that the patient is relaxed and the neck can be extended

E. Scan during the emergency only if the landmarks cannot be felt, bringing the machine in at that point

146. A 35-year-old man in a cervical collar after trauma is in a cannot intubate, cannot oxygenate emergency. The anesthesiologist stands at the head of a low bed with a flexed arm and bent wrist. The saturation is 70 percent, a deep neuromuscular block is in place, and an assistant is available. Which of the following changes best improves the conditions for a front-of-neck procedure?

A. Remain at the head of the bed and cut through the collar with the neck flexed forward for access

B. Remove the collar completely and release all stabilization to allow full neck extension now

C. Lower the bed further so that the neck is at eye level for the operator at the head of the bed

D. Stand at the dominant-side shoulder and open the collar while in-line stabilization is held

E. Keep the collar fully fastened and make the incision below it at the sternal notch instead

147. During planning for a front-of-neck procedure, ultrasound with color Doppler in a 60-year-old man shows an anterior jugular vein crossing the midline at the intended track. The airway then becomes an emergency before the vessel can be avoided. He has a thick neck, and the planned midline track has been marked on the skin. Which of the following is the best approach?

A. Abandon the procedure until a surgeon can come to ligate the vein and expose the field

B. Proceed in the midline within the membrane, ready with pressure if the vessel bleeds

C. Make the incision lateral to the vein, away from the midline, so as to avoid the vessel entirely

D. Switch to a narrow-bore cannula so as to avoid cutting the vessel in the midline track

E. Place an additional high skin incision and ligate the vein before the procedure continues

148. An anesthesiologist performing a scalpel cricothyroidotomy finds that the 6.0 cuffed tube will not pass through the membrane after the bougie is in the trachea. The saturation is 65 percent. The incision has been made, the blade turned, and the bougie lies in the trachea with clicks felt. Which of the following is the best next step?

A. Abandon the bougie and apply firm, steady force to push the tube through the membrane in one movement

B. Remove the bougie and switch to a narrow-bore cannula connected to a jet ventilator at once

C. Continue attempts to push the same tube while rotating the bougie within the trachea repeatedly

D. Remove the bougie and pass the same tube again without it in place, using a firmer push

E. Exchange for a smaller cuffed tube and, if needed, widen the incision with a gentle turn of the blade

149. A 49-year-old man is in a cannot intubate, cannot oxygenate emergency, and the leader announces the intention to perform a front-of-neck procedure. The assistant is opening the kit. The patient is paralyzed, the saturation is 70 percent, and the neck has been prepared. Which of the following should occur in parallel with preparation for the cut?

A. Stop all oxygen delivery to avoid blowing gas into the soft tissues during the incision itself

B. Continue 100 percent oxygen by every route still open and keep the neuromuscular block deep

C. Allow the neuromuscular block to wear off so that the patient can resume breathing on their own

D. Pause preparation to allow further attempts at laryngoscopy before the incision is made

E. Remove the neuromuscular block with sugammadex so that spontaneous breathing resumes during the cut

150. After a scalpel cricothyroidotomy, a cuffed tube is advanced over the bougie in a 54-year-old man. The anesthesiologist pushes the tube in as far as it will go. The saturation then rises briefly and the breath sounds are heard only on the right. The tube is lubricated and the cuff syringe is attached. Which of the following is the most likely cause?

A. The tube was advanced too deeply and entered the right mainstem bronchus

B. The cuff herniated over the tip and obstructed the lumen

C. The tube lies in the pretracheal tissue in front of the trachea

D. The bougie remains inside the tube and blocks the right lung

E. The bougie has perforated the posterior tracheal wall and is lying in the esophagus

151. During a scalpel cricothyroidotomy in a 62-year-old woman, blood obscures the field and the bougie route is not visible. The saturation is 70 percent. The anesthesiologist is considering stopping to control the bleeding. The assistant is calling out the saturation every ten seconds. The skin incision has been made in the midline, and the neck is swollen from earlier attempts. Which of the following is the best action?

A. Stop and pack the wound with gauze before proceeding with any further step in the procedure

B. Abort the front-of-neck approach and return to laryngoscopy once the field has cleared

C. Request a surgeon to control the bleeding before the bougie is placed in the airway

D. Continue, guiding the bougie by feel, because oxygen matters more than blood at this point

E. Stop and apply cautery to every visible vessel before the bougie is passed through the incision

152. An anesthesiologist's first attempt at a scalpel cricothyroidotomy in a 50-year-old woman has failed to place a tube, and the saturation continues to fall. The anesthesiologist is considering what to do next. A different operator is available and a smaller tube is on the tray. The first incision was short, the membrane could not be felt, and no tube has passed. Which of the following is the best approach?

A. Switch to a cannula and jet ventilator, since this is quicker than a second scalpel attempt

B. Return to laryngoscopy while waiting for a surgeon to arrive and take over the airway

C. Pause to reassess the landmarks with ultrasound before any further action is taken

D. Remove the blade and reattempt at the same site with an identical technique and the same tube

E. Start a second approach at once: a longer incision, another operator, or a smaller tube

153. A trainee asks why a standard intravenous cannula placed through the cricothyroid membrane cannot simply be connected to the anesthesia circuit and ventilated with the reservoir bag. The trainee has just seen a cannula connected to a jet ventilator and is unfamiliar with the resistance of narrow tubing. Which of the following is the best explanation?

A. The cannula is too short to reach the trachea, so gas would collect in the neck tissues

B. The standard circuit connector cannot attach to any catheter, whatever its internal diameter

C. The narrow lumen resists flow too much for a bag, so a high-pressure source is needed

D. A bag can deliver oxygen this way only if the upper airway is completely obstructed above the cannula

E. The circuit alarm will not permit ventilation through any catheter smaller than a tracheal tube

154. A manual jet ventilator is used through a narrow-bore cannula. The anesthesiologist has a regulated high-pressure oxygen source and a cannula that is confirmed in the trachea, and the upper airway is partly open for exhalation. The anesthesiologist is holding the cannula with one hand while the other controls the valve. Which of the following settings reflects the recommended practice?

A. Inspiration for about four seconds and expiration for one second, at a high pressure for rapid inflation of the lungs

B. Inspiration for about one second and expiration for one second, at a pressure giving vigorous chest rise on every breath

C. Continuous gas flow without an expiratory pause, at the lowest pressure that keeps the saturation up on the monitor

D. Inspiration for about one second and expiration for at least four seconds, at the lowest pressure giving chest rise

E. Inspiration for about two seconds and expiration for two seconds, at a pressure chosen by the clinician for each patient

155. Narrow-bore cannula jet ventilation is started in a 45-year-old man. After the first jet of oxygen, the neck swells and crepitus is felt in the tissues, while the chest barely moves and the saturation does not rise. The cannula was placed under pressure by a resident, and air was not aspirated to confirm entry. Which of the following is the most likely explanation?

A. The cannula has entered the esophagus, and the oxygen is passing into the stomach instead

B. The cannula tip is in a bronchus, and the oxygen is inflating only one lung

C. The cuff of the cannula has herniated over the tip and the jet is blocked completely

D. A tension pneumothorax on the left accounts for the swollen neck and the crepitus

E. The cannula lies in the pretracheal tissue, so the oxygen is inflating the neck

156. A department stocks a wide-bore cannula kit that uses a needle on a syringe, a guidewire, and a dilator. A trainee asks about the principle of this kit compared with a narrow-bore cannula. The department also stocks scalpel kits and trains on them every year. The trainee has memorized the steps and asks how the kit differs in principle. Which of the following statements is correct?

A. A lumen of 4 mm or more takes a conventional circuit, but the many steps invite error

B. The lumen is under 2 millimeters, so a high-pressure oxygen source is still required to ventilate

C. The kit has fewer steps than the scalpel technique, so it is quicker in every setting

D. The kit removes the risk of posterior wall puncture because the wire guides the cannula

E. The kit has no risk of a kinked wire because the dilator is rigid while it lies in the airway

157. A hospital considers stocking a flow-controlled ventilation device that delivers oxygen through a narrow cannula and then actively draws gas back out. The chief of anesthesia asks how this device fits into the front-of-neck strategy. The chief of anesthesia has read case reports and wonders whether the training budget should be redirected. Which of the following is the most accurate statement?

A. It replaces the scalpel technique, since case reports show better outcomes than any surgical method

B. It is an exhalation adjunct with limited evidence, and it does not replace the scalpel

C. It is unsuitable in every patient because it needs a fully open upper airway for exhalation

D. It requires no training because it works automatically after the cannula has been inserted

E. It is the standard of care in children under 8 years and should replace the cannula there

158. A 14-year-old adolescent and a 4-year-old child are each in separate cannot intubate, cannot oxygenate emergencies. The anesthesiologist asks about the approach to each. Neither has had a successful supraglottic airway, both are being oxygenated poorly, and the surgeon is delayed. Each child has had failed laryngoscopy, a failed supraglottic airway, and failed mask ventilation, and a technique must be chosen for each. Which of the following correctly describes the guidance?

A. Both are managed with a scalpel and a cuffed tube of the same useful size

B. The adolescent is managed as an adult, and the 4-year-old with a cannula technique

C. Both are managed with a cannula technique, since all children have a tiny membrane

D. The adolescent is managed with a cannula, and the 4-year-old is managed like an adult

E. The adolescent is managed with oxygen insufflation, and the 4-year-old is managed with a scalpel and bougie

159. A 57-year-old man has had a successful front-of-neck airway after a prolonged period with a saturation below 70 percent. Oxygen now flows through the tube. The team asks what complication to anticipate during or soon after the rescue. The surgeon is asking whether he can leave for the ward soon. Which of the following is the best answer?

A. No complications beyond local bleeding, since oxygenation has been restored and the airway is secure

B. Delayed hypertension that requires immediate treatment with a vasodilator and urgent imaging

C. Arrhythmia, bradycardia, or arrest, with early checks of temperature, glucose, and blood pressure

D. Isolated hoarseness that will resolve without further assessment or specific monitoring

E. Hypotension from a vagal reflex only, which resolves without monitoring of rhythm or glucose

160. A 46-year-old woman has been stabilized in the intensive care unit after an emergency scalpel cricothyroidotomy. The anesthesiologist is completing the handover and the record. The surgeon asks whether the details need to be recorded now or can be left to the discharge summary. Which of the following should be included for the benefit of the next anesthesiologist?

A. The final tube size only, because other details are not needed by the next anesthesiology team

B. The surgeon's opinion about the technique, with no description of the anatomy found at the neck

C. A statement that no airway alert is needed because the access was temporary and has been removed

D. Decision time, attempts at each step, technique, tube size, capnography, complications, and anatomy

E. Only the time of the procedure, because the remaining details belong in the surgical record alone

QUESTION GROUP 9

Laryngospasm, Bronchospasm, and Aspiration

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Questions 161 to 180. Choose the single best answer for each.

161. A 3-year-old child with asthma and a recent cold is undergoing mask induction with a pungent volatile agent for a minor procedure. Stridor develops immediately after an oral airway is inserted at a light plane. The capnography trace is flat and the saturation has dropped to 90 percent. Which of the following best explains the mechanism?

A. Paralysis of the recurrent laryngeal nerve by the volatile agent, leaving the cords fixed in adduction

B. A mucosal reflex via the superior laryngeal nerve that adducts the cords, strongest at light planes

C. Histamine release that causes smooth muscle contraction in the lower bronchi and obstructs expiration

D. Relaxation of the pharyngeal muscles that lets the tongue fall back against the posterior wall

E. Activation of the carotid sinus reflex, which slows the heart and closes the glottis in response

162. A 25-year-old woman develops what appears to be complete airway obstruction during emergence, with no air movement. The anesthesiologist performs a gentle jaw thrust with positive pressure, and the airway immediately opens and ventilation becomes easy. She is a smoker and was extubated at a light plane of anesthesia. What does this response most likely indicate?

A. The cords are in complete spasm and will require succinylcholine to resolve completely

B. The obstruction was from bronchospasm that has now responded to the positive pressure

C. The cords were not fully shut, and the obstruction was easier to relieve than it first appeared

D. The obstruction was from a foreign body that has been dislodged by the jaw thrust

E. The obstruction was from anaphylactic laryngeal edema, which has been reversed by positive pressure

163. A 7-year-old child has laryngospasm with a falling saturation. Positive pressure and jaw thrust have not opened the airway, and the anesthesiologist has applied firm bilateral pressure at the laryngospasm notch. The child has intravenous access. The child weighs 20 kg and anesthesia is being maintained with sevoflurane. Which of the following is the first drug to give?

A. A small bolus of propofol to deepen anesthesia and relax the cords

B. Atropine alone, to treat the bradycardia that is expected with hypoxia in a child

C. A large dose of an inhaled beta-agonist, to relax the lower airway smooth muscle

D. Naloxone, to reverse the opioid given at induction and restore spontaneous breathing

E. Hydrocortisone, because inflammation of the cords is the cause of the closure

164. A 34-year-old man has persistent laryngospasm after extubation. Propofol has been given without effect. His saturation is 70 percent and falling, and he has an intravenous line. The anesthesiologist is deciding on the succinylcholine dose. Atropine is drawn up and the airway equipment is ready. The airway is still obstructed after propofol, and an intravenous line is secure. Which of the following is the best approach?

A. A reduced dose of 0.1 to 0.5 mg/kg may suffice, but in hypoxia give enough to be certain

B. Withhold succinylcholine and wait for the propofol to take effect before any further drug is given

C. Give only a fixed small dose and avoid repeating it, whatever the saturation does afterward

D. Give the full dose of 1 to 1.5 mg/kg as the only acceptable dose in laryngospasm

E. Give atropine first and withhold succinylcholine until bradycardia is established on the monitor

165. A 28-year-old man has laryngospasm after a short anesthetic. He has not received a muscle relaxant, and his cords are closed. A junior colleague wants to intubate immediately to secure the airway. A laryngoscope and a tracheal tube are at the bedside. The laryngospasm has lasted 60 seconds. Which of the following is the best response?

A. Intubate at once, since a tracheal tube is the most reliable way to bypass the spasm of the cords

B. Intubate only if the capnogram shows a waveform, which would confirm that the cords are open

C. Do not intubate until a relaxant is given, since attempts through shut cords provoke more reflex

D. Intubate after rapid, forceful ventilation against the closed cords to open the supraglottic tissues

E. Intubate only after propofol has deepened anesthesia, without giving a relaxant of any kind

166. A 4-year-old child with a recent upper respiratory infection has been anesthetized for a minor procedure. The anesthesiologist plans the emergence to reduce the risk of laryngospasm. The surgery has ended and the surgeon is applying the dressing. He had a mild cough and a runny nose last week. Which of the following is the best approach?

A. Extubate in the transition between unconscious and awake, when the reflexes have just returned

B. Extubate immediately after the last surgical stimulus, regardless of the depth of anesthesia

C. Stimulate the child vigorously during extubation to speed the return of the airway reflexes

D. Extubate once the first cough returns, since this signals that the airway reflexes are intact

E. Extubate fully awake with intact reflexes, or deep if the airway is easy, but not in between

167. A 38-year-old man with moderate asthma has a tracheal tube placed under light anesthesia. Peak airway pressure rises, and there is wheeze on the left but absent breath sounds on the right. Suction catheter passes easily. He was intubated minutes ago, and the operation has not yet begun. Which of the following is the most appropriate next step?

A. Give epinephrine immediately without further examination of the tube position or the circuit

B. Auscultate both sides and withdraw the tube slightly, since a bronchial tube can mimic bronchospasm

C. Administer magnesium sulfate at once for presumed refractory bronchospasm in this asthmatic patient

D. Deepen anesthesia with desflurane and wait for the peak pressure to fall over several breaths

E. Remove the tube and proceed with mask ventilation until the wheeze has resolved in the airway

168. A 45-year-old man with reactive airways is being induced for surgery. The anesthesiologist selects agents to minimize the risk of bronchospasm. He uses beta-blocker eye drops for glaucoma and has had wheeze after previous anesthetics. The operation is a laparoscopic cholecystectomy under general anesthesia. Which of the following is the best choice for the induction agent and the volatile agent?

A. Thiopental for induction and desflurane as the volatile agent for maintenance

B. Propofol or ketamine for induction, and sevoflurane as the volatile agent

C. Etomidate for induction and desflurane as the volatile agent for maintenance

D. Propofol for induction and morphine to deepen anesthesia before intubation

E. Atracurium for relaxation and desflurane as the volatile agent for maintenance

169. A 60-year-old woman with chronic obstructive disease has bronchospasm during surgery. The anesthesiologist delivers albuterol through the circuit. Delivery is poor because flow is limited. A resident asks about the usual approach to inhaled therapy. The peak pressure remains high, and she is also being given a deeper volatile concentration. Which of the following is correct?

A. Give albuterol once and do not repeat it, since repeated doses are known to increase mortality

B. Replace albuterol with ipratropium because it acts faster than albuterol on bronchial smooth muscle

C. Stop inhaled therapy if there is no response after one dose, and use corticosteroids instead

D. Replace albuterol with systemic corticosteroids, because inhaled drugs cannot reach the lower airway

E. Repeat albuterol as needed, add ipratropium for a second mechanism, and watch rate and potassium

170. A 52-year-old man with severe bronchospasm has not improved after deepening anesthesia, repeated albuterol, ipratropium, and epinephrine boluses. The saturation remains low. The anesthesiologist considers further therapy. He is on 100 percent oxygen at a slow ventilatory rate, and the surgeon has paused. Peak pressures are 45 cm H2O. Which of the following is an appropriate next addition?

A. Aminophylline as a first-line agent because of its strong and safe bronchodilation

B. Neostigmine to increase bronchial tone and clear secretions from the lower airway

C. A beta-blocker to reduce the heart rate and the oxygen consumption of the patient

D. Increase the tidal volume and respiratory rate to remove the carbon dioxide faster

E. Magnesium sulfate over 20 minutes, and an epinephrine infusion titrated to effect

171. A 33-year-old woman with asthma and wheeze on examination this morning is scheduled for elective surgery in two hours. She uses an inhaler daily. Her lung examination shows diffuse wheeze, and her peak flow is below her usual baseline. She has been well controlled on an inhaled steroid and a rescue inhaler for years. Which of the following is the best plan?

A. Proceed, since regular inhaler use makes active wheeze acceptable under general anesthesia

B. Proceed and give a nonselective beta-blocker to reduce airway reactivity before induction

C. Delay elective surgery and optimize with bronchodilators and sometimes a short steroid course

D. Proceed after stopping her inhalers on the morning of surgery to avoid drug interactions

E. Proceed and give hydrocortisone alone, since steroids work within minutes of the dose being given

172. A 40-year-old healthy man is scheduled for an elective hernia repair. He drank a glass of water 90 minutes ago, and had a light breakfast of toast four hours ago. He has no medical history and takes no regular medication. Under current American Society of Anesthesiologists guidance for healthy adults having elective procedures, which of the following is the best statement?

A. Clear liquids until two hours and a light meal until six hours, so the meal is not yet allowed

B. Clear liquids are allowed until four hours before, and a light meal needs eight hours

C. Clear liquids are allowed until two hours before, and a light meal is allowed after four hours

D. Clear liquids are allowed up to the time of transport, and a light meal needs six hours

E. Clear liquids are allowed until one hour before, and a light meal needs four hours

173. A 56-year-old woman with abdominal discomfort is being evaluated for urgent surgery after a recent meal. The anesthesiologist performs gastric ultrasound with the patient in the right lateral decubitus position, with the probe in a sagittal plane near the left lobe of the liver and the aorta. The antrum shows a mixed echogenic pattern with a frosted glass look. Which of the following does this indicate?

A. A collapsed empty antrum, so a standard induction is appropriate here

B. Clear fluid in the antrum, which appears as a dark distended structure

C. An artifact from air, which should be disregarded when planning the airway

D. Solid gastric contents, so manage the airway as for a full stomach

E. Scarring from earlier surgery, which makes the study uninterpretable

174. A 30-year-old woman at term is scheduled for cesarean delivery under general anesthesia. The anesthesiologist considers pharmacologic aspiration prophylaxis. She has no gastrointestinal symptoms, an epidural was not possible, and the airway examination is reassuring. Her last meal was eight hours ago. Which of the following is the best choice before induction?

A. An H2 blocker alone, since it neutralizes gastric acid immediately after intravenous dosing

B. Metoclopramide alone, since it prevents regurgitation completely during the induction

C. Sodium citrate, which acts within minutes, before every cesarean general anesthetic

D. No prophylaxis, since it is not indicated in pregnancy at any stage of gestation

E. A proton pump inhibitor given just before induction, since it acts within minutes

175. A 45-year-old man with a bowel obstruction undergoes rapid sequence induction. He has had nothing by mouth for ten hours but has vomited three times today. The plan is preoxygenation followed by an induction agent and a rapid-acting neuromuscular blocker, and a colleague asks about positioning and cricoid pressure. Which of the following statements is correct?

A. Cricoid pressure has strong evidence of benefit and is applied with the head kept flat during induction

B. Mask ventilation is given routinely before laryngoscopy to prevent desaturation during the induction

C. Rocuronium is contraindicated because it cannot be reversed in an emergency situation in any patient

D. Rapid sequence induction requires the head to be kept flat to improve the laryngeal view for the operator

E. Head-up positioning lengthens apnea time and cuts regurgitation; cricoid pressure has limited evidence

176. A 62-year-old man aspirates gastric contents during induction. He is intubated, suctioned, and ventilated. A bronchoscopy is being considered. The chest radiograph is pending, and he is on 100 percent oxygen with PEEP. Gastric contents with food fragments were suctioned from the pharynx. Which of the following findings would be the strongest indication for bronchoscopy, and what should be avoided?

A. Any episode of aspiration, with saline lavage performed to clear the lungs of acid and particles

B. Fever on the second day, with corticosteroids started at the same time as the procedure

C. Suspected solid particles or lobar collapse, with no saline lavage since it spreads the material

D. Wheeze alone, with prophylactic antibiotics given before the procedure begins in the airway

E. Any hypoxemia in the first hour, with antibiotics started before the procedure is carried out

177. A 55-year-old woman has a small witnessed aspiration at induction. The operation is urgent. She remains without respiratory symptoms for two hours after the event. She is stable on room air, with a saturation of 97 percent and a clear chest. Which of the following is the best interpretation?

A. She is unlikely to develop complications, and proceeding with the urgent surgery is reasonable

B. She needs intensive care for at least 48 hours because acute respiratory distress syndrome is inevitable

C. She should receive routine broad-spectrum antibiotics because pneumonia always develops

D. She should have the surgery postponed in every circumstance, including urgent cases

E. She needs repeat chest radiographs and routine intensive care admission after any aspiration, and surgery must be postponed

178. A 26-year-old athletic man has had a prolonged episode of obstruction during emergence from a hernia repair because he bit his laryngeal mask. After the obstruction is relieved, he develops pink frothy sputum, hypoxemia, and bilateral infiltrates with a normal heart size. Which of the following is the most likely mechanism of his hypoxemia?

A. Regurgitation of gastric contents into the right lower lobe during the long period of obstruction at emergence

B. Forceful inspiration against the obstruction created large negative intrathoracic pressure that injured capillaries

C. Left ventricular failure from an acute myocardial infarction caused by the preceding hypoxia and tachycardia

D. Fluid overload from intraoperative crystalloid given in excess during the hernia repair and recovery

E. Severe airway edema from anaphylaxis that causes a protein-rich capillary leak in the alveoli of both lungs

179. A 12-year-old child with severe tonsillar hypertrophy and chronic obstruction undergoes adenotonsillectomy. An hour after the airway is cleared, the child develops tachypnea, pink sputum, and hypoxemia. The anesthesiologist recognizes the pattern and explains it to the recovery nurse. He has had loud snoring and mouth breathing for years, and the airway was cleared uneventfully at the end of the case. Which of the following best describes this event?

A. Type I negative pressure pulmonary edema, which always appears within seconds of the acute obstruction

B. Cardiogenic pulmonary edema from sudden afterload reduction after the adenotonsillectomy is completed

C. Aspiration pneumonitis, which develops only after witnessed vomiting during the operation itself

D. Type II negative pressure pulmonary edema after relief of chronic obstruction, which can appear late

E. Reexpansion pulmonary edema after drainage of a pneumothorax in the operating room earlier today

180. A 24-year-old man with negative pressure pulmonary edema after laryngospasm is hypoxemic. The resident wants to give furosemide for the pink frothy sputum. His blood pressure is normal, and he shows no signs of volume overload. The chest radiograph shows bilateral infiltrates with a normal heart size, and he is receiving oxygen by mask. Which of the following is the best response?

A. Give furosemide at once, since pulmonary edema of any cause is due to excess volume in the lungs

B. Give corticosteroids, since they hasten recovery from the capillary injury in this condition

C. Give a large fluid bolus to replace the intravascular volume that has been lost into the alveoli

D. Diuretics are usually unnecessary, as the problem is leakiness; give oxygen and positive pressure

E. Give a vasodilator to reduce preload, since the right heart is overloaded by the edema fluid

QUESTION GROUP 10

Airway Obstruction, Fire, and Extubation Failure

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Questions 181 to 200. Choose the single best answer for each.

181. A 66-year-old woman is scheduled for excision of a skin cancer on the cheek under monitored anesthesia care with sedation. Oxygen will flow by nasal cannula beneath the drapes, and the surgeon plans to use a cautery pencil near the nose. At the time-out, which of the following is the most appropriate assessment of the fire risk?

A. Low risk, because no tracheal tube is in place and the airway is not being instrumented

B. High risk, because an oxidizer and an ignition source will both be near the surgical site

C. Low risk, because low-flow oxygen from a nasal cannula is too dilute to support combustion

D. High risk only if a laser is used, because cautery is an uncommon ignition source

E. Uncertain, so the risk should be judged after the first use of the cautery pencil

182. During preparation for laser surgery of the larynx, a resident asks why the cuff of the laser-resistant tube is filled with saline tinted with methylene blue instead of air. Which of the following is the best explanation?

A. The saline absorbs the laser energy and makes the tube wall unable to ignite

B. The dye allows the cuff pressure to be read without a manometer or leak test

C. The colored fluid stops the tube from burning if the cuff itself catches fire

D. A breach of the cuff becomes visible as blue fluid and gives early warning of damage

E. The saline is incompressible and keeps the cuff from shifting toward the vocal cords

183. An otolaryngologist plans an airway operation with a laser that differs from the one used in the previous case. The room has a laser-resistant tube labeled for a carbon dioxide laser. The anesthesiologist is asked whether it is acceptable. Which of the following is the best action before the first pulse?

A. Proceed, because a laser-resistant tube is designed to protect against laser wavelengths in general

B. Confirm that the tube is rated for the wavelength of the laser in use and agree the settings

C. Switch to a metallic tube, because metal does not reflect the beam onto adjacent tissue

D. Use a silicone tube, because silicone does not burn when it is struck by a laser beam

E. Wrap a standard polyvinyl chloride tube in wet gauze, which gives equivalent protection

184. A fire starts on the drapes during a neck operation under general anesthesia in a 55-year-old man. The gas flow has been stopped, the burning drapes have been removed from the patient, and saline has been applied, but flames persist on the equipment table beside the field. The patient is not harmed at present. Which of the following is the most appropriate next step?

A. Use a carbon dioxide extinguisher on the burning drapes and activate the fire alarm

B. Resume ventilation with 100 percent oxygen to keep the patient well oxygenated during the response

C. Smother the flames with dry drapes and continue the operation once the visible fire is out

D. Continue using saline alone, because a carbon dioxide extinguisher leaves a harmful residue

E. Wait for the surgeon to finish before the room is evacuated or the alarm is raised

185. After an airway fire, a 45-year-old man has been reintubated. Bronchoscopy shows mild mucosal erythema with small soot deposits, which were removed. He is hemodynamically stable and oxygenating well. The surgeon asks where he should be managed. Which of the following is the most appropriate disposition?

A. Recovery room for two hours, then the ward, because the bronchoscopic findings were only mild

B. Same-day discharge with oral steroids, because the visible injury was limited to the mucosa

C. Return to the operating room to complete the surgery, now that the airway is clear and the fire is out

D. Intensive care unit, because airway edema may progress for hours and ventilation may be prolonged

E. General ward with intermittent saturation checks, because extubation is planned within the hour

186. A 2-year-old boy has had recurrent pneumonia in the right lung for three weeks. His mother recalls that he choked on peanuts a month ago and then seemed well. A foreign body in a bronchus acting as a ball valve is suspected. Which of the following findings on an expiratory chest film is characteristic?

A. Hyperinflation of the lung beyond the obstruction from trapped air

B. Loss of volume in the affected lung with the mediastinum drawn toward it

C. Symmetric hyperlucency of both lungs with flattened diaphragms

D. Complete opacification of the affected hemithorax with the mediastinum shifted away

E. A thickened subglottic outline with a narrowed tracheal air column

187. A stable 3-year-old with a peanut in the left main bronchus is scheduled for rigid bronchoscopy. The team is deciding between spontaneous breathing under sevoflurane and controlled ventilation with a short-acting relaxant. Which of the following is the main advantage of keeping the child breathing spontaneously?

A. It gives complete immobility and eliminates coughing during manipulation of the object

B. It removes the need for topical lidocaine because laryngospasm cannot occur in this setting

C. It allows the most precise control of carbon dioxide and of anesthetic depth during retrieval

D. It prevents the object from moving, so the lung beyond it is protected from air trapping

E. It preserves airway tone and avoids pushing the object deeper with positive pressure

188. A 70-year-old man collapses unconscious in the preoperative area after choking on a meal. He is not breathing, and his lips are cyanotic. An anesthesiologist arrives with a laryngoscope and sees a bolus of meat in the pharynx. Which of the following is the most appropriate action?

A. Continue abdominal thrusts for several more cycles before any attempt at laryngoscopy

B. Perform front-of-neck access immediately, without any attempt to remove the visible object

C. Sweep blindly through the mouth with a finger to hook out the bolus

D. Deliver back blows only, because they are the treatment for an unconscious adult

E. Remove the object under direct vision with Magill forceps, then ventilate with oxygen

189. A resident asks why induction of general anesthesia can cause sudden airway collapse in an adult with a large anterior mediastinal mass who breathes comfortably while awake. Which of the following is the best explanation?

A. Induction drugs cause bronchospasm that narrows the intrathoracic airway below the level of the mass

B. Anesthesia increases venous return, which engorges the mass and displaces the carina to one side

C. Lost negative pleural pressure, lower lung volume, and relaxed airway tone let the mass compress the airways

D. The supine position alone shortens the trachea and kinks it, whatever the depth of anesthesia

E. Positive pressure ventilation compresses the mass against the sternum and closes the larynx at once

190. A 35-year-old man with a large anterior mediastinal mass, orthopnea, and superior vena cava syndrome needs general anesthesia for a therapeutic procedure that cannot be done awake. Alternatives to anesthesia have been excluded. Which of the following preparations is the most appropriate?

- **A.** Induce with propofol and a relaxant for controlled ventilation, and arrange extracorporeal support only if collapse occurs
- **B.** Perform a rapid sequence induction to protect the airway, and turn the patient prone if collapse follows
- **C.** Keep spontaneous ventilation, avoid relaxants, and place femoral bypass access under local anesthesia
- **D.** Use a short-acting relaxant to allow a smaller tube, and keep the rigid bronchoscope in the next room
- **E.** Induce supine with the head down to improve venous return, and avoid femoral cannulation to limit bleeding

191. A 62-year-old man has a supraglottic mass causing hoarseness and stridor on exertion, but not at rest. He is scheduled for biopsy and imaging shows a lumen of a few millimeters at the narrowest point. The surgeon will be present with a rigid bronchoscope. Which of the following is the best approach to securing the airway?

- **A.** Propofol induction and a test of mask ventilation, followed by rocuronium only if ventilation proves easy
- **B.** Rapid sequence induction with succinylcholine and direct laryngoscopy using the smallest tube available
- **C.** Deep sedation with midazolam and fentanyl until the patient is unresponsive, then direct laryngoscopy
- **D.** Inhalational induction with deep anesthesia followed by an immediate relaxant dose to secure the tube
- **E.** Awake fiberoptic intubation under topical anesthesia, with the neck kit open and backup options agreed

192. A 64-year-old man taking an angiotensin-converting enzyme inhibitor develops swelling of the lips and tongue with early stridor. He has been given epinephrine and intravenous steroids without improvement. An anesthesiologist is called. Which of the following is the most appropriate next step?

- **A.** Repeat epinephrine and steroids at higher doses and observe for response over several hours
- **B.** Rapid sequence induction with a relaxant to relieve the muscular component of the obstruction
- **C.** Awake fiberoptic intubation with preparation for a surgical airway, without waiting for more drugs
- **D.** Give icatibant and delay airway management until the swelling begins to resolve in the next hour
- **E.** Admit to the ward on humidified oxygen and reassess the airway at the next morning ward round

193. A 4-year-old girl is brought to the emergency department with fever, drooling, a muffled voice, and a tripod posture. Epiglottitis is suspected. She is calm in her father's arms. Which of the following is the most appropriate approach?

- **A.** Keep her calm and move to the operating room for induction by experienced hands with surgical backup
- **B.** Examine the throat with a tongue depressor to confirm the diagnosis before any transfer is arranged
- **C.** Send her unaccompanied for a lateral neck film to confirm the diagnosis in the radiology suite
- **D.** Give midazolam premedication to prevent crying, then place an intravenous line in the emergency department
- **E.** Start a nebulized epinephrine treatment and observe on the ward for improvement over several hours

194. Which of the following patients is the most suitable candidate for deep extubation to avoid coughing at emergence?

- **A.** A patient with obstructive sleep apnea after an uncomplicated bariatric operation
- **B.** A patient with a full stomach after an emergency laparotomy for obstruction
- **C.** A patient with a known difficult airway after a long maxillofacial reconstruction
- **D.** A patient with a restricted airway access because the jaws were wired at surgery
- **E.** A fasted patient with reactive airways and an easy airway after hernia repair

195. A 58-year-old woman has just completed 9 hours of extensive neck and facial reconstruction with obvious pharyngeal and facial swelling. With the cuff deflated and the tube occluded, there is no leak around the tube. She is following commands. Which of the following is the best management?

- **A.** Extubate awake now, because the predictive value of the cuff leak test is limited and she obeys commands
- **B.** Leave the tube in place, transfer to intensive care, and consider steroids before reassessing extubation
- **C.** Replace the tube with a laryngeal mask under deep anesthesia, using the Bailey maneuver, to avoid coughing
- **D.** Give one dose of dexamethasone and extubate at once, since steroids relieve edema within minutes
- **E.** Extubate over an airway exchange catheter and remove the catheter after one hour if she is comfortable

196. A 120-kg man with severe obstructive sleep apnea has had a laparoscopic operation lasting 3 hours. The block is fully reversed. The anesthesiologist plans extubation and the recovery pathway. Which of the following is the best plan?

- **A.** Extubate deep in the supine position and place a nasopharyngeal airway to prevent coughing and obstruction
- **B.** Extubate awake and upright, apply CPAP or high-flow nasal oxygen, and monitor with capnography
- **C.** Extubate awake supine, give a long-acting opioid for comfort, and transfer to an unmonitored ward bed
- **D.** Extubate awake and give oxygen by simple face mask alone, with transfer to an unmonitored bed
- **E.** Extubate once a train-of-four count of four is seen, without any further assessment of the block

197. A patient with an airway exchange catheter in place needs reintubation after obstruction. The anesthesiologist uses a video laryngoscope, keeps the catheter taut, and advances a smaller-than-usual tube, but the tube catches at the laryngeal inlet. Which of the following is the best next step?

- **A.** Advance the tube with a firm, rapid thrust to overcome the resistance and seat it past the cords
- **B.** Remove the catheter and restart with direct laryngoscopy using a larger, stiffer tube
- **C.** Exchange the tube for a larger size, because a stiffer shaft will overcome the obstruction
- **D.** Withdraw the tube slightly and rotate it counterclockwise a quarter turn, then advance gently
- **E.** Pass a jet ventilation cannula through the catheter to oxygenate before another attempt is made

198. A patient has stridor after extubation, and endoscopy is not yet available. The stridor is heard throughout both inspiration and expiration. Which of the following sites of narrowing is most likely?

- **A.** A lesion at or above the glottis
- **B.** A subglottic or tracheal lesion
- **C.** An intrathoracic lesion below the carina
- **D.** Bronchospasm of the small distal airways
- **E.** A lesion confined to the nasopharynx

199. A 44-year-old woman has stridor with paradoxical respiratory effort immediately after extubation following total thyroidectomy. Her voice is normal, the neck is flat and soft, and quantitative monitoring showed a train-of-four ratio of 0.95 before extubation. Which of the following is the most likely cause?

- **A.** Unilateral recurrent laryngeal nerve injury with one mobile cord
- **B.** Residual neuromuscular block causing weakness of the pharyngeal muscles
- **C.** Compression of the airway from an expanding hematoma in the neck
- **D.** Laryngospasm that will settle with simple positive pressure alone
- **E.** Bilateral vocal cord paralysis, with the cords at the midline

200. A 60-year-old man has a swollen, blood-stained airway after neck surgery and has stridor and a saturation of 85 percent after extubation. He is breathing spontaneously, and mask ventilation is difficult. The team must reintubate. Which of the following is the best plan?

A. Give rocuronium at once to improve laryngoscopy conditions, because hypoxemia makes speed the priority here

B. Place the patient supine and let the most junior operator use a standard tube, so that experience is gained

C. Repeat the same laryngoscope several times, because the first intubation was uneventful and straightforward

D. Preoxygenate head-up with high-flow nasal oxygen; best operator, video laryngoscope, bougie, small tube; no relaxant

E. Sedate heavily so that the patient accepts a tight mask, then attempt intubation after he is unresponsive

QUESTION GROUP 11

Malignant Hyperthermia: Genetics and Pathophysiology

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Questions 201 to 220. Choose the single best answer for each.

201. A 34-year-old man with a known pathogenic RYR1 variant needs an emergency appendectomy and ate a meal two hours ago. The machine has been prepared for a trigger-free anesthetic. Which of the following induction plans is appropriate?

A. Propofol and succinylcholine, with total intravenous maintenance afterward

B. Thiopental and succinylcholine, with the vaporizers removed from the machine

C. Sevoflurane mask induction without a relaxant, followed by a nasogastric tube

D. Propofol and rocuronium for a rapid sequence induction, with sugammadex available

E. Etomidate and succinylcholine, with nitrous oxide and oxygen for maintenance

202. A 29-year-old woman with a genetically confirmed susceptibility to malignant hyperthermia is at term and needs a cesarean delivery. She has no contraindication to neuraxial block. Which of the following is the best anesthetic plan?

A. General anesthesia with sevoflurane and succinylcholine for speed, with dantrolene given first

B. General anesthesia with propofol and succinylcholine, because the fetus is not affected by triggers

C. Spinal or epidural anesthesia, which is safe in susceptible patients and avoids all triggers

D. Local infiltration alone, because amide local anesthetics are triggering agents in susceptible patients

E. Delay the operation so that a general anesthetic can be given with prophylactic dantrolene

203. A medical student asks how the introduction of dantrolene in the late 1970s changed the outcome of malignant hyperthermia. Which of the following is the most accurate statement?

A. Mortality exceeded 70 percent before dantrolene and is now under 5 percent with early treatment

B. Mortality was about 10 percent before dantrolene and is now close to zero in well-resourced hospitals

C. Mortality exceeded 70 percent before dantrolene and remains above 50 percent today despite treatment

D. Mortality was about 30 percent before dantrolene and is under 5 percent with cooling alone

E. Mortality was about 50 percent before dantrolene and is now about 20 percent with early treatment

204. The clinical incidence of malignant hyperthermia is estimated at between 1 in 10,000 and 1 in 100,000 anesthetics, yet many more people carry a relevant variant. Which of the following best explains the difference?

A. Most variants are recessive, so carriers are affected only when both copies are altered

B. The disease is expressed only in men, so half of all carriers can never react

C. Variants in the ryanodine receptor gene are lost when the muscle is exposed to a trigger

D. Many carriers never declare the disease, because penetrance and expressivity vary

E. Carriers who receive a volatile agent react within the first minutes of exposure

205. A student asks how a nerve impulse normally causes calcium release in skeletal muscle. Which of the following best describes excitation-contraction coupling?

A. Cav1.1 in the T tubule is mechanically linked to the ryanodine receptor, which opens and releases calcium

B. Depolarization opens the ryanodine receptor in the cell membrane, and extracellular calcium then enters the fiber

C. Calcium entering through the transverse tubule channel triggers further release from the store, as in cardiac muscle

D. Calcium binds troponin C first, and troponin C then opens the ryanodine receptor to sustain release

E. The sarco-endoplasmic reticulum ATPase releases calcium from the store into the myoplasm whenever ATP is plentiful

206. A 20-year-old man has recovered from a confirmed malignant hyperthermia episode. Before discharge, the team discusses what he should be told. Which of the following is the best advice?

A. Precautions apply only to general anesthesia lasting more than an hour, so dental and emergency care are unaffected

B. Carry a letter and an alert card, and tell every clinician, including ambulance teams who may give succinylcholine

C. Carry oral dantrolene tablets and take them before every procedure to prevent a reaction from occurring

D. Inform only the anesthesiologist, because surgeons, dentists, and emergency teams do not use triggering drugs

E. Precautions end after about a year, because the susceptibility fades once a reaction has been treated

207. Which of the following sequences best explains the high potassium and dark urine seen in a fulminant malignant hyperthermia reaction?

A. Sustained contraction depletes ATP, the membrane fails, and potassium, myoglobin, and creatine kinase leak out

B. Primary tubular injury of the kidney prevents excretion of potassium and myoglobin while the muscle remains intact

C. Heat causes hemolysis of red cells, which releases potassium and the hemoglobin that colors the urine

D. Succinylcholine receptor upregulation drives potassium out of normal muscle independent of cellular energy

E. Respiratory alkalosis shifts potassium out of the cells and increases the muscle production of myoglobin

208. The parents of a 7-year-old who recovered two weeks ago from a suspected malignant hyperthermia crisis ask when the contracture test can be done. Which of the following statements is most accurate?

A. It must be done within 48 hours of the episode, while the muscle injury is still evolving and easy to measure

B. The biopsy can be frozen and shipped to any hospital laboratory, where it can be tested several weeks later

C. It is delayed after the acute episode and uses fresh muscle tested within hours at a specialized center

D. It is no longer used anywhere, because genetic testing now detects every person who is susceptible

E. It can be done at the bedside by giving the child a small test dose of halothane under observation

209. During a discussion of malignant hyperthermia, a colleague proposes giving a calcium channel blocker because the disorder involves excess calcium. Which of the following is the best reason this would not be the specific treatment?

- **A.** Calcium channel blockers bind the ryanodine receptor but with lower affinity than dantrolene
- **B.** The cardiac calcium channel is the origin of the disorder, so skeletal muscle is unaffected by the drugs
- **C.** These drugs lower the threshold of the ryanodine receptor and would therefore trigger further calcium release
- **D.** They act too slowly in an emergency, because their half-life is much longer than that of dantrolene
- **E.** The excess calcium comes from the sarcoplasmic reticulum, not through the membrane channels these drugs block

210. A 25-year-old woman reacts to sevoflurane during surgery. The vaporizer is switched off, but the end-tidal carbon dioxide continues to rise for several minutes. Which of the following best explains why the reaction persists after the trigger is removed?

- **A.** Residual neuromuscular blockade from the relaxant is keeping the muscle contracted
- **B.** The reaction at this stage is independent of calcium and driven by heat alone
- **C.** The vaporizer is the only source of volatile agent and must be physically removed
- **D.** Tissues keep releasing volatile agent, and calcium is already loose in the muscle
- **E.** Dantrolene is needed to wash the volatile trigger out of the lungs and the circuit

211. A 36-year-old woman is scheduled for knee arthroscopy. She has had three uneventful general anesthetics. During the interview she reports dark urine after strenuous exercise, and that a cousin developed a very high fever during surgery. Which of the following is the best plan?

- **A.** Plan a trigger-free anesthetic while this is clarified, because one positive answer is enough
- **B.** Proceed with a volatile anesthetic, because her three uneventful anesthetics are reassuring
- **C.** Proceed with sevoflurane and give prophylactic dantrolene to cover the possibility of susceptibility
- **D.** Cancel the surgery permanently, because susceptibility cannot be managed safely in any hospital
- **E.** Measure creatine kinase and proceed with a volatile agent if the resting value is normal

212. A 3-year-old boy of Native American descent has hypotonia, cleft palate, and scoliosis, and is scheduled for palate repair. His condition is attributed to a variant in STAC3. Which of the following is the best anesthetic plan?

- **A.** Sevoflurane induction with rocuronium, because the risk is related only to succinylcholine
- **B.** Desflurane maintenance, because it is a less potent trigger than the other volatile agents
- **C.** Nitrous oxide with low-concentration sevoflurane, because nitrous oxide protects against a reaction
- **D.** A standard volatile technique, because the risk is confined to adults with myopathy
- **E.** A trigger-free technique with total intravenous anesthesia and no volatile agent or succinylcholine

213. A 38-year-old woman with myotonic dystrophy is given succinylcholine for induction. She develops sustained generalized stiffness that makes ventilation and intubation very difficult. Which of the following statements is most accurate?

- **A.** The stiffness is diagnostic of malignant hyperthermia, and the cause is the same ryanodine receptor defect
- **B.** Succinylcholine can cause sustained contracture in myotonia that mimics malignant hyperthermia but is distinct
- **C.** The stiffness is the normal fasciculation response and will not interfere with ventilation or intubation
- **D.** The stiffness indicates hyperkalemic periodic paralysis, which has no relation to myotonic disease
- **E.** The stiffness proves that the patient is safe to receive volatile agents, since the problem is a drug effect

214. A 22-year-old soldier had exertional heat stroke with rhabdomyolysis last summer and now needs knee ligament repair. He and his family have no known anesthetic problems. Which of the following is the most appropriate preoperative approach?

- **A.** Proceed with a standard volatile anesthetic, because exertional illness is unrelated to anesthetic reactions
- **B.** Check creatine kinase and proceed with sevoflurane if the value is normal
- **C.** Take a careful anesthetic history, discuss genetic testing, and plan a trigger-free anesthetic
- **D.** Give prophylactic dantrolene and use a volatile agent at a low concentration
- **E.** Refuse anesthesia until a muscle biopsy has been done at a specialized center

215. A surgeon asks the anesthesiologist to give prophylactic dantrolene to a susceptible patient who will receive total intravenous anesthesia. Which of the following is the best response?

- **A.** Give oral dantrolene the night before surgery to reduce the risk of a reaction
- **B.** Prophylactic dantrolene is not recommended; use the trigger-free technique and keep dantrolene available
- **C.** Give dantrolene intravenously at induction, because trace volatile in the machine can still trigger a reaction
- **D.** Give dantrolene only if the case lasts longer than two hours, because short cases are unlikely to trigger
- **E.** Give dantrolene to every susceptible patient receiving a regional technique but not a general anesthetic

216. A technician asks why simply turning the vaporizer dial to zero is not enough to prepare an anesthesia machine for a susceptible patient. Which of the following is the best answer?

- **A.** Zero on the dial eliminates all agent, so only the vaporizer needs a label to warn the staff
- **B.** Residual agent is harmless below the anesthetic range, so flushing is only needed for long cases
- **C.** Only the breathing circuit retains agent, so replacing the reservoir bag and tubing alone makes the machine safe
- **D.** Residual agent leaches from vaporizers, absorbent, and rubber circuit parts, so remove them, change the circuit, and flush
- **E.** Residual agent clears from any machine within one minute at high flow, so no other preparation is required

217. A 5-year-old boy is scheduled for strabismus surgery. His parents have read that all children with strabismus are at high risk for malignant hyperthermia. He has no weakness, and there is no family history of anesthetic problems. Which of the following is the best approach?

- **A.** Treat every child with strabismus as susceptible and give prophylactic dantrolene before every operation
- **B.** Proceed with a routine volatile technique without questions, since no child with strabismus is at risk
- **C.** Ask about weakness, hypotonia, and family history, and go trigger-free only if the answers are uncertain
- **D.** Order a muscle contracture test in every child before any strabismus surgery at a specialized center
- **E.** Cancel the surgery until genetic testing of the whole family has been completed and reviewed

218. A 30-year-old man has a transient rise in end-tidal carbon dioxide to 48 mm Hg, a heart rate of 110, and sweating after one hour of sevoflurane. The changes resolve within 10 minutes after the vaporizer is turned off and propofol is started. There is no rigidity. Which of the following is the best subsequent management?

- **A.** Dismiss the episode as light anesthesia, since the changes resolved, and take no further action
- **B.** Treat as a possible abortive reaction, check creatine kinase, and arrange testing and counseling
- **C.** Discharge the patient at once, because abortive reactions cannot occur in the absence of rigidity
- **D.** Start oral dantrolene for 24 hours and discharge home once the patient has eaten and walked
- **E.** Plan another volatile anesthetic for the next procedure, because the reaction resolved without treatment

219. A registrar asks about the clinical grading scale for malignant hyperthermia published by Larach and colleagues. Which of the following statements about it is correct?

A. It is a bedside tool that must be completed in the room before dantrolene is given in a crisis

B. It gives a definitive diagnosis, which makes the contracture test and genetic testing both unnecessary

C. It ranks likelihood from 0 to 100 percent, using only the creatine kinase and the peak temperature

D. It scores rigidity, muscle breakdown, acidosis, and temperature after the event, ranking likelihood 1 to 6

E. It is used only to decide the dose of dantrolene in children who have had a suspected reaction in theater

220. A 41-year-old woman had a fulminant malignant hyperthermia reaction. Sequencing of RYR1 and CACNA1S finds no known pathogenic variant. Clinical suspicion remains high. Which of the following is the best next step?

A. Conclude that she is not susceptible, and clear her for volatile agents in all future anesthetics

B. Repeat the same genetic panel in six months, and clear her if it is negative again

C. Measure creatine kinase, and clear her for volatile agents if the resting value is normal

D. Test her relatives for a single variant, because the same variant must exist in the family

E. Refer for contracture testing, and manage her and relatives as susceptible until then

QUESTION GROUP 12

Treating a Malignant Hyperthermia Crisis

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Questions 221 to 240. Choose the single best answer for each.

221. A 45-year-old woman is anesthetized with sevoflurane for laparoscopic cholecystectomy. Within 10 minutes of carbon dioxide insufflation the end-tidal carbon dioxide rises from 36 to 52 mm Hg. The heart rate is 82, the jaw is relaxed, the temperature is unchanged, and the absorbent is fresh. Which of the following is the best next step?

- **A.** Give dantrolene at once, because any rise in carbon dioxide under a volatile agent is malignant hyperthermia
- **B.** Convert to total intravenous anesthesia and start active cooling, because the diagnosis is presumed to be certain
- **C.** Reduce minute ventilation to allow permissive hypercapnia, because absorbed carbon dioxide is harmless here
- **D.** Stop the operation and replace the anesthesia machine, because equipment failure is the only alternative
- **E.** Suspect absorbed insufflation gas, increase minute ventilation, and watch for tachycardia, rigidity, or fever

222. An anesthesiologist suspects malignant hyperthermia in a 70-kg man whose minute ventilation is currently at the usual level. The vaporizer has been turned off. Which of the following ventilation and gas settings is most appropriate?

- **A.** Maintain the usual minute ventilation, to avoid the vasoconstriction that comes with hypocapnia
- **B.** Reduce the fresh gas flow to conserve oxygen and keep the circuit warm and humid for the patient
- **C.** Hyperventilate at two to three times usual minute ventilation with high-flow 100 percent oxygen
- **D.** Hyperventilate with a 50:50 mixture of oxygen and nitrous oxide at a high fresh gas flow
- **E.** Hyperventilate on a low fresh gas flow so that the volatile agent is rebreathed and diluted

223. During a malignant hyperthermia crisis, the attending anesthesiologist is directing treatment from the head of the bed. A nurse asks who should telephone the national hotline. Which of the following is the best response?

- **A.** A second team member, such as a nurse or colleague, so the leader stays at the head of the bed
- **B.** The leader, before dantrolene is ordered, in order to confirm the diagnosis by telephone
- **C.** No one until the crisis has ended, because the call would only distract the team
- **D.** The surgeon, once the patient has been transferred to the intensive care unit
- **E.** The pharmacist, once the full dose of dantrolene has been prepared and given to the patient

224. During treatment of malignant hyperthermia, the vaporizer is off and dantrolene is being mixed. The patient has been given a relaxant and is hypermetabolic and rigid. The resident asks how anesthesia should be maintained meanwhile. Which of the following is the best answer?

- **A.** Propofol and an opioid infusion, because a paralyzed hypermetabolic patient may be aware
- **B.** Nitrous oxide at high concentration alone, because it is safe and needs no infusion
- **C.** A reduced concentration of sevoflurane until the end-tidal carbon dioxide begins to fall
- **D.** No anesthetic drugs, because depth no longer matters when the patient is rigid and paralyzed
- **E.** Additional rocuronium to relieve the rigidity, which will also keep the patient asleep

225. During a laparotomy for a perforated appendix, a 22-year-old man shows features of malignant hyperthermia. The surgeon wants to finish the operation before the team changes anything. Which of the following is the best response?

A. Allow the operation to proceed unchanged and treat the reaction afterward, because the surgery is the priority

B. Pack the abdomen and leave it open, because wound closure is not possible during treatment of the crisis

C. Switch to desflurane so that surgery can continue under a deeper, more stable volatile anesthetic

D. Ask the surgeon to stop or finish quickly, and continue with total intravenous anesthesia during treatment

E. Ask the surgeon to continue and give half the dantrolene dose to avoid interrupting the case

226. A 22-year-old man has malignant hyperthermia at a free-standing ambulatory surgery center that stocks dantrolene. After dantrolene and cooling, his heart rate is 90, the end-tidal carbon dioxide is normal, and the temperature is 37.5 degrees Celsius. He feels well and asks to go home. Which of the following is the best management?

A. Observe in the center for 6 hours and discharge him if there is no recurrence

B. Transfer to a hospital with an intensive care unit, whatever the apparent response

C. Discharge after 2 hours with written instructions and a prescription for oral dantrolene

D. Keep him overnight in the center's recovery area with dantrolene at the bedside

E. Discharge to the care of a relative after a normal creatine kinase is documented

227. A 20-kg child in a malignant hyperthermia crisis has a potassium of 7.0 mmol/L with peaked T waves, and calcium has been given. Which of the following insulin and dextrose doses is appropriate?

A. Regular insulin 10 units with 50 mL of 50 percent dextrose, as for adults

B. Regular insulin 0.1 units/kg alone without dextrose, to limit the fluid load

C. Regular insulin 0.1 units/kg (2 units) with 25 percent dextrose 2 mL/kg

D. Regular insulin 2 units with only 4 mL of 25 percent dextrose in total

E. Regular insulin 0.5 units/kg with 25 percent dextrose 2 mL/kg given together

228. A 70-kg adult in a malignant hyperthermia crisis is to receive the first dose of nanocrystalline dantrolene, supplied as a 250 mg vial reconstituted in 5 mL of sterile water (50 mg/mL). Using 2.5 mg/kg, which of the following volumes should be given?

A. 1.75 mL, which is 87.5 mg

B. 3.5 mL, which is 175 mg

C. 7 mL, which is 350 mg

D. 5 mL, which is 250 mg

E. 35 mL, which is 1750 mg

229. A 60-kg woman is receiving the first dantrolene bolus for malignant hyperthermia through a small cannula in the dorsum of the hand. A colleague notes the solution is strongly alkaline. Which of the following is the best approach?

A. Give the drug slowly over an hour through a small hand vein to reduce irritation of the tissue

B. Mix the dantrolene in the same syringe as sodium bicarbonate to neutralize the alkalinity of the solution

C. Dilute the drug in normal saline to reduce the pH and give it through any available peripheral vein

D. Use a secure large or central vein, watch the site, and give it by a dedicated route without mixing

E. Inject it into the subcutaneous tissue near the muscle, to bring the drug close to the site of action

230. A hospital pharmacy committee asks what quantity of conventional dantrolene a facility providing anesthesia should hold and where it should be stored. Which of the following is the commonly recommended answer?

A. About 6 vials in central pharmacy, delivered on request once a reaction is suspected in a room

B. About 12 vials in the main operating suite only, with remote sites relying on transfer to hospital

C. About 36 vials or the equivalent, reachable from any anesthetizing location within 10 minutes

D. About 36 vials in central pharmacy, delivered to the anesthetizing location within 30 minutes

E. About 72 vials in the pharmacy, with a single vial kept in each operating room as a first dose

231. An 80-kg man has responded to dantrolene for malignant hyperthermia and is admitted to the intensive care unit. The intensivist asks for the maintenance dose. Which of the following is the best order?

A. 2.5 mg/kg intravenously once daily for the next three days

B. 1 mg/kg by mouth every 24 hours until the creatine kinase falls

C. 1 mg/kg intravenously every 4 to 6 hours for at least 24 hours

D. 0.1 mg/kg intravenously every hour for six doses and then stop

E. 10 mg/kg as a single intravenous dose, repeated only if the temperature rises

232. A 150-kg woman with a malignant hyperthermia crisis has a calculated ideal body weight of 55 kg. The team debates which weight to use for dantrolene dosing, and the process of mixing is already slow. Which of the following is the best approach?

A. Use actual body weight and round up, because under-dosing leaves the reaction untreated

B. Use ideal body weight to avoid an overly large dose and a long preparation time

C. Use lean body weight, because dantrolene acts only in skeletal muscle and not in fat

D. Give a fixed dose regardless of weight, because the drug is titrated to effect anyway

E. Delay dosing until the patient has been weighed on a bed scale in the operating room

233. A 28-year-old man has responded well to dantrolene for malignant hyperthermia. The carbon dioxide and temperature are normal, and the team considers extubating him in the operating room. Which of the following is the best consideration?

A. Dantrolene can cause weakness lasting hours, so support ventilation until he is clearly strong

B. Extubate now, because a normal end-tidal carbon dioxide shows that the respiratory muscles are normal

C. Give neostigmine to reverse the weakness that dantrolene causes at the neuromuscular junction

D. Give additional nondepolarizing relaxant to prevent shivering before the tube is removed

E. Extubate now, because the weakness occurs only with chronic oral dantrolene use and not intravenous use

234. A 32-year-old woman at 30 weeks of gestation develops fulminant malignant hyperthermia during an appendectomy. The surgeon is worried about fetal effects of dantrolene. Which of the following is the best response?

A. Withhold dantrolene until the baby is delivered and use cooling alone in the meantime

B. Give half the usual dose to limit placental transfer to the fetus until delivery

C. Use a calcium channel blocker instead, because it does not cross the placenta to the fetus

D. Give dantrolene only after delivery by cesarean section has been completed and the baby is stable

E. Give dantrolene to the mother, because an untreated reaction is the greater risk to the fetus

235. During a malignant hyperthermia crisis in a hypotensive patient, the axillary temperature is 37.4 degrees Celsius while an esophageal probe reads 39.6 degrees. Which of the following should guide cooling decisions?

A. The axillary reading, because it is noninvasive and is not affected by shock

B. The rectal reading, because it responds fastest to changes in core temperature

C. The skin temperature of the forehead, because it is the most accessible site

D. The esophageal reading, because a core site reflects the trend most rapidly

E. The average of the axillary and esophageal readings, to reduce measurement error

236. A 15-year-old boy in malignant hyperthermia received calcium, insulin with dextrose, and dantrolene for a potassium of 7.2 mmol/L. Twenty minutes later he is improving and the team considers the problem solved. Which of the following monitoring plans is most appropriate?

A. Repeat potassium once in 6 hours, because the patient is improving clinically and stable

B. Repeat potassium every 15 to 30 minutes at first, and glucose hourly while insulin is active

C. Repeat potassium every 4 hours and check glucose only if symptoms of hypoglycemia occur

D. Measure creatine kinase every hour in place of repeating potassium, as it tracks muscle injury

E. Stop frequent measurements once the electrocardiogram has normalized and the rhythm is stable

237. A 40-year-old man in malignant hyperthermia has a pH of 7.12 with a large base deficit and a high carbon dioxide tension. Sodium bicarbonate is given. Which of the following measures should accompany it?

A. Reduce ventilation to avoid overshoot alkalosis once bicarbonate has been given in full

B. Withhold bicarbonate if the pH is below 7.2, because it will paradoxically worsen the acidosis

C. Keep ventilation high to clear bicarbonate-derived carbon dioxide, and repeat the blood gas

D. Give bicarbonate repeatedly at fixed intervals without a repeat blood gas in between

E. Give calcium chloride in the same line to prevent the bicarbonate from precipitating

238. Thirty minutes after dantrolene, cooling, and correction of potassium and acidosis in a malignant hyperthermia crisis, a 35-year-old man remains in sinus tachycardia at 135 beats per minute. A colleague proposes a beta-blocker. Which of the following is the best response?

A. Give a beta-blocker to bring the heart rate below 100 before cooling is stopped, to protect the heart

B. Give verapamil, because it is the usual choice for persistent supraventricular tachycardia in this setting

C. Stop dantrolene, because the persistent tachycardia shows that the drug has failed to work in this patient

D. Give a vagal maneuver and observe, because a sinus rhythm at this rate is benign during recovery

E. Titrate dantrolene to effect, and use beta-blockers with caution because they can mask recurrence

239. Eight hours after a malignant hyperthermia crisis, a 27-year-old man in the intensive care unit has oozing from intravenous sites and the surgical wound, and his thighs are tense and swollen. Which of the following is the best next step?

A. Give vitamin K and observe, attributing the thigh swelling to excess intravenous fluid alone

B. Start therapeutic heparin for presumed disseminated coagulation and continue dantrolene unchanged

C. Check platelets, PT, fibrinogen, and D-dimer, transfuse by results, and request surgical review of the thighs

D. Attribute the oozing to dantrolene phlebitis at the infusion site and apply local pressure only

E. Defer all further investigation until the creatine kinase has peaked at 12 to 24 hours

240. Fourteen hours after an initial response to dantrolene in the intensive care unit, a 30-year-old woman has a rising temperature of 38.4 degrees Celsius, increasing end-tidal carbon dioxide, muscle rigidity, and a heart rate climbing to 125. Which of the following is the best response?

A. Treat as postoperative infection with antibiotics and acetaminophen, and recheck in an hour

B. Wait for a second set of readings, because a recurrence rarely occurs after the first day

C. Give a benzodiazepine for the rigidity, and continue observation without changing therapy

D. Resume a volatile agent at a low concentration to deepen the anesthetic and relax the muscle

E. Treat as recurrence: repeat dantrolene, cool, correct metabolic abnormalities, and reassess

QUESTION GROUP 13

Hypermetabolic Mimics

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Questions 241 to 260. Choose the single best answer for each.

241. A 45-year-old man taking sertraline received tramadol for pain after an open reduction of the wrist. Four hours later in the recovery unit he is agitated and sweating, with inducible ankle clonus, hyperreflexia, a temperature of 38.6 degrees Celsius, and a heart rate of 118 beats per minute. Lorazepam has settled the agitation only partly. He has a nasogastric tube in place. Which additional drug is most appropriate for moderate disease?

A. Cyproheptadine given by slow intravenous injection

B. Cyproheptadine given through the nasogastric tube

C. Bromocriptine given through the nasogastric tube

D. Acetaminophen given every six hours to lower the fever

E. Propranolol given in small intravenous doses

242. A resident asks which mechanism underlies the lead-pipe rigidity, fever, and autonomic instability that develop in a patient two days after a dose increase of an antipsychotic drug. Which explanation is most accurate?

A. Excess serotonergic stimulation of 5-HT2A receptors in the central nervous system

B. Uncontrolled calcium release from the sarcoplasmic reticulum of skeletal muscle

C. Excess catecholamine release from a tumor of chromaffin tissue

D. A raised hypothalamic temperature set point driven by circulating pyrogens

E. Blockade or loss of dopaminergic signaling in the hypothalamus and basal ganglia

243. A 58-year-old man scheduled for urgent surgery is examined before induction. He has been taking haloperidol, with the dose increased three days ago. He is now rigid, confused, and febrile, with a creatine kinase of 6,000 units per liter. No anesthetic drug has been given. His blood pressure is 150 over 90 mm Hg and he is oriented to person only. Which feature best favors neuroleptic malignant syndrome over malignant hyperthermia?

A. Onset over days after a dopamine antagonist, before anesthesia

B. An elevated creatine kinase level on the first blood sample

C. Fever with tachycardia and fluctuating blood pressure on arrival

D. Generalized lead-pipe muscle rigidity on examination

E. Metabolic acidosis with a raised lactate and hyperkalemia

244. A 72-year-old woman with Parkinson disease has just had a cystoscopy under general anesthesia. In the recovery unit she is nauseated and retching. Her morning dose of levodopa was given by mouth before surgery. She has Parkinson disease and takes carbidopa-levodopa three times daily; her last dose was this morning. The anesthesia team wants to avoid worsening her condition. Which antiemetic is the most appropriate choice?

A. Ondansetron

B. Metoclopramide

C. Droperidol

D. Haloperidol

E. Promethazine

245. A 66-year-old man is receiving linezolid for a prosthetic joint infection and returns from theater with moderate postoperative pain. The team wants to avoid precipitating serotonin syndrome. He has normal renal function and no history of opioid intolerance, and a pharmacist has flagged the interaction risk on the medication list. He is otherwise stable, with a heart rate of 80 beats per minute and normal oxygen saturation. Which analgesic is the safest choice?

A. Meperidine

B. Tramadol

C. Morphine

D. Methadone

E. Dextromethorphan

246. A 55-year-old woman taking phenelzine, a monoamine oxidase inhibitor, develops hypotension after induction of general anesthesia for a hysterectomy. Fluids have been given. The team must choose a vasopressor. Her heart rate is 62 beats per minute, the capnogram is normal, and the surgeon is waiting to begin. No opioid or antiemetic has been given yet. Which drug should be avoided because of the risk of a hypertensive reaction?

A. Ephedrine

B. Phenylephrine

C. Norepinephrine

D. Hydromorphone

E. Morphine

247. A 40-year-old man with neuroleptic malignant syndrome has a creatine kinase of 38,000 units per liter, and his potassium is 6.9 mmol/L with new peaked T waves. Urine output is falling and the urine is dark. He has been rigid and febrile for two days, has received intravenous fluids at a modest rate, and has a blood pressure of 118 over 70 mm Hg. Which is the best immediate management?

A. Start hemodialysis at once and defer other measures until it begins

B. Give fluids alone and repeat the potassium in six hours

C. Give a beta-blocker to shift potassium into the cells

D. Give dantrolene alone and wait for the potassium to fall

E. Give calcium, then insulin with glucose, and increase ventilation

248. A 35-year-old woman in thyroid storm has a heart rate of 150 beats per minute, a temperature of 39.5 degrees Celsius, and a blood pressure of 100 over 60 mm Hg. The team gives propranolol rather than another beta-blocker. She has no history of asthma or heart block. She is intubated and sedated, and an arterial line is in place. Which additional property of propranolol explains this choice?

A. It blocks the release of stored hormone from the thyroid gland

B. It stops the synthesis of new thyroid hormone within the gland

C. It displaces thyroid hormone from its plasma binding proteins

D. It reduces conversion of thyroxine to triiodothyronine

E. It binds thyroid hormone in the gut and prevents its reabsorption

249. A 50-year-old man in thyroid storm after emergency surgery is hypotensive and tachycardic. A senior colleague recommends adding hydrocortisone to the beta-blocker, thionamide, and iodine. He had a perforated viscus repaired six hours ago and has no known adrenal disease. Glucose and electrolytes are being replaced. Which rationale is most accurate?

A. It blocks the synthesis of new hormone within the thyroid gland

B. It removes thyroid hormone from the circulation through the gut

C. It displaces bound hormone from carrier proteins to speed its clearance

D. It reduces hormone conversion and covers adrenal insufficiency

E. It lowers the hypothalamic set point that is driving the fever

250. A 47-year-old woman is under general anesthesia for a thyroid lobectomy. Three hours into the case her heart rate rises to 140 beats per minute, her temperature climbs to 39.2 degrees Celsius, and end-tidal carbon dioxide rises modestly. The jaw and limbs are flaccid. The preoperative history notes that she stopped her antithyroid medication two weeks ago. Which finding most strongly supports thyroid storm over malignant hyperthermia?

A. A rising end-tidal carbon dioxide value despite increased ventilation

B. Tachycardia with hypotension that persists after fluids

C. Flaccid muscles with a history of a stopped antithyroid drug

D. Metabolic acidosis with a raised lactate on the blood gas

E. A steadily rising core temperature over the past hour

251. A 52-year-old man with a newly found adrenal mass and episodic headaches is scheduled for laparoscopic adrenalectomy in eight days. The endocrine team asks the anesthesiologist about optimal preparation. Imaging shows a 4 cm mass and plasma metanephrines are markedly elevated; he has labile hypertension and has been losing weight. He is otherwise healthy and takes no other regular medicines. Which plan best reduces the intraoperative catecholamine surge?

A. A beta-blocker started first, with an alpha blocker added on the morning of surgery

B. A beta-blocker alone for one to two weeks, with fluids restricted to prevent heart failure

C. Alpha blockade for one to two weeks, with salt and fluid restriction to limit volume overload

D. No preoperative drugs, with vasodilators held ready in the operating room

E. Alpha blockade for one to two weeks, then a beta-blocker, with salt and fluids

252. During adrenalectomy for pheochromocytoma, the surgeon ligates the tumor vein and the blood pressure falls from 150 over 85 to 68 over 34 mm Hg with a heart rate of 100 beats per minute. The patient is vasodilated and volume depleted. Blood loss has been small. The arterial line trace is falling and the urine output has dropped over the last ten minutes. Which is the best management?

A. Give phentolamine to treat the residual catecholamine effect

B. Give fluid and norepinephrine, and check the glucose level

C. Give esmolol to control the reflex tachycardia

D. Restrict fluids because of the risk of pulmonary edema

E. Give glucagon to raise the blood pressure and glucose

253. A 38-year-old woman is admitted with fever of 40.8 degrees Celsius, encephalopathy, and shock. Norepinephrine is escalated for presumed sepsis, but her pressure swings between severe hypertension and collapse. Echocardiography shows a stress cardiomyopathy. A relative mentions years of episodic headaches, sweating, and palpitations. Her lactate is 7 mmol/L and her creatinine is rising. Cultures were drawn and broad-spectrum antibiotics were started without improvement. Which is the best next step?

A. Increase the norepinephrine dose and add vasopressin to restore the blood pressure

B. Stop catecholamine pressors, start alpha blockade, cool, and support organs

C. Give dantrolene for presumed malignant hyperthermia, repeating if no response

D. Give esmolol first to treat the tachycardia and the fever before other drugs

E. Give iodine and a thionamide at once for presumed thyroid storm with steroid cover

254. A 61-year-old man emerges from a long abdominal operation. He is shivering intensely and his core temperature is 36.1 degrees Celsius after a period of rapid rewarming. End-tidal carbon dioxide has risen, the inspired value is zero, and muscle tone is normal between shivers. Her skin is cool and she has no urticaria or rash. A blood gas is pending, and the surgeon reports an uneventful operation. Which is the best approach?

A. Give dantrolene at once and begin aggressive active cooling

B. Give a thionamide and a beta-blocker for presumed thyroid storm

C. Give broad-spectrum antibiotics and fluids for presumed sepsis

D. Treat the shivering, warm gently, and follow the temperature trend

E. Give phentolamine for a presumed catecholamine surge in this patient

255. A 63-year-old woman develops severe hypotension and fever of 39 degrees Celsius two hours into a bowel resection. The skin is warm, the muscles are flaccid, potassium and creatine kinase are normal, and end-tidal carbon dioxide is unchanged. Blood gas shows a pure metabolic acidosis with a lactate of 6 mmol/L. She is hemodynamically stable on arrival in the recovery area. Which diagnosis does this laboratory pattern best support?

A. Malignant hyperthermia crisis

B. Neuroleptic malignant syndrome

C. Serotonin syndrome from drugs

D. Rebreathing from an equipment fault

E. Sepsis or catecholamine excess

256. An anesthesia machine is used for the first case on Monday morning after fresh gas flow was left running over the weekend. Soon after sevoflurane is started, the patient has airway irritation and the absorber canister feels hot to the touch. The machine was serviced last week and the absorbent color looks unchanged. The first case of the day is a short elective procedure. Which is the most likely explanation?

A. Early malignant hyperthermia raising the temperature of the whole breathing circuit

B. Volatile agent over desiccated absorbent, generating heat and carbon monoxide

C. Exhausted absorbent that has reverted to its original color after resting

D. A stuck inspiratory unidirectional valve causing rebreathing of exhaled gas

E. Inadequate fresh gas flow in a circuit that depends on adequate flow

257. During laparoscopic cholecystectomy under general anesthesia, the end-tidal carbon dioxide suddenly falls from 38 to 12 mm Hg and the blood pressure drops to 60 over 30 mm Hg with a rising heart rate. Insufflation had just begun. She was stable until insufflation began and has no known cardiac or lung disease. The pulse oximeter waveform amplitude has fallen sharply. Which explanation best fits?

A. Peritoneal absorption of insufflated carbon dioxide

B. Early malignant hyperthermia raising carbon dioxide production

C. Venous carbon dioxide embolism or tension pneumothorax

D. Extraperitoneal gas with subcutaneous emphysema

E. Exhausted carbon dioxide absorbent causing rebreathing

258. A 32-year-old man has a fever of 39.4 degrees Celsius recorded by an axillary probe 30 minutes into a general anesthetic, with a normal heart rate and unchanged end-tidal carbon dioxide. The team wants a reliable measurement before acting. No warming device is in contact with the probe. Which site is the most appropriate for confirming the true core temperature?

A. A probe in the axilla, arm adducted

B. A skin strip on the forehead

C. A skin probe taped over the chest wall

D. A probe in the lower esophagus

E. A probe taped to the thigh

259. Thirty minutes into transfusion of a red cell unit during a hip revision, a 70-year-old woman develops a temperature of 38.7 degrees Celsius, a blood pressure of 72 over 40 mm Hg, red-brown urine, and oozing from the surgical field. The unit was checked by two staff members before it was started, and no rash is present. Which is the best immediate action?

A. Stop the transfusion, return the bag to the blood bank, and treat shock

B. Slow the transfusion rate and give acetaminophen to control the fever

C. Continue the unit while giving fluids and recheck vital signs in an hour

D. Give dantrolene for presumed malignant hyperthermia and cool actively

E. Give antibiotics only and continue the unit through a second line

260. A 28-year-old man has had sevoflurane and succinylcholine for an appendectomy. Twenty minutes in, end-tidal carbon dioxide has risen to 62 mm Hg despite increased ventilation, the heart rate is 128 beats per minute, and the temperature is 37.9 degrees Celsius. The inspired carbon dioxide is zero. A colleague suggests waiting until the temperature exceeds 39 before giving dantrolene. Which is the best course?

A. Wait for a higher temperature before giving dantrolene, since fever is the defining sign

B. Send thyroid function tests and give dantrolene only if they are abnormal

C. Rule out an equipment fault completely before any drug is given

D. Treat for serotonin syndrome and avoid dantrolene until clonus is excluded

E. Stop the volatile, hyperventilate, give dantrolene, and look for mimics in parallel

QUESTION GROUP 14

Local Anesthetic Pharmacology and the Mechanism of Toxicity

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Questions 261 to 280. Choose the single best answer for each.

261. A student asks why a rapidly firing nerve or heart is blocked more strongly by a local anesthetic than the same tissue at rest. She asks why a rapid heart rate worsens the cardiac effects after an inadvertent intravascular injection. The explanation concerns how the drug binds to the channel in different states of the cycle. Which property of the drug-channel interaction best explains this use-dependent block?

A. Higher affinity for channels in the resting state than in the open state

B. Binding to the outer vestibule of the channel from the extracellular side

C. Higher affinity for open or inactivated channels than for resting channels

D. Direct inhibition of potassium channels that lengthens repolarization

E. Reduced diffusion of the drug into tissues that fire at higher rates

262. A pharmacology resident is asked how a local anesthetic reaches its binding site in the sodium channel pore. The student has read that the drug acts on the inner side of the channel pore and wonders how a charged molecule gets there. Which sequence is correct?

A. The uncharged form crosses the membrane, and the charged form binds inside

B. The charged form crosses the membrane and binds from the outside surface

C. The uncharged form binds the channel directly from the extracellular fluid

D. The charged form is carried across by a transporter and then loses its charge

E. The uncharged form blocks the channel only after being trapped by alkalosis

263. A patient receiving an unintended intravascular dose of lidocaine first reports perioral numbness and a metallic taste, becomes restless and talkative, and then has a generalized seizure. He had perioral numbness and tinnitus about 30 seconds before the seizure, and his heart rate rose to 120 beats per minute with a blood pressure of 170 over 95 mm Hg. Which mechanism best explains this early excitation rather than early depression?

A. Excitatory cortical neurons are blocked before inhibitory neurons

B. Sodium channels in the heart are blocked before those in the brain

C. Mitochondrial injury raises cerebral energy supply at low concentrations

D. Local anesthetics act as direct agonists at excitatory receptors

E. Inhibitory cortical neurons are blocked before excitatory neurons

264. A 40-year-old man has a generalized seizure after an intravascular injection of bupivacaine during a block. He is apneic and cyanotic, and a second seizure begins within a minute even though the injection has stopped. His oxygen saturation has fallen to 82 percent. The seizure was witnessed by two nurses. Which statement best explains why the next seizure is easier to provoke?

A. Hypocarbia and alkalosis raise the free drug fraction in the plasma

B. Hypoxemia alone increases the clearance of local anesthetic by the liver

C. Muscle activity during the seizure lowers the plasma concentration of drug

D. Hypercarbia and acidosis raise brain delivery and trap the charged drug

E. Acidosis converts the drug to its uncharged form so it leaves neurons

265. A 70-year-old man receives a large dose of a long-acting local anesthetic for a plexus block and is placed on continuous monitoring. He has no symptoms. The block was performed with ultrasound guidance and aspiration was negative. He weighs 85 kg and has no cardiac history. Which electrocardiographic finding in the first half hour should most concern the anesthesiologist as an early sign of channel blockade?

A. A narrow QRS complex with a rising sinus rate

B. Isolated atrial premature beats at a stable rate

C. A shortened PR interval with a normal QRS duration

D. A widening QRS complex with a slowing sinus rate

E. A new right bundle branch pattern present before the block

266. A resident states that all local anesthetics cause vasodilation at every concentration. The attending corrects this. He points out that toxic collapse is marked by profound hypotension that responds poorly to fluid. Which statement about the vascular effects of local anesthetics is most accurate?

A. Low concentrations constrict vessels and higher concentrations dilate them

B. All concentrations cause vasodilation, with ropivacaine the most potent dilator

C. All concentrations cause vasoconstriction, so toxicity never lowers pressure

D. Bupivacaine is more vasoconstrictive than ropivacaine at clinical concentrations

E. Vascular effects are absent at any dose unless epinephrine has been added

267. A 34-year-old woman has a documented reaction with rash and swelling after dental chloroprocaine. She needs a peripheral nerve block, and the team wants an agent in a different chemical family. She has no other allergies, and the surgeon wants a prolonged block for a hand procedure. Her vital signs are stable. Which agent is the best choice from a chemical standpoint?

A. Procaine, an amino-ester

B. Ropivacaine, an amino-amide

C. Tetracaine, an amino-ester

D. Benzocaine, an amino-ester

E. Chloroprocaine again at a lower dose

268. A department plans to choose between two long-acting amides for a postoperative catheter. Pharmacologists point out that three physicochemical properties explain most clinical differences among agents. The team is deciding among agents for a two-day catheter infusion, and a pharmacist notes that bupivacaine is both highly lipid soluble and highly protein bound. Which property chiefly determines the duration of the block?

A. Lipid solubility of the base

B. The dissociation constant

C. Degree of protein binding

D. Molecular size of the base

E. Plasma pseudocholinesterase activity

269. A 42-year-old woman has tumescent infiltration with lidocaine for liposuction. She is discharged home, and eight hours later she is confused and complains of perioral numbness and lightheadedness. She was cheerful at discharge with normal vital signs and has no history of heart or liver disease. Her blood pressure is 128 over 74 mm Hg on arrival. Which explanation best accounts for the late presentation?

A. Slow absorption from tumescent infiltration delays the peak

B. An inadvertent intravascular injection during the procedure

C. Rapid hydrolysis of lidocaine by plasma pseudocholinesterase

D. Accumulation of para-aminobenzoic acid from ester metabolism

E. Methemoglobinemia caused by metabolites of the infiltrated lidocaine

270. A 56-year-old man is having a procedure in which an inadvertent intravenous injection of a local anesthetic may occur. The team wants the agent with the shortest systemic effect and the lowest chance of prolonged toxicity. The patient weighs 80 kg and has stable vital signs. He has normal liver and heart function. Which agent is hydrolyzed within about a minute in adult plasma?

A. Bupivacaine

B. Ropivacaine

C. Chloroprocaine

D. Levobupivacaine

E. Mepivacaine

271. A 60-kg (lean body weight) patient is to receive a block with plain bupivacaine. The syringe contains 40 mL of a 0.5 percent solution. The attending asks the resident to state the dose in milligrams before the block begins, and the patient has normal liver and heart function. Using the commonly cited maximum for plain bupivacaine, which statement is correct?

A. The syringe holds 400 mg, which exceeds the limit of 150 mg

B. The syringe holds 100 mg, which is within the limit of 150 mg

C. The syringe holds 200 mg, which is within the limit of 225 mg

D. The syringe holds 200 mg, which exceeds the limit of 150 mg

E. The syringe holds 20 mg, which is far below the limit of 150 mg

272. An anesthesiologist plans to inject bupivacaine at 60 percent of its maximum dose for a paravertebral block. Before that, the surgeon infiltrated lidocaine totaling about 50 percent of its own maximum. The patient weighs 70 kg and has normal hepatic function, and the surgeon is infiltrating the wound at the same time. Which statement best describes the total local anesthetic exposure?

A. Toxicity is additive and the combined fractions of 1.1 exceed the safe total

B. Toxicity is independent for each agent, so each limit applies separately

C. The two agents cancel each other because they bind different sodium channel sites

D. The total is safe because lidocaine is less potent than bupivacaine

E. The total is safe because the two drugs are given at different sites

273. A 64-year-old woman is receiving an intravenous lidocaine infusion for bowel function after colectomy. The surgical team requests a large-volume plane block with bupivacaine in the recovery area. She has mild hepatic dysfunction. Her last aspartate aminotransferase was mildly raised, and the infusion has run for 12 hours dosed on ideal body weight. Which action best reduces the risk of toxicity?

A. Continue the infusion and give the full standard dose of the block now

B. Double the infusion rate to cover the added pain of the surgery

C. Give the block first and review the infusion the next morning

D. Reduce the block volume but ignore the lidocaine infusion

E. Stop the infusion before the block, or count its dose in the total

274. A resident adds epinephrine to a long-acting local anesthetic expecting a major reduction in the peak plasma concentration, as is seen with lidocaine. He asks whether the epinephrine will make a large difference to the peak level after a high-volume block. The patient weighs 75 kg and has normal liver function. Which statement about epinephrine as an additive is correct?

A. It markedly lowers peak levels of long-acting agents more than short-acting agents

B. It reduces peak levels by slowing absorption, less so for long-acting agents

C. It eliminates any risk of late toxicity as drug leaves the injected depot site

D. It increases absorption by dilating the vessels around the injection site

E. It has no effect on the peak plasma concentration of any local anesthetic

275. A 9-month-old infant under general anesthesia receives a caudal block with a long-acting amide. Ten minutes later the electrocardiogram shows a sudden widening of the QRS complex with ventricular ectopy. There is no movement or seizure. The infant weighs 8 kg and was hemodynamically stable before the block. Oxygen saturation is 99 percent. Which statement best explains why the first sign was an arrhythmia?

A. Infants cannot develop seizures, so toxicity always appears first in the heart

B. Caudal blocks are absorbed slowly, so toxicity occurs only through cardiac effects

C. Anesthesia masks early signs, and infants bind less drug and clear it slower

D. Infants clear amide agents faster than adults, concentrating the effect in the heart

E. The infant heart is affected by local anesthetics earlier than any other tissue

276. A 45-year-old man is to have intravenous regional anesthesia of the forearm. A colleague proposes using bupivacaine for its longer duration. The patient is otherwise healthy, and the surgeon expects the operation to last about 45 minutes. A resident points out that the cuff can leak or fail. He weighs 75 kg and the tourniquet equipment has been checked. Which statement about this technique is most accurate?

A. Bupivacaine is contraindicated since tourniquet failure releases drug systemically

B. Bupivacaine is preferred because its high protein binding keeps it in the limb

C. Any amide is safe because an inflated tourniquet makes systemic absorption impossible

D. Only long-acting agents are appropriate because they remain bound to the limb tissue

E. The cuff may be deflated at once because the drug is already fixed in the tissue

277. A 50-year-old woman takes fluvoxamine for depression. She is started on a continuous ropivacaine infusion after thoracotomy. On the second day she becomes drowsy with perioral tingling. Her liver function tests are normal and she takes no other interacting drugs. The pain team increased the infusion rate that morning, and her vital signs are stable. Which mechanism most likely contributed?

A. Induction of CYP3A4 speeding the breakdown of bupivacaine

B. Inhibition of plasma pseudocholinesterase slowing ester hydrolysis

C. Displacement of the drug from alpha-1 acid glycoprotein by fluvoxamine

D. Inhibition of CYP1A2 by fluvoxamine slowing ropivacaine clearance

E. Increased renal excretion of ropivacaine caused by acidosis

278. A 68-year-old man with heart failure is receiving a lidocaine infusion. His plasma levels rise unexpectedly after the cardiac output falls. His ejection fraction is 25 percent and he takes a beta-blocker. The infusion was started 24 hours ago and has not been changed. His serum albumin is normal. Which pharmacokinetic property explains the sensitivity of lidocaine clearance to liver blood flow?

A. A low hepatic extraction ratio with capacity-limited clearance

B. A high hepatic extraction ratio with flow-dependent clearance

C. Complete hydrolysis by plasma pseudocholinesterase

D. Exclusive renal elimination of the parent drug

E. Negligible protein binding in the plasma

279. A 62-year-old woman on postoperative day two continues a ropivacaine infusion with no sign of toxicity, although the same infusion rate might have caused concern in the operating room. She has normal liver and kidney function, and the pain team reviews the infusion every eight hours. She is alert and comfortable, with normal vital signs. Which physiologic change best explains this tolerance?

A. A postoperative fall in alpha-1 acid glycoprotein raises the free drug fraction

B. A postoperative rise in alpha-1 acid glycoprotein lowers the free fraction

C. Albumin becomes the sole binding protein for the drug after surgery

D. Plasma pseudocholinesterase activity rises and hydrolyzes the amide drug

E. Hepatic blood flow doubles after surgery and speeds ester clearance

280. A 40-year-old man weighing 150 kg with a lean body mass of 80 kg is scheduled for a large infiltration with lidocaine. The resident calculates the dose from total body weight. He has normal liver and kidney function and takes no interacting drugs. The surgeon plans to infiltrate a large wound area. He is otherwise healthy and the lean weight was measured. Which statement is most accurate?

A. Total body weight is correct because fat absorbs and stores the drug safely

B. Dosing should be based on age alone, since weight does not matter

C. Total weight overestimates the safe dose, so lean weight is more prudent

D. Obesity reduces the risk of toxicity, so the maximum may be exceeded

E. Lean weight should be used only for pediatric patients and never adults

QUESTION GROUP 15

Recognizing and Treating Local Anesthetic Systemic Toxicity

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Questions 281 to 300. Choose the single best answer for each.

281. A 72-year-old man taking a beta-blocker receives an epidural test dose containing a marker dose of epinephrine before a top-up of bupivacaine. No tachycardia occurs. The resident concludes that the catheter cannot be intravascular. The monitor shows a regular rhythm at 70 beats per minute. The catheter was placed an hour ago and aspirated clear of blood. Which statement is most accurate?

A. A negative marker excludes intravascular placement in every patient tested

B. Beta-blockers can blunt the marker, so a negative result is unreliable

C. The marker is more reliable in older adults than in young healthy adults

D. The marker is more reliable under general anesthesia than in awake patients

E. A brief tachycardia after injection can be dismissed as simple anxiety

282. A 55-year-old man receives an ultrasound-guided supraclavicular block with sedation. Twenty-five minutes later he becomes bradycardic and hypotensive. The nurse suggests a vasovagal reaction and plans only to observe. The cause is not clear. He has no history of heart disease or allergy, and the block itself was uneventful. Atropine has not yet been given. Which approach is most appropriate?

A. Treat local anesthetic toxicity as the leading cause until excluded

B. Treat as a vasovagal reaction and do not prepare any rescue drugs

C. Wait for a plasma local anesthetic level before starting any treatment

D. Assume myocardial infarction and defer consideration of other causes

E. Assume anaphylaxis and treat only the rash and the bronchospasm

283. A 48-year-old woman becomes hypotensive 10 minutes after a nerve block. Her heart rate is 100 beats per minute and her saturation is 94 percent. Which finding would most strongly favor anaphylaxis over local anesthetic systemic toxicity?

A. A widening QRS complex with sinus slowing

B. Perioral numbness and a metallic taste

C. A generalized seizure after a brief agitated phase

D. Urticaria with bronchospasm and facial angioedema

E. Ventricular ectopy with a progressively longer PR interval

284. A 65-year-old man develops a wide-complex bradycardia and hypotension after a large-volume block. The resident wants to send a plasma bupivacaine level and start treatment only after it returns. Lipid emulsion is available in the room. The patient is on a monitor and oxygen has been started. The team debates whether a level should guide treatment. Which statement is most accurate?

A. A plasma level returns quickly enough to guide the first minutes of care

B. A level in the toxic range is required before lipid emulsion is allowed

C. A normal level always excludes toxicity because harm is dose dependent

D. Treatment may begin only once a level exceeds a fixed toxic threshold

E. The diagnosis is clinical, and treatment never waits for the level

285. A 59-year-old woman is sedated for a shoulder block. During monitoring after the injection, the waveform on the electrocardiogram shows a taller T wave, a longer PR interval, and a slight widening of the QRS complex, while the heart rate and saturation numbers look normal. Her blood pressure is 118 over 70 mm Hg. She has not had a seizure. Which interpretation is best?

A. Normal findings after a block that need no further action

B. Artifact from the nerve stimulator or monitoring leads

C. Early signs of toxicity that can precede overt bradycardia

D. A sign of anaphylaxis that requires epinephrine now

E. Evidence of a high spinal block that requires only observation

286. A 33-year-old man has a generalized seizure after an intravascular injection of local anesthetic. His saturation is 88 percent, and he is apneic between convulsions. No airway device is in place. He weighs 80 kg, has no known cardiac disease, and the block was performed 90 seconds earlier. Suction and a bag-mask are at the bedside. Which is the best initial airway management?

A. Bag-mask ventilation with an oral airway and 100 percent oxygen

B. Immediate intubation without any medication during the seizure

C. Waiting for the seizure to stop before providing oxygen

D. Nasal cannula oxygen at low flow while the seizure is treated

E. Immediate surgical airway placement as the first step

287. A 45-year-old man continues to seize despite repeated benzodiazepine doses after a local anesthetic injection. Acidosis and hypoxemia are worsening, and the airway equipment is ready. His heart rate is 122 beats per minute, the blood pressure is 160 over 90 mm Hg, and a bag-mask is being used with 100 percent oxygen. Lipid emulsion has been requested. Which next step is most appropriate?

A. Give a large bolus of propofol regardless of the blood pressure

B. Give an intravenous lidocaine bolus as an anticonvulsant

C. Observe until the seizure stops on its own, then reassess

D. Give a small dose of succinylcholine, with the airway secured

E. Give a phenytoin load and wait for it to take effect

288. A 62-year-old woman has bradycardia at 38 beats per minute with a widened QRS complex after a plexus block. Atropine has produced no response. A trainee asks why atropine often fails here. The block was done under sedation. Her blood pressure is 78 over 40 mm Hg and the rhythm is a slow wide-complex bradycardia. A benzodiazepine is available. Which explanation is best?

A. Atropine is inactivated by circulating local anesthetic in the plasma

B. The bradycardia is caused entirely by excess vagal tone at the node

C. Local anesthetics block muscarinic receptors irreversibly in the nodes

D. Atropine is effective only after lipid emulsion has been completed

E. Sodium channel block causes conduction block atropine cannot reverse

289. A hospital audits its block rooms after a simulation exposed delays in a rescue. Hospital policy requires that equipment be available but does not specify where. Which arrangement for the lipid emulsion rescue kit is most appropriate?

A. Stored in the central pharmacy and issued only on a written request form

B. Stored in the operating room manager's office with no label or dose card

C. Stored as a single adult bag in the block room with no written dose instructions

D. Stored on a hospital ward with the resuscitation drugs for the neonatal unit

E. Stored where blocks are done, with large syringes, tubing, and a dose card

290. During a rescue, 20 percent lipid emulsion is flowing very slowly through a 22-gauge peripheral catheter in the hand while the patient remains hypotensive. Her blood pressure is 70 over 40 mm Hg and the QRS complex is wide. She is in the block room with the team present. Which action is most appropriate?

A. Dilute the emulsion with an equal volume of saline to speed flow

B. Use a larger vein or pressure bag because the emulsion is viscous

C. Stop the emulsion and give an alternative lipid such as propofol

D. Reduce the planned dose to match the slow flow rate through the line

E. Wait until the flow rate improves on its own over several minutes

291. A trainee asks how lipid emulsion works in local anesthetic systemic toxicity. The trainee has read that fatty acids and a lipid compartment in plasma are both relevant. Which statement best summarizes the proposed mechanisms?

A. It blocks the sodium channel binding site and prevents the drug from acting

B. It raises the plasma pH and speeds the hepatic metabolism of the drug

C. It acts only as a volume expander that raises central venous pressure

D. It displaces the drug from alpha-1 acid glycoprotein to speed its excretion

E. It extracts lipophilic drug from tissue and gives the heart fatty acids

292. After a successful rescue, a blood sample from a patient who received lipid emulsion returns with spurious hemoglobin, electrolyte, and creatinine values, and the laboratory reports the sample as lipemic. The laboratory notes that the plasma looks milky and cloudy. The team is reviewing the sequence of the rescue. Which statement best explains the problem and how to avoid it?

A. Lipid emulsion lowers true hemoglobin and creatinine, so results should be accepted

B. Lipemia distorts many assays, so draw samples before lipid if it causes no delay

C. The values are accurate and reflect acute renal failure caused by the lipid

D. Samples after local anesthetic toxicity are never useful for any purpose

E. Samples should be taken only after the lipid has fully cleared from the plasma

293. A 57-year-old woman with refractory collapse after a block is being considered for extracorporeal support after lipid emulsion has been given. The cardiac surgeon asks about any special concern. The perfusion team is preparing the circuit, and the call to the center was made early in the resuscitation. Compressions are continuing. Which point should be raised with the perfusion team?

A. Lipid emulsion makes extracorporeal circulation unnecessary

B. Lipid emulsion prevents any use of heparin in the circuit

C. Lipid emulsion is contraindicated in a patient already on bypass

D. Lipid can clog the oxygenator of an extracorporeal circuit

E. Lipid emulsion would shorten the circuit's lifespan only in children

294. A resident asks why lipid emulsion is recommended for cardiovascular toxicity when there are no randomized trials in humans. He asks for the rationale in plain language, and notes that a skeptical colleague has pointed out that the evidence is weak. Which answer best reflects the reasoning of most authorities?

A. Plausible mechanism, animal data, recoveries after failed care, and low harm

B. Large randomized trials in humans have shown a clear mortality benefit

C. Animal models fully reproduce clinical toxicity, so human trials are unnecessary

D. Case reports are free of publication bias, so they are fully conclusive here

E. The treatment is completely harmless, so very little evidence is required

295. A 29-year-old woman has a brief generalized seizure after a block, which stops after small increments of midazolam. She is now awake with oxygen saturation of 99 percent, a normal QRS duration, a normal blood pressure, and a heart rate of 80 beats per minute. She received oxygen by mask throughout and has no history of heart disease. Which approach to lipid emulsion is most appropriate?

A. Give no lipid and release from monitoring after about ten minutes

B. Give the full bolus and infusion immediately as a prophylactic measure

C. Observe closely with emulsion ready, and treat any hemodynamic change

D. Withhold lipid even if the QRS complex widens or hypotension develops

E. Delay the emulsion until a full cardiac arrest has actually occurred

296. A 60-year-old man has a continuous interscalene catheter infusion running on the first postoperative day. He becomes drowsy and then develops a new ventricular arrhythmia. His blood pressure is 92 over 54 mm Hg and the pump settings have not changed since placement. He takes no sedative medication. He is rousable but sleepy, and the ECG is being attached. Which is the most appropriate action?

A. Stop the infusion and treat as toxicity until proven otherwise

B. Increase the infusion rate to relieve the pain causing the agitation

C. Reduce the infusion by half and observe on the ward

D. Flush the catheter with more local anesthetic to confirm its position

E. Attribute the drowsiness to opioids and continue the infusion

297. A laboring woman has had an uneventful epidural for two hours. The midwife repositions her, and the anesthesiologist is called for a top-up of bupivacaine. She is comfortable, her vital signs are stable, and the fetal heart tracing is reassuring. The catheter was checked at placement. The attending is called to the bedside. Which precaution is most appropriate before the injection?

A. Skip aspiration because the catheter worked well earlier in the day

B. Give the entire dose as one bolus to save time in a busy room

C. Aspirate and repeat a test dose because the catheter may have moved

D. Use the highest available strength of bupivacaine for a faster onset

E. Attribute any later tingling to anxiety and continue the dose

298. A 56-year-old man who recovered from cardiovascular local anesthetic toxicity had a period of low flow before lipid emulsion was given. He is now awake on a monitored bed with normal heart rate and blood pressure. The intensive care team asks what monitoring and testing are needed. His troponin has not yet been checked. Which follow-up evaluation is most appropriate?

A. Electrolytes, gas, lactate, creatine kinase, troponin, and echo if needed

B. No further testing, since recovery is complete once the rhythm normalizes

C. A repeat plasma local anesthetic level alone, repeated every hour for a day

D. Echocardiography alone, and only if the patient later reports chest pain

E. A neurologic examination only if a seizure recurs after the first recovery

299. A 48-year-old woman had local anesthetic toxicity from an excessive dose during a plane block and recovered fully. Six months later she needs another operation for which a regional technique would be ideal. The surgeon is concerned that a general anesthetic carries its own risk for her, and the anesthetist reviews the event record. She is otherwise healthy. Which statement is most accurate?

A. She is permanently barred from every form of regional anesthesia in future

B. She may have the same dose and technique as before, without extra monitoring

C. Only general anesthesia is acceptable, with no further evaluation of the cause

D. Regional anesthesia is acceptable only after a full year without symptoms

E. She may have a block once the cause is known, using lower, slower doses

300. A department is reviewing a recent case of local anesthetic toxicity. The manager proposes disciplining the operator and filing no report. The manager notes that the patient survived and asks that the matter not be recorded. A senior anesthesiologist disagrees. The department has a patient safety committee that meets monthly. The anesthesiologist involved has requested a review. Which response best reflects the recommended approach?

A. Keep the event confidential to avoid harming the reputation of the department

B. Assign responsibility to the single operator and close the case promptly

C. Review only the dose calculation and ignore how the team responded afterward

D. Report to a registry and review dose, technique, and response without blame

E. Report the case to a registry only if the patient died or was disabled

QUESTION GROUP 16

Preventing Toxicity and Managing Block and Neuraxial Complications

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Questions 301 to 320. Choose the single best answer for each.

301. A 44-year-old man is having a continuous femoral nerve catheter placed for analgesia after knee replacement. The catheter has several side holes. The anesthesiologist aspirates gently, sees no blood, and prepares to give the full planned bolus in one push because the aspirate was negative. Which statement best describes the meaning of the negative aspiration?

A. It confirms that the catheter lies outside any vessel, so one rapid bolus is acceptable

B. It excludes intravascular placement whenever ultrasound was used to place the catheter

C. It is a useful habit but cannot exclude a vessel, so inject incrementally

D. It is informative only for single-shot needles and means nothing for multiorifice catheters

E. It is reliable provided the aspiration is repeated once before the first injection

302. A 35-year-old woman is lightly sedated and awake during a supraclavicular block. After the second increment of local anesthetic she reports a metallic taste and tingling around her mouth. Her monitors are unchanged, and about half of the planned volume has been given. The needle is in a good position on ultrasound. Which is the best next action?

A. Finish the remaining volume slowly, since symptoms are mild and monitors are stable

B. Give additional sedation so that she tolerates the rest of the injection

C. Reassure her and continue in smaller increments with longer pauses

D. Stop injecting, call for help, and prepare to treat local anesthetic toxicity

E. Move the needle tip slightly and resume the injection once symptoms settle

303. A 29-year-old man with a body mass index of 31 is scheduled for an infraclavicular block before hand surgery. The resident plans deep sedation with propofol and an opioid because the patient is anxious. The block will be done near the axillary vessels with standard monitoring in place. A colleague asks what is lost by this choice of sedation. Which is the most important safety cost?

A. Local anesthetic absorption is slowed, which delays onset and raises the dose needed

B. Ultrasound views of the vessels are degraded by the sedative effect on blood flow

C. Injection pressure rises, which increases the risk of intrafascicular placement

D. Aspiration becomes unreliable because sedation causes the veins to collapse

E. The patient cannot report early symptoms, so a monitor change may be the first sign

304. A pressure monitor is in line during an interscalene block in a 52-year-old woman who is awake with light sedation. At the start of injection the opening pressure reads above about 15 psi, although she reports no pain and the nerve appears unchanged on ultrasound. Which interpretation and action are most appropriate?

A. The reading reflects dense perineural tissue, so continue injecting slowly while watching the nerve

B. The pressure may signal intrafascicular placement, so stop injecting and reposition the needle

C. The absence of pain confirms extraneural placement, so continue at the same rate

D. The reading suggests an intravascular tip, so give an epinephrine test dose

E. The reading is expected with interscalene needles, so it can be ignored if there is no paresthesia

305. A hospital safety committee reviews a near miss in the labor unit. A nurse almost connected an intravenous antibiotic to a patient's epidural catheter because both lines looked similar and the labels were small. No harm resulted, but the committee wants the single most reliable system-level safeguard against wrong-route errors of this kind. Which measure best meets that goal?

A. A poster in every obstetric unit reminding staff to confirm the route

B. Mandatory use of an epinephrine test marker with every neuraxial injection

C. Restriction of neuraxial catheter handling to the most experienced anesthesiologists

D. Use of larger syringes for all neuraxial drugs so that mistakes are easier to see

E. Small-bore connectors for neuraxial use that cannot join intravenous tubing

306. A 30-year-old woman receives hyperbaric intrathecal local anesthetic for cesarean delivery and is placed head-down immediately after injection to improve her blood pressure. Ten minutes later she reports numb arms, difficulty speaking, and nausea, and her pressure falls. She has no fever or back pain. Which factor most plausibly contributed to this picture?

A. Cephalad spread of the hyperbaric solution favored by head-down positioning

B. Intravascular injection producing a rapid onset of seizures

C. Subdural injection with a late, patchy block that spares the sacral segments

D. Allergic reaction to the local anesthetic with bronchospasm and urticaria

E. Epidural hematoma from placement, producing back pain and leg weakness

307. A 40-year-old man receives a spinal block that spreads to the cervical segments. He becomes unresponsive and apneic with a falling blood pressure, and the team begins ventilation. A resident states that the apnea is caused by paralysis of the diaphragm, because the phrenic nerve roots have been blocked. Which statement about the mechanism of the apnea is most accurate?

A. It is mainly from direct destruction of the phrenic nerve by the local anesthetic

B. It is mainly from pulmonary edema caused by the fluid given for hypotension

C. It is mainly from medullary hypoperfusion and drug effect, not diaphragm palsy

D. It is mainly from intercostal weakness with a normally active diaphragm

E. It is mainly from respiratory depression caused by the intrathecal opioid alone

308. A 78-year-old woman is having cataract surgery under a retrobulbar block. Three minutes after injection she becomes unresponsive and apneic. Her pupils are dilated, the heart rate has fallen to 48 beats per minute, the blood pressure is 70 over 38 mm Hg, and the oxygen saturation is falling. She received no other drugs. Which is the best management?

A. Give intravenous lipid emulsion first for presumed local anesthetic toxicity

B. Give an antihistamine and corticosteroid for presumed anaphylaxis, then observe

C. Obtain urgent imaging of the brain and spine before beginning any treatment

D. Control the airway, ventilate, and support circulation until the effect fades

E. Observe closely, since the effect is minor and resolves without airway support

309. A 31-year-old woman at term has a total spinal after a catheter top-up for urgent cesarean delivery. She is apneic and pulseless. The airway is secured, compressions are under way with left uterine displacement, and epinephrine has been given. About four to five minutes have passed without return of circulation, and the neonatal team has arrived. Which is the most appropriate next step?

A. Continue compressions alone until the block has receded and then reassess

B. Deliver the fetus emergently to improve the effectiveness of compressions

C. Give lipid emulsion to treat the possibility of local anesthetic systemic toxicity

D. Delay delivery until fetal heart tones can be confirmed and documented

E. Transfer the patient to the radiology suite for imaging of the spine

310. A 70-year-old woman on enoxaparin develops new leg weakness after removal of an epidural catheter. Emergency imaging confirms a spinal epidural hematoma. The surgical team asks how long the neurologic deficit can be present before the chance of recovery is often lost. Which is the best answer?

A. Beyond about 8 to 12 hours after the onset of the deficit

B. Beyond about 48 to 72 hours after the onset of the deficit

C. Beyond about 5 to 10 days after the onset of the deficit

D. Beyond about 10 to 14 days after the onset of the deficit

E. Beyond about 4 to 6 weeks after the onset of the deficit

311. A 70-year-old woman on enoxaparin develops new leg weakness after removal of an epidural catheter placed for knee replacement. Emergency magnetic resonance imaging confirms a spinal epidural hematoma, and the surgical team is being assembled. The anesthesiologist explains to the family that timing matters. After how long from the onset of the deficit is decompression often too late to prevent permanent injury?

A. Dilute solutions remove the risk of hematoma by lowering the volume infused

B. Preserved motor function allows new weakness from a hematoma to be noticed

C. Dilute solutions prevent catheter migration into the subarachnoid space

D. Dilute solutions allow the catheter to be removed without regard to anticoagulant timing

E. Dilute solutions make scheduled neurologic checks unnecessary in anticoagulated patients

312. A 66-year-old man has a thoracic epidural infusion after thoracotomy and receives prophylactic anticoagulation. The ward nurse notes that the team uses a dilute local anesthetic solution with scheduled motor checks rather than a concentrated solution that gives a dense block. The nurse asks why. Which reason is most relevant to the early detection of a neuraxial complication?

A. Dilution removes the risk of hematoma by lowering the volume infused

B. Dilution prevents catheter migration into the subarachnoid space

C. Preserved motor function lets new weakness from a hematoma be noticed

D. Dilution allows catheter removal without regard to anticoagulant timing

E. Dilution makes scheduled neurologic checks unnecessary in anticoagulated patients

313. A 68-year-old woman with reduced renal function takes a recently started anticoagulant and needs a spinal anesthetic for hip surgery. The anesthesiologist cannot recall how long to wait after the last dose and knows the interval may depend on the agent, the dose, and renal function. Which is the best approach to setting the interval before the block?

A. Apply the interval remembered for a different anticoagulant in the same class

B. Consult the current society guideline for that agent and document the plan

C. Use one standard interval for all anticoagulants regardless of dose

D. Proceed with the block and rely on postoperative neurologic checks

E. Select the interval from the platelet count alone, ignoring the drug

314. A 62-year-old woman with diabetes has had an epidural catheter for 6 days after a laparotomy. She now has fever, local back pain, and mild leg weakness, and magnetic resonance imaging with contrast confirms an epidural abscess. The team is choosing empiric therapy and asks about the likely organism. Which organism is most common, and which empiric coverage is appropriate?

A. Oral streptococci from the operator's mouth, with oral penicillin alone

B. Candida species from the skin, with systemic antifungal treatment alone

C. Staphylococcus aureus, covering resistant staphylococci and gram-negative rods

D. Escherichia coli from the bowel flora, with narrow-spectrum gram-negative therapy only

E. Pseudomonas aeruginosa from water, with a single antipseudomonal agent

315. A 35-year-old woman develops fever, severe headache, and neck stiffness one day after a spinal anesthetic for a minor gynecologic procedure. Cerebrospinal fluid culture grows oral streptococci. The department reviews its practice to prevent a recurrence. Which preventive measure would most likely have avoided this complication?

A. Use of a smaller gauge spinal needle for all patients

B. Routine antibiotic prophylaxis before every spinal anesthetic

C. Injecting the local anesthetic more slowly into the subarachnoid space

D. The operator wearing a face mask during the procedure

E. Replacing the skin preparation with a water-based antiseptic solution

316. On the first day after elective aortic surgery with an epidural catheter in place, a 72-year-old man has painless weakness of both legs. Vibration sense is preserved. He has no back pain, fever, or coagulopathy, and the infusion has been stopped so that he can be examined. Which diagnosis best fits this pattern of findings?

A. Epidural abscess with fever and progressive deficit

B. Transient neurologic symptoms with buttock pain

C. Cauda equina syndrome from concentrated lidocaine

D. Anterior spinal artery syndrome with cord ischemia

E. Epidural hematoma with severe back pain

317. A 32-year-old man has back pain radiating to both buttocks on the day after a spinal anesthetic for knee arthroscopy. He walks without difficulty, is afebrile, and has full strength and normal sensation on examination. He has no bowel or bladder symptoms and takes no anticoagulant. Which is the most likely diagnosis?

A. Transient neurologic symptoms after spinal anesthesia

B. Epidural abscess with spreading infection and fever

C. Spinal epidural hematoma with early cord compression signs

D. Anterior spinal artery syndrome with cord ischemia

E. Cauda equina syndrome with bladder sphincter loss

318. A 54-year-old woman has persistent numbness and weakness of the hand on the second day after shoulder surgery under an interscalene block with ultrasound guidance. The block was uneventful, and she had no paresthesia during injection. The arm is in a sling with a tight dressing. Which is the most appropriate first step in the evaluation?

A. Reassure her that every deficit resolves within days and end follow-up

B. Order electromyography immediately, since it is most informative at once

C. Attribute the deficit to local anesthetic neurotoxicity and give corticosteroids

D. Repeat the block under ultrasound to see whether the deficit improves

E. Seek treatable causes such as a tight dressing, document, and refer early

319. A 58-year-old man had a paravertebral block this morning and is now under general anesthesia with positive-pressure ventilation and nitrous oxide for another procedure. He develops hypotension, rising airway pressure, falling oxygen saturation, and absent breath sounds on the blocked side. Which is the best management? The surgeon is waiting for a decision.

A. Immediate needle decompression, then a chest drain, without waiting for imaging

B. Obtain a chest radiograph first to confirm the diagnosis before any intervention

C. Increase tidal volume and add positive end-expiratory pressure to the ventilator

D. Treat as bronchospasm with inhaled albuterol and observe the response

E. Give a large fluid bolus and reassess in fifteen minutes before intervening

320. A 24-year-old man with a tibial fracture has a popliteal sciatic catheter infusion after surgery. Eight hours later he reports severe, escalating pain despite the infusion. The calf is tense, and passive stretch of the toes produces marked pain. His heart rate is rising. Which is the best response?

A. Give a bolus of local anesthetic through the catheter and reassess in the morning

B. Arrange urgent assessment for compartment syndrome, not another bolus

C. Increase the infusion concentration to achieve a denser block

D. Elevate the limb, apply a compressive wrap, and add an oral analgesic

E. Stop all limb checks until the catheter infusion has been removed

QUESTION GROUP 17

Perioperative Hypersensitivity: Mechanisms

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Questions 321 to 340. Choose the single best answer for each.

321. At a morbidity conference, a resident describes a perioperative reaction as anaphylactoid because skin tests were negative and no specific IgE was found. The patient, a 47-year-old woman, had hypotension and bronchospasm during induction and was treated with epinephrine and fluids. The chair asks which description reflects current terminology and practice.

A. Anaphylactoid remains preferred for non-IgE reactions because treatment differs

B. Anaphylactoid is reserved for reactions beginning more than an hour after exposure

C. The term is retired; classify reactions by mechanism, and treatment is the same

D. Non-IgE reactions are called intolerance and do not require epinephrine

E. Nonallergic reactions are limited to the skin and never cause hypotension

322. An allergist is explaining a positive skin test to a 34-year-old man who reacted to a drug at his second anesthetic. She notes that a reaction mediated by immunoglobulin E is usually expected on a second or later exposure and not on the first. The patient asks why a first exposure is typically uneventful. Which statement best explains this?

A. Mast cells must first mature under stem cell factor in the days after the first dose

B. Complement stores must be depleted by first exposure before a reaction can occur

C. Histamine granules are only formed after the first contact with the antigen

D. Tryptase must first accumulate in plasma to a threshold level over several days

E. Antigen-specific IgE must first be made and bound to mast cells and basophils

323. A 55-year-old man develops flushing and hypotension one minute after a rapid injection of atracurium, which appears to be his first exposure to a neuromuscular blocker. There is no bronchospasm. Intradermal testing weeks later with the drug is negative. The allergist asks about the most plausible mechanism. Which mechanism could best explain this reaction?

A. Cross-linking of preformed IgE on mast cells after earlier sensitization to the drug

B. Immune complexes of IgG and the drug activating macrophages through Fc receptors

C. Dose-dependent activation of MRGPRX2 on mast cells without prior sensitization

D. Accumulation of bradykinin because of impaired breakdown by converting enzyme

E. Beta-2 receptor blockade removing the inhibition of mast cell mediator release

324. A 48-year-old woman has hypotension, flushing, and back discomfort within minutes of the first infusion of a liposomal drug. She has no prior exposure to the product, and specific IgE testing is negative. The allergist wants to explain the reaction in terms of mast cell biology. Which mechanism is most likely?

A. Cross-linking of drug-specific IgE bound to mast cells and basophils

B. Complement activation producing C3a and C5a, which trigger degranulation

C. Direct histamine release triggered by a morphine-like opioid chemical structure

D. Accumulation of bradykinin from inhibition of its enzymatic breakdown

E. Immediate sensitization by a cross-reacting cosmetic ammonium compound

325. A basic science review describes how mast cells arise. These tissue cells are derived from bone marrow progenitors and mature in the tissues near vessels, nerves, and surfaces that meet the outside world. A student asks which signal drives this tissue maturation. Which is the best answer?

A. Stem cell factor acting on its KIT receptor on the cell surface

B. Immunoglobulin E binding to its high-affinity receptor on the cell

C. Complement fragment C5a acting on its receptor on the cell surface

D. Histamine acting on H2 receptors in the surrounding tissue

E. Platelet-activating factor acting on its receptor in the circulation

326. A 40-year-old man has a suspected anaphylactic reaction under anesthesia, and the first serum tryptase sample is drawn as soon as treatment allows. It is reported as normal. The recovery nurse asks how to schedule further samples and what a normal first value means. Which is the best answer?

A. Take one sample at 24 hours only, since tryptase has a half-life of several days

B. A baseline is unnecessary, because any value above a fixed number confirms the diagnosis

C. A normal first value excludes anaphylaxis, so no further samples are needed

D. Repeat in one to two hours and add a later baseline; normal does not exclude it

E. Repeat only at one week, when the level has reached a stable plateau

327. A lecturer contrasts the mediators stored in mast cell granules with those made only after the cell is activated. Histamine, tryptase, chymase, carboxypeptidase, heparin, and tumor necrosis factor are preformed. Which of the following is synthesized after activation from membrane lipids and thought to contribute strongly to severe hypotension and bronchospasm?

A. Chymase, a granule protease

B. Heparin, a granule proteoglycan

C. Carboxypeptidase, a granule enzyme

D. Histamine, the granule amine

E. Platelet-activating factor

328. A 59-year-old man with no known coronary disease develops severe anaphylaxis during surgery. The electrocardiogram shows ST elevation, and troponin rises afterward. Coronary angiography later shows normal coronary arteries. The cardiologist asks which mechanism best explains the cardiac findings in the setting of the allergic reaction.

A. Mast cell mediators in the heart causing coronary vasospasm

B. Right ventricular strain produced by bronchospasm alone

C. Plaque erosion caused by the platelet effect of the antibiotic

D. Hyperkalemia produced by release of potassium from mast cells

E. Pericardial inflammation caused by deposition of immune complexes

329. A 33-year-old woman is under general anesthesia with drapes covering her chest and arms and a ventilator delivering positive-pressure breaths. A reaction begins within minutes after a drug is given intravenously. A resident asks which findings are most likely to be the earliest evidence in this setting.

A. Urticaria across the chest with swelling of the lips

B. Complaints of itching of the palms and a sense of doom

C. Hypotension, tachycardia, and a rise in airway pressure

D. Fever and chills beginning within minutes of the injection

E. Hoarseness and abdominal cramping reported by the patient

330. A 44-year-old woman in the preoperative clinic had an unexplained severe reaction during a previous anesthetic, and her baseline serum tryptase is persistently raised on two occasions. She takes no regular medication. The anesthesiologist considers the underlying cause before planning her next anesthetic. Which condition should be considered?

A. Deficiency of immunoglobulin A with anti-IgA antibodies

B. Inherited glucose-6-phosphate dehydrogenase deficiency

C. Sensitization to pholcodine from a cough preparation

D. Clonal mast cell disease, often from a KIT gene mutation

E. Hereditary C1 esterase inhibitor deficiency with angioedema

331. A medical student on the operating room rotation watches a team treat anaphylaxis with epinephrine first, while the antihistamine and corticosteroid are drawn up later. The student asks why epinephrine is more effective than an H1 antihistamine plus a corticosteroid in this setting. Which is the best explanation?

A. It blocks the H1 receptor more strongly than any antihistamine can

B. It acts on alpha and beta receptors and limits mediator release

C. It destroys histamine and tryptase already released into the plasma

D. It prevents IgE from binding to mast cells and basophils in tissue

E. It inhibits the synthesis of leukotrienes in the airway only

332. A 50-year-old man recovers his blood pressure after a reaction to rocuronium, and the surgeon proposes sending him from recovery to the ward after one hour of normal readings. The anesthesiologist wants him observed longer in a monitored setting. Which immunologic point best supports longer monitored observation?

A. Epinephrine causes rebound degranulation of every mast cell as it clears

B. Ongoing release of lipid mediators and cytokines can cause relapse hours later

C. Specific IgE concentrations double during the hour after the first reaction

D. Complement activation begins only after the first hour and always causes relapse

E. Tryptase becomes active in plasma only after an hour, which causes relapse

333. An allergist reviews the intradermal test of a 41-year-old woman who reacted during anesthesia. The test was positive for a neuromuscular blocker at a high concentration, while the skin prick test was negative and the history is unclear. The allergist considers how far to trust the result. Which statement about interpreting the test is most accurate?

A. Skin tests are always positive in IgE-mediated reactions, so a positive is definitive

B. Specific IgE assays are reliable for every neuromuscular blocker, so skin tests are unneeded

C. Skin tests are unaffected by drug concentration and need no specialist interpretation

D. The basophil activation test has no role in evaluating drug reactions at all

E. Intradermal tests at high concentration can be falsely positive from histamine release

334. Five minutes after sugammadex is given to reverse rocuronium at the end of a case, a 57-year-old man becomes hypotensive with rising airway pressure. He has received sugammadex uneventfully in the past. The resident attributes the deterioration to residual neuromuscular blockade. Which statement is most accurate?

A. Sugammadex reactions occur only at first exposure, so earlier use excludes it

B. Sugammadex can cause anaphylaxis within minutes, even after prior exposure

C. Cyclodextrins cannot trigger immune reactions, so sugammadex is unlikely

D. Only neostigmine is known to cause reactions at the end of an anesthetic case

E. Reactions after reversal are caused by surgical stimulation and not by drugs

335. A 63-year-old man has a severe reaction with hypotension and bronchospasm minutes after protamine is given to reverse heparin following cardiopulmonary bypass. The team reviews his history for factors known to raise the risk of this specific reaction. Which history is recognized to increase the risk?

A. Previous long-term exposure to NPH insulin

B. A history of allergy to egg

C. A history of allergy to shellfish

D. A label of allergy to penicillin

E. A history of allergy to a sulfonamide drug

336. A 36-year-old woman had hypotension and bronchospasm within two minutes of rocuronium at induction, and a later skin test was positive. She will need another anesthetic next year. A colleague proposes that she avoid all neuromuscular blockers for life without testing. Which advice is best?

A. Avoidance of all neuromuscular blockers without testing, since cross-reactivity is universal

B. Free use of cisatracurium, since a different chemical class guarantees safety

C. Repeat rocuronium with steroid and antihistamine premedication to confirm tolerance

D. Use of any other relaxant, since only one drug was involved in the reaction

E. Testing to find safe alternatives, since cross-reactivity is common but not universal

337. A 28-year-old woman has a chart label of morphine allergy, which she describes as flushing and itching after a rapid intravenous dose, without breathing difficulty or swelling. She now needs postoperative opioid analgesia after surgery. The nurse asks how to handle the label. Which approach is best?

A. Avoid all opioids permanently, labeling her allergic to the whole opioid drug class

B. Give codeine instead, since it is structurally unrelated to morphine

C. Clarify the history, recognize histamine release, and use a synthetic opioid

D. Premedicate with a corticosteroid and give a rapid bolus of morphine

E. Skin test with morphine at high concentration to confirm the diagnosis

338. A 50-year-old man had a mild reaction of unknown cause during an earlier anesthetic and now presents for elective surgery. A colleague proposes giving a corticosteroid and an antihistamine before induction to prevent anaphylaxis and suggests that the rescue kit can then be simplified. Which response is most accurate?

A. They reliably prevent severe immediate reactions and should be given routinely

B. They prevent reactions only to neuromuscular blockers and should be used for that

C. They can replace epinephrine in the rescue kit if given before induction

D. No proof they prevent severe reactions, and they may mask early signs

E. They prevent reactions only if given at least one hour before incision

339. A 52-year-old woman with a shellfish allergy needs an iodinated contrast study and povidone-iodine skin preparation for a procedure. She is worried that iodine is the shellfish allergen and asks whether both exposures should be avoided. Which statement is most accurate?

A. Iodine is the shellfish allergen, so both exposures are contraindicated

B. Contrast is safe only after steroid and antihistamine premedication

C. Povidone-iodine is safe, but iodinated contrast must be avoided permanently

D. Shellfish allergy does not predict reaction to contrast or povidone-iodine

E. Povidone-iodine is contraindicated, but iodinated contrast may be given

340. A 35-year-old woman has a childhood label of penicillin allergy based on a remote mild rash, without hives, airway involvement, or hypotension. She will need surgical prophylaxis again within the year, and the surgeon prefers a first-line agent. Which practical step in the preoperative clinic is most useful?

A. Keep the label and use a second-line antibiotic for all future procedures

B. Give a penicillin at induction under antihistamine cover to test her tolerance

C. Add a note that she must never receive any beta-lactam antibiotic again

D. Replace the label with a diagnosis of anaphylaxis to all antibiotics

E. Arrange supervised testing or an oral amoxicillin challenge to delabel her

QUESTION GROUP 18

Recognizing and Treating Anaphylaxis Under Anesthesia

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Questions 341 to 360. Choose the single best answer for each.

341. A 33-year-old woman is under general anesthesia for an elective procedure. Moments after rocuronium is given, the peak airway pressure rises sharply, the capnogram slopes upward like a shark fin, and expiration is slow. Blood pressure and heart rate are still normal, and the skin cannot be seen under the drapes. Which interpretation is best at this point?

A. Bronchospasm, possibly early anaphylaxis, so search for the trigger

B. Esophageal intubation, so remove the tube and reintubate at once

C. Rebreathing from exhausted absorbent, so change the canister

D. Opioid-induced chest wall rigidity, so raise the inspired oxygen fraction

E. Hyperventilation by the ventilator, so lower the respiratory rate

342. Four minutes after an antibiotic is hung, a 52-year-old man under general anesthesia has a shrinking plethysmographic waveform, a falling blood pressure, and a rising heart rate. The end-tidal carbon dioxide falls while the peak airway pressure climbs. The surgeon says the field is dry. Which interpretation of this combination is most appropriate?

A. Hyperventilation by the ventilator, so reduce the minute ventilation

B. Light anesthesia, so deepen the volatile concentration to blunt the response

C. A circuit disconnection, so inspect and reconnect the breathing circuit

D. An esophageal intubation, so remove the tube and reintubate the patient

E. Falling output with airway narrowing, as in anaphylaxis, so treat promptly

343. A 60-year-old man who takes no beta-blocker has a severe reaction under anesthesia. His heart rate was 130 beats per minute earlier. After an epinephrine bolus it falls to 52 beats per minute while his blood pressure is 50 over 20 mm Hg and the plethysmographic waveform is poor. Which interpretation is best?

A. Late bradycardia signals imminent arrest, so escalate and prepare compressions

B. A sign of recovery as the heart rate normalizes, so reduce the epinephrine

C. A vagal response to surgical stimulation, so treat with atropine and observe

D. A consequence of deep anesthesia, so lighten the anesthetic and observe closely

E. A normal baroreflex to a rising blood pressure, so continue current therapy

344. A 42-year-old woman becomes suddenly hypotensive and tachycardic five minutes after induction. The anesthesiologist lifts the drape and finds the chest and arms clear of urticaria and flushing. A colleague states that anaphylaxis is therefore excluded and suggests looking for hemorrhage instead. Which response is most accurate?

A. Agree, because anaphylaxis under anesthesia always produces a rash

B. Skin signs are unreliable, and some severe reactions never produce a rash

C. Agree unless the tryptase is also elevated at the bedside

D. Clear skin indicates direct histamine release rather than an allergic reaction

E. Absence of rash means hemorrhage, so give blood before epinephrine

345. A 28-year-old woman in the preoperative holding area has no intravenous access yet when she develops stridor, wheeze, and hypotension after a medication. Help has arrived and the anaphylaxis kit is open. The nurse asks which epinephrine route, site, and dose should be used first. Which is most appropriate?

A. 0.5 mg intramuscularly into the deltoid muscle

B. 0.5 mg subcutaneously into the upper arm

C. 50 mcg intramuscularly into the anterolateral thigh

D. 0.5 mg intramuscularly into the anterolateral thigh

E. 0.5 mg of the 1 mg/mL preparation slowly intravenously

346. A 70-kg adult under general anesthesia develops hypotension and bronchospasm shortly after an antibiotic is given, and the team decides to treat anaphylaxis. The assistant offers a vial of 1 mg/mL (1:1000) epinephrine and a prefilled syringe of 100 mcg/mL (1:10,000). Intravenous access is working. Which is the best choice and dosing?

A. Use the 1 mg/mL vial undiluted, giving 0.5 mg intravenously as a single bolus

B. Use the 100 mcg/mL syringe, giving 1 mg intravenously as a single bolus

C. Use the 100 mcg/mL syringe, giving 50 mcg intravenously every 10 minutes

D. Use the 100 mcg/mL syringe, giving 50 mcg intravenously every 1 to 2 minutes

E. Dilute the 1 mg/mL vial into a liter bag and infuse it over about an hour

347. A 20-kg child under general anesthesia develops hypotension and bronchospasm shortly after an antibiotic. Intravenous access is in place, and a diluted epinephrine syringe has been prepared in advance. The anesthesiologist calculates the first titrated intravenous bolus by weight. Which dose is appropriate?

A. 2 mcg

B. 200 mcg

C. 20 mcg

D. 500 mcg

E. 50 mcg

348. A 64-year-old woman has an arterial line during surgery and develops anaphylaxis. She has received three epinephrine boluses and two liters of crystalloid. Her mean arterial pressure rises briefly after each bolus and falls back to 50 mm Hg. A second anesthesiologist has arrived. Which is the best next step?

A. Give hydrocortisone and an antihistamine and wait for their effect

B. Switch to a phenylephrine infusion and stop epinephrine

C. Start an epinephrine infusion titrated to blood pressure

D. Continue identical boluses indefinitely and defer any infusion

E. Give a second vasopressor first and leave epinephrine unchanged

349. A 38-year-old man has been stabilized after anaphylaxis with epinephrine and fluid. Blood pressure is 85 over 50 mm Hg and urticaria persists. A resident wishes to push chlorpheniramine rapidly intravenously now to treat the rash. Which response is best?

A. Give it as a rapid push, since H1 blockers reverse vasodilation and restore pressure

B. Wait until stable and give it slowly, since it does not treat shock

C. Give it in place of the next epinephrine dose to treat the urticaria

D. Give intravenous promethazine instead because it is faster and non-sedating

E. Give it with hydrocortisone before any further epinephrine

350. An intubated 25-year-old woman has severe bronchospasm during anaphylaxis, with a falling blood pressure and rising airway pressures despite epinephrine. The ventilator is set to a rate of 16 breaths per minute with an inspiratory to expiratory ratio of 1 to 2, and the expiratory flow does not return to zero before the next breath. Which adjustment is most appropriate?

A. Raise the respiratory rate to correct the hypercapnia more quickly

B. Increase the tidal volume to overcome the high airway pressure

C. Lengthen the expiratory time, for example by lowering the rate

D. Add a high level of positive end-expiratory pressure to stent the airways

E. Shorten the inspiratory time and raise the rate to keep minute ventilation

351. A 45-year-old man has persistent bronchospasm during anaphylaxis despite epinephrine and nebulized albuterol through the circuit. His blood pressure is now acceptable on an epinephrine infusion, and a volatile agent is already running. Which is the most appropriate addition for resistant bronchospasm?

A. Hydrocortisone, expecting rapid bronchodilation within minutes

B. Magnesium sulfate 2 g intravenously over 20 minutes

C. A rapid intravenous antihistamine bolus for the wheeze

D. Esmolol to slow the heart rate and reduce oxygen demand

E. A further large crystalloid bolus as the only wheeze treatment

352. A 56-year-old man has profound vasoplegia after anaphylaxis that has not responded to epinephrine, norepinephrine, and vasopressin. Methylene blue, which inhibits the nitric oxide and guanylate cyclase pathway, is considered as a rescue. The pharmacist asks which patient characteristic is the most important caution before it is given.

A. A label of penicillin allergy dating from childhood

B. Long-term treatment with a statin for cholesterol

C. A history of mild intermittent asthma, well controlled

D. Current treatment with a serotonergic antidepressant

E. Chronic use of a low-dose inhaled corticosteroid

353. A 55-year-old man with poor ventricular contractility has refractory hypotension in anaphylaxis despite epinephrine. A resident proposes adding a phenylephrine infusion as the second vasopressor. The attending asks which second agent is preferred and why. Which response is most accurate?

A. Phenylephrine, since a pure alpha agonist carries no arrhythmia risk at all

B. Neither, since phenylephrine is contraindicated as a direct mast cell trigger

C. Phenylephrine, since it improves contractility in a failing left ventricle

D. Neither, since vasopressin receptors are downregulated in anaphylaxis

E. Norepinephrine, since phenylephrine lacks inotropy and may cause bradycardia

354. A 47-year-old woman develops hypotension and wheeze minutes after a drug is given under general anesthesia. The team is debating anaphylaxis, tension pneumothorax, and a high spinal, and a senior colleague wants to wait for more information before treating. Which course is best?

A. Wait for a serum tryptase result to return before giving any epinephrine

B. Give hydrocortisone and an antihistamine first while the diagnosis is clarified

C. Obtain a chest radiograph and decide on treatment after it is reviewed

D. Treat the wheeze with albuterol alone and reserve epinephrine for arrest

E. Give titrated epinephrine now, since it is low risk if the diagnosis differs

355. A 46-year-old man has refractory shock after anaphylaxis, with a mean arterial pressure of 45 mm Hg despite epinephrine and norepinephrine infusions, vasopressin, and fluids. The hospital has an extracorporeal membrane oxygenation team available. The surgeon asks whether to wait. Which is the best action?

A. Wait until conventional measures have failed for at least an hour before calling

B. Avoid extracorporeal support, because anaphylactic arrests are never reversible

C. Transfer the patient to the ward for observation once the infusions are stable

D. Notify the extracorporeal team early while resuscitation continues

E. Give a second dose of corticosteroid and wait for the extracorporeal decision

356. A 50-year-old woman has been resuscitated from grade III anaphylaxis in the operating room, and her pressure is recovering on a small epinephrine infusion. She is being transferred to the intensive care unit. The handover team asks what the plan should be for the hours that follow. Which plan is most appropriate?

A. Stop all infusions at once, since myocardial dysfunction does not follow anaphylaxis

B. Give a diuretic prophylactically to prevent pulmonary edema from large fluid volumes

C. Check ECG and troponin, repeat echocardiography, and monitor for pulmonary edema

D. Remove monitoring once the blood pressure is normal, since the reaction is complete

E. Wean every infusion simultaneously within minutes to test hemodynamic stability

357. A 39-year-old man has a reaction to an antibiotic during surgery with a moderate fall in blood pressure and tachycardia, without severe bronchospasm, shock, or arrest. The team wants to apply the Ring and Messmer scale to decide on escalation and on the operation. Which grade best describes this reaction?

A. Grade II, a multisystem reaction that is not life threatening

B. Grade I, skin and mucosal signs only, with no organ involvement

C. Grade III, life threatening with shock, severe bronchospasm, or both

D. Grade IV, cardiac or respiratory arrest needing resuscitation

E. Grade V, death despite full resuscitation efforts

358. A team debriefs after a perioperative anaphylaxis in a 48-year-old man who recovered. The resident asks which feature of the first event best identifies a patient at higher risk of a biphasic reaction and who therefore needs longer monitored observation. Which is the best answer?

A. Presence of urticaria at the time of the first signs of the reaction

B. Prompt recovery after a single small dose of intravenous epinephrine

C. Reaction to an antibiotic rather than to a neuromuscular blocker

D. Need for repeated epinephrine doses or an infusion in the episode

E. Female sex of the patient together with a past history of atopy

359. A 44-year-old woman is awake and recovering after a grade III reaction during surgery. Her husband asks what caused it. The anesthesiologist suspects rocuronium, but investigation has not begun and tryptase samples have been sent. Which approach to communication is best?

A. Record rocuronium as her allergy now and tell her to avoid it for life

B. Say nothing until the outpatient visit, to avoid causing her any anxiety

C. Say a reaction occurred and will be investigated, naming no culprit

D. Tell her she is allergic to all anesthetic drugs, to be on the safe side

E. Tell the family only, and leave the patient unaware until discharge

360. A 30-year-old woman at 28 weeks of pregnancy has hypotension, wheeze, and fetal bradycardia on the monitor during anaphylaxis under anesthesia. The obstetric team asks about priorities while the anesthesiologist is preparing epinephrine. Which approach is best?

A. Withhold epinephrine to protect uterine blood flow and give crystalloid alone

B. Treat as a nonpregnant adult, adding left uterine displacement, without delay

C. Proceed immediately to cesarean delivery before beginning maternal treatment

D. Treat only after the fetal heart rate has been confirmed and documented on a monitor

E. Give a large antihistamine dose alone to avoid fetal effects of vasopressors

QUESTION GROUP 19

After the Event: Diagnosis and Future Anesthetics

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Questions 361 to 380. Choose the single best answer for each.

361. A 44-year-old man develops bronchospasm and hypotension after induction of anesthesia and is treated successfully. During the debrief, a colleague asks why the team sent serum tryptase and not a plasma histamine level, since histamine is the mediator responsible for the symptoms. Which of the following is the best explanation?

A. Tryptase stays raised for hours, whereas histamine returns to baseline within about 30 minutes

B. Histamine is released only in IgE-mediated reactions and would miss direct mast cell activation

C. Tryptase comes only from circulating basophils, so a raised level identifies the culprit drug

D. Histamine assays are valid only when a baseline sample is taken a week after full recovery

E. Tryptase rises faster than histamine and so allows an immediate diagnosis in the room

362. In the recovery unit, a 39-year-old woman develops wheeze and hypotension forty minutes after emergence. The anesthesiologist draws three tubes over two hours and sends them together without comment. The next day the laboratory reports a single result and explains that the other two tubes looked like duplicates and were discarded. Which of the following would most likely have prevented this problem?

A. Drawing all three tubes at the same moment so that they could be pooled into one sample for analysis

B. Sending the tubes only after the baseline value had been obtained several weeks later by the clinic

C. Using plasma histamine tubes in place of tryptase tubes for the later two draws after the first

D. Asking the ward staff to order the later tubes after the patient had left the recovery unit

E. Labeling every tube with draw time and onset time and telling the laboratory it is a timed series

363. A 29-year-old woman has sudden urticaria, bronchospasm, and hypotension minutes after an antibiotic infusion and responds promptly to epinephrine. A tryptase drawn at ninety minutes is reported as normal, and the baseline is not yet available. A resident states that anaphylaxis has been excluded. Which of the following is the best response?

A. A normal tryptase does not exclude it, as some reactions occur without a measurable rise

B. Agreed, because mast cell mediators are always raised at ninety minutes in true anaphylaxis

C. Agreed, unless a histamine level drawn the following day is also found to be normal

D. Disagreed, because a normal tryptase proves that the reaction was IgE-mediated in nature

E. Disagreed, because a normal value identifies the antibiotic as the culprit by exclusion

364. A raised tryptase is documented after a reaction in a 52-year-old man who received rocuronium, propofol, fentanyl, cefazolin, and chlorhexidine skin preparation. The surgeon asks the anesthesiologist to record rocuronium in the allergy field as the proven cause, because tryptase confirms a neuromuscular blocker reaction. Which of the following is the best response?

A. Explain that tryptase shows mast cell activation but cannot name the trigger

B. Record rocuronium, since a raised tryptase after a relaxant is diagnostic of that drug

C. Record cefazolin, since antibiotics account for most reactions with a raised tryptase

D. Record all five exposures as confirmed allergies until the baseline sample is available

E. Leave the allergy field blank, since tryptase cannot separate allergic from other mechanisms

365. A 31-year-old man undergoing elective knee arthroscopy develops flushing, urticaria, and a fall in systolic pressure to 70 mm Hg within two minutes of his first known exposure to a neuromuscular blocker. He is treated successfully with epinephrine and fluids. A tryptase drawn at one hour is clearly raised, but later skin tests and specific IgE assays are unremarkable. Which of the following mechanisms best explains these findings?

A. Complement-mediated reaction to a colloid given during induction of anesthesia

B. Delayed T cell reaction to the skin preparation applied before the incision

C. Direct mast cell activation through MRGPRX2 without prior sensitization

D. Hereditary angioedema caused by deficiency of C1 esterase inhibitor in the plasma

E. Bacterial contamination of the drug ampule causing endotoxin release into the circulation

366. A 41-year-old woman has a persistently mildly raised baseline tryptase of 11 ng/mL measured well after a severe perioperative reaction. She has no skin lesions, and her kidney function and blood count are normal. Her sister has had a similarly high value. Which of the following is the most likely explanation?

A. Systemic mastocytosis, which any value above the usual reference range confirms

B. Hereditary alpha tryptasemia caused by extra copies of the alpha tryptase gene

C. Residual mast cell activation from the reaction that will clear within days

D. Laboratory error, since inherited conditions do not change the baseline value

E. Chronic kidney disease, which is the usual cause of any raised baseline

367. A 66-year-old man receives cefazolin and rocuronium at induction for a bowel resection. Within three minutes his blood pressure is unrecordable and the airway pressure rises, and a cardiac arrest follows. Anaphylaxis is suspected and compressions are under way. A team member asks whether to wait until the patient is stable before drawing the first tryptase. Which of the following is the best advice?

A. Wait one hour so that the peak can be captured in a single well-timed sample

B. Skip the first sample, since values during arrest cannot be read against a baseline

C. Draw it only after the pH, lactate, and potassium have returned to normal values

D. Draw it from the line that delivered the suspected drug, to maximize the level

E. Draw the first sample now; a sample taken during resuscitation is still valuable

368. A 48-year-old woman recovers after a grade III reaction during hip surgery. On rounds the next day, the surgical resident says the surgical team will mention the event to the primary care physician and that no further action is needed from anesthesia. Which of the following is the best course for the anesthesiologist who managed the event?

A. Leave the referral to the surgical team, who know her best after the operation

B. Wait for her to request testing when she next needs another operation or procedure

C. Personally refer her to a drug allergy clinic, sending the record and tryptase results

D. Refer her only if the culprit can be identified from the history alone, without testing

E. Document the event in the chart and advise avoidance of all future anesthetics

369. A 57-year-old man had anaphylaxis during cholecystectomy. The anesthetic record lists rocuronium, propofol, fentanyl, and cefazolin, but omits the skin disinfectant, a blue dye used for cholangiography, and the type of gloves. The allergy clinic prepares its skin test panel from the record. Which of the following is the most likely consequence?

A. The clinic will automatically test every product used in the hospital that day

B. Substances missing from the record may go untested, leaving the culprit unidentified

C. Testing will be unaffected, since the culprit is always a neuromuscular blocker

D. The clinic will test only the omitted items, since the listed drugs are presumed safe

E. A repeat tryptase level can substitute for testing the missing items

370. A 35-year-old woman is tested in the allergy clinic eight weeks after a perioperative reaction. The technician performs intradermal testing with morphine at a concentration higher than the validated table, and a large wheal appears within minutes. The control wheals are appropriate, and the result is recorded as a positive allergy. Which of the following best describes the problem?

A. Opioids and some relaxants cause wheals through direct histamine release above validated doses

B. Opioids cause wheals only in patients who have been sensitized by earlier IgE-mediated exposure

C. Wheals at excessive concentration are false negatives that hide a true underlying allergy

D. The result should have been read after one hour and not at fifteen to twenty minutes

E. Intradermal testing should be done before prick testing whenever opioids are being tested

371. A 60-year-old man has had perioperative anaphylaxis and is referred for testing. He takes a beta-blocker for severe angina, and his cardiologist states that it cannot be stopped. He is otherwise well. The allergy clinic is planning skin testing with the drug thought most likely to have been the culprit. Which of the following is the best reason to modify the plan?

A. Beta-blockers suppress the wheal and make every skin test unreadable in practice

B. A systemic reaction to the test would be harder to treat on a beta-blocker

C. Beta-blockers cause false-positive wheals at all tested concentrations of each drug

D. Beta-blockers permanently deplete the mast cell tryptase stores in the skin

E. Beta-blockers are contraindicated for six months after any allergic reaction

372. A 72-year-old man had anaphylaxis during surgery, with a clearly raised tryptase. The allergy service considers skin testing unsafe because of his severe cardiac disease and orders specific IgE assays instead. The assay for succinylcholine returns negative, and the surgeon asks whether the drug can be used at his next operation. Which of the following is the best interpretation?

A. The negative result clears succinylcholine and allows its use in all future anesthetics

B. The negative result proves that the reaction was not allergic in mechanism at all

C. The negative result clears every neuromuscular blocker, since they share common epitopes

D. The negative result does not clear it, as sensitivity is incomplete for several drugs

E. The negative result should be repeated daily for a week before it is accepted as valid

373. A 45-year-old woman had a severe reaction when rocuronium and sugammadex were both given at the end of a long operation. Intradermal tests performed six weeks later are equivocal for both drugs, and she will need further surgery within the year. A specialized laboratory is available to the allergy service. Which of the following additional investigations is most appropriate?

A. A graded provocation challenge with rocuronium under observation

B. A plasma histamine level taken on the morning of the next operation

C. A twenty-four-hour urine collection for complement fragments

D. A repeat intradermal test at a tenfold higher concentration

E. A basophil activation test performed at a specialized center

374. An allergist is planning the final steps of evaluation for two patients. Patient A needs an antibiotic for a future operation, and skin testing with it was negative. Patient B had severe anaphylaxis, and one drug is now considered the probable culprit on history, tryptase, and skin testing. Which of the following best describes the appropriate use of graded drug provocation?

A. Use it for patient A to confirm tolerance, not to confirm the culprit in patient B

B. Use it for patient B to confirm the culprit, but not for patient A with a negative test

C. Use it for both patients, since a negative skin test is unreliable in every case

D. Use it for neither patient, since it is reserved for latex allergy testing alone

E. Use it for patient A only after specific IgE has been repeated weekly for a month

375. A 50-year-old patient arrives for a preoperative visit carrying a discharge note from another hospital that reads allergic to anesthesia, with no drug names and no mention of testing. He is anxious about the planned operation, and the anesthesiologist cannot tell which drugs should be avoided or which techniques are acceptable. Which of the following is the most appropriate action?

A. Avoid all general anesthetic drugs and offer only a regional technique for the operation

B. Accept the note as a record of allergy to every neuromuscular blocker in clinical use

C. Disregard the note, since it contains nothing that can guide the anesthetic plan

D. Ask the allergy service for a letter classifying each agent and listing alternatives

E. Proceed with the usual drugs after steroid and antihistamine premedication has been given

376. A 34-year-old woman with a documented egg allergy since childhood is scheduled for hysteroscopy under general anesthesia. She has had no previous anesthetic reactions. During the preoperative check, the nurse asks whether propofol must be replaced with another induction agent because of the allergy label. Which of the following is the best answer?

A. Propofol is contraindicated, because its lipid emulsion contains the egg allergen

B. Propofol may be used only after steroid and antihistamine premedication

C. Propofol may be used only if skin testing for soy protein is negative

D. Propofol may be used, since egg and soy allergy are not contraindications to it

E. Propofol may be used only after a graded challenge in the preoperative unit

377. A 63-year-old man with a shellfish allergy is scheduled for a contrast CT and, the following week, an operation with a povidone-iodine skin preparation. He is convinced that iodine is the allergen in shellfish and asks that both be avoided. He has never reacted to either product. Which of the following is the best response?

A. Iodine is the shellfish allergen, so contrast and povidone-iodine are both contraindicated

B. Shellfish allergy does not predict reactions to contrast or povidone-iodine

C. Contrast is contraindicated, but povidone-iodine is safe because it is applied topically

D. Both are safe only when a steroid and an antihistamine are given first as premedication

E. Povidone-iodine is contraindicated, but contrast is safe because it is intravenous

378. A 55-year-old woman with confirmed anaphylaxis to a beta-lactam is scheduled for urgent arthroplasty after a fracture. The allergy service is not available, and the team chooses vancomycin for surgical prophylaxis because it is effective against the likely organisms. The anesthesiologist is asked about administration. Which of the following is the most important precaution?

A. Give it as a rapid bolus before incision to reach a peak level quickly

B. Give it with a corticosteroid and an antihistamine to prevent anaphylaxis

C. Infuse it slowly, over at least an hour, to avoid histamine release

D. Give it through the same line that carried the neuromuscular blocker

E. Give it only after a negative specific IgE assay for vancomycin

379. A 40-year-old woman with confirmed neuromuscular blocker anaphylaxis, and no tested alternative agent yet, needs a brief diagnostic hysteroscopy next week. She has no other medical problems and a normal airway examination. Which of the following is the best anesthetic plan?

A. Use rapid sequence induction with succinylcholine, since it is chemically different

B. Use an untested benzylisoquinolinium relaxant, since cross-reactivity within the class is rare

C. Use rocuronium with sugammadex ready, since reversal removes the risk of reaction

D. Use a steroid and antihistamine premedication and then any relaxant that is available

E. Avoid relaxants, using a supraglottic airway with propofol and a short-acting opioid

380. A 68-year-old man dies despite full resuscitation after perioperative anaphylaxis during a hip operation. His wife and daughters arrive and ask for an explanation while the team is still in the room. The cause has not been established. Which of the following best describes the obligations of the anesthesia team?

A. Wait for the autopsy report before speaking to the family, and discard the used syringes and tubing

B. Delegate the conversation to the surgeon and avoid meeting the family until the review has ended

C. Give an honest account, notify risk management, preserve drugs and tubing, and sample tryptase

D. Disclose only the cause of death and leave out any uncertainty about the culprit drug or event

E. Name the most likely drug to the family at once to demonstrate transparency and honesty

QUESTION GROUP 20

Hemorrhagic Shock Physiology

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Questions 381 to 400. Choose the single best answer for each.

381. A 34-year-old woman is two hours after a vaginal delivery and continues to bleed. Her heart rate is 112 beats per minute and her systolic pressure is 102 mm Hg. The midwife notes that the blood pressure is normal and suggests waiting. Which of the following findings best supports the need for early transfusion?

A. A shock index, heart rate over systolic pressure, above 0.9 and rising

B. A systolic pressure above 100 mm Hg, which excludes significant hemorrhage

C. A hemoglobin concentration that is unchanged from the value on admission to the unit

D. A heart rate below 120 beats per minute, which indicates a class I loss

E. A shock index below 0.6, which is typical for a healthy adult at rest

382. A previously healthy 6-year-old child arrives after a fall from a height. The blood pressure is 100 over 60 mm Hg, the heart rate is 150 beats per minute, capillary refill is four seconds, and the extremities are cool. A resident says that the normal pressure shows there is no significant hemorrhage. Which of the following is the best response?

A. Hypotension appears early in children, so a normal pressure reliably excludes significant blood loss at this age

B. Tachycardia in a child reflects pain and fear and carries no real information about volume status

C. Cool extremities in a child reflect the ambient temperature and do not suggest hypovolemia in this setting

D. Capillary refill becomes slow only after the systolic pressure has already fallen well below the normal range

E. Pressure holds until a quarter to a third of volume is lost, so tachycardia and poor perfusion come first

383. A 74-year-old man with long-standing hypertension, whose usual pressure is about 170 over 90 mm Hg, is under general anesthesia for repair of a femoral fracture. The blood pressure is 105 over 60 mm Hg and the heart rate is 78 beats per minute. The surgeon says the pressure is acceptable. Which of the following is the most appropriate interpretation?

A. The pressure is normal in absolute terms, so no further evaluation is needed

B. The pressure is low relative to his baseline, so look for bleeding and consider volume

C. The heart rate of 78 beats per minute excludes ongoing hemorrhage at his age

D. The pressure is an expected finding in older adults and requires no action

E. The pressure should be left alone until the hemoglobin falls below 7 grams per deciliter

384. A 30-year-old man who is bleeding steadily from a liver laceration appeared well for twenty minutes, with a heart rate of 100 and a normal pressure, and then deteriorated within a few minutes to a systolic pressure of 60 with confusion. The surgical trainee asks why the decline was so abrupt after such a reassuring period. Which of the following is the best explanation?

A. Hemoglobin concentration fell early in the case, which warned of the coming collapse

B. Anaerobic metabolism began at once with the first loss of any blood volume

C. Cardiac output rose in proportion to the blood lost until it failed abruptly

D. Vasoconstriction protected all of the organs equally until it was overwhelmed

E. Oxygen extraction rose from about 25 toward 70 percent until the reserve ran out

385. A 25-year-old man arrives by ambulance after a shooting with hypotension and a heart rate of 130. No intravenous fluid was given before arrival. The first blood sample shows a prolonged prothrombin time, and the viscoelastic trace shows rapid clot breakdown despite a normal temperature. Which of the following best explains the coagulopathy?

A. Dilution of clotting factors by the prehospital crystalloid and the first blood products

B. Citrate binding of ionized calcium after rapid infusion of stored blood products

C. Hypothermia alone, with an otherwise normal response of the clotting cascade enzymes

D. Protein C activation and glycocalyx loss cause anticoagulation and hyperfibrinolysis

E. Heparin contamination of the arterial line flush used for the first blood sample

386. A 55-year-old man bleeding from a pelvic fracture has been resuscitated for ninety minutes. The core temperature is 33 degrees Celsius, and there is diffuse oozing from every surface. The surgeon asks the anesthesiologist how the low temperature contributes to the bleeding. Which of the following is correct?

A. Fibrinogen synthesis increases enough to offset the loss of clotting factors

B. Platelet adhesion is impaired and clotting enzymes slow below about 35 degrees

C. The oxygen dissociation curve shifts to the right and improves tissue oxygen unloading and delivery

D. Myocardial contractility increases and the risk of arrhythmia falls substantially

E. Shivering restores adequate heat production in anesthetized, paralyzed patients

387. During control of a bleeding aortic injury, a 62-year-old man has an arterial pH of 7.15 and a mean arterial pressure of 55 mm Hg despite blood products, and there is diffuse oozing. A resident proposes sodium bicarbonate to correct the coagulopathy and improve the response to vasopressors. Which of the following is the best response?

A. Give bicarbonate first, since it reliably corrects the clotting defect at this pH

B. Give large volumes of normal saline to dilute the acid load and restore pH

C. Withhold further blood products until the pH has been corrected above 7.3 by bicarbonate

D. Restore perfusion and stop the bleeding; bicarbonate does not correct the coagulopathy

E. Give bicarbonate to restore catecholamine responsiveness before any other step

388. During a laparotomy for trauma, a 36-year-old man has been receiving massive transfusion for two hours. The core temperature is 33.8 degrees Celsius, the arterial pH is 7.18, and there is clinically evident coagulopathy with oozing from the raw surfaces. The surgeon wishes to complete a lengthy definitive vascular repair. Which of the following is the best statement from the anesthesiologist?

A. The values are acceptable for now, so definitive repair can continue as planned

B. The damage control thresholds are met, so shorten the operation and pack

C. Definitive repair can continue once calcium has been replaced and fluids warmed

D. Damage control is justified only if the pH falls below 6.9 and the lactate exceeds 10

E. Damage control is justified only if the temperature falls below 30 degrees Celsius

389. After six units of red cells in thirty minutes for a ruptured spleen, a 55-year-old man's blood pressure is unresponsive to volume and phenylephrine. The electrocardiogram shows a prolonged QT interval, the surgical field is oozing, and the hemoglobin is 9 grams per deciliter. The blood gas analyzer is at hand. Which of the following is the best next step?

A. Send a serum total calcium and withhold treatment until it returns

B. Give a larger phenylephrine bolus and repeat it until the pressure responds

C. Give sodium bicarbonate to correct the prolonged QT interval

D. Slow the red cell infusion because of the risk of hyperkalemia

E. Measure ionized calcium on the blood gas and replace calcium

390. A 40-year-old man is two hours after a gunshot wound to the abdomen and is still bleeding in the operating room. Two units of red cells have been given. The viscoelastic trace shows rapid clot breakdown. A resident suggests waiting for the fibrinogen result before giving tranexamic acid. Which of the following is the best response?

A. Wait for all laboratory values, since tranexamic acid is effective at any time after injury

B. Give tranexamic acid only after three hours, once the fibrinolytic phase is complete

C. Give platelets alone, since rapid clot breakdown reflects platelet failure and not fibrinolysis

D. Give tranexamic acid now, since hyperfibrinolysis responds to it and benefit falls with time

E. Avoid antifibrinolytic drugs in trauma, and give cryoprecipitate alone instead

391. A 62-year-old man is scheduled for an elective liver resection for a large tumor near the hepatic veins, with a high likelihood of major bleeding. The anesthesia resident is preparing the room and asks when warming measures should begin, since the patient is normothermic and awake in the holding area. Which of the following is the best answer?

A. Once the core temperature falls below 35 degrees Celsius in the operating room

B. After the first blood product has been requested and issued by the blood bank

C. When the patient begins to shiver or the forehead feels cool to the touch

D. At closure of the abdomen, to avoid heat stress on the patient during the dissection

E. At induction, with forced-air warming, warmed fluids, and a core temperature probe

392. A 55-year-old man on controlled ventilation with a tidal volume of 8 mL/kg, sinus rhythm, and a closed chest becomes hypotensive during open repair of an aortic injury. The arterial waveform shows a pulse pressure variation of 18 percent with ventilation. The heart rate is 105 beats per minute and there is no right heart disease. Which of the following is the best interpretation?

A. He is likely to respond to volume, since variation above about 13 percent predicts it

B. Volume will not help, since the variation indicates fluid overload and congestion

C. The index is invalid because the tidal volume is 8 mL/kg of ideal body weight

D. The variation indicates right heart failure and calls for a vasopressor alone

E. The variation is expected with positive pressure ventilation and has no clinical meaning

393. During a spinal fusion, the surgeon estimates blood loss at 300 mL. The suction canister holds 1,400 mL, of which the circulating nurse reports about 900 mL is irrigation fluid, and the sponges and drapes appear soaked. The anesthesiologist is concerned that the blood pressure is drifting down. Which of the following is the best way to estimate the loss?

A. Weigh the sponges, subtract the irrigation, and remember estimates tend to be low

B. Accept the surgeon's visual estimate, since visual estimates are usually reliable

C. Use the full canister volume alone, ignoring the irrigation and the sponges

D. Rely on the hemoglobin concentration measured before incision at the start of the case

E. Rely on the urine output measured over the previous hour of surgery alone

394. After transfusion of two units of red cells during a myomectomy, a 48-year-old woman's blood pressure appears normal and the heart rate has settled. Arterial lactate was 2.5 mmol/L on the first gas and has risen to 4.1 mmol/L on the second gas thirty minutes later. Urine output is low. Which of the following is the best interpretation?

A. An expected effect of transfusion and warming that needs no action

B. A laboratory artifact caused by delayed processing and warming of the sample

C. Ongoing hypoperfusion, so look for bleeding and reassess resuscitation

D. A sign that the volatile anesthetic is too deep for the patient

E. Evidence that resuscitation has been adequate and perfusion restored

395. Transesophageal echocardiography in an anesthetized trauma patient with a pressure of 80 over 45 mm Hg shows a small, underfilled left ventricle with hyperdynamic contraction, walls nearly touching at end systole, and an inferior vena cava that collapses with ventilation. There is no pericardial effusion. Which of the following is the best conclusion?

A. Cardiac tamponade from an unrecognized pericardial effusion

B. Cardiogenic shock from severe left ventricular systolic failure

C. Massive pulmonary embolism with a dilated right ventricle

D. Fixed outflow obstruction from severe calcific aortic stenosis

E. Hypovolemia, although vasodilatory states can look similar

396. A 58-year-old man with uncontrolled abdominal bleeding from a ruptured aneurysm is hypotensive, and norepinephrine is started by the team while the surgeon is still gaining control. The mean arterial pressure reaches 65 mm Hg, but the lactate continues to rise and urine output falls. Which of the following best explains this finding?

A. Vasopressors can mask a volume deficit, so bleeding control and volume are needed

B. Norepinephrine is contraindicated in hemorrhage, so crystalloid should replace it

C. Urine output always recovers once the pressure reaches 65 mm Hg regardless of volume

D. Vasopressin should replace norepinephrine as the first-line agent in hemorrhage

E. A pressure of 65 mm Hg proves that tissue perfusion is adequate in every patient

397. A 50-year-old man with a bleeding abdominal aortic injury is hypotensive at 85 over 50 mm Hg with a heart rate of 130 beats per minute, and needs urgent intubation for laparotomy. One large-bore line is in place, and the surgeon is ready to prep and drape. Which of the following is the best approach to induction?

A. Give a full induction dose of propofol quickly and treat any hypotension afterward with fluid

B. Have blood ready, draw up a vasopressor, and give a reduced dose of a stable agent slowly

C. Induce first and request blood from the blood bank once the pressure begins to fall

D. Give a full dose of thiopental to ensure amnesia and blunt the stress response of intubation

E. Delay induction until the blood pressure has returned to normal without giving blood

398. A 36-year-old man in profound shock from multiple injuries is receiving high-dose norepinephrine. The radial arterial pressure reads low, yet the carotid pulses are strong, the extremities are warm, and the readings do not fit the clinical picture. The team asks how to obtain a reliable pressure. Which of the following is the best next step?

A. Increase the vasopressor dose until the radial mean pressure exceeds 80 mm Hg

B. Place a cuff on the same arm and treat the cuff value as the true arterial pressure

C. Consider a femoral arterial line, as radial pressure may underestimate central pressure

D. Ignore the arterial waveform and rely on the measured central venous pressure alone

E. Remove the arterial line, since pressure cannot be measured reliably in shock by any method

399. A team prepares for rapid infusion of blood through standard sets for a patient with a ruptured aneurysm, with pressure bags placed on the blood containers and a rapid infusion device nearby. A nurse asks what the most important safety precaution is when using pressurized systems. Which of the following is the best answer?

A. Remove air from bags and tubing, and never leave a pressure bag on an empty container

B. Use pressure bags only on crystalloid and never on blood products, colloids, or platelets

C. Use pressure bags only when a central venous catheter or a rapid infuser is in place

D. Avoid blood warmers because they generate air bubbles in the tubing and the filter

E. Raise the bag pressure to the maximum to ensure continuous flow through the catheter

400. A 45-year-old man in hemorrhagic shock after trauma has a central venous pressure of 12 mm Hg and a blood pressure of 78 over 40 mm Hg. A resident argues that the central venous pressure shows adequate filling and proposes withholding blood. Which of the following is the best response?

A. Central venous pressure above 8 mm Hg reliably excludes hypovolemia in bleeding patients

B. Central venous pressure is more accurate than arterial pressure for guiding blood replacement

C. Central venous pressure predicts fluid responsiveness better than pulse pressure variation

D. Central venous pressure is useful only in spontaneously breathing patients and in children

E. Central venous pressure predicts volume status poorly and should not guide this decision

QUESTION GROUP 21

Massive Transfusion and Balanced Resuscitation

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Questions 401 to 420. Choose the single best answer for each.

401. A trauma anesthesiologist is called to the operating room for a 30-year-old man with a ruptured liver after a motor vehicle collision. A resident says he will activate the massive transfusion protocol only after ten units of red cells have been given in 24 hours, because that is the definition of massive transfusion. Which of the following is the best response?

A. That definition is correct, and earlier activation only wastes blood and staff effort in practice

B. That definition is retrospective, so activate on predicted need from mechanism and physiology

C. That definition is correct, but five units is a safer trigger in clinical practice

D. Activate only when the hemoglobin falls below 7 grams per deciliter on two samples

E. Activate only when the fibrinogen concentration falls below 100 mg/dL on repeat testing

402. A 33-year-old driver with blunt abdominal injury has a heart rate of 104 beats per minute and a systolic pressure of 88 mm Hg. The focused ultrasound examination of the abdomen is positive. The emergency physician says no scoring tool applies because the mechanism was blunt. Which of the following is the best response?

A. The score is 1, which is below the threshold, so activation should wait for laboratory results

B. The score is 3, because a heart rate above 100 beats per minute counts toward it

C. The score is 0, because a blunt mechanism excludes the patient from scoring

D. The Assessment of Blood Consumption score is 2, which predicts massive transfusion

E. The score is 4, because all four features are present in this patient

403. The massive transfusion protocol has just been activated for a bleeding trauma patient, and the blood bank calls back to confirm the order. A new team member asks the anesthesiologist what the first cooler should contain and what will happen afterward if the bleeding continues. Which of the following is the best answer?

A. Twelve red cell units, with plasma and platelets sent only after the laboratory results return

B. Six red cell units and a liter of crystalloid, with further coolers sent on individual request only

C. Six red cells, six plasma, and one apheresis platelet unit, with later coolers sent until canceled

D. Six plasma units and six platelet units, with red cells released only after crossmatching

E. Two units of each component, with every later cooler requiring a new call to the blood bank

404. Two hours into a massive transfusion, the surgeon reports that bleeding is controlled. The 41-year-old patient is hemodynamically stable with a fibrinogen of 220 mg/dL, a platelet count of 90,000 per microliter, and a prothrombin time ratio of 1.3. Another cooler is on its way. Which of the following is the best action?

A. Continue the coolers as a precaution until the patient reaches the intensive care unit

B. Deactivate the protocol, return unused products, and move to goal-directed therapy

C. Continue fixed-ratio transfusion until the hemoglobin exceeds 10 grams per deciliter

D. Stop all further blood products for the rest of the admission

E. Keep the coolers coming until the viscoelastic trace is completely normal

405. A 34-year-old woman has a postpartum hemorrhage after cesarean delivery, with ongoing uterine bleeding and a blood pressure of 95 over 55 mm Hg. The fibrinogen concentration returns at 180 mg/dL. A colleague says this is acceptable because it is above the lower limit for a nonpregnant adult. Which of the following is the best interpretation?

A. It is acceptable, because it exceeds the lower limit for a nonpregnant adult

B. It reflects hemodilution from crystalloid that will correct spontaneously with red cells

C. It is low for pregnancy, where fibrinogen normally rises, and signals consumption

D. It is falsely low because of citrate anticoagulant carried over in the sample tube

E. It reflects platelet dysfunction from the hemorrhage and not a fibrinogen problem

406. A cooler of red cells for a bleeding patient was taken out of its transport container and left on the counter for ninety minutes while the team stabilized the patient and dealt with a monitor failure. The nurse now asks whether the units can be given. Which of the following is the best management of these red cells?

A. Return or discard them, since the limit outside a validated cooler is often 30 minutes

B. Transfuse them, since red cells remain safe for several hours at room temperature on the counter

C. Transfuse them after warming them briefly in a microwave to body temperature

D. Transfuse them if the labels have been checked by a second person at the bedside

E. Transfuse them if the units look normal on visual inspection of the bag and color

407. During a journal review, a reviewer notes that early military studies of plasma to red cell ratios were weakened by a methodological problem, which a later prospective observational study of major trauma transfusion tried to address by accounting for time. Which of the following was the problem?

A. Exclusive enrollment of children, whose physiology differs from that of adults

B. Use of whole blood in the control group, which distorted the comparison of ratios

C. Survivor bias, since longer survivors were more likely to receive more plasma

D. Absence of any plasma administration in either of the groups that were compared

E. Follow-up limited to the first hour after arrival, which missed late deaths

408. A trainee reviewing the first cooler for a trauma patient notices six red cell units, six plasma units, and a single apheresis platelet unit. The trainee worries that the cooler is unbalanced because only one platelet unit accompanies six red cell units, and asks whether more should be requested at once. Which of the following is the best explanation?

A. Platelets are intentionally given at one-sixth of the ratio of red cells in all protocols

B. The platelet unit is added only to replace the fibrinogen lost with the plasma units

C. The cooler is incomplete and further platelets should be requested immediately from the blood bank

D. One apheresis unit equals roughly six pooled units, so the cooler approximates a 1:1:1 ratio

E. The platelet dose is based on body weight and is unrelated to the red cell ratio in the cooler

409. A student reading about balanced resuscitation asks about the composition of blood reconstructed from one unit each of red cells, plasma, and platelets in the usual proportions, compared with fresh whole blood. The instructor wants the student to understand why patients still develop coagulopathy. Which of the following best describes the reconstituted product?

A. A hematocrit of about 45 percent, a platelet count of about 250,000, and factor activity near normal

B. A hematocrit of about 38 percent, a platelet count of about 150,000, and factor activity of about 90 percent

C. A hematocrit of about 29 percent, a platelet count of about 80,000, and factor activity of about 65 percent

D. A hematocrit of about 20 percent, a platelet count of about 30,000, and factor activity of about 40 percent

E. A hematocrit of about 33 percent, a platelet count of about 120,000, and factor activity of about 50 percent

410. A 38-year-old man with a traumatic brain injury and ongoing massive bleeding from a pelvic fracture has a platelet count of 70,000 per microliter after several balanced coolers. A colleague notes that the usual target of 50,000 has been met and suggests that no more platelets are needed. Which of the following is the best response?

A. Withhold platelets, since the usual target of 50,000 has already been reached

B. Give platelets only if the count falls below 50,000 on the next measurement

C. Give cryoprecipitate instead, since it corrects the platelet count in massive bleeding

D. Give platelets only to a target of 20,000 in order to limit donor exposure

E. Give platelets, since above 100,000 is advised in brain injury or massive bleeding

411. A 48-year-old man was injured four hours ago in a fall from a roof and is still bleeding from a pelvic injury. Crystalloid has been given, but tranexamic acid has not. A resident proposes giving it now because it is part of the protocol. Which of the following is the best response?

A. Do not start it now, since benefit falls with time and it may harm beyond three hours

B. Give it now, since the benefit is the same at any time after the injury has occurred

C. Give it now at double the usual dose to offset the delay in starting the treatment altogether

D. Give it now, but only after the fibrinogen has been replaced with cryoprecipitate

E. Give it only after the bleeding has stopped, in order to prevent later rebleeding

412. After the initial coolers, a 44-year-old woman with a liver injury continues to bleed. The point-of-care results are fibrinogen 210 mg/dL, platelets 42,000 per microliter, prothrombin time ratio 1.3, and hemoglobin 8.0 grams per deciliter. Using targets from European guidance, which of the following is the abnormal parameter requiring replacement?

A. The fibrinogen, which is below the target of 150 to 200 mg/dL

B. The platelet count, which is below the target of 50,000 per microliter

C. The prothrombin time ratio, which is above the target of 1.5

D. The hemoglobin, which is below the target of 10 grams per deciliter

E. The prothrombin time ratio, which should be below 1.0 before surgery continues

413. A hospital transfusion committee considers replacing much of its plasma with viscoelastic-guided fibrinogen concentrate and prothrombin complex concentrate, as used in many European centers. The committee asks the anesthesiologist among its members to summarize the strength of the evidence for such a change. Which of the following best summarizes the evidence?

A. It is proven superior, since a multicenter trial showed lower mortality than conventional care

B. It is harmful, since concentrates caused more thrombosis than plasma in every trial reported

C. It replaces all clotting factors found in plasma, so no plasma is ever needed in any bleeding patient

D. It is a reasonable alternative, since a multicenter trial found no difference in outcome

E. It applies only to pediatric patients and has not been studied in adults or in trauma

414. A 57-year-old woman in a massive transfusion after a ruptured ectopic pregnancy has an ionized calcium of 0.85 mmol/L and a falling blood pressure. Only a peripheral intravenous catheter is available, and central access will take several minutes to establish. Which of the following is the best treatment?

A. Calcium chloride through the peripheral vein, because it has more elemental calcium per volume

B. Calcium gluconate peripherally, in larger volume because it has less elemental calcium

C. Withhold calcium until central access has been safely established in the neck

D. Sodium bicarbonate through the peripheral vein to raise the ionized fraction of calcium

E. Observe, since hypocalcemia produces clinical symptoms only in awake patients

415. During a massive transfusion in a 45-year-old man, a nurse connects a calcium chloride infusion at a Y-site on the same tubing that is carrying citrated red cells, and within minutes the line clots and the flow stops. The anesthesiologist is asked how this could have been prevented. Which of the following would have prevented this problem?

A. Mixing the calcium directly into the red cell bag before the infusion starts

B. Switching to calcium gluconate in the same tubing, since it does not interact with citrate

C. Reducing the infusion rate of the red cells while the calcium is being given

D. Warming the calcium solution to body temperature before giving it in the same tubing

E. Infuse the calcium through a separate port, or flush the tubing between the two

416. A 3-kg neonate requires large volumes of blood after a cardiac laceration during a surgical procedure. The blood bank offers units of different ages, and a central line has been placed with its tip in the right atrium. The team asks for the best approach to transfusion in this infant. Which of the following is the best approach?

A. Use the oldest irradiated units first so that fresher units are saved for others

B. Give red cells as a rapid bolus through the central line with its tip in the right atrium

C. Withhold calcium until the low ionized calcium produces visible clinical symptoms

D. Use fresher or washed red cells, give calcium early, and infuse slowly peripherally

E. Treat the neonate as a small adult, since the metabolic risks are the same in both

417. Six hours after a massive transfusion for a ruptured aneurysm, a 52-year-old man in the intensive care unit has an arterial pH of 7.52 with a high bicarbonate concentration and a normal lactate. He is hemodynamically stable and the bleeding has stopped. Which of the following is the most likely cause?

A. Early bicarbonate given for acidosis, which has been retained

B. Lactate accumulation from persistent hypoperfusion and tissue ischemia

C. Potassium leaked from the older stored red cells that were transfused

D. Hyperchloremic acidosis from large volumes of normal saline infusion

E. Metabolism of citrate from transfused blood to bicarbonate

418. A hospital transfusion service is considering adding low-titer group O whole blood to its protocol for patients whose blood type is unknown. A physician asks why low anti-A and anti-B titers matter when the product may be given to patients of other blood groups. Which of the following is the best explanation?

A. The low titer reduces the citrate load and lowers the calcium requirement

B. The low titer removes platelets and so lowers the risk of bacterial contamination

C. The low titer allows storage at room temperature for several weeks

D. The low titer eliminates the risk of Rh sensitization in women of childbearing age

E. The low titer reduces the risk of hemolysis when the recipient is not group O

419. A 62-year-old man is undergoing open aortic repair with cell salvage in use. The surgeon places thrombin-soaked microfibrillar collagen on an oozing suture line and continues to use the same suction that feeds the salvage reservoir. The anesthesiologist notices this. Which of the following is the best action?

A. Continue salvage, since the washing step removes these agents completely

B. Continue salvage, since the leukocyte depletion filter traps these agents

C. Continue salvage, with extra heparin added to the reservoir to prevent clotting

D. Pause salvage or use a separate suction device while these agents are in use

E. Continue salvage, with antibiotic solution added to the reservoir to prevent infection

420. After a prolonged resuscitation with citrated products for a ruptured aneurysm, a 49-year-old man has persistent ventricular ectopy and an incomplete response to calcium, although the ionized calcium has been corrected. The blood gas analyzer also reports glucose and ionized magnesium. Which of the following additional abnormalities should be sought?

A. A high magnesium level from the magnesium content of stored plasma

B. A low magnesium level from citrate binding and intracellular loss

C. A high phosphorus level from the additive solution in the stored red cells

D. A high sodium level from the plasma component of the transfused products

E. A low chloride level from the large volume of saline that was given

QUESTION GROUP 22

Coagulopathy and Hemostatic Therapy

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Questions 421 to 440. Choose the single best answer for each.

421. A 58-year-old man oozes diffusely during an aortic operation. The prothrombin time and activated partial thromboplastin time, returned after an hour, are only mildly abnormal, yet the whole-blood viscoelastic tracing shows weak clot strength and rapid dissolution of the clot. The surgeon asks why the laboratory results did not explain the oozing. Which of the following is the best explanation?

A. They reflect only the early plasma phase of clotting, not platelets or fibrinolysis

B. They are run at a temperature lower than normal, which falsely corrects the factor activity

C. They measure the final strength of the clot but are insensitive to a factor deficiency

D. They estimate the fibrinogen concentration directly, which was normal in this patient

E. They are affected mainly by the platelet count, which was adequate in this patient

422. A 60-year-old man with cirrhosis and an INR of 2.2 is scheduled for an elective abdominal procedure. He has no active bleeding, and his platelet count and fibrinogen are acceptable. A resident proposes giving plasma to bring the INR below 1.5 before the incision. He takes no anticoagulants and has had no prior bleeding with procedures. Which of the following is the best response?

A. Give plasma, since a raised INR reliably predicts surgical bleeding in this setting

B. Give plasma, since it carries little risk of volume overload in patients with cirrhosis

C. Give vitamin K intravenously, since the raised INR reflects a warfarin effect

D. Withhold prophylactic plasma, since the INR predicts bleeding poorly in cirrhosis

E. Give cryoprecipitate, since the fibrinogen is certain to be low when the INR is raised

423. An 81-year-old woman with heart failure and chronic kidney disease receives plasma rapidly during bleeding in a revision hip arthroplasty. Forty minutes later she becomes dyspneic, with hypertension, tachycardia, distended neck veins, bilateral crackles, and a markedly raised natriuretic peptide. Her oxygen saturation has fallen to 88 percent on supplemental oxygen, and she has no urticaria or wheeze. Which of the following is the best treatment?

A. Slow the transfusion, give a loop diuretic, and support oxygenation

B. Give epinephrine for presumed anaphylaxis to donor plasma proteins

C. Start broad-spectrum antibiotics for presumed contamination of the product

D. Give a vasopressor for presumed hypotension from an acute hemolytic reaction

E. Give an antihistamine and resume the transfusion at the usual rate

424. A 49-year-old woman receives apheresis platelets for oozing during a liver resection. Near the end of the infusion she develops rigors, a temperature of 39.2 degrees Celsius, tachycardia, and falling blood pressure. There is no rash or wheeze, and the urine is clear. The unit was issued four hours earlier from room temperature storage, and cultures are not yet available. Which of the following is the best management?

A. Give an antihistamine and an antipyretic, then resume the infusion at a slower rate

B. Give a diuretic and slow the infusion, since circulatory overload is the likely cause

C. Stop the infusion, culture patient and product, give antibiotics, and notify the blood bank

D. Give epinephrine and complete the infusion, since anaphylaxis is the likely cause

E. Give a vasopressor and continue the infusion, since hypotension is expected under anesthesia

425. A 35-year-old woman with known IgA deficiency and anti-IgA antibodies had bronchospasm and hypotension with a previous red cell transfusion. She now needs elective surgery with possible transfusion. Which of the following is the best product strategy?

A. Leukoreduced units, which remove the plasma proteins responsible for the reaction

B. Washed cellular products or units from IgA-deficient donors if available

C. Irradiated units, which destroy the antibodies responsible for the reaction

D. Standard crossmatched units given after prophylactic antihistamine and steroid

E. Fresh whole blood units, which contain less IgA than stored red cells

426. A 72-year-old man with renal impairment has a generalized seizure on arrival in the intensive care unit after cardiac surgery with cardiopulmonary bypass, during which a high-dose tranexamic acid regimen was used. Glucose, sodium, and calcium are normal. He has been hemodynamically stable, the electrocardiogram shows sinus rhythm, and a head computed tomography scan shows no hemorrhage or infarct. Which of the following is the most likely mechanism?

A. Cerebral arterial thrombosis caused by profound inhibition of fibrinolysis

B. Hyponatremia from impaired free water excretion produced by the drug

C. Heparin rebound after bypass causing microhemorrhage in the cortex

D. Antagonism of inhibitory receptors in the brain by the drug itself

E. Local anesthetic toxicity from a regional block placed before the operation

427. A 30-year-old man is brought to the operating room four hours after a motor vehicle collision with continuing hemorrhage from a pelvic fracture. No antifibrinolytic has been given, and the surgeon asks that tranexamic acid be started now to improve the odds. His hemoglobin is falling, and the team has already given red cells, plasma, and platelets in balanced amounts. Which of the following is the best response?

A. Give it now, since benefit is greatest when the drug is started late in the course

B. Give the infusion alone now, since the initial dose adds little in late hemorrhage

C. Give it once lysis has been shown on a viscoelastic tracing, whatever the elapsed time

D. Give a reduced dose only, since late administration retains benefit but raises toxicity

E. Withhold it, since benefit is lost and harm is possible beyond three hours

428. A 64-year-old man bleeds after aortic valve surgery. The Clauss fibrinogen is 110 milligrams per deciliter and the platelet count is adequate. Cryoprecipitate would take 20 to 30 minutes to thaw, and the pharmacy has fibrinogen concentrate. The surgeon is reluctant to wait, the temperature is 36.8 degrees Celsius, and the ionized calcium is normal. Bleeding is diffuse and no surgical source is evident. Which of the following is the best choice?

A. Fibrinogen concentrate, since it mixes in minutes and cryoprecipitate is noninferior

B. Plasma, since it contains enough fibrinogen to correct this level quickly

C. Cryoprecipitate after thawing, since concentrate has proved inferior in cardiac surgery

D. Apheresis platelets, since the fibrinogen deficit reflects platelet dysfunction

E. Recombinant activated factor VII, since it replaces the missing fibrinogen

429. A 33-year-old woman with sepsis from a perforated viscus oozes from every puncture site. Her fibrinogen and platelet count are low, and source control has been achieved. The viscoelastic tracing shows weak clot strength without increased lysis. Her temperature and calcium are corrected, and she is receiving broad-spectrum antibiotics and norepinephrine. Which of the following is the most appropriate hemostatic approach?

A. Give tranexamic acid first to stop the consumption of clotting factors

B. Give recombinant activated factor VII to stop the diffuse oozing

C. Replace fibrinogen, platelets, and factors as needed and withhold an antifibrinolytic

D. Give vitamin K and observe, since the oozing should resolve with time

E. Give prothrombin complex concentrate, since it replaces the consumed factors in a small volume

430. A 45-year-old man is bleeding after trauma laparotomy. The surgeon has controlled the source, and the patient is now warm, with a normal pH and a normal ionized calcium. Viscoelastic testing shows a low fibrinogen amplitude, and the clotting time is mildly prolonged. No antifibrinolytic has been given. The platelet count is 160,000 per microliter. Which of the following is the best next step?

A. Give prothrombin complex concentrate first to correct the prolonged clotting time

B. Give recombinant activated factor VII before any other hemostatic product is tried

C. Give plasma at 10 to 15 milliliters per kilogram and defer fibrinogen replacement

D. Give tranexamic acid and bring fibrinogen to target before other factor products

E. Give apheresis platelets first, since the extrinsic amplitude is likely to be low

431. A 52-year-old woman has refractory bleeding after cardiac surgery. The surgical source has been excluded, and she is warm, with a normal pH, normal ionized calcium, a fibrinogen above target, and an adequate platelet count. The surgeon requests recombinant activated factor VII as a last resort. Which of the following is the most important risk to discuss before giving it?

A. Hyponatremia with seizures from antidiuretic activity of the drug

B. Severe hypertension from pulmonary vasoconstriction in the lungs

C. An increase in arterial thrombotic events such as stroke

D. Neutralization of the drug by anti-IgA antibodies in the plasma

E. Hemolysis from complement activation by the product given

432. A 66-year-old man on day 7 of heparin after vascular surgery has a fall in platelet count from 280,000 to 90,000 per microliter and a new deep venous thrombosis. His central line is being flushed with heparinized saline. He has been receiving heparin for prophylaxis, and the platelet count had been normal on the first day. The team asks about the plan for anticoagulation. Which of the following is the best management?

A. Reduce the heparin dose and give platelets to restore the count

B. Stop all heparin, including flushes, and start a non-heparin agent

C. Change to low-molecular-weight heparin, which does not cause the condition

D. Give protamine and resume heparin once the platelet count recovers

E. Continue heparin and add warfarin to treat the new thrombosis

433. A 57-year-old man with end-stage kidney disease and a missed dialysis session has persistent surgical-site oozing. The platelet count, INR, and viscoelastic fibrinogen amplitude are normal. Uremic platelet dysfunction is suspected. Which of the following is the most appropriate drug?

A. Protamine, watching for hypotension and pulmonary vasoconstriction

B. Fibrinogen concentrate, watching for hypofibrinogenemia and bradycardia

C. Idarucizumab, watching for rebound anticoagulation and hypokalemia

D. Desmopressin, watching for hyponatremia and tachyphylaxis

E. Tranexamic acid, watching for seizures and thrombosis

434. A 70-year-old woman received enoxaparin 5 hours ago and now has significant bleeding requiring urgent laparotomy. The anesthesiologist considers reversal. Which of the following is the best approach?

A. Protamine at 1 milligram per milligram of enoxaparin, expecting complete reversal of the effect

B. Protamine at 1 milligram per milligram of enoxaparin, accepting only partial reversal

C. Intravenous vitamin K, expecting reversal of the anti-factor Xa activity within minutes

D. Idarucizumab, expecting the antibody fragment to bind enoxaparin and restore clotting

E. Plasma at 10 to 15 milliliters per kilogram, expecting neutralization of the anti-factor Xa effect

435. A 76-year-old man taking apixaban, with the last dose 3 hours ago, has a life-threatening gastrointestinal hemorrhage and needs emergency surgery. The hospital pharmacy confirms that andexanet alfa is no longer marketed in the United States and that four-factor prothrombin complex concentrate is stocked. His creatinine is 1.8 milligrams per deciliter and he weighs 82 kilograms. Which of the following is the best reversal plan?

A. Idarucizumab as the specific antidote for the factor Xa inhibitors

B. Protamine sulfate to neutralize the factor Xa inhibition directly

C. Plasma alone, since it completely reverses the direct oral anticoagulants

D. Intravenous vitamin K, which restores the factors inhibited by the oral Xa inhibitors

E. Four-factor prothrombin complex concentrate according to the local protocol

436. A 72-year-old man taking aspirin has an acute spontaneous intracerebral hemorrhage. Before neurosurgical review, a resident asks to transfuse platelets to reverse the antiplatelet effect. Which of the following is the best response?

A. Transfuse platelets, since aspirin inhibits the platelets for their whole lifespan

B. Give protamine, since it neutralizes the platelet inhibition from aspirin

C. Do not transfuse routinely, since a randomized trial showed harm in this setting

D. Give four-factor prothrombin complex concentrate to restore platelet function

E. Give recombinant activated factor VII to restore platelet function in this setting

437. A 68-year-old woman received alteplase for a massive pulmonary embolism two hours ago and now has an expanding retroperitoneal hematoma after a femoral puncture, requiring urgent surgery. The fibrinogen level is very low. The infusion is still running. The heart rate is 124 beats per minute, the hemoglobin is falling, and the platelet count and temperature are acceptable. Which of the following is the best management?

A. Wait for the short half-life of alteplase to clear before giving any treatment

B. Give protamine to neutralize the thrombolytic and restore normal clotting

C. Stop the infusion, give cryoprecipitate, and give an antifibrinolytic

D. Give vitamin K and plasma to replace the depleted vitamin K dependent factors

E. Give four-factor prothrombin complex concentrate to replace the depleted plasminogen

438. A 74-year-old man on rivaroxaban received four-factor prothrombin complex concentrate for a gastrointestinal bleed 6 hours ago and is now stable. He needs hip fracture fixation, and the surgeon requests a spinal anesthetic because the anticoagulant has been reversed. He is hemodynamically stable with a hemoglobin of 9.4 grams per deciliter and a normal creatinine. The surgeon states that the drug effect is gone and that the case is urgent. Which of the following is the best response?

A. The reversal restored hemostasis, so a spinal anesthetic may proceed immediately

B. A normal INR confirms the absence of drug effect, so a spinal may proceed

C. An epidural catheter is safer than a single injection after reversal of the drug

D. Proceed with a spinal after a delay of four hours from the reversal agent

E. Reversal does not shorten the interval, so general anesthesia is often safer

439. A 69-year-old man is hemodynamically stable on postoperative day 2 after hip arthroplasty. He has no chest pain, electrocardiographic changes, or other symptoms, and the hemoglobin is 8.8 grams per deciliter. The intern requests two units of red cells to bring the hemoglobin above 10. His hemoglobin was 9.0 a day earlier and has not fallen, and his heart rate and blood pressure are normal. Which of the following is the best response?

A. Do not transfuse, since a restrictive threshold of 7 to 8 applies in stable patients

B. Transfuse two units, since a threshold of 10 lowers mortality after orthopedic surgery

C. Transfuse one unit, since the circulating hemoglobin lags behind loss in this setting

D. Transfuse one unit, since the level will fall further without any active bleeding

E. Transfuse one unit, since asymptomatic anemia after orthopedic surgery impairs wound healing

440. A 63-year-old man on aspirin and clopidogrel bleeds after coronary bypass. The surgeon sees no surgical source. The standard viscoelastic tracing, with normal clotting time and amplitude, was reported as normal. Platelet inhibition from the drugs is suspected. The chest tube output is 250 milliliters over the past hour, the platelet count is 150,000 per microliter, and the fibrinogen is normal. Which of the following is the best next step?

A. Accept the tracing as proof that platelet function is normal and look only for a surgical source

B. Repeat the standard tracing in a freshly filled tube to unmask the drug effect

C. Send an anti-factor Xa level to quantify the antiplatelet effect

D. Give protamine, since residual heparin after bypass is the likeliest cause

E. Run a platelet mapping assay, since standard assays miss P2Y12 inhibitor effects

QUESTION GROUP 23

Obstetric and Trauma Hemorrhage

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Questions 441 to 460. Choose the single best answer for each.

441. A 26-year-old woman delivered vaginally 20 minutes ago and is calm and quiet. Her heart rate is 114 beats per minute and her blood pressure is 106 over 62 mm Hg. Weighed drapes and counted sponges give a quantified blood loss of 800 milliliters. A nurse notes that the pressure is normal and no action is needed. Which of the following is the best interpretation?

A. Early hemorrhage; a shock index above one is a warning, so call for help and treat now

B. Stable physiology, since hypotension must be present before hemorrhage can be diagnosed

C. Overestimated loss, since weighed drapes tend to exceed the true amount of bleeding

D. Anxiety with normal blood loss, to be managed with reassurance and observation

E. Normal postpartum finding, since the heart rate stays elevated for days after delivery

442. A 31-year-old woman has bright red bleeding after a forceps delivery of a large infant. The uterus is firm and well contracted, the placenta is intact, and the bleeding is steady despite an oxytocin infusion. Her heart rate is 112 beats per minute and the quantified blood loss is 900 milliliters and rising. Which of the following is the most appropriate next step?

A. Give carboprost for presumed uterine atony

B. Give a second rapid oxytocin bolus

C. Place an intrauterine balloon for tamponade

D. Give misoprostol rectally for presumed uterine atony

E. Examine the lower genital tract for lacerations

443. A 29-year-old woman has uterine atony after delivery. After 1.5 liters of loss, her heart rate is 125 beats per minute and her blood pressure is 88 over 50 mm Hg. A colleague prepares to push a large dose of oxytocin as a rapid intravenous bolus. Two large-bore lines are in place and blood is being requested. Which of the following is the best response?

A. Give a small dose slowly, then an infusion, since a bolus can cause hypotension

B. Proceed with the bolus, since a faster injection produces more rapid uterine tone

C. Hold the oxytocin until volume resuscitation has normalized the blood pressure

D. Replace the oxytocin with a rapid intravenous bolus of methylergonovine

E. Switch to carboprost intramuscularly, since oxytocin is contraindicated when hypotension is present

444. A 35-year-old woman with severe preeclampsia and a blood pressure of 168 over 108 mm Hg has uterine atony after delivery that persists despite oxytocin and uterine massage. She has no asthma. Which of the following drugs should be avoided?

A. Carboprost given intramuscularly

B. Misoprostol given sublingually

C. Tranexamic acid given intravenously

D. Methylergonovine given intramuscularly

E. Oxytocin given by slow intravenous infusion

445. A 38-year-old woman has atony that persists after oxytocin, methylergonovine, carboprost, and tranexamic acid. Blood loss is 2.5 liters, the blood pressure is 82 over 45 mm Hg despite transfusion, and the uterus remains boggy. Interventional radiology is available in the building. A second anesthesiologist is present, the massive transfusion protocol is active, and the obstetrician is available for surgery. Which of the following is the best next step?

A. Transfer her to interventional radiology for uterine artery embolization

B. Repeat the same uterotonic drugs in sequence for another hour

C. Place an intrauterine balloon while the obstetrician prepares surgical options

D. Continue uterine massage alone and observe until she is stable

E. Give additional crystalloid and wait for the fibrinogen result before escalating

446. A 32-year-old woman with a labor epidural has brisk postpartum bleeding. Her heart rate is 130 beats per minute, the systolic pressure is 85 mm Hg, and the fibrinogen is falling. The obstetrician plans uterine exploration and repair. She is awake and comfortable, the platelet count is 110,000 per microliter, and the operating room is ready with blood and a warmer. Which of the following is the best anesthetic plan?

A. Extend the epidural with a full-dose local anesthetic bolus to reach a high sensory level

B. Convert early to general anesthesia, with a reduced-dose ketamine or etomidate induction

C. Continue the epidural alone to avoid any airway risk in the bleeding patient

D. Add a spinal injection to extend the block quickly through the existing catheter

E. Convert early to general anesthesia using a full-dose propofol induction and fentanyl bolus

447. During a cesarean delivery for suspected placenta accreta, a trainee objects to cell salvage because the blood may contain amniotic fluid. The anesthesiologist considers whether the device may be used. Which of the following is the most accurate statement?

A. Cell salvage is not acceptable because washing cannot remove amniotic fluid

B. Cell salvage is acceptable only when the patient refuses donor blood products

C. Cell salvage is acceptable only once the hemoglobin has fallen below seven

D. Cell salvage is acceptable only when no filter is used on the return line

E. Cell salvage is acceptable in obstetrics with a leukocyte-depletion filter

448. A 36-year-old woman at 33 weeks with two prior cesarean deliveries presents to a small hospital with vaginal bleeding. Ultrasound suggests placenta previa with features of placenta percreta. The hospital has a limited blood supply and no interventional radiology or intensive care unit. She is hemodynamically stable. She has had no labor and no contractions. Which of the following is the best plan?

A. Proceed to immediate cesarean at the local hospital and remove the placenta

B. Admit her locally and wait for spontaneous labor before planning delivery

C. Discharge her with bleeding precautions and schedule delivery at term locally

D. Stabilize her and arrange transfer to a center with a multidisciplinary team

E. Start oxytocin and induce labor to deliver before the bleeding worsens

449. During cesarean hysterectomy for placenta percreta, blood loss reaches 3.5 liters, and the patient is hypotensive despite rapid infusion. Transfusion is behind the loss, and the surgeon is dissecting the bladder. Cell salvage is in use, and a massive transfusion protocol has been active for an hour. The room is warm and calcium has been given. Which of the following is the best request from the anesthesiologist?

A. Ask the surgeon to attempt manual removal of the placenta to speed hemostasis

B. Ask the surgeon to continue and treat the hypotension with a vasopressor infusion alone

C. Ask the surgeon to finish the dissection before any further products are given

D. Ask the surgeon to pause and pack the field while volume is caught up

E. Ask the surgeon to speed up and complete the hysterectomy before more blood is lost

450. A 30-year-old woman at 36 weeks who uses cocaine has severe abdominal pain, a rigid uterus, and modest vaginal bleeding. The fetal tracing is currently reassuring. Her fibrinogen is 130 milligrams per deciliter and her platelet count is 140,000 per microliter. She requests a labor epidural. She is anxious, in pain, and eager for analgesia. Which of the following is the best plan?

A. Place the epidural now, since the platelet count is acceptable for neuraxial blockade

B. Place the epidural after plasma alone, since the fibrinogen level is only borderline

C. Avoid neuraxial blockade while coagulopathic and replace fibrinogen early

D. Place a spinal instead, since a smaller needle lowers the risk of a hematoma

E. Place a combined spinal-epidural to allow rapid conversion to general anesthesia

451. A 33-year-old woman with a prior myomectomy is in labor when she develops sudden severe abdominal pain, loss of the fetal heart rate tracing, and hypotension with tachycardia. Which of the following is the most likely diagnosis?

A. Uterine rupture

B. Placenta previa

C. Amniotic fluid embolism

D. Uterine inversion

E. Uterine atony

452. A trauma surgeon asks the anesthesiologist what the large randomized trial comparing plasma, platelets, and red cells in a 1 to 1 to 1 ratio with a 1 to 1 to 2 ratio found. Which of the following is the most accurate summary?

A. A significantly lower 30-day mortality with the 1 to 1 to 1 ratio, driven by fewer deaths from sepsis

B. Significantly higher mortality with the 1 to 1 to 1 ratio, driven by transfusion reactions and overload

C. No mortality difference, but better hemostasis and fewer early exsanguination deaths with 1 to 1 to 1

D. More hemostasis and fewer exsanguination deaths with the 1 to 1 to 2 ratio at the same mortality

E. No difference in hemostasis or exsanguination deaths between the two ratios, with equal mortality

453. A 27-year-old man with a stab wound to the abdomen and a suspected iliac vein injury has hypotension. The only functioning access is a large catheter in the femoral vein, and blood is being infused through it. The surgeon has just opened the abdomen and found a large retroperitoneal hematoma, and the patient is cold and tachycardic. Which of the following is the best action?

A. Continue infusing through the femoral line, since flow is greatest in large leg veins

B. Add large-bore access above the diaphragm and use a warmed rapid infuser

C. Place a second femoral line to double the infusion rate to the circulation

D. Place a tibial intraosseous line to give blood while keeping the leg accessible

E. Rely on a central line placed through the femoral vein since it spares the arm veins

454. A 44-year-old man with multiple gunshot wounds to the abdomen has been packed by the surgeon after 50 minutes. His temperature is 34 degrees Celsius, the pH is 7.12, and diffuse microvascular bleeding persists. The surgeon wishes to continue with bowel anastomoses and definitive repairs. The surgical team is ready to continue. Which of the following is the best response from the anesthesiologist?

A. Support the long operation and plan to correct the coagulopathy once repairs are done

B. Advocate temporary closure and transfer to the intensive care unit for resuscitation

C. Support definitive repair, since contamination of the abdomen is worse than bleeding

D. Request that the patient stay in the operating room until the temperature normalizes

E. Continue the definitive repairs while giving calcium and warm fluids to offset the triad

455. A 24-year-old man with a stab wound to the chest has lost cardiac activity in the trauma bay a few minutes after arrival, and the injury is penetrating. The trauma surgeon is considering resuscitative thoracotomy. His airway has been secured and compressions are being given. The injury occurred minutes ago. Which of the following best describes its role?

A. It is indicated for blunt trauma after prolonged absence of a pulse

B. It replaces blood transfusion in a patient with refractory hypotension

C. It is indicated mainly for stable patients with pelvic bleeding

D. It is reserved for penetrating injury with recent loss of cardiac activity

E. It is indicated for any patient with a pulseless rhythm regardless of the mechanism

456. A 36-year-old woman with blunt trauma has a blood pressure of 70 over 40 mm Hg. The team is preparing blood, and a colleague suggests starting a norepinephrine infusion as the main treatment for the hypotension. She is cold and tachycardic with a hemoglobin of 7.2 grams per deciliter, and the operating room is five minutes away. Blood is being warmed at the bedside. Which of the following is the best approach?

A. Use a continuous vasopressor infusion as the main therapy and defer blood products

B. Use a vasopressor only as a short bridge during induction while blood is given

C. Withhold vasopressors and blood until the source of bleeding has been located

D. Use a vasopressor infusion to reach a normal pressure before any blood is given

E. Start vasopressin in place of blood, since it has fewer effects on tissue perfusion

457. A 25-year-old man has a cervical spinal cord injury after a fall, with no other source of bleeding found. His blood pressure is 78 over 40 mm Hg and his heart rate is 48 beats per minute despite fluid administration. Imaging shows a C5 fracture, and the abdominal ultrasound and chest radiograph show no free fluid or hemothorax. Which of the following is the best next step?

A. Give repeated large crystalloid boluses until the heart rate rises

B. Accept a systolic pressure near 80 to avoid disrupting clot

C. Start a vasopressor, since neurogenic shock requires more than volume

D. Activate the massive transfusion protocol for presumed hemorrhagic shock

E. Give atropine alone and avoid vasopressors to protect the injured cord

458. A 78-year-old man taking metoprolol falls and fractures his pelvis. His heart rate is 72 beats per minute and his blood pressure is 105 over 60 mm Hg. His lactate is 5.1 millimoles per liter, the base deficit is large, and the pulse pressure on the arterial line is narrowing. His extremities are cool. Which of the following is the best interpretation?

A. Probable occult shock; the heart rate is unreliable, so use lactate and base deficit

B. Stable physiology, since the heart rate and pressure are within acceptable limits

C. Beta-blocker effect without shock, so transfusion should be withheld for now

D. Early sepsis, so antibiotics should be given before any blood products are considered

E. Pain response only, so give analgesia and recheck the vital signs in an hour

459. A 40-year-old man with a penetrating abdominal injury in a rural area faces a helicopter transfer of about three hours to a trauma center. His systolic pressure is 85 mm Hg. The paramedic asks whether to keep permissive hypotension for the entire transfer. He has had one unit of red cells and is receiving oxygen, and there is no head injury. Which of the following is the best advice?

A. Maintain a systolic pressure of 80 for the whole transfer, since clot protection outweighs hypoperfusion

B. Give crystalloid to reach a systolic pressure of 120 to prevent an oxygen debt

C. Give blood products and adjust the target upward, since prolonged hypoperfusion is unsafe

D. Withhold all fluids and blood until arrival, since any infusion would disrupt the clot

E. Use vasopressors continuously to hold the pressure at 80 during the transfer

460. A 50-year-old man with a liver laceration has been packed by the surgeon, and bleeding is now controlled. His systolic pressure remains near 80 mm Hg, his mean pressure is 55, and the lactate is rising. The surgeon suggests keeping the pressure low for another hour as a precaution. Urine output has been 10 milliliters over the past hour. Which of the following is the best response?

A. Continue the low target for the rest of the case, since a lower pressure limits the risk of rebleeding

B. Maintain the target until the lactate falls on its own, whatever the pressure

C. Wait for the intensive care unit team to decide, since the target is theirs to change

D. Raise the pressure only after the patient reaches the intensive care unit

E. Restore normal perfusion now, since the bleeding is controlled and hypotension is harming him

QUESTION GROUP 24

Intraoperative Cardiac Arrest: Causes and Response

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Questions 461 to 480. Choose the single best answer for each.

461. A 56-year-old man in the lateral position loses his arterial waveform and plethysmograph trace, and the end-tidal carbon dioxide collapses. The monitor shows ventricular fibrillation, and the defibrillator is charged. Alcohol-based skin preparation has pooled beneath the dependent chest, and oxygen is flowing across the field from the circuit. Which of the following is the best action just before the shock?

A. Postpone the shock until the preparation solution has dried and the drapes are removed

B. Shock at once, since fire risk applies only during use of monopolar electrocautery

C. Give epinephrine and amiodarone first so that the pooled solution can evaporate from the skin

D. Wipe away the pooled solution, move oxygen off the chest, clear everyone, and shock

E. Switch to synchronized cardioversion at a low energy to avoid igniting the drapes

462. An adult is receiving chest compressions on a soft operating table. The compressor has worked for three minutes, the compressions look shallow, and the end-tidal carbon dioxide is 8 mm Hg. The team hesitates to pause because the end-tidal carbon dioxide has been low and the arterial diastolic pressure is poor. Which of the following is the best way to improve the resuscitation?

A. Slow the rate to 60 per minute to allow fuller ventricular filling between each compression

B. Compress deeper than two and four-tenths inches at a rate of 140 to overcome the mattress

C. Keep the same compressor for five minutes to avoid pauses for a change of position

D. Rely on the arterial line to guide depth and keep the same compressor until tired

E. Place a backboard, change compressors, and coach two inches deep at 100 to 120 per minute

463. During resuscitation of an intubated patient, a resident ventilates with large breaths at about 20 per minute. The arterial line shows low diastolic pressure during compressions. Which of the following is the best change in ventilation?

A. Two breaths after every thirty compressions with a pause for each breath

B. One breath about every six seconds with no pause in compressions

C. Thirty breaths per minute to wash out carbon dioxide and raise the pH

D. Hold ventilation during compressions to maximize venous return to the heart

E. One breath every ten seconds with a ten-second pause in compressions

464. A 70-year-old woman arrests shortly after induction of anesthesia. Asystole is confirmed in a second lead with the gain turned up. Compressions have begun and intravenous access is working. When should the first dose of epinephrine be given?

A. As soon as it can be prepared, then every three to five minutes

B. After the second rhythm check, then every ten minutes

C. Only after the cause has been identified and corrected

D. After two full cycles of compressions, to avoid provoking arrhythmia

E. After the first rhythm check, to confirm that the rhythm is truly non-shockable

465. A 42-year-old woman receives a large-volume erector spinae block with bupivacaine. Within minutes she has a seizure, then a wide-complex bradycardia, and loses her pulse. Compressions are begun. Which of the following is the most appropriate cause-directed therapy?

A. Dantrolene with active surface cooling

B. Calcium chloride with sodium bicarbonate

C. Lipid emulsion with reduced epinephrine doses

D. Standard-dose epinephrine alone every three to five minutes

E. Naloxone with supportive ventilation

466. A 24-year-old man under sevoflurane anesthesia develops end-tidal carbon dioxide of 85 mm Hg, masseter rigidity, and a rising temperature to 40.1 degrees Celsius, followed by ventricular fibrillation and loss of pulse. Compressions and a shock are given. The anesthesia team has followed the standard sequence, the airway is secured, and the surgeon has stopped operating. Which of the following is the most appropriate cause-directed therapy?

A. Give lipid emulsion with reduced epinephrine doses

B. Stop the volatile agent; give dantrolene with active cooling

C. Give lipid emulsion bolus with reduced epinephrine doses

D. Give a systemic thrombolytic with early extracorporeal support

E. Give calcium chloride followed by insulin with glucose infusion

467. A 47-year-old woman receives rocuronium and cefazolin at induction. Within two minutes she has bronchospasm, profound hypotension, and then loses her pulse with an organized rhythm on the monitor. Compressions are started. The airway is secured, a large-bore line is in place, and the surgeon has stopped operating. No local anesthetic or nerve block was performed in this case. Which of the following is the most appropriate cause-directed therapy?

A. Lipid emulsion with reduced epinephrine doses

B. Dantrolene with active surface cooling

C. Needle decompression of the left chest

D. Epinephrine with rapid intravenous volume

E. Calcium chloride followed by insulin with glucose

468. During compressions for pulseless electrical activity, the end-tidal carbon dioxide rises abruptly from 12 to 42 mm Hg, and the arterial line shows a pulsatile waveform. The next scheduled rhythm check is 90 seconds away. Compressions are being delivered at an adequate rate by a rested compressor, and ventilation is one breath every six seconds. Which of the following is the best action?

A. Pause briefly to check the rhythm, the arterial waveform, and a pulse

B. Continue compressions until the scheduled check, since the value is unreliable

C. Give an additional dose of epinephrine, since the value suggests low output

D. Increase the compression rate to maintain the improved carbon dioxide level

E. Give sodium bicarbonate, since the value suggests acidosis

469. A 63-year-old man under general anesthesia for colectomy has a heart rate of 32 beats per minute and a blood pressure of 52 over 30 mm Hg. A pulse is still palpable, and the arterial line shows a waveform. The surgeon is retracting the mesentery, the end-tidal carbon dioxide is 28 mm Hg, and the anesthetic infusion has been paused. Which of the following is the best management?

A. Give a full arrest dose of epinephrine immediately, then reassess the pulse

B. Give small epinephrine boluses and treat the cause, without waiting for a flat line

C. Begin chest compressions now, since a rate this slow is arrest by definition

D. Wait for the rhythm to flatten before starting any treatment of the underlying cause

E. Give atropine alone and observe for ten minutes before further treatment

470. A 71-year-old man with an implantable defibrillator in the left upper chest has ventricular fibrillation during surgery, and the device has not delivered a shock. The pads were placed directly over the generator. The surgeon has stopped, the airway is secured, compressions are under way, and the end-tidal carbon dioxide has collapsed. Which of the following is the best action?

A. Leave the pads over the generator, since current delivery is best over the device

B. Delay the shock until a device representative can reprogram the implanted defibrillator

C. Move the pads about a hand's breadth from the generator, and do not delay the shock

D. Rely on the implanted device to treat the rhythm and withhold an external shock

E. Place the pads in an anterior-posterior position with one directly over the device

471. Ventricular fibrillation is confirmed in an anesthetized patient, and the defibrillator is charged. The surgeon asks whether to deliver three stacked shocks with rhythm checks between them, as in older teaching. The pads are well placed on a dry chest, the area is clear of oxygen flow, and no pooled solution is present. Pad contact has been confirmed. Which of the following is the best approach?

A. Deliver three stacked shocks, with a rhythm check and pulse check after each one

B. Deliver one shock and check for a pulse for thirty seconds before compressions

C. Deliver one shock and recheck the rhythm after thirty seconds of compressions

D. Deliver two shocks in succession, then resume compressions after a pulse check

E. Deliver one shock, then resume compressions at once for a full two minutes

472. A 59-year-old woman has no palpable pulse and a flat arterial line trace after a major vascular case. Transesophageal echocardiography shows organized but weak cardiac motion, and the electrocardiogram shows a normal-looking narrow rhythm. Capnography shows an end-tidal carbon dioxide of 12 mm Hg during compressions, and the pulse oximeter trace is absent. Epinephrine has been given once. Which of the following is the best interpretation?

A. Pseudo-pulseless electrical activity, which needs the cause treated as well

B. Normal circulation with a damped arterial line, which needs no specific treatment now

C. Shockable ventricular tachycardia, which needs immediate defibrillation at once

D. Massive pulmonary embolism, which needs thrombolysis before further assessment

E. Asystole, which needs confirmation in a second lead before epinephrine is given

473. A 29-year-old woman has a refractory arrest from a suspected drug overdose during surgery at a hospital with an extracorporeal support program. She has had 12 minutes of good compressions without return of circulation and has no contraindication to anticoagulation. The surgeon has controlled the surgical field, the airway is secure, and a defibrillator and an echocardiography probe are available. Which of the following is the best step?

A. Wait until 45 minutes of conventional resuscitation have failed before calling

B. Defer cannulation because candidates are limited to patients older than 60 years

C. Call only after the cause has been excluded as drug toxicity

D. Call the extracorporeal support team now, since the decision must come early

E. Defer the call until return of circulation has been attempted with a vasopressor

474. A patient regains circulation after a short resuscitation during a long case. The volatile agent and infusions were stopped at the start of the arrest. She is paralyzed and her pressure is 85 over 50 mm Hg, with a rising heart rate. Intravenous access is working, and the surgeon is closing the abdomen while the team discusses the plan. Which of the following is the best action?

A. Restart the volatile agent at full concentration immediately

B. Give no further anesthetic, since the arrest has erased awareness

C. Give a full induction dose of propofol to ensure unconsciousness

D. Give a small dose of an amnestic agent once perfusion is stable

E. Reverse the neuromuscular blocker and wait for the patient to move

475. A 68-year-old man regains a pulse after a prolonged arrest at the end of a long case under volatile anesthesia. He is profoundly vasodilated with a blood pressure of 60 over 30 mm Hg despite norepinephrine and volume. Echocardiography shows adequate filling and good contractility, and urine output has been minimal for the past hour. Which of the following is the most reasonable addition?

A. Vasopressin, since it works when acidosis blunts catecholamine response

B. Epinephrine 1 milligram every three to five minutes as in cardiac arrest

C. Routine sodium bicarbonate to correct the pH and restore vascular tone

D. Magnesium sulfate to treat vasodilation after return of circulation

E. Phenylephrine alone, since it is a pure vasoconstrictor with no cardiac effect

476. During laparoscopic cholecystectomy, a patient arrests with pulseless electrical activity. The abdomen is insufflated with the instruments in place. Meanwhile the anesthesia team is performing compressions and ventilating with pure oxygen, and a reader is calling out the reversible causes. The surgeon is scrubbed in at the table. Which of the following is the most appropriate first duty of the surgeon?

A. Remove the insufflation, release retraction, and control bleeding

B. Continue the dissection to finish the operation as quickly as possible

C. Increase the insufflation pressure to tamponade any bleeding vessel

D. Leave the room to give the resuscitation team more space to work

E. Convert to an open procedure before releasing the pneumoperitoneum

477. A team has given three doses of epinephrine and run three cycles for pulseless electrical activity. The leader notices that the team keeps repeating the same actions while no cause has been identified, and the room has grown quiet. A reader is available, an ultrasound probe is on the cart, and no one has yet looked at the heart or the lungs. Which of the following is the best step?

A. Shorten the interval between epinephrine doses to every minute until the rhythm changes

B. Pause at the next check to restate the working diagnosis and read the causes aloud

C. Defer the search for a cause until circulation has returned to a stable level

D. Ask the surgeon alone to decide the working diagnosis and tell the team

E. Add a second leader who directs the nurses and technicians in the room

478. A 36-year-old man arrests from hyperkalemia during a massive transfusion. After 20 minutes, ventricular fibrillation recurs intermittently, and calcium and insulin with glucose have been given. A junior colleague suggests stopping because out-of-hospital arrests are usually ended at about 20 minutes. The airway is secure, the team is rested, and compressions are of good quality. The surgeon is available to assist. Which of the following is the best response?

A. Stop, since 20 minutes is the accepted fixed limit for perioperative arrest

B. Stop, since any arrest beyond 15 minutes has a uniformly poor neurologic prognosis

C. Continue, since a reversible cause is being treated and the rhythm is shockable

D. Stop, since the end-tidal carbon dioxide is 12 mm Hg despite compressions

E. Stop, since the rhythm did not respond to the first three attempted shocks

479. During splenectomy for trauma, a 55-year-old man arrests from massive hemorrhage. After transfusion and surgical control, his circulation returns and he is stable on a modest vasopressor infusion. The surgeon asks what should happen to the operation. The surgeon has packed the abdomen and controlled the source, and the patient is now warm, with a normal ionized calcium and a hemoglobin of 9 grams per deciliter. Which of the following is the best decision?

A. Stop immediately and close, leaving the bleeding source untreated at that point

B. Defer the remainder of the operation to the next day whatever the bleeding

C. Leave the decision to the anesthesiologist alone, who can state it later to the team

D. Continue the full planned operation, since the arrest has resolved without sequelae

E. Complete an abbreviated operation, a decision made jointly and stated aloud

480. A 66-year-old woman dies after an unsuccessful intraoperative resuscitation. Her family asks the anesthesiologist whether a mistake was made during surgery. The formal review has not yet been done. Which of the following is the best approach?

A. Wait until the formal review is complete before saying anything to the family about the event

B. Give a definite cause of the arrest so that the family has a clear and final answer

C. Speak soon, plainly, and in person with the surgeon, without speculating or assigning blame

D. Refer the family to the risk management office for a written statement of events

E. Tell the family that the arrest was unavoidable and that nothing could have been done

QUESTION GROUP 25

Arrest in Special Positions and Populations

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Questions 481 to 500. Choose the single best answer for each.

481. A 48-year-old man is prone on a spinal frame for a lumbar fusion, with his head in pins, when ventricular fibrillation appears and the arterial line goes flat. The stretcher is on its way, but the surgeon is still releasing the head fixation. A defibrillator with adult pads is at the table. The team decides to deliver the first shock without waiting for the turn. Which of the following pad positions is acceptable?

A. Both pads side by side over the lower lumbar spine and the sacrum

B. One pad over the right scapula and the other at the left mid-axillary line

C. One pad over the occiput and the other over the mid-thoracic spine

D. Both pads overlapping at the midline over the posterior thorax

E. Both pads over the buttocks, where access is easiest and the frame is absent

482. A department is writing a protocol for prone spine surgery with head pins. In a simulation the arrest team lost several minutes because nobody knew who should unlock the head fixation, who should protect the head and neck during the turn, and who should remove the frame. The medical director asks what should be agreed before the first incision. Which of the following is the best arrangement?

A. The circulating nurse releases the pins while the surgeon first completes the wound closure and dressing

B. Roles are assigned at the time of the arrest by the first clinician who reaches the table

C. Roles are assigned during the arrest by whichever clinician reaches the table first and takes charge

D. The surgeon releases the pins, one named person protects the head, and another removes the frame

E. The pins stay in place and the team gives compressions prone until the surgeon finishes the operation

483. A 44-year-old man is prone for a lumbar discectomy. His blood pressure was stable for the first hour. Over two minutes it falls steadily while the heart rate rises, and then he becomes pulseless with pulseless electrical activity. The wound is nearly dry, the tube is well placed with equal ventilation, and the end-tidal carbon dioxide trace is present but very low. Which of the following is the most likely cause to seek and treat alongside resuscitation?

A. Concealed retroperitoneal hemorrhage from injury to a great vessel

B. Venous air entry because the surgical site lies well below the level of the heart

C. Kinking of the tracheal tube with complete airway obstruction

D. Reduced venous return from abdominal compression by the frame alone

E. Tension pneumothorax of the dependent lung from positive pressure ventilation

484. A 58-year-old man is in the right lateral decubitus position for a left thoracotomy. The chest is open, the lung is retracted, and the surgeon is at the table. His arterial pressure becomes unrecordable, the electrocardiogram shows ventricular fibrillation, and the pulse is absent. Which of the following is the best way to provide circulation while the defibrillator is prepared?

A. The team rolls the patient supine and starts external compressions before any other step

B. The surgeon closes the chest temporarily so that external compressions can be given

C. The team starts external compressions in the lateral position with a second rescuer supporting the back

D. The team waits until the bean bag is removed and the table is flat before any compressions

E. The surgeon begins internal cardiac massage through the open chest at once, before any other step

485. A 66-year-old woman is in the left lateral position for a thoracoscopic lobectomy with one-lung ventilation through a double-lumen tube. The chest is closed apart from the ports. She suddenly loses her pulse, the capnography trace disappears, and the saturation falls. Until this moment she had been hemodynamically stable, and the surgeon reports no major bleeding in the field. The anesthesiologist suspects a mechanical cause. Which of the following is the most appropriate next action?

A. Turn her supine, begin compressions, and return to two-lung ventilation with 100 percent oxygen

B. Keep her lateral and give compressions over the dependent chest wall while the tube is checked

C. Turn her supine and begin compressions, but continue one-lung ventilation with 100 percent oxygen

D. Turn her supine and begin compressions, delaying the oxygen increase until the tube position is confirmed

E. Keep her lateral and give a vasopressor first, starting compressions if there is no response in two minutes

486. A 55-year-old man is in the sitting position for a posterior fossa craniotomy. The end-tidal carbon dioxide suddenly falls, the blood pressure drops, and he loses his pulse. Venous air entry is suspected. The head is held in a rigid head holder. Moments earlier the surgeon had reported that the operative field was well above the level of the heart, and his hemodynamics had been stable throughout. Which of the following is the best immediate response?

A. Keep him sitting to protect cerebral perfusion and start compressions in that position

B. Give a fluid bolus and a vasopressor first, and start compressions only if there is no response

C. Lower him supine or head-down, flood the field with saline, and start compressions

D. Increase the minute ventilation and observe the carbon dioxide trace before any other step

E. Aspirate the central line and delay compressions until the air has been withdrawn completely

487. Six hours after aortic valve replacement, a 64-year-old man in the intensive care unit has ventricular fibrillation. Three consecutive shocks have failed, compressions have begun, and the surgeon is preparing for emergency resternotomy. A nurse asks which drug approach to use. The monitored rhythm was recognized within seconds, the defibrillator is charged and ready, and a resternotomy tray has been brought to the bedside. Which of the following is the most appropriate?

A. Give 1 mg of epinephrine every 3 to 5 minutes, as in a standard adult arrest

B. Give calcium chloride and magnesium routinely with every cycle of compressions

C. Give no drug of any kind until the chest has been reopened and the heart inspected

D. Give amiodarone and use small-dose epinephrine or none, on the surgeon's advice

E. Give atropine to counter vagal tone, then repeat the same three shocks

488. Three hours after coronary artery bypass grafting, a 70-year-old man has mediastinal drainage that abruptly stops after a period of steady bleeding. Within ten minutes the arterial pressure falls, the central venous and pulmonary artery diastolic pressures rise, and he becomes pulseless. Bedside ultrasound images are poor. Which of the following is the best next step?

A. Obtain a formal echocardiogram and reopen only if tamponade is confirmed

B. Proceed to emergency reopening of the chest without waiting for imaging

C. Give a large fluid bolus and continue external compressions for fifteen minutes

D. Transfer him to the scanner for contrast imaging of the mediastinum

E. Strip the chest tubes repeatedly and observe for ten minutes before any other action

489. Four hours after mitral valve surgery, a 60-year-old woman has asystole on the monitor and no pulse. Epicardial pacing wires are in place and an external pacemaker is available but is currently switched off. The wires were tested at the end of the operation and capture was reliable at low output. The cardiac surgeon is scrubbing, and pads are also available. Which of the following is the best use of the wires?

A. Leave the pacemaker off to unmask any underlying ventricular fibrillation

B. Pace at 60 beats per minute at low output to avoid injury to the myocardium

C. Give a full-dose epinephrine bolus and defer pacing until the chest is reopened

D. Use transcutaneous pads in place of the wires because they are more reliable

E. Connect them to the external pacemaker and pace at a high rate at maximum output

490. During emergency resternotomy for refractory ventricular fibrillation, the surgeon holds paddles directly on the heart. The anesthesiologist is asked which energy setting to select for the first internal shock. Two external shocks were unsuccessful, the surgeon opened the chest within minutes, and the heart is fibrillating in the field. The device is a biphasic unit capable of internal discharge. Which of the following is the most appropriate?

A. 100 joules

B. 150 joules

C. 20 joules

D. 200 joules

E. 360 joules

491. A 30-year-old woman at 34 weeks of gestation has a cardiac arrest during a nonobstetric operation. Compressions are under way with manual left uterine displacement. She has one peripheral intravenous catheter, placed in a saphenous vein at the ankle. Epinephrine is ready to be given. Her blood pressure had been falling for several minutes before the arrest, and the obstetric team is on the way. Airway management and compressions are being handled by others. Which of the following is the best approach?

A. Give the drug through the ankle line, since flow below the diaphragm is unaffected

B. Place new venous access above the diaphragm and give the drug through it

C. Place a femoral central venous catheter, since it is the largest accessible vein

D. Give the drug through the endotracheal tube first and place access afterward

E. Hold all drugs until the infant has been delivered and caval compression is relieved

492. A 34-year-old woman at 37 weeks of gestation arrests during induction of general anesthesia for a nonobstetric procedure. She is being ventilated by mask, and compressions with uterine displacement have started. The most experienced anesthesiologist has arrived. She has no history of a difficult airway, and the obstetric team is on its way to the room. A video laryngoscope and several tube sizes are on the cart. Which of the following is the best plan for the airway and ventilation?

A. Intubate with a smaller tube using a video laryngoscope and ventilate at 10 breaths per minute

B. Intubate with a standard tube using direct laryngoscopy and ventilate at 20 breaths per minute, matching her faster rate

C. Continue bag-mask ventilation until delivery and then secure the airway afterward with a standard tube

D. Intubate with a larger tube using direct laryngoscopy and ventilate at 15 breaths per minute

E. Place a supraglottic device only and avoid intubation because of aspiration risk

493. A 28-year-old woman is receiving a magnesium sulfate infusion after cesarean delivery for severe preeclampsia. She is oliguric. She becomes unresponsive with absent reflexes and shallow breathing, develops bradycardia with a widened QRS, and then loses her pulse. Resuscitation has begun. Her serum creatinine was elevated at the last measurement, and the infusion has been running for 14 hours at the standard rate. Which of the following is the most specific additional treatment?

A. Reduce the magnesium infusion by half and give atropine for the bradycardia

B. Stop the magnesium infusion and give naloxone for the respiratory depression

C. Continue the magnesium infusion and give lipid emulsion for presumed local anesthetic toxicity

D. Stop the magnesium infusion and give dantrolene for presumed malignant hyperthermia

E. Stop the magnesium infusion and give 1 g of calcium chloride intravenously

494. A 26-year-old woman at 37 weeks of gestation has a generalized seizure in the delivery suite. Her blood pressure is 176/112 mm Hg and the saturation is 88 percent. She has not received an epidural or any local anesthetic. Her fetus is at term, the obstetric team is present, and an intravenous line is in place. A nurse asks what should be done first. Which of the following is the best initial management?

A. Give lipid emulsion for local anesthetic toxicity and then observe

B. Proceed to immediate cesarean delivery before the seizure or the airway is treated

C. Support the airway, give magnesium sulfate, and control the blood pressure

D. Obtain a head computed tomography scan before any treatment is started

E. Give a phenytoin load and delay airway support until the seizure ends

495. A 31-year-old woman at 14 weeks of gestation arrests during a laparoscopic appendectomy. The uterus is palpable just above the pubic symphysis and well below the umbilicus. Compressions and ventilation have begun. She had no medical history before this event. Compressions are good and the airway is secured with a tube. The team asks how the pregnancy changes their approach. Which of the following statements about uterine management is correct?

A. Manual displacement is indicated, and hysterotomy should begin at four minutes of arrest

B. Table tilt of thirty degrees is indicated, and hysterotomy should begin at four minutes of arrest

C. Hysterotomy is indicated at four minutes, but displacement is unnecessary at this gestation

D. Manual displacement is indicated, and hysterotomy should wait until ten minutes of arrest

E. Neither displacement nor hysterotomy is indicated, because the fundus is below the umbilicus

496. A 6-month-old infant with congenital heart disease and pulmonary hypertension is under anesthesia for a hernia repair. After tracheal suction the saturation falls sharply, the systemic pressure drops, and the heart rate slows. Pulmonary hypertensive crisis is suspected. A cardiac anesthesiologist is not in the room. The infant had been stable and the surgery was proceeding well until this event, which occurred just after the airway was suctioned. Which of the following is the best treatment?

A. Reduce the inspired oxygen and allow mild hypercapnia to balance pulmonary and systemic flow

B. Give a rapid crystalloid bolus and continue the current ventilator settings

C. Begin chest compressions and defer any ventilator change until circulation returns

D. Give a beta blocker to slow the heart rate and relieve outflow obstruction

E. Give 100 percent oxygen, hyperventilate, and give a pulmonary vasodilator

497. A 12-kg toddler has a pulseless cardiac arrest under anesthesia. Compressions are under way and intravenous access is secure. Epinephrine is available as a 0.1 mg/mL solution. Intravenous access is established and a nurse has brought the drug chart and a length-based tape. The team leader asks the pharmacist to confirm the dose before it is drawn up. Which of the following is the correct volume for the first dose?

A. 1.2 mL

B. 12 mL

C. 0.12 mL

D. 0.6 mL

E. 2.4 mL

498. During laryngoscopy in a 4-kg infant, the heart rate falls to 50 beats per minute with weak pulses. Ventilation with 100 percent oxygen is effective and the tube is confirmed, but the bradycardia persists and a vagal cause is suspected. The calculated atropine dose at 0.02 mg/kg is 0.08 mg. The tube position has been confirmed by capnography and the infant has a pulse. The surgeon has paused the operation, and the trainee has calculated the dose by weight. Which of the following should be given?

A. 0.08 mg, exactly as calculated by weight

B. 0.1 mg, because a minimum single dose applies

C. 0.04 mg, half the calculated dose, to avoid tachycardia

D. 0.2 mg, double the calculated dose, to ensure an effect

E. 0.4 mg, the usual adult dose for bradycardia

499. A 6-year-old boy has a cardiac arrest during induction for dental surgery. Two rescuers are present, compressions are under way, and he is being ventilated with a bag and mask because an advanced airway is not yet in place. The monitor shows no pulse, and the second rescuer has taken over compressions. The anesthesiologist asks the team to confirm the ratio before proceeding. Which of the following is the correct compression-to-ventilation ratio?

A. 30 compressions to 2 breaths

B. 3 compressions to 1 breath

C. 5 compressions to 1 breath

D. 15 compressions to 2 breaths

E. Continuous compressions with asynchronous breaths

500. A 30-week preterm infant is delivered by emergency cesarean under general anesthesia and is limp. The anesthesiologist is resuscitating the infant in a cool operating room. The infant weighs about 1.3 kilograms, and the neonatal team has not yet arrived. The anesthesiologist asks the circulating nurse what should be prepared. Which of the following measures best addresses the infant's particular risks of heat loss and hypoglycemia?

A. Dry the infant with cotton towels on the open table and defer glucose until admission to the nursery

B. Use a radiant warmer and plastic wrap, and check the glucose early in the resuscitation

C. Raise the room temperature alone and check the glucose after transfer

D. Cover the infant with a warm wet blanket and give dextrose before measuring glucose

E. Place the infant on the mother's chest under the drapes and check the glucose at one hour

QUESTION GROUP 26

Peri-Arrest Arrhythmias, Pacing, and Cardioversion

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Questions 501 to 520. Choose the single best answer for each.

501. A 74-year-old woman with a prior anterior myocardial infarction is having hip fracture surgery under spinal anesthesia. The electrocardiogram shows fixed PR intervals with sudden dropped beats and a wide QRS. The ventricular rate is 44 beats per minute, the blood pressure is 98/60 mm Hg, and she is alert and comfortable. Which of the following is the best management?

A. Observe only with routine monitoring, because this is a benign block that arises within the node

B. Give repeated atropine boluses as the sole treatment, because the block lies within the node

C. Start an esmolol infusion to reduce myocardial oxygen demand after the old infarction

D. Apply pacing pads and prepare to pace, because the block may progress without warning

E. Give diltiazem to slow the atrial rate and reduce the number of dropped beats in the block

502. A 38-year-old woman presents for emergency laparotomy after ingesting an unknown quantity of her father's metoprolol. The heart rate is 38 beats per minute, the blood pressure is 68/40 mm Hg, and repeated doses of atropine have produced no change. Pacing pads are in place. She is drowsy and pale, with cool extremities and a prolonged capillary refill. Which of the following should be added to pacing?

A. Atropine doses beyond the usual maximum until the rate responds

B. Digoxin-specific antibody fragments and magnesium

C. Sodium bicarbonate with hyperventilation to narrow the QRS

D. Naloxone followed by flumazenil to exclude a coingested sedative

E. Glucagon, high-dose insulin with dextrose, and calcium

503. At the end of a laparoscopic cholecystectomy, a 61-year-old man receives sugammadex. Two minutes later the heart rate falls from 72 to 28 beats per minute and the blood pressure drops. The surgical stimulus has ended, and the team suggests that this is part of emergence. He had been hemodynamically stable throughout the case, and the train-of-four had shown a deep block before the drug was given. Which of the following is the best response?

A. Attribute the slowing to emergence and observe closely for five minutes before acting

B. Treat the bradycardia promptly with atropine and have epinephrine ready

C. Give neostigmine to reverse the remaining block more completely and protect the airway

D. Give a second dose of sugammadex to complete the reversal of the deep block

E. Give adenosine to interrupt the nodal conduction block and restore the rate

504. A 4-year-old boy receives succinylcholine for laryngospasm. Five minutes later a second dose is given without an anticholinergic, and the heart rate falls to 40 beats per minute with a blood pressure of 60/30 mm Hg. He has no cardiac history and weighs 18 kilograms. The saturation is 91 percent, and ventilation by mask is possible but difficult. Which of the following is the best management?

A. Observe, because bradycardia after succinylcholine resolves without treatment

B. Deliver a synchronized shock to restore sinus rhythm

C. Start an epinephrine infusion at the adult rate before checking oxygenation

D. Confirm oxygenation, ventilate with 100 percent oxygen, and give atropine

E. Give neostigmine to hasten recovery from the neuromuscular block

505. A 22-year-old woman with congenital long QT syndrome and a history of exertional syncope is scheduled for an elective laparoscopic cholecystectomy. She takes a beta-blocker. The anesthesiologist is planning her perioperative care. Her last electrocardiogram showed a corrected QT interval of 510 milliseconds, and her potassium and magnesium are at the low end of normal. Her father died suddenly at a young age. Which of the following is the best plan?

A. Stop the beta-blocker, give ondansetron prophylactically, accept a low normal potassium, and have lidocaine ready

B. Continue the beta-blocker, give droperidol for nausea prophylaxis, keep electrolytes in range, and have magnesium ready

C. Hold the beta-blocker, avoid QT-prolonging antiemetics, keep electrolytes in range, and have atropine ready

D. Continue the beta-blocker, avoid QT-prolonging antiemetics, keep electrolytes in range, and have magnesium ready

E. Continue the beta-blocker, avoid QT-prolonging antiemetics, aim for light anesthesia and rapid emergence, and have amiodarone ready

506. A 68-year-old man develops atrial fibrillation at 168 beats per minute two days after lobectomy. His blood pressure is 70/40 mm Hg, he is confused, and a pulse is palpable. Pads are applied. He had a recent coronary revascularization, the potassium is normal, and defibrillator pads are already on his chest. He has just been sedated for the procedure. Which of the following is the best next step?

A. Synchronized cardioversion starting at 50 joules biphasic, escalating only if the first shock fails

B. Unsynchronized defibrillation at 200 joules biphasic, as in ventricular fibrillation

C. Synchronized cardioversion starting at 200 joules biphasic, with the pads already placed

D. An esmolol bolus followed by an infusion to control the ventricular rate

E. Adenosine as a rapid intravenous push followed by a saline flush

507. During a spinal procedure, a 56-year-old man develops a rapid wide-complex rhythm with a changing QRS morphology. No pulse can be felt, the arterial line is flat, and the monitor shows no organized electrical activity that the defibrillator can synchronize to. The patient is in the lateral position, no cause has yet been identified, and the code team has arrived with the defibrillator and is waiting for instructions. Which of the following is the best treatment?

A. Immediate unsynchronized defibrillation

B. Synchronized cardioversion at 100 joules

C. Intravenous magnesium over 30 minutes before any shock

D. Intravenous amiodarone 150 mg over 10 minutes

E. Rapid intravenous adenosine as a diagnostic test

508. A 63-year-old woman taking dipyridamole develops a stable, regular narrow-complex tachycardia at 180 beats per minute under general anesthesia. Her blood pressure is 104/62 mm Hg. Adenosine is chosen as the first drug. She has no history of asthma or heart transplantation, and the electrocardiogram shows no preexcitation. The tachycardia began suddenly without any change in anesthetic depth. Which of the following is the best adjustment?

A. Give a reduced dose, because dipyridamole potentiates the effect of adenosine

B. Give a larger than usual dose to overcome receptor blockade

C. Give the usual dose, because dipyridamole does not interact with adenosine

D. Avoid adenosine altogether and give verapamil as the first drug

E. Choose synchronized cardioversion, since adenosine is contraindicated here

509. A 59-year-old man has a regular narrow-complex tachycardia at 150 beats per minute with a sawtooth baseline in the inferior leads. He is stable. Adenosine produces a brief pause with flutter waves, after which the rapid rhythm resumes. He has no signs of ischemia or heart failure, and the electrolytes are normal. The surgeon asks whether the operation can continue. Which of the following is the best next step?

A. Repeat adenosine at progressively higher doses until the rhythm converts to sinus

B. Give atropine to reduce the degree of atrioventricular block and clarify the flutter rate

C. Recognize flutter and control the ventricular rate with esmolol or diltiazem

D. Give calcium chloride to stabilize the cardiac membrane before any further drug

E. Deliver an unsynchronized shock at 100 joules to terminate the flutter at once

510. A 71-year-old man is found to have atrial fibrillation at 130 beats per minute after cardiac surgery. The rhythm has lasted about three days by the nursing record. His blood pressure is 118/70 mm Hg and he has no symptoms. A colleague proposes immediate cardioversion. He has no history of bleeding and takes no anticoagulant, and the surgical site is dry. Which of the following is the best response?

A. Perform immediate synchronized cardioversion, since duration does not alter embolic risk

B. Give adenosine to convert the rhythm, since it is effective for atrial fibrillation

C. Give diltiazem and amiodarone together for chemical conversion without anticoagulation

D. Control the rate and plan anticoagulation before any attempt at cardioversion

E. Give magnesium alone and observe, since this rhythm needs no rate control

511. A 64-year-old man undergoing noncardiac surgery develops new ST depression in lead V5 with a heart rate of 118 beats per minute, a blood pressure of 82/46 mm Hg, and a hemoglobin of 6.9 g/dL after blood loss. He has no known coronary stent. The arterial line shows a pressure that falls each time the heart rate rises. The anesthetic depth has been constant, and the surgeon has not changed the plan. Which of the following is the most appropriate initial management?

A. Start a nitroglycerin infusion to lower preload and relieve coronary spasm

B. Give an esmolol bolus to lower the heart rate despite the pressure

C. Give aspirin and a heparin bolus before discussing the bleeding risk

D. Observe the tracing, since perioperative ST changes are rarely clinically significant

E. Restore blood pressure and hemoglobin, slow the heart rate, and give oxygen

512. Before a pulmonary artery catheter is floated in a 67-year-old man with known left bundle branch block, the anesthesiologist reviews the risks. Passage through the right ventricle may injure the right bundle. The catheter is for a high-risk cardiac operation. The electrocardiogram shows a typical left bundle branch pattern, and the pacing options on the unit have not yet been checked. Which of the following is the best precaution?

A. Give prophylactic lidocaine to suppress ventricular ectopy as the catheter passes

B. Have pacing ready, because passage may produce complete heart block

C. Float the catheter with the balloon deflated to reduce septal contact

D. Give a procainamide infusion to stabilize conduction before advancing the catheter

E. Place defibrillator pads but leave the device switched off until needed

513. A 58-year-old man with a pacemaker, who is not pacemaker-dependent, is having an inguinal hernia repair under general anesthesia. The surgeon plans to use monopolar electrosurgery. The device was interrogated four months ago. He has an intrinsic rhythm at 70 beats per minute and no history of syncope. The surgeon asks whether anything special is required. Which of the following statements is correct?

A. Interference is unlikely at this distance, but the pulse should be monitored by plethysmography

B. Interference is certain with any monopolar cautery, so the device must be turned off beforehand

C. Bipolar cautery causes more interference than monopolar cautery, whatever the site of surgery

D. The risk is highest below the umbilicus because the return pad lies closer to the generator

E. Interference occurs only with ultrasonic scalpels and never with monopolar electrosurgery

514. A 72-year-old man with a biventricular implantable defibrillator is pacemaker-dependent. He is scheduled for neck surgery with monopolar electrocautery. A colleague suggests simply placing a magnet over the generator. The generator is on the left chest, and the surgeon expects to use the cautery often near it. The device clinic is open today, and the anesthesiologist wants a reliable plan. Which of the following is the best plan?

A. Place a magnet over the generator, which will give asynchronous pacing and suspend shocks

B. Ask the device team to program asynchronous pacing and suspend tachycardia therapies

C. Rely on the magnet to maintain pacing, and omit the external pads to keep the field clear

D. Leave the device unchanged, because modern defibrillators are fully shielded from cautery

E. Place a magnet and remove external pads to avoid interference with the cautery current

515. A 66-year-old woman with a left-sided pacemaker generator develops ventricular fibrillation during a laparotomy. Pads are available. The anesthesiologist asks where to place them. The pads are placed anterolaterally and the surgeon has paused. The anesthesiologist knows that a shock may alter the device settings and wants to shock quickly. Which of the following is the best placement?

A. Directly over the generator for the best current delivery

B. Anywhere on the chest, but only after the device has been interrogated

C. Anterolateral, with the apex pad overlying the generator

D. At least 8 cm from the generator, preferably anterior-posterior

E. Anterolateral at half the usual energy to protect the device

516. A 69-year-old man had his defibrillator therapies suspended with a programmer for shoulder surgery that used monopolar cautery near the chest. The operation is complete and he is in the recovery room with pads in place. He is alert, and the heart rate and blood pressure are normal. The electrophysiology service is on call for the hospital. Which of the following is the best plan?

A. Transfer to the ward, because therapies return automatically once the cautery has ended

B. Defer interrogation to the routine visit in about twelve months, since he has no symptoms

C. Confirm therapies are restored and interrogate the device before he leaves monitoring

D. Interrogate only if he reports palpitations, dizziness, or a shock from the device

E. Leave therapies off until the clinic review next week, keeping pads at the bedside

517. A 66-year-old man on a diuretic, who had a bowel preparation, develops frequent ventricular ectopy and runs of nonsustained ventricular tachycardia during surgery. The potassium is 2.8 mmol/L. Despite a potassium infusion the level barely rises over an hour. He is otherwise stable, with a normal blood pressure and no signs of ischemia. The arterial blood gas shows a normal pH. Which of the following is the best next step?

A. Give insulin with dextrose to shift potassium into the cells and speed the correction

B. Give sodium bicarbonate to correct acidemia and so suppress the ventricular ectopy

C. Check and replace magnesium, because low magnesium can prevent potassium correction

D. Give calcium chloride to stabilize the membrane and suppress the ventricular tachycardia

E. Start an amiodarone infusion because the potassium cannot be corrected at this rate

518. A 45-year-old woman has received 14 units of red cells during a spine operation. The blood pressure is falling, contractility looks poor on the echocardiogram, and the QT interval is prolonged. The ionized calcium is 0.8 mmol/L. The arterial blood gas shows a normal pH, and a rapid red cell infusion is running through a warmer. Which of the following is the most appropriate treatment?

A. Calcium chloride via a large vein, titrated to the ionized calcium

B. Calcium chloride in a fixed dose guided by the total calcium alone

C. Magnesium sulfate as the only treatment, to shorten the prolonged QT interval

D. Sodium bicarbonate to alkalinize the blood and improve contractility

E. Atropine to counter the vagal slowing and raise the falling blood pressure

519. A 78-year-old man with renal failure takes digoxin and is having an emergency repair of a hip fracture. The electrocardiogram shows bidirectional ventricular tachycardia with a pulse and a blood pressure of 80/50 mm Hg. The potassium is 2.9 mmol/L, the magnesium is low, and the digoxin level is high. Which of the following is the best treatment?

A. Synchronized cardioversion at 200 joules as the sole treatment of the arrhythmia

B. Amiodarone loading as the sole treatment, followed by observation of the rhythm

C. Verapamil to slow the ventricular rate and terminate the tachycardia

D. Procainamide to suppress the ventricular tachycardia while the level is rechecked

E. Digoxin-specific antibody fragments, with potassium and magnesium correction

520. A 34-year-old man arrives in the operating room after a prolonged exposure outdoors, with a core temperature of 28 degrees Celsius. He is pulseless in ventricular fibrillation. Three shocks have failed to restore a rhythm. Cardiopulmonary bypass is not available in the room. His clothing is wet, and compressions are in progress. Which of the following is the best next step?

A. Deliver repeated shocks at escalating energies without rewarming, until the rhythm converts

B. Rewarm actively and continue resuscitation, because shocks may fail until he is warm

C. Stop resuscitation, since ventricular fibrillation at this temperature is unresponsive

D. Give amiodarone and epinephrine at the usual intervals and keep repeating shocks

E. Switch to transcutaneous pacing at 80 beats per minute and stop compressions

QUESTION GROUP 27

Post-Arrest Care and Mechanical Support

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Questions 521 to 540. Choose the single best answer for each.

521. A 70-year-old man is undergoing emergency laparotomy for perforated bowel when he arrests from hemorrhage. Circulation returns after resuscitation and the bleeding is controlled. He remains hypotensive on a vasopressor. The surgeon says the resection is almost ready and wants to finish the operation. The anesthesiologist notes that the blood pressure remains low and that the team has been working for several hours. Which of the following is the best approach?

A. Complete the planned resection, because the incision is already open and closing would waste the exposure

B. Stop at once and transfer him to the intensive care unit with the abdomen open, without discussion

C. Let the surgeon decide alone, since the operative plan is a surgical matter and not a physiologic one

D. Continue the operation until the vasopressor requirement falls, and only then decide about closure

E. Decide together and explicitly, favoring damage control closure if he cannot tolerate more surgery

522. Return of circulation has occurred after an unexplained intraoperative arrest in a 52-year-old woman, and the cause is not yet known. She is still on the table and the surgical team is waiting. An electrocardiogram, blood gas analyzer, and ultrasound machine are available in the room. The anesthesiologist wants a structured approach to the first hour. Which of the following sets of investigations best covers the most common reversible causes in the first hour?

A. Serum troponin and a portable chest radiograph, with the rest of the cause search deferred until she is settled in the intensive care unit

B. Head computed tomography and electroencephalography first, because the brain is the injured organ and neurologic outcome comes first

C. Coagulation studies and blood cultures with a broad-spectrum antibiotic, because sepsis is the commonest cause of intraoperative arrest

D. Blood gas with potassium, glucose, lactate, and hemoglobin, a twelve-lead electrocardiogram, an echocardiogram, and a drug review

E. Cardiac catheterization in every patient, before any other investigation, because coronary occlusion is the commonest cause

523. A 36-year-old woman has return of circulation after an arrest caused by hemorrhage during a trauma laparotomy. She is comatose and still has ongoing coagulopathic bleeding with a core temperature of 36 degrees Celsius. The intensive care team proposes deliberate cooling to 33 degrees. Her base deficit is large, fibrinogen is low, and surgical hemostasis is incomplete after packing of the abdomen. Which of the following is the best temperature strategy?

A. Cool to 33 degrees for 24 hours, since neuroprotection takes priority over bleeding

B. Cool to 32 degrees to reduce metabolic demand and the transfusion requirement

C. Apply no temperature control until the core temperature exceeds 38.5 degrees

D. Hold 34 degrees for 72 hours using deep sedation and neuromuscular blockade

E. Aim for normothermia and prevent fever, because hypothermia would worsen clotting

524. A 57-year-old man was cooled to 34 degrees Celsius after an intraoperative arrest, and after 12 hours the team proposes to switch off the cooling device and let him warm quickly to whatever temperature he reaches. He has had no fever or seizures so far, and his hemodynamics are stable. The cooling device is in place and working. Which of the following is the best response?

A. Agree, because 12 hours of cooling gives the full neurologic benefit and further cooling adds risk

B. Continue cooling for 6 hours more, then stop temperature monitoring and let him rewarm naturally

C. Rewarm quickly to 38 degrees to restore normal metabolism and shorten the sedation period

D. Maintain temperature control for at least 36 hours, rewarm slowly, and prevent fever

E. Maintain cooling for 7 days, since longer cooling gives progressively better outcomes

525. A comatose 62-year-old woman is being cooled after an intraoperative arrest and begins to shiver. A neuromuscular blocker infusion is started to prevent shivering from defeating the cooling. The intensive care physician asks what the main disadvantage of blockade is. Her core temperature is measured by a bladder probe, and the cooling device is working. The nurse is concerned about the examination later. Which of the following is the best answer?

A. It raises intracranial pressure in all patients through histamine release and cerebral vasodilation

B. It abolishes the cooling effect of the surface device by causing peripheral vasodilation

C. It prevents accurate measurement of the core temperature by a bladder or esophageal probe

D. It causes a false-positive loss of cortical somatosensory responses during prognostic testing

E. It hides seizures and clouds the examination, so doses and times must be recorded

526. A comatose 66-year-old man remains unresponsive 72 hours after an intraoperative arrest. Sedation has been off for two days, normothermia has been restored, and no confounders remain. The team wishes to use the most specific single finding available to support a poor prognosis. The family has asked which test is the most reliable, and the team agrees that a firm prediction will need more than one test. Which of the following is the most appropriate finding?

A. Bilaterally absent cortical responses on somatosensory evoked potentials

B. A reactive continuous background on electroencephalography at 72 hours

C. Absent pupillary reflexes recorded at 12 hours while still under sedation and cooling

D. A single serum neuron-specific enolase value drawn at 6 hours after arrest

E. Myoclonus appearing within the first hour after arrest, before sedation was reduced

527. A 52-year-old woman is cooled after an intraoperative arrest. A resident orders prophylactic anticonvulsants for all comatose survivors. The nurse then reports brief jerking of the limbs. The sedation is propofol and there is no neuromuscular blockade. Her blood pressure and temperature are on target, and the electroencephalography service is available today. She has no history of epilepsy. Which of the following is the best next step?

A. Start prophylactic anticonvulsants as ordered and treat any jerking as seizure activity without further testing

B. Add propofol to burst suppression to abolish every epileptiform discharge on the tracing for 24 hours

C. Obtain electroencephalography to separate myoclonus from seizure, and treat seizures promptly

D. Ignore the jerking and obtain no tracing, since myoclonus carries no prognostic significance at all

E. Add three antiseizure drugs at once, because nonconvulsive status is always present after arrest

528. An hour after return of circulation, a comatose 58-year-old man has a glucose of 260 mg/dL. A resident proposes intensive insulin therapy aiming for 80 to 110 mg/dL. He has no history of diabetes. His blood pressure and temperature are on target, and the nurse asks about the glucose plan for the next day. Which of the following is the best glucose strategy?

A. Insulin infusion to keep the glucose between 70 and 180 mg/dL, with frequent checks

B. Intensive insulin therapy to 80 to 110 mg/dL, because tight control is neuroprotective

C. No insulin, because hyperglycemia is protective and should be left untreated

D. Insulin only if the glucose exceeds 300 mg/dL, with checks every six hours

E. Dextrose infusion to keep the glucose above 200 mg/dL to feed the brain

529. A 175-cm, 120-kg man is ventilated after an intraoperative arrest complicated by aspiration. His predicted body weight is about 70 kg. The anesthesiologist is choosing ventilator settings for transport to the intensive care unit. Chest radiography shows bilateral infiltrates and the oxygenation is poor. The respiratory therapist asks which tidal volume to set. Which of the following is the best approach?

A. A tidal volume of 10 to 12 mL/kg of predicted body weight to recruit the aspirated lung

B. A tidal volume of 6 to 8 mL/kg of actual body weight, with high plateau pressures accepted

C. A tidal volume of 4 mL/kg of actual body weight with permissive hypercapnia to a PaCO2 of 60 mm Hg

D. A tidal volume of 6 to 8 mL/kg of predicted body weight, with low plateau pressures

E. Pressure support alone during transport, with the tidal volume left unmeasured

530. A 45-year-old woman with a massive pulmonary embolus has return of circulation after intraoperative arrest. The right ventricle is dilated and hypokinetic on echocardiography and the mean arterial pressure is 54 mm Hg. She is ventilated with a high positive end-expiratory pressure. She is on a norepinephrine infusion, the saturation is 92 percent, and there is no active bleeding. Which of the following is the best strategy?

A. Support the pressure with a vasopressor, give fluid sparingly, and keep positive end-expiratory pressure low

B. Give repeated large fluid boluses to fill the dilated right ventricle and raise the output

C. Raise the positive end-expiratory pressure further to improve the oxygenation and recruit lung

D. Start an intravenous vasodilator infusion to reduce the pulmonary and systemic afterload on the right ventricle

E. Allow hypercapnia to develop in order to improve right ventricular filling and contractility

531. One hour after return of circulation, a 63-year-old man has a mean arterial pressure of 68 mm Hg on norepinephrine. His lactate has risen from 4.0 to 6.5 mmol/L over the hour, and urine output is minimal. The pressure target has been met. His urine output is minimal and his skin is cool. The team is surprised that the number is rising despite the target being met. Which of the following is the best next step?

A. Reassure the team, since the pressure target has been reached and the trend will settle

B. Look for low cardiac output, ongoing bleeding, or an undrained source of sepsis

C. Raise the mean arterial pressure to 100 mm Hg, without further assessment of the cause

D. Give sodium bicarbonate to lower the lactate and correct the metabolic acidosis

E. Stop the vasopressor to see whether the lactate clears on its own over the next hour

532. A 60-year-old man with renal impairment needs sedation after resuscitation. The team wants the earliest reliable neurologic examination after cooling ends. A neuromuscular blocker was used earlier for shivering. He will be cooled for a day and a half, and the family expects an early neurologic assessment. The nurse asks which sedatives to start. Which of the following is the best sedation and monitoring plan?

A. A long-acting benzodiazepine infusion for the full period, with the examination timed by the clock alone

B. A morphine infusion titrated to comfort, since its metabolites clear quickly in renal impairment

C. Continuous neuromuscular blockade for the whole duration of cooling, with sedation as needed

D. Deep sedation with several agents, with doses recorded only after the examination is done

E. Short-acting agents such as propofol and remifentanil, with a train-of-four before assessment

533. Four patients have arrested during surgery in the same week, and an extracorporeal cardiopulmonary resuscitation team is available on a single phone call. The team can support only one patient at a time. On the basis of the information below, which of the following patients is the best candidate for extracorporeal cardiopulmonary resuscitation?

A. An older patient with an unwitnessed asystolic arrest, a cause not yet known, and 45 minutes of compressions

B. A young patient with a witnessed shockable arrest from massive pulmonary embolus and 10 minutes of low flow

C. A patient with end-stage disease and a limited life expectancy, with a witnessed arrest of 5 minutes

D. A patient with an irreversible cause, a nonshockable rhythm, and an unknown period without flow

E. A patient with uncontrolled exsanguinating hemorrhage from an unsurvivable injury and a witnessed arrest

534. At a departmental meeting, a colleague claims that extracorporeal cardiopulmonary resuscitation improves survival in all cardiac arrests and should be started in every patient. You are asked to summarize the evidence. The team includes cardiac surgeons and intensivists who disagree, and the department wants a written position. Which of the following is the best summary?

A. Randomized trials have shown clear benefit in all types of arrest, so it should be used indiscriminately in every patient

B. Observational data from in-hospital arrest are consistently unfavorable, so it should not be used in any patient

C. Benefit is limited to selected patients treated quickly by experienced teams; large out-of-hospital trials were not positive

D. The large Prague and INCEPTION trials proved a major survival benefit over conventional resuscitation in out-of-hospital arrest

E. One single-center trial was negative, while the larger multicenter trials proved a significant benefit in survival

535. A 45-year-old woman is placed on peripheral venoarterial extracorporeal support through femoral cannulas during an arrest. Once full flow is established, the end-tidal carbon dioxide falls from 20 to 6 mm Hg while the arterial pressure and lower limb saturation are good. The anesthesiologist is worried about the airway. Which of the following is the best interpretation?

A. The circuit carries most of the output, so a low end-tidal value is expected

B. The tracheal tube has been displaced into the esophagus and must be replaced immediately

C. The cannulas have become dislodged and circuit flow should be stopped at once

D. A recurrent pulmonary embolus is the likeliest explanation and needs thrombolysis

E. The oxygenator has failed and the whole circuit should be changed at once

536. A 68-year-old man who had cardiac surgery earlier today is placed on extracorporeal support after a postoperative arrest. The chest was just closed and there is ongoing oozing from the mediastinal drains. The team asks how to manage anticoagulation and monitoring. He is on aspirin, the INR is normal, and the platelet count is falling. Which of the following is the best approach?

A. Give the standard full-dose heparin bolus regardless of the surgical bleeding or the falling platelets

B. Withhold all anticoagulation permanently, since a modern heparin-coated circuit never clots

C. Check the coagulation profile once a day, since the circuit consumes nothing and the oozing will settle

D. Give heparin by a fixed schedule independent of clotting measurements, to keep dosing simple and safe

E. Individualize heparin, possibly lower or delayed, and check platelets, fibrinogen, and hemoglobin

537. A 56-year-old man is being transferred from the operating room to the intensive care unit after resuscitation from an intraoperative arrest. The anesthesiologist prepares the verbal handover. He is on a transport ventilator, and a defibrillator and infusion pumps have been prepared. The intensive care nurse is waiting at the bedside. Which of the following sets of information is most useful to the receiving team?

A. The last rhythm, the last blood pressure, and the current ventilator settings, and nothing more

B. Arrest time, suspected cause, drugs given, cooling plan, and sedatives that cloud the examination

C. The time circulation returned, the ventilator settings, and the vasopressor doses, and nothing more

D. The steps of the operation, the surgical findings, and the total blood loss, and nothing more

E. The names of the staff present, the equipment used, and the order in which they arrived

538. The day after a resuscitation, an anesthesiologist reviewing the record notices that the time of the first shock was entered incorrectly by a few minutes. A colleague suggests quietly changing the entry so that the record is consistent with the monitor data. The surgeon and the nurse have both noticed the error. The medical record has already been scanned, and the colleague is worried about the review. Which of the following is the best action?

A. Overwrite the original entry with the monitor time so the record reads cleanly

B. Delete the entry and rewrite the entire note from memory, with the correct first shock time

C. Leave the error uncorrected to avoid appearing inconsistent in the scanned record

D. Add a dated addendum explaining the correction, leaving the original entry unaltered

E. Ask the surgeon or the nurse to edit the entry directly in the anesthesia record

539. Five days after an intraoperative arrest, a 62-year-old woman has remained comatose. Her neurologic examination, electrophysiology, imaging, and time all point to a poor outcome, and the team and family have agreed on goals of care. Organ donation becomes a possible consideration. The intensive care team is preparing for a family meeting. Which of the following is the most appropriate action?

A. Have the anesthesiologist raise donation with the family at the first prognosis conversation

B. Have the bedside nurse raise donation before the family has been told the prognosis, to save time

C. Notify the organ procurement organization per policy, and let trained specialists raise it

D. Delay notification until after withdrawal of therapy has been completed and death declared

E. Avoid mentioning donation to the organ procurement organization unless the family asks first

540. A 61-year-old man has return of circulation after an intraoperative arrest. The cause was hypovolemia from hemorrhage. His electrocardiogram shows sinus tachycardia without ST elevation, and he is hemodynamically stable after transfusion. Cardiology suggests immediate coronary angiography for every post-arrest patient. His transfusion requirement is falling, his hemoglobin is stable, and his lactate is clearing. Which of the following is the best response?

A. Treat the cause; immediate angiography has shown no benefit without ST elevation

B. Take him for emergency angiography at once, because every arrest has a coronary cause

C. Give systemic thrombolysis to treat a presumed coronary occlusion causing the arrest

D. Start a heparin infusion immediately, without discussing the surgical bleeding

E. Defer all investigation of the cause until neurologic recovery is confirmed

QUESTION GROUP 28

Embolic Crises

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Questions 541 to 560. Choose the single best answer for each.

541. A 58-year-old man is undergoing a hepatic resection under general anesthesia with a nitrous oxide and oxygen mixture. A hepatic vein tear allows air entrainment, and a transesophageal probe shows bubbles in the right atrium. A resident asks why nitrous oxide is especially hazardous in this situation. Which of the following is the best explanation?

A. It is less soluble in blood than nitrogen, so it leaves the bubble and collapses the pulmonary vasculature around it

B. It is a potent pulmonary vasoconstrictor that directly raises right ventricular afterload independent of bubble size

C. It lowers the nitrogen content of the bubble and so prolongs the time the air remains in the right atrium

D. It is more soluble in blood than nitrogen, so it enters the bubble faster than nitrogen leaves it

E. It displaces oxygen from hemoglobin and so worsens the hypoxemia caused by the air lock

542. A 64-year-old woman is anesthetized for a high-risk pulmonary embolism that was diagnosed on the ward and now requires emergency surgical embolectomy. She is hypotensive despite a norepinephrine infusion, and the right ventricle is dilated on bedside echocardiography. The anesthesiologist plans induction. Which of the following is the most appropriate approach?

A. Rapid induction with a standard dose of propofol, relying on later vasopressor titration

B. Slow inhalation induction with sevoflurane to preserve spontaneous ventilation

C. Induction with ketamine or etomidate, with a vasopressor infusion immediately available

D. Induction after a large fluid bolus to fill the right ventricle and prevent hypotension

E. Awake intubation with heavy sedation and topical anesthesia, avoiding any induction agent

543. A 71-year-old man has a sudden fall in end-tidal carbon dioxide and a blood pressure of 70 over 38 after a tourniquet is released during knee arthroplasty. Echocardiography shows a dilated right ventricle and a bowed septum. Which vasopressor is the best choice to support the circulation?

A. Phenylephrine, which raises systemic pressure through alpha stimulation and also lowers the pulmonary resistance

B. Dopamine at a high dose, which selectively dilates the pulmonary circulation and raises cardiac output

C. Norepinephrine, which raises systemic pressure and right coronary flow while sparing pulmonary resistance

D. Dobutamine alone, which lowers pulmonary resistance and has no effect on systemic vascular resistance

E. Isoproterenol, which relaxes pulmonary arterioles and increases right ventricular filling pressure

544. A 45-year-old woman with massive pulmonary embolism is intubated in the operating room. Minutes later the blood pressure falls further and the central venous pressure rises. The ventilator is delivering a large tidal volume with high positive end-expiratory pressure, and the end-tidal carbon dioxide is 20 mm Hg. Which of the following adjustments is most appropriate?

A. Increase positive end-expiratory pressure to recruit atelectatic lung and improve oxygenation

B. Reduce minute ventilation and accept permissive hypercapnia to lower airway pressures

C. Prolong the inspiratory time to improve gas exchange across the obstructed lung

D. Reduce tidal volume, PEEP, and inspiratory time while keeping oxygenation and normocapnia

E. Add a larger tidal volume to wash out the dead space created by the embolus

545. A 67-year-old woman in the sitting position has a central venous catheter removed in the recovery area after a long operation. She is hypovolemic. Which of the following measures best prevents air entry during and after catheter removal?

A. Remove the catheter with the patient upright and breathing deeply to flush the tract free of clot

B. Place the patient head down for removal and apply an occlusive dressing afterward

C. Remove the catheter quickly and apply dry gauze with light pressure, changed after an hour

D. Leave the hub open to air during removal to allow the tract to drain freely before pressure is applied

E. Remove the catheter during a spontaneous deep inspiration to lower the venous pressure gradient

546. A 39-year-old woman is resuscitated after a large volume of air enters a central vein from a pressurized rapid infuser during hemorrhage control. The team asks which feature of the event most strongly determines whether the circulation will stop. Which of the following is correct?

A. The total volume of air, because only large cumulative volumes affect the pulmonary arterioles

B. The temperature of the infused fluid, because air in cold fluid dissolves less readily and persists longer

C. The rate of entry, because a rapid bolus can form an air lock in the right ventricular outflow tract

D. The patient's hemoglobin concentration, because anemia determines how long a bubble persists

E. The route of the infusion, because only a central venous route allows air to reach the right heart

547. A 31-year-old woman in labor with preeclampsia, induced for term pregnancy, becomes acutely short of breath and unresponsive, with a seizure, hypotension, and fetal bradycardia. Within minutes, blood fails to clot at the intravenous sites. A colleague asks how the diagnosis of amniotic fluid embolism is made. Which of the following is correct?

A. It is confirmed by a bedside capnographic pattern that is specific to amniotic fluid

B. It is a clinical diagnosis of exclusion, because no test is reliable at the bedside

C. It is confirmed by a positive D-dimer, which is specific to the condition

D. It requires transesophageal echocardiography showing fetal squames in the right heart

E. It is excluded when the patient has any identifiable obstetric risk factor

548. A 36-year-old woman at 34 weeks of gestation suffers cardiac arrest from a presumed amniotic fluid embolism before delivery. Compressions are in progress and the fetus has not been delivered. Which of the following steps best improves the chance of maternal return of circulation?

A. Delay delivery until the obstetric team has completed full fibrinogen and platelet replacement

B. Place the patient in a full left lateral position and hold compressions until delivery

C. Wait for a repeat echocardiogram and a cardiology opinion before deciding on delivery

D. Reserve delivery for the benefit of the fetus only, and only if the maternal rhythm is shockable

E. Left uterine displacement, with fetal delivery within about five minutes if arrest persists

549. A 52-year-old man in the intensive care unit has a massive pulmonary embolism and arrests despite epinephrine, and fibrinolysis has just been given. Compressions are in progress. A nurse asks how long to continue. Which of the following is the best guidance?

A. Stop CPR after two cycles because fibrinolysis restores circulation immediately

B. Continue CPR for 60 to 90 minutes after fibrinolysis to allow the drug to act

C. Stop CPR once the first epinephrine dose has been given a second time

D. Continue only until the first rhythm check shows asystole

E. Continue compressions for exactly 10 minutes and then reassess for termination

550. An 80-year-old man has a cardiac arrest from massive pulmonary embolism during a hemiarthroplasty. Compressions, epinephrine, and fibrinolysis have not restored circulation after prolonged efforts, and a hospital with an extracorporeal program is in the building. Which of the following is the most appropriate next step?

A. Place an intra-aortic balloon pump to augment coronary perfusion pressure and support the failing circulation

B. Begin venovenous support to improve oxygenation of the blood returning to the obstructed lung

C. Alert the extracorporeal team early for venoarterial support to bypass the obstructed pulmonary circulation

D. Insert a percutaneous left ventricular assist device to unload the failing left ventricle

E. Terminate resuscitation, because extracorporeal support is contraindicated in embolic arrest

551. During a spinal fusion in a patient with severe chronic obstructive pulmonary disease, the end-tidal carbon dioxide has always been 12 mm Hg below the arterial value. After a sudden hypotensive episode, the end-tidal value falls by a further 10 mm Hg. A trainee says the baseline gradient makes the change uninterpretable. Which of the following is the best response?

A. The change should be ignored because the baseline gradient is already abnormal

B. Only the absolute end-tidal value is useful, and the baseline gradient adds nothing

C. Only an arterial sample is valid, and end-tidal values are unreliable in lung disease

D. A sudden change from the known baseline remains significant despite the abnormal gradient

E. The fall reflects hyperventilation, which always lowers the end-tidal value gradually

552. In a patient with suspected pulmonary embolism under general anesthesia, the end-tidal carbon dioxide falls sharply and the waveform is intact. A different team member notes that the value briefly rose after a drug was given and then fell again. Which of the following drugs transiently raises the end-tidal value by generating carbon dioxide?

A. Sodium bicarbonate

B. Norepinephrine

C. Vasopressin

D. Rocuronium

E. Heparin

553. A 60-year-old man sustains a pulseless arrest from suspected pulmonary embolism. After fibrinolysis and 10 minutes of compressions, the end-tidal carbon dioxide rises from 8 to 28 mm Hg. Which of the following is the best interpretation?

A. It indicates the compressions are too deep and that the depth should be reduced at once

B. It indicates a measurement artifact and should be disregarded during arrest

C. It indicates the embolus has worsened and the right ventricle has failed further

D. It indicates hyperventilation and the ventilation rate should be lowered at once

E. It is among the earliest signs that compressions or reperfusion therapy are working

554. A 55-year-old man is scheduled for total hip arthroplasty and is at high risk for venous thromboembolism. The patient and surgeon ask whether the choice of anesthetic can influence this risk. Which of the following statements is most accurate?

A. General anesthesia reduces venous thrombosis because it eliminates the stress response

B. Neuraxial anesthesia lowers venous thrombosis after hip and knee surgery

C. Anesthetic technique has no measurable effect on thromboembolic risk after orthopedic surgery

D. Neuraxial anesthesia increases venous thrombosis by reducing limb blood flow and venous return

E. Regional techniques remove the need for mechanical or pharmacologic prophylaxis

555. A 68-year-old woman becomes acutely dyspneic and tachycardic in the recovery room after a long abdominal cancer operation. Saturation is 86 percent on supplemental oxygen. The resident wants to order a D-dimer to decide whether pulmonary embolism is possible. Which of the following is the best response?

A. A normal D-dimer would exclude embolism even in this setting, so it should be ordered first

B. A scoring system designed for the emergency department is the most reliable guide here

C. D-dimer is nearly always elevated after surgery, so clinical suspicion should drive imaging

D. Embolism should be considered only after atelectasis and aspiration are ruled out by imaging

E. The D-dimer is highly specific for embolism after surgery, and a high value confirms the diagnosis

556. A 26-year-old man had a femoral shaft fracture treated with reamed intramedullary nailing. Which of the following findings most strongly points to fat embolism syndrome rather than another cause of postoperative hypoxemia?

A. Hypoxemia and fever within minutes of tourniquet inflation, with no rash or confusion

B. Wheezing and urticaria developing within minutes after a new intravenous antibiotic

C. Hypoxemia with a sudden fall in end-tidal carbon dioxide on day seven after surgery

D. Hypoxemia with a tender calf and unilateral leg swelling, without any rash or confusion

E. Confusion and hypoxemia on day two with a petechial rash on the chest and axillae

557. During hip hemiarthroplasty, a patient becomes hypotensive and hypoxic within minutes of cementing, and the heart rate falls. The surgeon has stopped cementing and oxygen is at 1.0. Blood pressure remains low despite fluids. Which of the following agents is often the most effective for the collapse?

A. A large bolus of crystalloid alone

B. Intravenous heparin to prevent further clot formation

C. Systemic steroids to reduce the inflammatory response

D. Nitrous oxide to supplement the anesthetic depth

E. Epinephrine in small intravenous boluses

558. A patient with suspected massive pulmonary embolism is hypotensive and has an unremarkable right ventricle, a normal septum, and a collapsed inferior vena cava on bedside echocardiography. Which of the following is the best next step?

A. Widen the search to hemorrhage, tamponade, or anaphylaxis, as the findings argue against embolism

B. Start systemic fibrinolysis at once, because echocardiography can miss a pulmonary embolus

C. Give a large fluid bolus and assume that the diagnosis of embolism is confirmed by the hypotension

D. Repeat the study only after sending a D-dimer and a troponin to exclude embolism

E. Begin an inhaled pulmonary vasodilator before any other measure, to offload the right ventricle

559. Intraoperative transesophageal echocardiography in a patient undergoing a pelvic cancer operation shows a mobile, serpiginous mass in the right atrium that prolapses toward the tricuspid valve. Blood pressure is stable. Which of the following is the most appropriate action?

A. Continue the operation unchanged and plan elective imaging in the morning

B. Give a large fluid bolus to flush the clot into the pulmonary arteries

C. Reassure the team because a stable patient has a low risk of further embolism

D. Remove the transesophageal probe at once to avoid dislodging the clot into the lungs

E. Announce the finding at once so the surgical and cardiac teams can adjust plans

560. A 54-year-old man in the operating room has a right ventricle that is distended and a left ventricle that is small and hyperdynamic. The right ventricular free wall is hypokinetic while the apex contracts normally. Which of the following echocardiographic findings does this describe?

A. The D-shaped left ventricle of left heart failure

B. Diastolic collapse of the right atrium seen in tamponade

C. The McConnell sign, supporting acute pulmonary embolism

D. The mill-wheel flow pattern of venous air in the coronary sinus

E. Systolic anterior motion of the mitral valve in outflow obstruction

QUESTION GROUP 29

Obstructive and Pressure Crises

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Questions 561 to 580. Choose the single best answer for each.

561. A 47-year-old hypovolemic trauma patient with multiple rib fractures is intubated and ventilated. Soon after induction the peak airway pressure rises, the saturation falls, and the blood pressure drops to 66 over 34. Breath sounds are diminished on the left with hyperresonance. The neck veins are flat and the trachea is midline. A colleague says tension pneumothorax is unlikely because these signs are absent. Which of the following is the best response?

- **A.** Flat neck veins exclude tension pneumothorax, so the search should turn to tamponade or hypovolemia
- **B.** A midline trachea excludes a mediastinal shift, so decompression is unnecessary at this point
- **C.** Hyperresonance is a late sign of tension pneumothorax and is unreliable in ventilated patients
- **D.** Flat neck veins occur in hypovolemia and tracheal deviation is late, so the diagnosis remains likely
- **E.** A chest radiograph must be obtained first because the clinical findings are insufficient to justify decompression

562. A 29-year-old man on positive pressure ventilation has a tension pneumothorax and is hypotensive. The anesthesiologist is trained in chest drainage and wants the preferred method of decompression in a ventilated patient. Which of the following is the best choice?

- **A.** Finger thoracostomy in the fifth intercostal space anterior to the midaxillary line, then a chest tube
- **B.** Needle aspiration through the second intercostal space, midclavicular line, using a small caliber needle only
- **C.** Tube placement in the seventh intercostal space posterior to the scapular line without opening the pleura
- **D.** Observation after switching to spontaneous ventilation, because a small leak will seal on its own
- **E.** Percutaneous drainage with a small pigtail catheter placed through the subxiphoid approach

563. A muscular 35-year-old man weighing 130 kg develops a tension pneumothorax after a failed interscalene block, and needle decompression is chosen because no one is trained to perform thoracostomy. Which of the following best describes the technique?

- **A.** Use a long, large-bore catheter, because failure is most common in muscular and obese patients
- **B.** Use a short small-gauge catheter, because deeper insertion risks injury to the great vessels
- **C.** Use a catheter of any length, because chest wall thickness does not affect the chance of success
- **D.** Insert at the sixth intercostal space posterior to the midaxillary line to avoid the muscle
- **E.** Delay the procedure until imaging identifies the exact depth and size of the pneumothorax

564. During a laparoscopic gastric operation, the end-tidal carbon dioxide and airway pressure rise together and the saturation falls. A lung ultrasound shows absent lung sliding on the right and no comet-tail artifacts. Which of the following is the best reason to avoid diagnosing pneumothorax from absent sliding alone?

- **A.** Absent sliding is specific on its own, because a lung point is never found at the edge of a collapse
- **B.** Lung sliding is always absent over a pneumothorax, so no other sign is ever needed
- **C.** Absent sliding is more reliable than the airway pressure, which rises only with a pneumothorax
- **D.** A mainstem intubation can produce absent sliding on the nonventilated side
- **E.** Normal sliding on the contralateral side proves that the absent sliding is caused by pneumothorax

565. A 60-year-old woman suffers cardiac arrest during a difficult central line insertion in the operating room. She is being ventilated and compressions are in progress. The anesthesiologist suspects a reversible cause. Which of the following is the best approach?

A. Decompress only the side with the line, since a contralateral pneumothorax cannot occur

B. Wait for the chest radiograph to confirm the side before decompressing

C. Pause compressions for the ultrasound study before deciding on any chest procedure

D. Give epinephrine alone, which resolves obstructive shock without drainage

E. Decompress both sides of the chest during resuscitation as a reversible cause

566. A 56-year-old man is undergoing thoracic surgery with one-lung ventilation. The ventilated lung suddenly develops high airway pressure, the saturation falls to 70 percent, and the blood pressure drops. Breath sounds on the ventilated side are absent. Which of the following statements is most accurate?

A. The condition is expected and resolves when two-lung ventilation resumes without drainage

B. A pneumothorax in the ventilated lung is less dangerous than in the nonventilated lung

C. The finding is best managed by increasing tidal volume to reexpand the lung

D. A pneumothorax in the ventilated lung is an emergency because no other lung is available

E. The diagnosis can be excluded because the nonventilated lung is collapsed by design

567. A 63-year-old man on mechanical ventilation after major surgery has an unexplained sudden fall in blood pressure. A trainee notes that the pulsus paradoxus on the arterial line is not present and concludes there is no tamponade. Which of the following statements is most accurate?

A. Pulsus paradoxus is the most sensitive sign in ventilated patients and its absence excludes tamponade

B. Pulsus paradoxus is unreliable during positive pressure ventilation, when the pattern may reverse

C. The finding is defined as an expiratory fall in systolic pressure of more than ten millimeters of mercury

D. Beck's triad is present in most patients and is easy to detect under the surgical drapes

E. Electrical alternans with low-voltage complexes excludes tamponade when it is present on the monitor

568. A 70-year-old man is in the intensive care unit on the second day after valve replacement. The chest tubes drained briskly overnight and then stopped. The heart rate is rising, cardiac output is falling, and the right atrial pressure is rising. A transthoracic echocardiogram shows no significant pericardial fluid. Which of the following is the best next step?

A. Perform transesophageal echocardiography to look for a localized clot

B. Accept the negative study and treat for heart failure with a diuretic

C. Remove the chest tubes to prevent infection and repeat the study in 24 hours

D. Give a beta-blocker to slow the heart rate and improve filling

E. Obtain a repeat transthoracic study with the patient in the same position

569. A 66-year-old woman with a hemodynamically significant pericardial effusion is waiting for drainage. She is hypotensive and tachycardic, and the central venous pressure is high. While the team prepares the equipment, which of the following temporizing measures is most appropriate?

A. Give fluid and an inotropic agent, with a vasopressor, without delaying drainage

B. Give a beta-blocker to slow the heart rate and prolong ventricular filling time

C. Apply high positive end-expiratory pressure to support the failing right heart

D. Withhold fluids because the central venous pressure is already high and the heart is full

E. Give a vasodilator to reduce afterload and improve forward flow through the compressed ventricle

570. A 44-year-old man develops tamponade after a percutaneous cardiac procedure. Echocardiography shows a large effusion with organized clot filling the pericardial sac. A colleague suggests needle pericardiocentesis. Which of the following is the best response?

A. Needle pericardiocentesis is always sufficient for any traumatic or postoperative effusion

B. The effusion should be observed with serial imaging, because small volumes do not cause tamponade

C. Clot cannot be aspirated, so surgical drainage by subxiphoid window or sternotomy is needed

D. Repeat needle attempts are preferred over surgery in order to avoid the risks of general anesthesia

E. Pericardiocentesis is contraindicated in every case of tamponade after a percutaneous procedure

571. A 24-year-old man arrives with a stab wound near the left sternal border, hypotension, and fluid in the pericardium on focused ultrasound. He is taken to the operating room. Which of the following is the best anesthetic plan?

A. Induce slowly with a large dose of propofol and let the surgeon complete scrubbing meanwhile

B. Delay induction until a full set of laboratory studies and a computed tomogram are done

C. Induce with the surgeon scrubbed, blood and vasopressors running, and no delay to incision

D. Induce only after percutaneous pericardiocentesis has restored a normal blood pressure

E. Begin with a high-dose volatile agent by mask to blunt the sympathetic response

572. A 52-year-old man undergoing laparoscopic nephrectomy becomes hypotensive immediately after trocar insertion. There is no bradycardia, the abdomen is not yet distended, and the end-tidal carbon dioxide is stable. The surgeon sees no blood in the field. Which of the following is the most likely cause that must be excluded?

A. Vagal reaction to peritoneal stretch, which would cause bradycardia

B. Carbon dioxide absorption causing hypercapnic vasodilation

C. Pneumothorax from carbon dioxide tracking through the diaphragm

D. Major vessel injury with concealed retroperitoneal hemorrhage

E. Reverse Trendelenburg positioning reducing venous return

573. A 50-year-old man is in the intensive care unit after laparotomy for pancreatitis with massive resuscitation. The bladder pressure is repeatedly 15 mm Hg and he has no organ dysfunction. Which of the following best describes his condition?

A. Abdominal compartment syndrome, which is defined by a sustained pressure above 12 mm Hg alone

B. Intra-abdominal hypertension, a sustained pressure of 12 mm Hg or higher without organ dysfunction

C. A normal finding after massive resuscitation, because normal intra-abdominal pressure is 12 to 15 mm Hg

D. Abdominal compartment syndrome, which requires only a pressure above 5 mm Hg with oliguria

E. Normal pressure, with the repeated measurement falsely high because of bladder overfilling

574. A 42-year-old woman is receiving mechanical ventilation after a trauma laparotomy. The bladder pressure is to be measured. Which of the following describes the correct technique?

A. Instill a small volume of saline and read at end expiration, zeroed at the midaxillary line

B. Instill 500 mL of saline into the bladder and read at peak inspiration with the transducer at the symphysis

C. Read from the rectum with a balloon catheter at the end of a deep inspiration, zeroed at the symphysis

D. Read a central venous pressure and subtract 10 mm Hg to estimate the abdominal value

E. Measure the bladder pressure with the patient sitting upright and the transducer at the xiphoid

575. A 38-year-old man with abdominal compartment syndrome has a decompressive laparotomy. The blood pressure improves, but a few minutes later the monitor shows peaked T waves and wide complexes. Which of the following is the best immediate treatment?

A. Give a vasodilator to offset the expected reperfusion rise in blood pressure

B. Restrict all fluid because the abdomen is now open and the pressure is relieved

C. Give atropine because the rhythm is a vagal bradycardia after decompression

D. Return to closing the abdomen at once to restore the earlier intra-abdominal pressure

E. Give calcium chloride intravenously and prepare bicarbonate and volume

576. A 58-year-old woman is receiving pneumoperitoneum in the steep head-down position. She becomes hypoxemic, the peak pressure rises, and breath sounds are absent on the left. Which of the following is the most likely explanation?

A. Pneumoperitoneum has abolished the vagal tone to the left airways and caused them to collapse

B. Carbon dioxide absorption from the peritoneum has caused isolated left-sided bronchodilation

C. Cephalad movement of the carina has pushed the tube into the right mainstem bronchus

D. The diaphragm has been pushed caudad by the insufflation and has overexpanded the left lung

E. The tube cuff has been displaced out of the trachea above the cords by the insufflation

577. A 74-year-old man with severe aortic stenosis and heart failure requires laparoscopic cholecystectomy because open surgery is thought riskier. Which of the following approaches best reduces the circulatory cost of pneumoperitoneum?

A. Use the highest pressure for the best view, since a shorter operation matters most of all

B. Use extreme reverse Trendelenburg positioning to improve venous return and the surgical view

C. Avoid arterial monitoring because the hemodynamic effects are brief and well tolerated

D. Restrict fluids to prevent pulmonary edema before insufflation and keep the preload low

E. Use low insufflation pressure and a slow rate, with an arterial line and preloading

578. A 45-year-old woman with severe asthma is on mechanical ventilation, with expiratory flow that does not return to zero before the next breath. The anesthesiologist wants a numerical measure of trapped gas. Which of the following statements is most accurate?

A. An end-expiratory hold is always accurate, even in patients who are breathing actively

B. The peak airway pressure alone gives the exact trapped gas volume at end expiration

C. The set PEEP on the ventilator equals the intrinsic PEEP in all cases of airflow obstruction

D. An end-expiratory hold gives total PEEP but underestimates it in active breathers

E. Measurement is unnecessary because trapped gas cannot be quantified at the bedside by any method

579. A 32-year-old man with a status asthmaticus attack is intubated and develops auto-PEEP, which has been relieved by a brief disconnection. He is now stable. Which of the following ventilator strategies is most appropriate going forward?

A. Increase the respiratory rate and minute ventilation to keep the carbon dioxide normal

B. Lower the rate and tidal volume, shorten inspiratory time, and accept hypercapnia

C. Use an inverse ratio with a prolonged inspiratory pause to improve oxygenation and gas distribution

D. Raise the tidal volume to recruit the airways and reduce the dead space ventilation

E. Apply high external PEEP in every case to splint the airways open and prevent collapse

580. A 61-year-old man on positive pressure ventilation has a sudden fall in blood pressure. The peak airway pressure is low, the central venous pressure is high, the neck veins are distended, and there is no wheeze. Which of the following pairs of diagnoses is most likely?

A. Cardiac tamponade or a pulmonary embolus

B. Tension pneumothorax or auto-PEEP

C. Bronchospasm or a kinked endotracheal tube

D. Abdominal compartment syndrome or mainstem intubation

E. Mucus plug or anaphylaxis

QUESTION GROUP 30

Hemodynamic Collapse and Metabolic Emergencies

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Questions 581 to 600. Choose the single best answer for each.

581. A 67-year-old man is weaned from bypass after mitral valve repair. The mean arterial pressure is 52 mm Hg, the cardiac index is 1.7 L/min/m2, the pulmonary capillary wedge pressure is high, the extremities are cool, and the mixed venous oxygen saturation is poor. The rhythm is sinus. Which of the following best describes the hemodynamic state?

A. Vasoplegia, in which cardiac output is high and systemic vascular resistance is low, with warm extremities

B. Ventricular failure, a pump problem where an inotrope is reasonable once reversible causes are excluded

C. Hypovolemia, in which the filling pressures are low and rapid volume loading is the first treatment

D. Anesthetic vasodilation, in which a modest dose of vasopressor restores the pressure at once

E. Adrenal crisis, in which the pressure is refractory to vasopressors but cardiac output is high

582. A 55-year-old man with glucose-6-phosphate dehydrogenase deficiency develops refractory vasoplegia after bypass. He is taking no serotonergic drugs. A colleague proposes methylene blue as rescue therapy. Which of the following is the most important concern with this plan?

A. Methylene blue binds nitric oxide and so needs to be given only after a trial of hydroxocobalamin has failed

B. Methylene blue always lowers systemic vascular resistance and so worsens the vasoplegia in every patient

C. Methylene blue risks hemolysis in glucose-6-phosphate dehydrogenase deficiency, so another agent is better

D. Methylene blue is the first-line vasopressor for vasoplegia and must precede norepinephrine and vasopressin

E. Methylene blue interferes with the cortisol assay and delays the diagnosis of adrenal insufficiency

583. A 63-year-old man taking a beta-blocker is in low output failure after bypass. The team chooses an inotrope that acts independently of the beta receptor. A trainee asks about its key drawback. Which of the following is correct?

A. Milrinone has a very short half-life and needs frequent bolus doses to maintain effect

B. Milrinone is a long-acting vasodilator, so a vasopressor commonly accompanies it

C. Milrinone reliably raises systemic vascular resistance and blood pressure, so no pressor is needed

D. Milrinone needs a higher dose in renal impairment because clearance is increased

E. Milrinone acts only on the beta receptor and is ineffective in this patient

584. A 68-year-old man receives protamine to reverse heparin after bypass. Within a minute his pulmonary artery pressure rises sharply, the right ventricle dilates, and the systemic pressure falls. There is no rash or bronchospasm. Which of the following is the best interpretation and management?

A. Simple histamine release, which is treated by giving the remaining protamine faster

B. A transfusion reaction to the pump blood, which is treated by starting an infusion of calcium

C. Aspiration of gastric contents at induction, which is treated by urgent bronchoscopy and lavage

D. Hypovolemia from surgical bleeding, which is treated by a large fluid bolus and blood

E. Pulmonary vasoconstriction from protamine; stop it and support the right ventricle

585. A 45-year-old woman receives a volatile anesthetic and propofol for a long surgery and her mean pressure falls to 55 mm Hg. Cardiac output is normal on echocardiography and filling is adequate. The resident wants to start methylene blue for vasoplegia. Which of the following is the best response?

A. Start methylene blue because normal output with low resistance defines vasoplegia

B. Give hydroxocobalamin because anesthetic effect is rarely reversible

C. Lighten the anesthetic and give phenylephrine or norepinephrine as needed

D. Add a second inotrope to raise contractility and the cardiac output

E. Start hydrocortisone and withhold vasopressors until the cortisol returns

586. A 32-year-old man with a severe head injury has an intracranial pressure monitor showing high pressure. He is bradycardic with a heart rate of 44 beats per minute, hypertensive at 190 over 100 mm Hg, and has irregular breathing. Oxygenation is normal. Which of the following is the best interpretation?

A. A vagal reflex from airway suction that resolves with atropine alone

B. The Cushing reflex from raised intracranial pressure

C. Bezold-Jarisch reflex from an underfilled left ventricle

D. Hypoxia-induced bradycardia that resolves with oxygen alone

E. A drug effect from dexmedetomidine that requires pacing

587. A 28-year-old man is placed upright after a spinal anesthetic for knee surgery and becomes bradycardic and faint, with a heart rate of 40 beats per minute. A trainee asks which reflex best explains a sudden paradoxical bradycardia with vasodilation after a spinal. Which of the following is correct?

A. The Bezold-Jarisch reflex, triggered by mechanoreceptors in an underfilled left ventricle

B. The oculocardiac reflex, triggered by trigeminal afferent input from pressure on the globe

C. The Cushing reflex, triggered by brainstem compression from raised intracranial pressure

D. The baroreceptor reflex, triggered by stretch receptors during systemic hypertension

E. The Bainbridge reflex, triggered by stretch receptors during a rise in right atrial pressure

588. A patient has symptomatic bradycardia and hypotension during surgery and the anesthesiologist decides to give atropine. A resident prefers to give a small dose to avoid tachycardia. Which of the following statements is most accurate about the dosing?

A. Small doses are safer because tachycardia from atropine is always dangerous

B. Atropine is ineffective below 3 mg, which should be given in one bolus

C. Atropine paradoxically slows the rate only at doses above 1 mg

D. Doses below 0.5 mg may paradoxically slow the rate, so a full dose is given

E. The dose should be reduced in every patient who received a beta-blocker

589. A 40-year-old woman presents to the operating room after an intentional overdose of a beta-blocker. She is bradycardic and hypotensive despite standard atropine and fluids. Which of the following is the best additional therapy?

A. Higher doses of the same beta-blocker to overcome receptor binding

B. Glucagon, with calcium and high-dose insulin with dextrose

C. Digoxin immune antibody fragments as the specific antidote

D. Lipid emulsion as the first-line treatment before any other drug

E. Hydrocortisone to restore the sensitivity of the beta receptor

590. A 58-year-old man with renal failure has an unexplained arrhythmia during a long operation. A blood gas analyzer shows a potassium of 7.8 mEq/L. Which sequence best describes the electrocardiographic progression of worsening hyperkalemia?

A. Short QT interval, ST elevation, new Q waves, atrial flutter, then supraventricular tachycardia

B. Prolonged QT interval, U waves, flattened T waves, ST depression, then torsades de pointes

C. Peaked T waves, prolonged PR, loss of P waves, widened QRS, sine wave, then fibrillation or asystole

D. Loss of QRS voltage, electrical alternans, sinus tachycardia, narrow QRS, then pulseless activity

E. Delta waves, short PR interval, atrial fibrillation with rapid conduction, then ventricular tachycardia

591. A 49-year-old woman has widened QRS complexes and peaked T waves from hyperkalemia during surgery. Calcium chloride has been given and the electrocardiogram is improving. The anesthesiologist plans insulin and dextrose. Which of the following correctly describes the timing of these two treatments?

A. Calcium lasts many hours, so insulin and dextrose are rarely needed once the QRS has narrowed

B. Insulin and dextrose act within seconds and so make any further calcium unnecessary

C. Calcium removes potassium from the body, while insulin only stabilizes the membrane

D. Calcium acts in minutes for half an hour; insulin with dextrose acts in 15 to 30 minutes

E. Both agents act only after dialysis has started and potassium is actually removed

592. A 36-year-old man with extensive burns sustained three days ago needs urgent surgery with a rapid sequence induction. Which of the following is the best choice of neuromuscular blocker?

A. Rocuronium, with sugammadex available, to avoid a massive potassium rise

B. Succinylcholine, because the burn is not on the first day

C. Succinylcholine at a reduced dose, which prevents the potassium rise

D. Cisatracurium alone, which has a faster onset than rocuronium

E. No neuromuscular blocker, because rapid sequence induction is contraindicated

593. A 55-year-old man with suspected hyperkalemia from rhabdomyolysis arrests during surgery. Compressions are under way. Which of the following statements is most appropriate for the resuscitation?

A. Inhaled beta-agonists are the first-line treatment in arrest and should be given before calcium

B. Resuscitation should stop after the usual 20 minutes, because the cause is not reversible with drugs

C. Calcium must be avoided in hyperkalemic arrest because it worsens the hyperkalemia

D. Calcium, bicarbonate, and insulin with dextrose are given, but inhaled beta-agonists are not

E. Only dialysis is useful, so drug therapy is pointless during compressions in this setting

594. A 57-year-old woman with type 2 diabetes, who takes a sodium-glucose cotransporter 2 inhibitor, has an emergency laparotomy. She is lethargic and tachypneic. Her glucose is 160 mg/dL, but the pH is 7.12 with a high anion gap and positive ketones. Which of the following is the best explanation?

A. Metformin-associated lactic acidosis, which never causes ketosis

B. Respiratory acidosis from hypoventilation, which would show a normal anion gap

C. Hyperchloremic acidosis from saline, which would show a high anion gap

D. Pure diabetic ketoacidosis, which always requires glucose above 250 mg/dL

E. Euglycemic ketoacidosis from the sodium-glucose cotransporter 2 inhibitor

595. A 65-year-old man with severe metabolic acidemia and hypotension is under general anesthesia. The pH is 7.05. The anesthesiologist considers sodium bicarbonate. Which of the following is the most important consideration?

A. Bicarbonate generates carbon dioxide, so ventilation must be increased and a repeat gas obtained

B. Bicarbonate has no effect on intracellular pH and so is useless in all patients with severe acidemia

C. Bicarbonate should be given in large doses, since the carbon dioxide it generates is harmless

D. Bicarbonate treats the cause of the lactic acidosis and so makes other treatment unnecessary

E. Bicarbonate is preferred over increasing ventilation in respiratory acidosis from hypoventilation

596. A 72-year-old man receives a massive transfusion for a ruptured aneurysm, and the liver clearance of citrate is impaired by shock. The arterial pressure falls further, contractility is poor, and bleeding increases. Which of the following is the most likely contributor that should be treated?

A. Hypercalcemia from red cell breakdown in stored blood, which causes vasodilation and bleeding

B. Hypernatremia from the sodium load in stored blood, which reduces clot formation and contractility

C. Hypermagnesemia from the preservatives in stored blood, which causes tachycardia and impairs contractility

D. Hypocalcemia from citrate binding ionized calcium, impairing contractility and coagulation

E. Hypophosphatemia from the anticoagulants in stored blood, which reduces oxygen delivery

597. A 46-year-old woman has a metabolic acidosis with a pH of 7.20, a bicarbonate of 12 mEq/L, and a measured carbon dioxide of 38 mm Hg during general anesthesia. Using Winter's formula, the expected carbon dioxide is about 26 mm Hg. Which of the following is the best interpretation?

A. Appropriate compensation, because any carbon dioxide below 40 mm Hg is adequate

B. A primary respiratory alkalosis requiring a lower minute ventilation to normalize the pH

C. Inadequate ventilatory compensation that can be corrected with ventilator changes

D. A laboratory error, because the carbon dioxide cannot exceed the predicted value

E. A normal anion gap acidosis, which requires bicarbonate replacement in every case

598. A 60-year-old woman who takes prednisone daily for rheumatoid arthritis is hypotensive and refractory to fluid and vasopressors during surgery. A cortisol and corticotropin level are available but would take time. Which of the following is the best approach?

A. Wait for the cortisol level to return, because a treatment without confirmation is unsafe

B. Give hydrocortisone now, drawing blood first only if the dose is not delayed

C. Give a steroid only if the random cortisol is above the normal range for a stressed patient

D. Delay hydrocortisone until the end of surgery to avoid wound complications and hyperglycemia

E. Give dexamethasone and then wait for the cortisol result, because it does not affect the assay

599. A 52-year-old man undergoes resection of a pheochromocytoma. After the tumor vein is ligated, he becomes profoundly hypotensive despite adequate volume. Which of the following best explains the hypotension?

A. Rebound catecholamine release from the remaining tumor causing severe vasoconstriction and hypertension

B. Hypercalcemia from coexisting hyperparathyroidism causing a loss of vascular tone

C. Hypoglycemia from excess insulin, which causes a drop in vascular resistance

D. Isolated hypovolemia from blood loss at the ligation, which resolves without a vasopressor

E. Sudden loss of catecholamine excess with vasodilation and possible adrenal insufficiency

600. A 68-year-old man took a sulfonylurea and received dextrose for hypoglycemia during a procedure, but his glucose drifts low again over several hours despite repeated boluses. Which of the following is the best additional therapy?

A. Repeated intravenous glucagon alone, which is most effective in malnourished patients

B. Additional insulin to prevent rebound hyperglycemia after the repeated dextrose boluses

C. A single large bolus of concentrated dextrose, with no repeat glucose check

D. Hydrocortisone as the specific antidote for sulfonylurea toxicity and rebound hypoglycemia

E. Octreotide, because sulfonylurea overdose causes recurrent hypoglycemia

Answers and Explanations

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Group 1. The Anatomy of an Intraoperative Crisis

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1. Correct answer C. Holes in several successive defenses lined up and allowed a small trigger through

Serious events in anesthesia rarely have one cause. The layered defenses model holds that harm reaches the patient only when the weaknesses in successive layers, such as labeling, machine checks, and alarm settings, line up. Blaming one clinician ignores the latent weaknesses that sat in the system for weeks. Reserve works in the opposite direction, since a small margin makes a small trigger more dangerous. A choice between fast and slow thinking may matter in the moment but does not account for several system defects aligning.

Key principle: *Harm reaches the patient when holes in successive defenses align, usually after latent weaknesses have accumulated.*

2. Correct answer E. A pregnant woman with fever at induction for an urgent operation

The same trigger produces different outcomes because reserve differs. Functional residual capacity falls with pregnancy, obesity, the supine position, and abdominal distension, while oxygen consumption rises with fever, sepsis, shivering, and pregnancy. A patient with both a small lung volume and high consumption has a fraction of the apneic time of a healthy volunteer. The other patients have either normal consumption or normal lung volume, or have been preoxygenated, so their margin is larger.

Key principle: *Estimate reserve before the case: low lung volume plus high oxygen consumption means very little time.*

3. Correct answer C. Remaining in fast pattern recognition after the pattern stopped fitting

Fast pattern recognition is right most of the time, but the danger is staying in that mode when the evidence no longer fits. Here the probe explanation has not produced improvement, and other findings point elsewhere, so the clinician should shift to slow, deliberate analysis and look at the patient and the ventilation. The other choices describe different problems. Slow analysis is reserved for cases pattern recognition cannot settle, and nothing here suggests treatment was withheld pending a diagnosis.

Key principle: *Switch from fast to slow thinking when the pattern stops fitting.*

4. Correct answer B. Check ventilation and the airway now, since an early change costs little to check

A crisis can be interrupted at any stage, and the earlier the cheaper. A fall from 100 to 97 percent costs a few seconds of attention, while a fall to 70 costs the room. Waiting for a threshold gives away that advantage. Assuming artifact is a documented habit of clinicians who have silenced many false alarms, and it delays recognition. Letting the surgeon finish removes any chance that a surgical cause is found early.

Key principle: *Intervene at the first change on the monitor; the earlier a crisis is interrupted, the cheaper it is.*

5. Correct answer A. Room A rewards reading the trend over time, and room B rewards speed of action

Crises differ in tempo. Anaphylaxis and local anesthetic toxicity can turn a stable patient into a collapsing one within a minute and reward speed. A slow leak or gradual bleeding announces itself through trends visible for a long time to anyone who looks, and it rewards the clinician who reads the trend and not the single number. Waiting or seeking confirmation first gives up time in the fast crisis and ignores the evidence already available in the slow one.

Key principle: *Fast crises reward speed; slow ones reward reading the trend and not the number.*

6. Correct answer B. Around transitions such as induction, positioning, emergence, and handover

Human factors research shows errors are most likely when a task is unfamiliar, rushed, interrupted, or done at the end of a long day, and they cluster around transitions. Induction, positioning, emergence, and every handover of care deserve extra attention. A short pause before induction to check the airway plan, drugs, and suction costs under a minute and removes a share of avoidable events. Stable maintenance has fewer transitions, and neither the morning check nor trainee involvement is the pattern the research describes.

Key principle: *Errors cluster at transitions; add a deliberate pause before induction.*

7. Correct answer A. Build a small number of habits that make the correct check automatic

Monitors produce data and not meaning, and alarms sound for artifacts far more often than for emergencies, so the clinician is slower to believe the next one. Vigilance cannot be sustained, so the remedy is a small number of habits that make the correct check automatic. Lowering volume or widening limits removes the warning that protects the patient. Fatigue, multitasking, and the pull of the surgical field all narrow attention, which is why a habit works better than effort.

Key principle: *Replace sustained vigilance with automatic habits that fire when the numbers change.*

8. Correct answer D. Call for help, then give 100 percent oxygen at a high flow

The first minute follows a fixed order that does not depend on the diagnosis. Help is called first, because early help can be sent away and late help cannot be recalled. Oxygen at 100 percent and high flow is almost never harmful over a few minutes and almost always useful. Replacing a probe, drawing a blood gas, or recalibrating a monitor are diagnostic steps that come after the patient has been supported. Treat the problem before the diagnosis.

Key principle: *First minute order: help, then 100 percent oxygen, then confirm ventilation and circulation.*

9. Correct answer A. No ventilation or no circulation until proven otherwise; trust the waveform

Capnography is the most reliable single monitor in the first minute. A flat trace means no ventilation or no circulation until proven otherwise. The saturation lags by many seconds, so a normal reading does not show that gas is moving at this moment. Trust the waveform over the chest rise and over the saturation. Calling the trace artifact or waiting for a gas result delays the response, and chest rise can be misleading, so the airway and circuit must be checked while ventilating by hand.

Key principle: *A flat capnogram means no ventilation or no circulation until proven otherwise.*

10. Correct answer C. Feel for a carotid or femoral pulse and check whether a lead has come off

A lead that has come off can mimic cardiac arrest, and a pulse check takes two seconds and shows whether the heart is delivering blood. Here the oximeter waveform and capnogram argue against arrest, so a lead problem is likely. Starting compressions on a perfused patient causes harm. Waiting for a cuff cycle wastes time. Silencing the alarm without identifying the cause is a habit that should be avoided. Confirm surprising numbers by touch when they do not match the picture.

Key principle: *When numbers and the patient disagree, feel the pulse before acting.*

11. Correct answer B. Review aloud what was given, in what dose, and when, and name the likely trigger

Drugs given in the last few minutes are the most common explanation for a sudden change. Reviewing them aloud, including dose, labeling, and timing, helps the team and the clinician recall the recipe for a specific emergency, since antibiotics and neuromuscular blockers are known triggers. Dismissing drugs because induction was smooth is the kind of early closure that causes errors. A technical search first delays the likeliest answer, and waiting for incision adds no information.

Key principle: *Recent drugs are the commonest cause of a sudden change; review them aloud.*

12. Correct answer D. It points toward drugs, a vagal event, or a primary cardiac problem

The monitors form a map. A fall in blood pressure with a rising heart rate points toward loss of volume or vasodilation, while a fall with a slow rate points toward drugs, a vagal event, or a primary cardiac problem. The

pattern does not settle the diagnosis, but it tells the clinician where to look next. Hypermetabolic states raise carbon dioxide and heart rate, and a circuit leak affects tidal volume and the capnogram.

Key principle: *Hypotension with tachycardia suggests volume loss or vasodilation; with bradycardia suggests drugs, vagal, or cardiac causes.*

13. Correct answer D. Move to the algorithm for the leading diagnosis and stop exploring

A mental clock helps. At about a minute, ask whether the patient is stable, improving, or worse. If stable and improving, there is time to look for a cause. If worse, the clinician moves to the algorithm for the leading diagnosis and does not continue to explore. A broad search risks losing the patient while the diagnosis is sought. Simply repeating the assessment gives no new benefit, and waiting for a surgeon's diagnosis cedes the decision to someone not watching the circulation.

Key principle: *Check the trajectory at one minute: worse means commit to the leading algorithm.*

14. Correct answer E. The anesthesiologist, who should state plainly that the surgery must pause

If the patient cannot be stabilized with the operation continuing, the operation should stop, and brief surgical acts such as packing a wound, releasing retraction, or deflating a tourniquet buy time. The decision is joint, but the anesthesiologist knows how long the patient can wait and should state it. Leaving the decision to the surgeon alone, or to the nurse as an intermediary, loses that information. Waiting for stability before speaking defeats the purpose, since the surgery is part of what is preventing stability.

Key principle: *The anesthesiologist knows the patient's clock and should say when surgery must pause.*

15. Correct answer E. The most dangerous diagnosis is the one that cannot be afforded to miss

Anchoring, confirmation bias, and availability bias push clinicians toward the explanation that feels familiar. Listing the lethal explanations first is a countermeasure because the most dangerous diagnosis is the one that cannot be afforded to miss, even if it is less likely. Common diagnoses are not necessarily lethal, and lethal causes are not always easy to confirm. Seriousness does not reduce disagreement, and inviting disagreement from the least senior person is a separate, complementary step.

Key principle: *Consider lethal explanations first, because a missed dangerous diagnosis cannot be afforded.*

16. Correct answer A. It turns recall into reading and protects steps that judgment tends to skip

Memory fails under stress, and cognitive aids exist because of that. An emergency checklist turns the recall problem into a reading problem, and teams with a checklist at hand omit fewer steps in simulated crises. The aid does not replace judgment. It protects the steps that judgment tends to skip when the room is loud, such as repeating a dose or stopping the volatile agent. Communication still matters, and experienced clinicians are as susceptible to stress effects as trainees.

Key principle: *Cognitive aids protect against omissions under stress and do not replace judgment.*

17. Correct answer B. Rehearse the response in simulation so that it is available under stress

Stress narrows thinking in predictable ways, and a response that exists only as knowledge is not reliably available when attention collapses. The antidote is rehearsal: a response practiced in simulation can be recalled under pressure. Reading and silent recitation build knowledge but not performance under stress. Depending on a supervisor transfers the problem to one person who is subject to the same stress effects.

Key principle: *Rehearsal in simulation makes a response available when stress narrows attention.*

18. Correct answer B. Outcome bias makes decisions look worse than they were when the result is bad

Hindsight and outcome bias distort learning from events. A bad outcome makes the decisions that preceded it look worse than they were, and a good outcome makes poor decisions look sound. Reviews should focus on the quality of reasoning given what was known at the time. The questions worth asking are what was seen, what was thought, what was done, and what would have changed the thinking. The other biases describe errors in real time diagnosis, not in retrospective judgment.

Key principle: *Judge the reasoning given what was known, not only the outcome.*

19. Correct answer A. Assign tasks to people by name and name one person as recorder

A request to someone is received by no one, so roles should be assigned out loud and by name. Documentation belongs to a named person, because times, doses, and interventions are forgotten within minutes. The recorder also reads back what has been done and when, which helps the team notice that a dose has not been repeated. Louder speech and private memory do not fix either problem. Overloading one helper, or having the leader record, removes the leader's thinking capacity, and excluding the surgeon removes useful information.

Key principle: *Name the person for every task, and name a recorder.*

20. Correct answer C. State before the case that anyone who sees something should speak up

Hierarchy can silence the people who see the problem, and junior staff, nurses, and technicians often notice a falling pressure or pooling blood before the senior clinician does. A leader who invites comment before the event makes the later statement much easier. A team that has been told it is allowed to interrupt will do so, and many catastrophes have been avoided by a short sentence from the least expected voice. Delaying concerns or restricting monitoring to one person removes the extra eyes.

Key principle: *Invite comment before the event so that the quietest observer feels able to speak.*

Group 2. Monitoring That Warns Early

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21. Correct answer E. Required with a tracheal tube or supraglottic airway, and sought in moderate and deep sedation

Capnography tests so many links in the chain from production to the sensor that the standard requires continual end-tidal carbon dioxide analysis from the time a tracheal tube or supraglottic airway is placed. For moderate and deep sedation, every effort should be made to monitor exhaled carbon dioxide. The standard is not limited by case length, by the use of a relaxant, or by the result of auscultation. Waiting for auscultation first would also defeat the purpose, since the waveform is the more reliable confirmation.

Key principle: *Continual end-tidal carbon dioxide is required with a tracheal tube or supraglottic airway, and sought during sedation.*

22. Correct answer D. Draw one arterial sample, learn his gradient, and read the monitor with that offset

In a healthy anesthetized patient the end-tidal value runs only a few millimeters of mercury below the arterial value. The gradient grows in the elderly, in chronic lung disease, in lateral and prone positions, and in any state that reduces pulmonary blood flow. When the clinical picture and the monitor disagree, one arterial sample reveals the patient's own gradient, and the capnograph then becomes a reliable guide. Assuming the two values are equal ignores this gradient. Abandoning the monitor, or relying only on repeated gases, throws away its continuous information.

Key principle: *Learn the patient's gradient with one arterial gas, then use the capnograph with that offset.*

23. Correct answer A. Look for a partial disconnection or a leak around the cuff

A fall in end-tidal carbon dioxide with a persisting waveform has two broad causes. If the blood pressure is falling in step, the circulation is the problem, as with embolism, hemorrhage, or low cardiac output. If the pressure is normal, the clinician should look for a partial disconnection or a leak around the cuff that dilutes the sample. Hemorrhage should show hemodynamic change. Malignant hyperthermia and exhausted absorbent both raise the value and do not lower it.

Key principle: *A falling end-tidal value with a normal blood pressure suggests a leak or partial disconnection.*

24. Correct answer B. A hypermetabolic state such as malignant hyperthermia

A rise in end-tidal carbon dioxide with a normal baseline points to increased production or hypoventilation. Because the minute ventilation has been increased and the value continues to climb, production must be rising. Together with tachycardia and rigidity, this is among the earliest signs of malignant hyperthermia and often appears before the

temperature moves. Rebreathing would raise the baseline above zero. There is no insufflation in this case, and hypoventilation would respond to the higher minute ventilation.

Key principle: *A rising end-tidal value despite more ventilation, with a normal baseline, suggests hypermetabolism.*

25. Correct answer D. Obstruction to airflow from a kinked tube, secretions, or bronchospasm

A sloping upstroke with no plateau, often called a shark fin, signals airflow obstruction from bronchospasm, a kinked tube, or secretions. The rising peak airway pressure agrees with this. Rebreathing produces a raised, rounded baseline and not a missing plateau. Spontaneous effort against a partial block produces a notch in the plateau. Cardiogenic oscillations are small regular wiggles at the end of the plateau, and they are reassuring and not obstructive.

Key principle: *A shark fin capnogram means airflow obstruction.*

26. Correct answer C. The patient is making an inspiratory effort against a partial block

A notch in the plateau, called a curare cleft, shows that the patient is making an inspiratory effort against a partial neuromuscular block. It is a sign of returning respiratory drive and a partial block. Rebreathing raises the baseline and does not cut a notch into the plateau. Small regular oscillations at the end of the plateau are cardiogenic, and they look different from a cleft in the midplateau. A kinked sampling line produces a flat or distorted trace, not a notch on a normal waveform.

Key principle: *A curare cleft shows inspiratory effort against a partial neuromuscular block.*

27. Correct answer E. This may be return of spontaneous circulation, so pause to check the rhythm and pulse

During resuscitation the end-tidal value reflects the cardiac output being produced. A sudden jump to a normal or high value is often the first sign of return of spontaneous circulation, and it can appear before a pulse is felt. The right response is to pause briefly and check the rhythm and pulse. An esophageal tube would lower the value to near zero and not raise it. Hyperventilation lowers the value, and a jump to a normal range does not suggest overly deep compressions.

Key principle: *A sudden rise in end-tidal carbon dioxide during CPR is often the first sign of return of circulation.*

28. Correct answer C. Correct compressions, rescuer fatigue, or volume loss, and continue resuscitation

During chest compressions the end-tidal value reflects the cardiac output being produced. A low value tells the team that compressions are ineffective, the rescuer is tiring, or the patient has lost volume, and it is used to coach compressions. A persistently very low value after prolonged, high-quality resuscitation predicts a poor outcome, but it should never be the only reason to stop, and it has been only eight minutes here. A misplaced tube gives a near-zero trace, and the value does depend on the cardiac output.

Key principle: *A low end-tidal value in CPR is a cue to improve compressions and replace volume, not by itself a reason to stop.*

29. Correct answer D. Check the sampling line and water trap within seconds while continuing to ventilate

Sidestream sampling draws gas through a thin line, and water or secretions can occlude it and produce a flat trace that mimics apnea. Here the chest rises, breath sounds are bilateral, and blood pressure and saturation are normal, so occlusion of the line is likely. The clinician should check the line and water trap within seconds but keep ventilating while checking. Removing a working tube, stopping ventilation, or starting compressions are harmful responses to a monitor problem.

Key principle: *When a trace vanishes but the patient looks well, check the line and trap while continuing to ventilate.*

30. Correct answer A. Lung reserve was consumed before the reading fell, then the curve steepened

In an apneic patient the oxygen reserve in the lungs is consumed long before the oximeter shows it. The oxyhemoglobin dissociation curve is flat above 90 percent, so large falls in arterial oxygen tension produce only small changes in saturation, and below that point the curve steepens and a few millimeters of mercury cost many percentage points. The device reads arterial blood during pulsatile flow, and its averaging is over seconds, not minutes. By the time the number falls, the patient is already on the steep portion of the curve.

Key principle: *The oximeter gives little warning because the curve is flat above 90 percent and then steep.*

31. Correct answer B. A large shunt or a poor signal, so the reading says little about reserve

A saturation of 90 percent corresponds to an arterial oxygen tension of about 60 mm Hg. A patient breathing 100 percent oxygen who still reads 94 percent has a large shunt or a bad signal, and the reading says little about how much reserve remains. Because the curve is flat in this range, a high-looking number does not mean a large margin. Carboxyhemoglobin would make a reading look better than the truth, but nothing in the stem suggests it. A reading above 90 percent does not prove adequate tissue oxygenation.

Key principle: *A reading of 94 percent on 100 percent oxygen implies a large shunt or a bad signal.*

32. Correct answer C. An arterial blood gas with co-oximetry to measure methemoglobin

Methemoglobin pulls the reading toward about 85 percent whatever the true saturation, and a persistent value in this range, with chocolate-colored blood and no response to oxygen, should prompt a co-oximeter blood gas. The standard two-wavelength device cannot make the diagnosis, so a second oximeter will agree with the first. A chest film looks for lung disease that this comfortable patient does not have. A normal capnogram shows ventilation but says nothing about the type of hemoglobin.

Key principle: *A persistent reading near 85 percent with chocolate blood and no response to oxygen needs co-oximetry.*

33. Correct answer D. Use the trend, the capnogram, and the clinical picture, and send an arterial blood gas

Studies in hospitalized patients show that pulse oximeters overestimate arterial oxygen saturation in patients with darker skin, so occult hypoxemia occurs more often in Black patients. Until devices improve, a borderline reading should not be relied on. The clinician should use the trend, the capnogram, and the clinical picture, and send a blood gas when the situation is unclear or the stakes are high. There is no fixed correction factor, and discarding the oximeter altogether discards useful information.

Key principle: *Oximeters can overestimate saturation in darker skin; confirm borderline readings with a blood gas.*

34. Correct answer E. The patient is right, and the numbers may hide compensation that is about to fail

A normal number does not prove a normal patient. A saturation of 98 percent in a patient who is blue and struggling is false reassurance, and a normal blood pressure in a patient who is gray, sweating, and tachycardic may be compensation about to fail. When the patient and the screen disagree, the clinician should assume the patient is right until there is proof otherwise. The look of the patient, the feel of the pulse, and the sound of the breath are monitors that need no batteries. Waiting thirty minutes ignores a patient who is already telling the clinician something.

Key principle: *A normal number does not prove a normal patient; trust the patient over the screen.*

35. Correct answer A. Yes, in high-risk patients the alarm performed as well as a processed index

Large trials of awareness prevention found that an alarm based on the end-tidal anesthetic concentration performed as well as a protocol based on the processed index in high-risk patients, so the processed monitor is not an absolute requirement. It remains a useful additional signal, especially when deep neuromuscular blockade hides the clinical signs of light anesthesia. The end-tidal agent concentration is readily measured with volatile anesthetics. The processed index has limitations but is not unreliable in all elderly patients.

Key principle: *An end-tidal anesthetic concentration alarm is an acceptable awareness strategy; a processed index is helpful but not required.*

36. Correct answer B. Never silence an alarm before its cause is found, and fix the artifact or limit aloud

Alarm fatigue is a documented hazard, and an alarm that sounds often for nothing will be ignored on the one occasion it matters. Limits should be set deliberately at the start of each case, tailored to the patient's baseline. An alarm should never be silenced without identifying its cause. If it is an artifact, the clinician fixes the artifact or changes the limit and says so aloud. Silencing for convenience, widening limits, or disabling the alarm removes the safeguard.

Key principle: *Set alarm limits deliberately and never silence an alarm before finding its cause.*

37. Correct answer E. About 15 mm Hg

The transducer should be leveled to the heart at the fourth intercostal space on the midaxillary line and zeroed to atmospheric pressure. Raising or lowering the table without releveling shifts the reading by about 0.74 mm Hg for

each centimeter of height difference. Over 20 centimeters this is about 15 mm Hg. That is a clinically important error, enough to turn a normal pressure into an apparent hypotension or the reverse. Smaller estimates underestimate it, and a value of 30 would require a height difference of roughly 40 centimeters.

Key principle: *Relevel the transducer after every table height change; the error is about 0.74 mm Hg per centimeter.*

38. Correct answer B. It poorly predicts fluid responsiveness, so it should not decide on a bolus

Central venous access serves two purposes that are easy to confuse. It delivers vasopressors and irritant drugs safely, and it measures central venous pressure. The first purpose matters more, since the absolute value is a poor predictor of fluid responsiveness and should not decide whether a bolus is given. The waveform, on the other hand, is informative: for example, a waves disappear in atrial fibrillation and cannon a waves appear in atrioventricular dissociation. Dynamic indices predict responsiveness better in the right setting.

Key principle: *Do not give fluid on the basis of the absolute central venous pressure alone.*

39. Correct answer D. Aspirate through the central catheter to confirm and treat air embolism

If the end-tidal carbon dioxide falls suddenly in a patient with an open venous system, air embolism should be suspected, and aspiration through the central venous catheter can both confirm and treat it. A bronchodilator treats bronchospasm, which would produce a sloping waveform and not a sudden fall with hypotension. Reducing the anesthetic treats depth, which does not explain this pattern. Absorbent problems raise carbon dioxide through rebreathing and do not lower it.

Key principle: *A sudden fall in end-tidal carbon dioxide with an open venous system suggests air embolism; aspirate the catheter.*

40. Correct answer A. Perform echocardiography to distinguish among these causes within minutes

When instability is unexplained and persists, echocardiography settles the argument among hypovolemia, ventricular failure, tamponade, embolism, and valvular catastrophe within minutes. Transesophageal echocardiography has taken over much of the diagnostic role of the pulmonary artery catheter, which is now used far less and is kept for selected patients with severe pulmonary hypertension, right heart failure, or complex cardiac surgery. The central venous pressure cannot separate these causes. Leaving the operating room delays care for an unstable patient.

Key principle: *Echocardiography is the quickest way to sort out persistent unexplained instability.*

Group 3. Teams, Communication, and Cognitive Aids

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41. Correct answer C. Everything but this, in which the obvious diagnosis is dropped for exotic ones

Gaba described three types of fixation error. In everything but this, the clinician dismisses the obvious diagnosis in favor of exotic ones, which is what happened here. In this and only this, the clinician clings to a single diagnosis. In persistent failure to revise, the clinician continues a plan that has stopped working. Failure to project the course of events is a lapse in situational awareness and not a fixation error.

Key principle: *Everything but this means dismissing the obvious in favor of the exotic.*

42. Correct answer D. Perceiving the relevant elements, because nobody noticed the information

Situational awareness has three levels: perceiving the relevant elements, understanding what they mean together, and projecting where the situation will be in a few minutes. Most failures occur at the first level, because the information was never noticed, or at the third, because nobody asked what would happen if nothing changed. Here the pieces were held by different people and the anesthesiologist did not perceive them. The findings had not been combined or misread, and declaring an emergency is a separate step.

Key principle: *Most situational awareness failures occur at perception, when information is never noticed.*

43. Correct answer E. Whoever can see the whole room, because leadership is a function, not a rank

Leadership in a crisis is a function and not a rank. Whoever is best placed to see the whole room should lead, and in the first minutes that is usually the anesthesiologist at the head of the bed. A good leader states the working problem, assigns tasks by name, keeps a mental list of what has been done, and invites correction. Basing leadership on rank, on being the surgeon, or on familiarity with the patient does not guarantee that the leader can see and direct the whole room.

Key principle: *Leadership is a function: whoever can see the whole room should lead.*

44. Correct answer A. Assign tasks early, match each to the helper's skill, and state what is expected

A single clinician can perform about one task at a time, and a crisis generates many. Tasks should be handed out early, matched to the skill of the person receiving them, with a clear statement of what is expected, for example by asking a nurse to draw up and label a drug and report when it is ready. Overloading oneself feels like diligence and works like a delay. Waiting for stability postpones the help most needed, and unstructured self-selection leaves gaps and duplication.

Key principle: *Distribute tasks early, matched to skill, with clear expectations.*

45. Correct answer E. The first room is too flat and the second room is too steep

A steep authority gradient lets errors pass because those who notice them do not feel entitled to speak. A flat gradient lets decisions drift because nobody decides. The aim is a gradient steep enough that someone leads and flat enough that anyone can challenge. The first room has debate without a decision, which describes a gradient that is too flat. The second has silence in the face of authority, which describes one that is too steep.

Key principle: *Aim for a gradient steep enough that someone leads and flat enough that anyone can challenge.*

46. Correct answer E. State the observation, state the concern, propose an action, and ask for agreement

A useful structure for assertiveness is to state the observation, state the concern, propose an action, and ask for agreement. For example, she might say that the label on this bag shows another name, that she is worried it is the wrong patient, that it should be held until checked, and ask whether that is acceptable. If the first statement is not acknowledged, it is repeated once with more emphasis and then escalated. Delaying the statement until the case is over defeats the purpose.

Key principle: *Assertive structure: observation, concern, proposed action, request for agreement.*

47. Correct answer D. Call if the cause is unclear in a minute or two, or the first treatment fails

Calling for help is the single most reliable action in crisis management, and clinicians call later than they should because of pride, the hope that the problem will resolve, and the fear of seeming incompetent. A rule set in advance helps: if the cause is not clear within a minute or two, or the patient is not improving after the first treatment, call. Waiting for deterioration or for failure of multiple treatments sacrifices the early window, and calling at the end of the case is not a response to an emergency at all.

Key principle: *Call when the cause is unclear within a minute or two, or if the first treatment fails.*

48. Correct answer B. Each check takes seconds, stops a class of error, and carries no accusation

Cross-checking is the habit of verifying critical actions with a second person or a second source. Reading every drug label, confirming pump settings, comparing the monitor with the patient, and confirming blood products against patient identity each take seconds and stop a class of error that would otherwise reach the patient. Making the check routine in every case means it is not felt as an accusation when done in an emergency. It is not limited to emergencies, and it adds a second safeguard and does not shift blame.

Key principle: *Make cross-checking routine so that it is not felt as an accusation in a crisis.*

49. Correct answer C. Say each digit of numbers that sound alike, and give drug, amount, and route

Precision is a part of communication. Vague words such as some, a bit, and soon are heard as different amounts by different people. A request for a medication should give the drug, the concentration, the quantity, and the route, and the helper should read the label aloud before it is given. For numbers that sound alike, such as fifteen and fifty, each

digit should be spoken. Louder speech does not fix ambiguity, abbreviations that only one person understands add risk, and leaving the dose to the nurse invites error.

Key principle: *Say each digit of numbers that sound alike and state the drug, quantity, and route in full.*

50. Correct answer D. Announce what is thought to be happening, what has been done, and what is wanted next

A shared mental model means that everyone has roughly the same idea of what is wrong and what is being done. It is created with brief, regular announcements stating what is thought to be happening, what has been done, what worries the leader, and what is wanted next. The statement takes about fifteen seconds and prevents duplicated effort and contradictory actions, and it gives the team a chance to correct the leader. Private pictures, delaying reconciliation, and cutting communication all preserve the conflicting views.

Key principle: *Create a shared mental model with brief, regular announcements.*

51. Correct answer B. Reassign the task plainly, without blame, and brief the new holder on progress

Roles must be able to change. A rigid assignment fails when the person is overloaded, unqualified for the task that arises, or affected by stress. The leader should watch for signs of saturation, such as slowed speech, fixation on a single task, and repeated mistakes, and reassign the task plainly and without blame. A handoff of a role works only if the new holder is told what has been done. Insisting on the original assignment, exhorting more effort, or waiting for a request to be made all leave the patient exposed.

Key principle: *Watch for saturation signs and reassign roles without blame, with a clear handoff.*

52. Correct answer A. Use neutral words and a steady tone, and address the patient when appropriate

The awake or lightly sedated patient hears what is said. Phrases such as we are losing him can frighten a patient under regional anesthesia, and some patients thought to be unconscious later remember them. The team should use neutral language, a steady tone, and speak to the patient when appropriate. The same applies to interpreters, family members in the room, and trainees who learn from what they overhear. Silencing the patient or whispering does not remove fear, and assuming nobody hears is incorrect.

Key principle: *Remember that the awake or lightly sedated patient hears; use neutral language.*

53. Correct answer C. Name the hazard, the plan if it occurs, who does what, and the first sign of failure

A two-minute briefing before a high-risk case pays back many times over. It names the main hazard, the plan if it occurs, who will do what, and the first sign that the plan has failed. It should include the surgeon and the nurses, and it should say aloud where the rescue equipment is and who will fetch it. A team that has rehearsed the first minutes in words begins the real event with a head start. A generic history or a consent check does not prepare the team for the specific hazard.

Key principle: *Brief the team on the hazard, plan, roles, and the first sign that the plan has failed.*

54. Correct answer C. Simulations show fewer omitted steps with a checklist, but real-world outcome data are limited

The strongest evidence comes from simulated operating room crises. In a randomized trial, teams managing a range of emergencies with a checklist at hand failed to perform critical steps far less often than teams working from memory, and later simulation studies found similar benefits for malignant hyperthermia, anaphylaxis, local anesthetic toxicity, and cardiac arrest. Real-world outcome data are limited, though the direction of the evidence is consistent. It would be incorrect to claim proven mortality benefit in all emergencies or no benefit.

Key principle: *Simulation shows checklists reduce omissions; real-world outcome evidence is limited.*

55. Correct answer A. State the problem and first actions from memory, then open the manual

Before reaching for a list, the leader should take five to ten seconds to state the problem and the first actions, which can usually be done from memory. The manual is opened once the immediate steps are under way. This order prevents the opposite failure from forgetting, which is spending the first minutes of an arrest or anaphylaxis reading

while the patient is not being treated. The manual serves the second minute and beyond. Saying that the manual is being opened shows the room that this is routine and not a sign of panic.

Key principle: *Act on the first reflexes from memory, then open the manual.*

56. Correct answer D. Add a dated correction without altering the original entry, in factual terms

Documentation should be prompt and factual, with the time of onset, the findings, each intervention with time and dose, who was present, the response, and the final condition. Speculation, blame, and comments about other staff should be avoided. An existing record must not be altered. If an error is found, a dated correction should be added. Overwriting hides the correction, and leaving the error misleads the reviewer.

Key principle: *Never alter an existing record; add a dated correction and keep the note factual.*

57. Correct answer E. Check on him, offer relief, use peer support, and report the event

Involvement in a serious adverse event can cause guilt, anxiety, sleeplessness, and self-doubt, sometimes called the second victim phenomenon. Many clinicians suffer in silence and some leave practice. Colleagues should check on those involved, offer relief from the next case if needed, and use peer support programs. Reporting the event through the safety system turns private distress into organizational learning. Treating it as rare, suppressing discussion, or removing the clinician permanently does not address the problem and discourages reporting.

Key principle: *Support the clinicians involved and report the event through the safety system.*

58. Correct answer B. Give a verbal summary at the bedside and wait until the baseline is confirmed

Transfers outside the operating room need equal care. Before moving a high-risk patient, the airway, monitoring, drugs, oxygen supply, and transport team should be ready. The receiving unit should be given a verbal summary at the bedside with the written record, and the departing clinician should wait until the receiving team has attached the monitors and confirmed the baseline. Most deterioration after transfer is predictable, and the handover is the chance to say which problem to expect and what to do about it. Leaving early removes that chance.

Key principle: *At transfer, hand over at the bedside and stay until the receiving team confirms the baseline.*

59. Correct answer A. Invite questions and ask what the resident would do if the patient worsened

Trainees need a particular kind of handover. A resident may be reluctant to ask for clarification or to say that a plan is unclear. The faculty member should say explicitly that questions are welcome and should ask what the resident would do if the patient deteriorated in the next ten minutes, because the answer shows whether the plan was understood. A handover that ends with a question about the next problem is better than one that ends with a yes or no. Repeating the same words does not test understanding, and silence does not prove it.

Key principle: *End a trainee handover by asking what the trainee would do if the patient deteriorated.*

60. Correct answer B. State what is known plainly, avoid blame, express sympathy, and promise to return

Disclosure is an ethical duty and an accreditation requirement. The conversation should be prepared, in a private place, with the surgeon and, where possible, the patient's own physician. It begins with what is known, in plain language, without speculation about cause or blame. Sympathy is expressed and, if an error is clear, it is acknowledged with an apology. The family is told what care is planned and what is under investigation, offered one contact person, and promised more information.

Key principle: *Disclose promptly: facts, no speculation, sympathy, a plan, and a single contact.*

Group 4. Emergency Drug Physiology

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61. Correct answer C. Alpha-1 constriction raises aortic diastolic pressure and coronary perfusion pressure

At the doses used in arrest the alpha-1 effect predominates, constricting arteries and veins and raising aortic diastolic pressure, which is the driving pressure for coronary flow. The beta effects increase myocardial oxygen demand and can provoke arrhythmias, so they are the price of the drug rather than the reason for giving it. Bronchial dilation is

not the goal during compressions. Epinephrine does not act on V1a receptors, which belong to vasopressin, and it raises rather than lowers myocardial oxygen demand.

Key principle: *The bolus of epinephrine in arrest is given for its alpha effect on coronary perfusion pressure.*

62. Correct answer D. Continue epinephrine and add glucagon or vasopressin because beta-blockade blunts it

In a patient taking a beta-blocker the beta effects of epinephrine are blunted, so the response may be poor, and glucagon, vasopressin, and other agents form part of the refractory plan. Epinephrine remains first line because no other single drug reverses vasodilation, supports the heart, and relaxes bronchial muscle together. Phenylephrine alone treats only the vascular component and leaves bronchospasm untreated. Antihistamines do not reverse shock. Dobutamine depends on the same blocked beta receptors and would lower resistance further.

Key principle: *In refractory anaphylaxis on a beta-blocker, continue epinephrine and add glucagon or vasopressin.*

63. Correct answer C. Vasopressin, through V2 free water retention and constriction of digital vessels

Vasopressin acts at V2 receptors in the renal collecting duct to reduce free water excretion, so hyponatremia can develop during prolonged use. It also constricts skin and other vessels, and ischemia of the skin and digits has been reported. Norepinephrine does not act on V2 receptors and has little beta-2 effect. Vasopressin has no beta-1 activity. The practical response is to review the need for the drug and to watch the sodium and the digits during prolonged infusion.

Key principle: *Prolonged vasopressin can cause hyponatremia through V2 receptors and digital or visceral ischemia.*

64. Correct answer E. Repeated doses deplete the releasable stores of norepinephrine in nerve terminals

Ephedrine acts partly by releasing stored norepinephrine from nerve terminals, so each dose depletes the stores and the response falls, a phenomenon called tachyphylaxis. It also has direct receptor effects, but these are not confined to beta-2 receptors. It is not converted to phenylephrine and does not act on V1a receptors. The better choice in this situation is a direct-acting agent such as phenylephrine or norepinephrine, titrated to the pressure and the heart rate.

Key principle: *Ephedrine causes tachyphylaxis because it depends on depletable norepinephrine stores.*

65. Correct answer D. Restore volume, correct acidosis, and warm the patient, then reassess the need for more drug

Acidosis, hypothermia, hypovolemia, hypoxia, and adrenal insufficiency all blunt receptor responses, so a rising requirement should prompt a search for correctable causes. Raising the dose to overcome a correctable cause treats the number and leaves the cause in place. Ephedrine depends on depleted stores and is a poor choice during a prolonged catecholamine infusion. Stopping the infusion abruptly in an under-resuscitated patient risks collapse and organ ischemia.

Key principle: *A rising pressor requirement should prompt correction of acidosis, cold, and volume before more drug.*

66. Correct answer A. Stop the infusion at once, and infiltrate the affected area with phentolamine

Extravasation of a concentrated catecholamine causes tissue necrosis by intense local vasoconstriction. The infusion should be stopped, access obtained elsewhere, and the area infiltrated with phentolamine, an alpha-blocker. Continuing the infusion adds drug to the tissue. Epinephrine is another vasoconstrictor and would worsen ischemia, and observation alone risks necrosis. The site should be checked before a pressor is started and watched continuously afterward.

Key principle: *Treat catecholamine extravasation by stopping the infusion and infiltrating with phentolamine.*

67. Correct answer E. Methylene blue can precipitate serotonin syndrome in patients taking serotonergic drugs

Methylene blue inhibits nitric oxide signaling and is a rescue option for vasoplegia, but it can precipitate serotonin syndrome in patients taking serotonergic drugs. It also interferes with pulse oximetry and is hazardous in glucose-6-phosphate dehydrogenase deficiency. It does not block V1a receptors, does not meaningfully prolong neuromuscular blockade, and is not known for causing hyperkalemia. The medication list must be reviewed before use.

Key principle: *Review serotonergic drugs before methylene blue, and expect interference with pulse oximetry.*

68. Correct answer E. Milrinone, which vasodilates the vessels and accumulates in renal failure

Milrinone inhibits phosphodiesterase-3, raising output while lowering resistance, and it is cleared by the kidney, so it accumulates in renal failure and its vasodilating effect persists. Dobutamine stimulates beta-1 and beta-2 receptors and lowers resistance, but it does not constrict through alpha-1 receptors. Phenylephrine is a vasoconstrictor and does not deplete stores. Vasopressin raises resistance. Hypotension from milrinone is treated by lowering the dose and supporting the pressure with norepinephrine.

Key principle: *Milrinone vasodilates and accumulates in renal failure.*

69. Correct answer A. Phenylephrine, to restore afterload without raising the heart rate

In severe aortic stenosis, tachycardia and increased contractility worsen the obstruction and the oxygen balance, so the goal is to keep the preload, avoid a fast rate, and support afterload with a pure vasoconstrictor such as phenylephrine. Dobutamine and ephedrine raise rate and contractility and can tighten the outflow. Glycopyrrolate raises the rate, which is not wanted, and the rate here is already adequate. Perfusion pressure is needed to supply the hypertrophied ventricle.

Key principle: *In outflow obstruction, support afterload with phenylephrine and avoid tachycardia and inotropes.*

70. Correct answer D. Raise the target above 65 to suit his baseline and reassess perfusion

A mean pressure of 65 mm Hg is a common goal, but patients with chronic hypertension need higher values, and perfusion is the real aim. Mentation under regional anesthesia, urine output, lactate, and capillary refill are signs of flow to the organs, and here they suggest it is inadequate. Accepting a number that works for others ignores the signs. Lowering the target would worsen perfusion. A diuretic would not address the cause and could reduce the volume.

Key principle: *Set the pressure target to the patient and judge it by perfusion, not by the number alone.*

71. Correct answer B. Synchronized cardioversion with sedation if time allows

An unstable patient with a tachyarrhythmia needs synchronized cardioversion first. In atrial fibrillation with a preexcitation pathway, nodal blockade can speed conduction down the accessory pathway, so adenosine, diltiazem, and esmolol should be avoided. Esmolol and diltiazem are also negative inotropes, which is undesirable in a hypotensive patient. Cardioversion terminates the rhythm without relying on the atrioventricular node.

Key principle: *Avoid nodal blockers in preexcited atrial fibrillation, and cardiovert if the patient is unstable.*

72. Correct answer E. Give it over 10 minutes, because rapid injection causes hypotension and bradycardia

In a perfusing patient the amiodarone bolus must be given slowly, because the drug and its solvent cause hypotension and bradycardia. The recommended dose for perfusing ventricular tachycardia is 150 mg over 10 minutes. A rapid push is the technique for adenosine, whose half-life is seconds, not for amiodarone, which has a very long half-life. It is not contraindicated by the presence of a pulse. A prolonged QT interval is a reason to avoid the drug, not to give it more slowly.

Key principle: *Give amiodarone slowly in a perfusing patient to avoid hypotension and bradycardia.*

73. Correct answer C. Suspect local anesthetic toxicity and avoid adding a sodium channel blocker

Lidocaine is itself a local anesthetic and a sodium channel blocker, and a wide-complex rhythm after a large dose of local anesthetic should raise the suspicion of the toxicity syndrome. Adding another sodium channel blocker risks worsening the toxicity. Adenosine acts on the atrioventricular node and does not treat a ventricular rhythm or a toxic state. A rapid push of amiodarone causes hypotension and bradycardia and does not address the cause.

Key principle: *Think local anesthetic toxicity before adding a sodium channel blocker to a wide-complex rhythm.*

74. Correct answer A. Give magnesium, replace potassium, and stop drugs that prolong the QT interval

Torsades de pointes is treated with magnesium, which stabilizes the membrane even when the serum level is normal, together with correction of low potassium and withdrawal of QT-prolonging drugs; pulseless torsades is defibrillated. Amiodarone prolongs the QT interval and should be avoided. Calcium chloride is not a treatment for torsades. Esmolol

can slow the heart, but in bradycardia-dependent torsades a slower rate lengthens the QT interval, and pacing or isoproterenol would be used to shorten it.

Key principle: *Treat torsades with magnesium, potassium, and removal of QT-prolonging drugs, and avoid amiodarone.*

75. Correct answer E. Give a direct beta agonist such as epinephrine and ready transcutaneous pacing

A transplanted heart has no vagal innervation, so atropine is ineffective, and further doses of it or of glycopyrrolate, which is also a muscarinic blocker, will not help. Beta-agonists such as epinephrine or ephedrine act directly on the heart, and transcutaneous pacing is the mechanical alternative. Neostigmine increases acetylcholine and would not speed a denervated node. The cause of the bradycardia should still be sought.

Key principle: *Atropine fails in a transplanted heart; use a direct beta agonist or pacing.*

76. Correct answer B. Unopposed muscarinic effect of neostigmine; give an anticholinergic such as glycopyrrolate

Neostigmine inhibits acetylcholinesterase and raises acetylcholine at muscarinic as well as nicotinic sites, so it must be given with an anticholinergic to prevent bradycardia and secretions. Here the anticholinergic was omitted, and glycopyrrolate corrects the picture. More neostigmine would worsen the muscarinic effects and has a ceiling effect on reversal. Neostigmine is not a beta-blocker. Residual relaxant does not act on cardiac muscarinic receptors.

Key principle: *Give neostigmine with an anticholinergic to prevent bradycardia and secretions.*

77. Correct answer B. Give calcium through a large vein or central line and remeasure the ionized level

Citrate in transfused blood lowers the ionized calcium, which supports contractility and vascular tone, so calcium is an emergency drug after large transfusion. It should be given by a large vein or a central line, because calcium chloride can damage tissue if it extravasates, and the ionized level should be measured. Sodium bicarbonate has narrow indications and does not raise ionized calcium. Giving calcium chloride into a small hand vein risks necrosis. Vasopressors alone do not correct the deficit.

Key principle: *Replace calcium after large transfusion through a large vein, and measure the ionized level.*

78. Correct answer C. Express the concentration as mass per volume and use a premixed or pharmacy-prepared syringe

Ratio notations such as 1 in 1000 and 1 in 10,000 are easy to confuse, and the safer practice is to express concentration as mass per volume. Premixed or pharmacy-prepared infusions and syringes, and dilution of emergency drugs by a standard protocol, reduce the chance of a concentration error. Freehand dilution and reliance on memory are the conditions under which tenfold errors occur. Waiting to see the pressure response does not prevent an overdose that may be harmful.

Key principle: *State concentrations as mass per volume and prefer premixed syringes.*

79. Correct answer B. A slip, best addressed by separating look-alike drugs and changing storage

A slip is an action that deviates from the intention, such as reaching for the wrong ampule from a familiar place, and slips are provoked by similar packaging and names, distraction, and haste. They are addressed by changing the system, for example by separating look-alike drugs, and a just culture treats such a near miss as valuable information. A mistake is a flawed plan, which knowledge and cognitive aids address. Exhorting individuals to concentrate does not remove the hazard.

Key principle: *Slips are prevented by system design such as separating look-alike drugs, not by exhortation.*

80. Correct answer D. Stop, treat the effects, call for help, keep the vial and syringe, and disclose afterward

When a drug error is recognized the anesthesiologist should stop, assess the effect, treat its consequences, call for help, keep the vial and the syringe, tell the team, and then report and disclose. Concealing the event or discarding the evidence prevents learning and breaches the duty of disclosure. Delaying the report until a later meeting or until harm appears leaves the team unprepared. A just culture encourages reporting of human error so that the system can be improved.

Key principle: *After a drug error, treat the effects, keep the evidence, tell the team, then report and disclose.*

Group 5. Machine, Gas, and Power Failures

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81. Correct answer D. A working self-inflating resuscitation bag within reach of the bed head

The self-inflating bag is the lifeboat when the gas supply, the electricity, the ventilator, or the circuit fails, and its presence and function are the user's responsibility because a self-test cannot see them. It should be tested as part of the checkout and kept within reach. A full vaporizer is irrelevant if the machine cannot deliver gas. A charged battery helps only with a power problem, and scavenging protects staff and does not rescue the patient.

Key principle: *Confirm a working self-inflating bag at every checkout; it is the backup for any failure.*

82. Correct answer E. An open cylinder silently takes over if pipeline pressure falls, spending the reserve unnoticed

The cylinder is the reserve for pipeline failure, so it is opened to read the pressure and then closed so that it is not consumed while the pipeline works. If it is left open and the pipeline pressure falls below the cylinder regulator setting, the machine switches to it silently, the alarm may not sound, and the reserve can be exhausted before anyone notices. Gas does not flow back into the pipeline, the cylinder does not change the set oxygen fraction, and ordinary exposure does not damage the regulator.

Key principle: *Open the cylinder to check it, then close it so the reserve is not silently consumed.*

83. Correct answer A. Repeat the test with each vaporizer on, since a cap or mount leak shows only then

A leak at the vaporizer filler cap or mount is present only when the vaporizer is open, so the test must be done with each vaporizer in the on position, and it must be repeated whenever a vaporizer is changed. A machine that fails should not be used until the leak is found. A negative test with the vaporizers off does not exclude the problem. Raising the dial compensates for the leak while it continues to expose the staff, and the canister is not suggested as the source.

Key principle: *Leak test the low-pressure system with each vaporizer on, and repeat it after any change.*

84. Correct answer C. Squeezing the reservoir bag and watching the valves open and close

Squeezing the reservoir bag and observing the movement of the unidirectional valves checks that they open and close in the correct directions, and an incompetent valve produces rebreathing. Occluding the Y-piece and watching the pressure hold tests the circuit for leaks and not valve function. Reading the analyzer on room air checks calibration, and the cylinder gauge tests the supply. Each is part of the checkout but none tests the valves.

Key principle: *Test unidirectional valves by squeezing the bag and watching their direction of movement.*

85. Correct answer D. Liquid agent may pass downstream and deliver a dangerously high concentration

Vaporizers must be kept upright because tipping can send liquid agent downstream, which can deliver a dangerous concentration of anesthetic. The device should be taken out of service and checked. Failure to deliver agent is the opposite of the hazard. The proportioning system relates to oxygen and nitrous oxide and is unaffected by tipping. Carbon monoxide arises from dry absorbent reacting with agent, not from a vaporizer chamber.

Key principle: *A tipped vaporizer can deliver a dangerously high concentration; keep it upright.*

86. Correct answer A. Pass a suction catheter through the tracheal tube to test its patency

A bag that is easy to squeeze points to the machine or circuit, whereas a hard bag after disconnection points to the tube or the patient. A suction catheter passed through the tube tests patency and settles the question in seconds, and the causes to consider include kinking, a bitten or displaced tube, secretions, bronchospasm, and pneumothorax. Once the patient is off the machine, the circuit hoses, the scavenging hose, and the pressure-limiting valve are no longer relevant to the resistance in the bag.

Key principle: *A hard bag after disconnection points to the tube or the patient; test the tube with a catheter.*

87. Correct answer B. A heat and moisture exchanger clogged with secretions or drug

Heat and moisture exchangers and filters can be blocked by secretions, blood, or nebulized drugs, producing a rise in pressure and a fall in flow that mimics bronchospasm, and the exchanger on a long case is a frequent culprit. The patent tube and absence of wheeze point away from the patient. The solution is to replace the exchanger and watch the pressure. A pipeline problem would affect oxygen supply, exhausted absorbent causes rebreathing, and an incompetent valve also causes rebreathing, not a rising pressure.

Key principle: *A blocked heat and moisture exchanger can mimic bronchospasm; replace it and watch the pressure.*

88. Correct answer C. Disconnect the scavenging hose from the circuit and watch the pressure

If the waste gas outlet or the connection to suction is blocked, pressure builds in the circuit and cannot escape, and active scavenging systems have relief valves that may fail. Disconnecting the scavenging hose from the circuit and seeing whether the pressure falls tests the system directly, and the same test applies when the bellows fail to descend. More fresh gas flow adds to the pressure. Deflating the cuff does not test the machine, and closing the valve would raise the pressure.

Key principle: *Unexplained high pressure may be a blocked scavenging system; disconnect the hose to test it.*

89. Correct answer C. In the patient interface, such as the cuff or Y-piece fitting

When the Y-piece is occluded and the circuit pressurizes, the machine side is intact, so the leak is in the patient interface, the cuff, or the Y-piece connection. If the circuit does not pressurize, the leak is in the machine, and a second circuit or a second machine may be needed. A leaking cuff also produces a changing sound at the mouth and a falling expired volume. The hoses, canister, bellows, and supply connections are all part of the portion that held pressure.

Key principle: *A circuit that pressurizes with the Y-piece occluded puts the leak at the patient interface.*

90. Correct answer A. Replace the exhausted absorbent and check the valves for rebreathing

Exhausted absorbent allows carbon dioxide to be rebreathed, shown by a rising baseline on the capnogram and a rising end-tidal value that does not respond to more ventilation, and the canister color confirms the diagnosis. The absorbent should be changed and the valves checked. Deeper anesthesia would not reduce rebreathing. Pulmonary embolism lowers the end-tidal carbon dioxide and does not raise the baseline. Withdrawing the tube would not help and risks displacing it.

Key principle: *A rising capnogram baseline with a changed absorbent indicates rebreathing; replace the absorbent.*

91. Correct answer B. Lower the fresh gas flow, ventilate by hand or avoid oxygen drive gas, and avoid the flush

Oxygen can be conserved by reducing the fresh gas flow to the lowest rate that maintains the circuit, using manual ventilation or a ventilator that does not consume oxygen as drive gas, and avoiding the oxygen flush. Many ventilators use oxygen to drive the bellows, adding to the patient's and the fresh gas consumption, so the cylinder may last much less time than expected. High flows and repeated flushing drain a cylinder in minutes. Additional cylinders should be requested early.

Key principle: *Conserve oxygen by lowering flow, avoiding oxygen-driven ventilation, and avoiding the flush.*

92. Correct answer E. Alert the other rooms, the charge nurse, and the engineering department at once

Unexplained hypoxemia in more than one patient suggests a building-wide problem, and other patients may be at risk, so the other operating rooms, the charge nurse, and the engineering department must be told immediately. The pipeline should not be trusted until it has been tested, and a normal pressure gauge does not show that the gas is oxygen. Delaying notification endangers others. Recalibration seeks reassurance and does not address the supply.

Key principle: *Warn other rooms and engineering at once when a pipeline gas problem is suspected.*

93. Correct answer E. Remove the room from service, label the machine and outlets, and save samples and logs

After any oxygen supply event the room should be removed from service, the machine and outlets labeled, and gas samples and logs saved, and the biomedical engineers, the supplier, and facility management should be notified.

Reporting follows through the institutional system and the device reporting system where appropriate, and the patient is told as for any adverse event. Reusing the room or purging the evidence destroys the lessons that protect the next patient.

Key principle: *Preserve the evidence and secure the room after a gas supply event.*

94. Correct answer A. The equipment will run on its batteries, which last half an hour to a few hours

Emergency generators restore power within about ten seconds to the essential circuits, whose outlets are typically red. A machine plugged into an outlet that is not on the essential circuit relies on its batteries, whose duration varies from half an hour to a few hours depending on the equipment and its age. The generator does not supply every outlet. The batteries serve the machine as well as the monitors. The anesthesiologist should therefore plan for hand ventilation and a move to a red outlet.

Key principle: *Red outlets are on the essential circuit; other outlets leave equipment on battery power.*

95. Correct answer E. Switch to intravenous anesthesia with propofol, supplemented with an opioid

A vaporizer works only when gas flows through it, and some workstations that control flow electronically stop delivering gas when power is lost, so inhaled agent may be unavailable. Total intravenous anesthesia with propofol, supplemented with an opioid and given by a manual technique or a battery-powered pump, belongs in the plan for any prolonged outage. Waking the patient is unnecessary and unsafe in the middle of an operation. Electronic controls cannot be forced to work without power.

Key principle: *Plan intravenous anesthesia for a prolonged outage, because a vaporizer needs gas flow.*

96. Correct answer D. Reset and check the machine and alarms before resuming, not assuming the old settings

Once power is restored the machine should be reset and checked before it is used again, and the clinician should not assume that settings have returned to their pre-failure values. The time of the outage and the actions taken should be documented. Reconnecting on trust risks delivering the wrong mode or volume or having alarms disabled. Taking the machine out of service for the day is not required if it passes the check, and repeated restarts add risk without a check.

Key principle: *After a power loss, reset and check the machine instead of trusting the displayed settings.*

97. Correct answer C. Surgery above the xiphoid with an open oxygen source and an ignition source in use

The risk is high when surgery is performed above the xiphoid process, an open source of oxygen or nitrous oxide is present, and an ignition source will be used, because oxidizer, fuel, and ignition then coincide near the face and drapes. In such cases the surgeon, anesthesiologist, and nurse should agree beforehand how the fire will be prevented and who will do what if it occurs. The other scenarios lack one or more elements of the triangle or are remote from the head and neck.

Key principle: *High fire risk means surgery above the xiphoid, an open oxidizer, and an ignition source.*

98. Correct answer D. Remove the tube, disconnect the circuit, and flood the airway with saline

In an airway fire the tube is removed, the circuit disconnected, and the airway flooded with saline, as the specific airway algorithm directs, and the patient is then ventilated and assessed for burns and inhalation injury. Increasing oxygen feeds the fire, and continuing ventilation through a burning tube delivers flames and hot gas to the lower airway. Wetting the drapes does not extinguish a fire inside the tube. Advancing a burning tube deeper drives the injury further down the airway.

Key principle: *In an airway fire, remove the tube, disconnect the circuit, and flood the airway with saline.*

99. Correct answer A. Use a carbon dioxide extinguisher, activate the alarm, and evacuate if necessary

If a fire is not quickly controlled the team should use a carbon dioxide extinguisher, activate the alarm, and evacuate if necessary, and the patient is assessed for burns and inhalation injury once the fire is out. Continuing to pour saline wastes time when the fire is spreading. A dry drape adds fuel. Increasing oxygen feeds the fire and continuing the operation puts the patient at risk. Fire needs oxidizer, fuel, and ignition, and the response removes them.

Key principle: *A fire not quickly controlled needs a carbon dioxide extinguisher, the alarm, and evacuation.*

100. Correct answer E. Treat the alarm as real, check the flows and circuit, and use a second analyzer

The inspired oxygen analyzer is the only device that measures what the patient actually inhales, and every alarm should be treated as real until proved otherwise. With a mixture of oxygen and nitrous oxide at low flows, falling inspired oxygen can also reflect a circuit that is not being refreshed with oxygen, so the flows should be checked. The oximeter lags behind a falling inspired fraction. Silencing the alarm or raising nitrous oxide removes the warning or worsens the hypoxic risk.

Key principle: *Respond to every inspired oxygen alarm as real; the analyzer measures what the patient inhales.*

Group 6. Predicting and Preparing for the Difficult Airway

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101. Correct answer D. Bedside tests have modest sensitivity, so most unexpected difficulty follows a normal test

Bedside tests have modest sensitivity, and a meta-analysis found that the Mallampati class detects only about half of difficult laryngoscopies and misclassifies many easy ones. Most unexpected difficulty therefore occurs in patients who passed the examination. The purpose of the examination is to shift the odds, find the patient who needs another plan, and prompt the question of what to do if the plan fails. Difficulty is not confined to patients with a history of surgery.

Key principle: *Prepare for the failure of your own prediction; bedside tests miss about half of difficult airways.*

102. Correct answer C. Several abnormal findings raise the odds, so plan videolaryngoscopy or an awake approach

The tests work best in combination, and a high-risk patient usually has several abnormal findings. Two or three abnormal results should move the plan toward videolaryngoscopy or an awake technique. No test is reliable alone and none guarantees difficulty, so cancellation is unnecessary. Dismissing the other findings because each is weak ignores their combined effect. The Mallampati class is only one of the predictors and should not be used alone.

Key principle: *Combine predictors; two or three abnormal findings should shift the plan to videolaryngoscopy or an awake technique.*

103. Correct answer B. Identify the cricothyroid membrane with ultrasound and mark it before induction

Predictors of difficult front-of-neck access include a neck mass, previous neck radiation, and landmarks that cannot be palpated, and the cricothyroid membrane should be located before induction in every patient with predicted difficulty. Ultrasound can identify the membrane when palpation is unreliable. Searching for it during an emergency wastes time that the patient cannot spare. A plan that depends on a supraglottic airway alone is weak when it is likely to fail, and landmarks that are not palpable cannot be relied on.

Key principle: *Locate the cricothyroid membrane before induction, using ultrasound if palpation fails.*

104. Correct answer C. Videolaryngoscope with a bougie, the collar loosened, and an assistant holding the head

In-line stabilization reduces neck movement but worsens the laryngoscopic view, so the best approach is a videolaryngoscope with a bougie or a flexible scope, with the collar loosened as needed while an assistant holds the head. Tighter stabilization worsens the view further. Hyperextension moves a possibly unstable spine. The nasal route is unsuitable with a skull base fracture and in an uncooperative patient, so the other options are unsafe or unlikely to succeed.

Key principle: *In suspected cervical injury, use videolaryngoscopy with a bougie, loosen the collar, and hold the head by hand.*

105. Correct answer E. A good view does not ensure passage, so use a bougie or a stylet matched to the blade

Videolaryngoscopy improves the view in most patients, but a good view does not guarantee that the tube will pass, particularly with a hyperangulated blade, and preparing to pass the tube is as important as the view. A bougie or a stylet matched to the blade curve helps. Assuming the view is wrong or changing to a standard blade discards the

advantage. The cuff is not the issue at this stage. An awake technique is not required because the glottis is visible and the patient is anesthetized.

Key principle: *A good view on the videolaryngoscope does not ensure tube passage; prepare how the tube will go in.*

106. Correct answer D. Desaturation occurs within a minute or two of apnea, so the first attempt must be prepared

Small children have a high oxygen consumption and a low functional residual capacity, so they desaturate within a minute or two of apnea, whereas a well preoxygenated healthy adult can tolerate eight minutes or more. The first attempt must therefore be well prepared, with the equipment of correct sizes and a team trained in the pediatric airway. A child has a smaller, not larger, reserve, and preoxygenation remains important in children.

Key principle: *Children desaturate within a minute or two of apnea; prepare the first attempt carefully.*

107. Correct answer B. Noninvasive positive pressure ventilation delivered before induction

A large trial in critically ill adults found that noninvasive positive pressure preoxygenation reduced hypoxemia during intubation compared with a face mask. Positive end-expiratory pressure reduces atelectasis in obese or hypoxemic patients. Three breaths from a mask, nasal cannula alone, and room air do not provide this benefit, and the patient should be positioned upright. Preoxygenation targets an end-tidal oxygen above about 90 percent where this can be measured.

Key principle: *Noninvasive positive pressure preoxygenation reduces hypoxemia during intubation of the critically ill.*

108. Correct answer C. Avoid hypoxia and hypercarbia, support pressure with a vasopressor, and ready an inhaled vasodilator

In right ventricular failure the aim is to avoid hypoxia, hypercarbia, and high airway pressures, which raise pulmonary vascular resistance, to maintain systemic pressure with a vasopressor so the right ventricle stays perfused, and to have an inhaled pulmonary vasodilator available. Positive pressure ventilation reduces venous return and raises pulmonary vascular resistance. A standard dose of induction agent can precipitate collapse, a fluid bolus may overdistend the ventricle, and hypercarbia or high pressure worsens the afterload.

Key principle: *In right ventricular failure, avoid hypoxia, hypercarbia, and high airway pressure, and keep the systemic pressure up.*

109. Correct answer C. It can depress the heart when the catecholamine system is exhausted

Ketamine or etomidate in reduced doses may preserve hemodynamics, but ketamine can depress an exhausted catecholamine system, because its usual sympathetic stimulation fails when stores are depleted. Etomidate is the drug associated with transient adrenal suppression, not ketamine. Ketamine does not block alpha-1 receptors, and it does not interfere with apneic oxygenation. The dose should be reduced in shock and a bolus vasopressor should be drawn up.

Key principle: *Ketamine may depress the heart when catecholamines are exhausted; reduce the dose in shock.*

110. Correct answer E. A modified rapid sequence with gentle low-pressure mask ventilation, since hypoxia is worse

The tradeoff in a rapid sequence is that apnea begins with the induction drug and there is no time to ventilate by mask. A modified rapid sequence with gentle mask ventilation at low pressures is an accepted alternative in a patient with low reserve, because hypoxia is more dangerous than the small risk of insufflating the stomach. The approach chosen should be stated and explained. An inhalational induction in a full stomach is unsafe, and delaying the airway in a tiring hypoxemic patient is not an option.

Key principle: *In a full stomach with low reserve, a modified rapid sequence with gentle mask ventilation is acceptable.*

111. Correct answer D. Place an early epidural so that the block can be extended if surgery is needed

In obstetrics the airway changes rapidly, failed intubation is more common, and regional anesthesia should be favored, so early epidural placement in a laboring woman at risk of airway difficulty is justified. A working epidural can be extended for a cesarean and avoids an urgent general anesthetic. Waiting until surgery is needed removes the

option when time is shortest. Intravenous opioids alone do not provide a pathway to surgical anesthesia, and routine direct laryngoscopy relies on a poor prediction.

Key principle: *In a laboring woman at risk of airway difficulty, place the epidural early.*

112. Correct answer B. Confirm mask ventilation first, then give the relaxant to improve compliance and view

In patients where difficulty is predicted, the ability to ventilate with the mask should be checked after the patient is anesthetized and before a long-acting relaxant is given. Once ventilation is confirmed, giving the relaxant improves compliance and the laryngoscopic view, and evidence suggests that blockade improves mask ventilation and intubating conditions. Withholding it is not supported. Giving it unchecked removes the chance to wake the patient if ventilation fails, and waiting for a failed attempt delays a helpful step.

Key principle: *Check mask ventilation first in predicted difficulty, then give the relaxant.*

113. Correct answer B. Awake surgical airway under local anesthesia, planned with the surgeon early

Severe stridor with critical obstruction may call for an awake surgical airway under local anesthesia, and a patient with critical obstruction may need an awake tracheostomy planned with the surgeon from the start. Induction of anesthesia removes the tone and respiratory drive that maintain a narrow airway and may make rescue impossible. Heavy sedation removes the main benefit of an awake technique, and passing a scope through a critical narrowing may worsen the obstruction.

Key principle: *Critical airway obstruction may need an awake surgical airway planned with the surgeon.*

114. Correct answer A. Give it early, ideally 15 minutes or more before topicalization starts

A dry airway improves the topical effect and the endoscopic view, so an antisialagogue such as glycopyrrolate is given early, ideally 15 minutes or more before topicalization, at 0.2 to 0.4 mg intravenously. Giving it when secretions already obscure the view is too late for the drug to act. Giving it after intubation misses the purpose. It is not true that the drug is without effect. It should be used with caution in patients with a risk of urinary retention or glaucoma.

Key principle: *Give the antisialagogue early, ideally 15 minutes or more before airway topicalization.*

115. Correct answer B. 540 mg, counting all routes together and recording what has been given

The maximum topical lidocaine dose is 9 mg per kilogram of lean body weight, here 540 mg, and the limit covers all routes, nebulized, sprayed, gargled, and injected. It should be calculated and written down before starting, with a running record of what has been given, because mucosal absorption can cause systemic toxicity at lower doses than expected. Using actual weight overestimates the safe dose in an obese patient, and treating each route as having its own allowance would exceed the limit.

Key principle: *Calculate the lidocaine maximum from lean body weight, across all routes, before topicalization.*

116. Correct answer C. The oral route with an intubating conduit to prevent biting of the scope

The nasal route risks bleeding and is unsuitable with a skull base fracture or coagulopathy, and the signs here suggest a skull base injury. The oral route requires an intubating conduit to prevent biting of the scope. A vasoconstrictor reduces bleeding but does not remove the risk of passing a tube or scope through a fracture. Passing the scope orally without a conduit exposes it to damage. The route is chosen by anatomy and the operator's skill.

Key principle: *Avoid the nasal route in skull base fracture or coagulopathy; use the oral route with a conduit.*

117. Correct answer D. Rotate the tube counterclockwise to reorient the bevel, keeping the scope still

The tube is advanced over the scope with the scope kept straight, and if it hangs up at the arytenoids it is rotated counterclockwise, which reorients the bevel. Placement is confirmed with capnography before anesthesia is induced, and induction follows only after the cuff is inflated and the tube is secure. Forcing the tube risks injury. Rotating clockwise while withdrawing the scope removes the guide. Withdrawing the scope and inducing anesthesia abandons the awake technique before the airway is secure.

Key principle: *If the tube hangs up at the arytenoids, rotate it counterclockwise over the straight scope.*

118. Correct answer E. Stop and reassess, then change the device, operator, route, or plan

If an awake intubation fails, the operator should stop and think instead of trying again with the same plan, and the options include a different device, a different operator, a different route, a surgical airway under local anesthesia, or postponing the case. Because the patient is awake there is time and oxygen to carry these out calmly. Deeper sedation removes the main benefit of the technique. Inducing anesthesia or giving a relaxant removes airway tone and respiratory drive in a patient who has just shown the airway is difficult.

Key principle: *After a failed awake attempt, change the plan; do not repeat it or abandon wakefulness.*

119. Correct answer A. Return to mask ventilation and decide whether to wake the patient

In the strategy described, plan A is the primary technique for tracheal intubation with a limit on the number of attempts, plan B is a supraglottic airway for oxygenation and as a conduit if appropriate, plan C is a return to face mask ventilation and the decision to wake the patient, and plan D is front-of-neck access when oxygenation fails. Each plan names the person, the device, and the trigger for moving on. The other options describe plans D, B, and a variant of plan A.

Key principle: *Plan A intubate, plan B supraglottic airway, plan C mask and wake, plan D front-of-neck access.*

120. Correct answer B. Stop laryngoscopy and reoxygenate, since the threshold agreed before induction was reached

Thresholds for help and for a change of plan should be set before the first laryngoscopy, when judgment is not clouded by stress, for example stopping and reoxygenating at a saturation of 93 percent. Agreement of the team on these numbers turns a difficult moment into an executed plan and lets others hold the operator to them. Continuing past the threshold erodes the safety margin, since desaturation then accelerates. Deepening anesthesia does nothing to restore oxygenation.

Key principle: *Agree the saturation threshold aloud before laryngoscopy, and stop and reoxygenate when it is reached.*

Group 7. Failed Intubation and the Cannot Intubate, Cannot Oxygenate Pathway

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121. Correct answer A. Tracheal intubation, supraglottic airway, final face mask attempt, front-of-neck access

Plan A is tracheal intubation with its limit on attempts, Plan B is a supraglottic airway to restore oxygenation, Plan C is a final attempt at face mask ventilation with waking the patient as its exit, and Plan D is front-of-neck access. Placing the face mask before the supraglottic device ignores that the device is the first oxygenation rescue. Starting with the supraglottic airway skips the attempts at intubation that define Plan A. Putting front-of-neck access before the last mask attempt would cut before every less invasive route has been tried. The value of the sequence is that the clinician always knows the next decision.

Key principle: *The pathway runs intubation, supraglottic airway, final mask attempt, then front-of-neck access, moving down only on failure.*

122. Correct answer A. Repeated blade passes cause bleeding, edema, and secretions that worsen the view and the mask seal

Every pass of the blade bruises tissue, and blood, edema, and secretions accumulate, so the fourth view is usually worse than the first. The same swelling that worsens the view can make a patient who was easy to mask ventilate impossible to ventilate. This is the reason for a ceiling on the number of attempts. Neuromuscular blockade does not accumulate in that way and does not explain a worsening view. The larynx does not become more anterior or wider with handling. Lightening anesthesia is possible but is not the mechanism that the chapter identifies for this progression.

Key principle: *Each attempt injures the airway, so more attempts make both intubation and rescue ventilation harder.*

123. Correct answer E. Agree the saturation at which attempts stop and mask ventilation resumes, since the second may be the last safe attempt

Pregnancy, obesity, sepsis, and small size all shorten the safe apnea time to tens of seconds, so the second attempt may already be the last that is safe. The ceiling of three is an outer limit and never a quota. A spoken stopping saturation turns a later decision into a reminder, which is easier to act on in a tense room. Leaving it open hands the decision to the monitor and to momentum. Apneic oxygenation lengthens the time for each look but does not remove the need to stop and reoxygenate by mask.

Key principle: *In physiologically difficult patients, agree beforehand on the saturation at which attempts stop.*

124. Correct answer E. After the first failed laryngoscopy, and again aloud at the second

Help should be called after the first failed laryngoscopy and called again, aloud, at the second. The cost of an unneeded colleague is a few minutes of that person's time, while the cost of a needed one who arrives late is the airway. Waiting until the ceiling is reached or until the supraglottic device fails brings help when most of the available attempts are already spent. Tying the call to a saturation threshold hands the decision to the monitor. The team should also fetch the difficult airway cart and assign someone to count attempts.

Key principle: *Call for help early, after the first failed look, and again at the second.*

125. Correct answer C. Use the video laryngoscope first, since predicted difficulty makes it the better opening device, not a late rescue

Where video laryngoscopy is available it should be the first device in a patient with predicted difficulty, not the third. Beginning with direct laryngoscopy spends attempts proving what the first look already suggested. A hyperangulated blade needs a stylet shaped to match it, because the tube must follow a path the eye cannot see. Placing a supraglottic airway first confuses a rescue device with an opening move. Video laryngoscopy improves first attempt success in predicted difficulty but does not guarantee passage of the tube.

Key principle: *In predicted difficulty, make video laryngoscopy the first device, not a late rescue.*

126. Correct answer B. The bougie tip is in the trachea, crossing tracheal rings and meeting hold-up at the carina

With a poor view the bougie is passed under the epiglottis, and tracheal placement is suggested by clicks as the tip crosses the tracheal rings and by hold-up at the carina. The tube is then railroaded over the bougie, rotated counterclockwise, with the laryngoscope kept in place until the tube is through. Esophageal placement gives smooth passage without clicks or hold-up. The vallecula and pharynx would not give the series of clicks followed by resistance. Capnography after the tube passes remains the proof of placement.

Key principle: *Clicks over tracheal rings and hold-up at the carina show the bougie is in the trachea.*

127. Correct answer D. It ventilates at higher pressures and offers some protection against regurgitation through a drain channel

A second-generation supraglottic airway has a drain channel and a better seal, so it ventilates at higher pressures and offers some protection against regurgitation. Its aim is a patent route for oxygen, and success is still judged by the capnography waveform first, then chest movement, then tidal volumes at acceptable pressures. It does not remove the need for capnography. Intubation through a supraglottic device requires a bronchoscope and the skill to use it, not just the device. Insertion attempts remain limited to three, with something changed each time.

Key principle: *A second-generation supraglottic airway seals better, tolerates higher pressures, and drains the stomach.*

128. Correct answer A. Remove and reinsert the device after relubrication, using a full jaw thrust or a different size

Supraglottic airways fail in recognizable ways, such as a tip that folds back, a device that sits too high or too deep, a cuff that leaks around an unopened mouth, or a larynx that closes against it. High peak pressures, a flat trace, and an audible leak are the signs, and the remedy is to remove the device, relubricate it, open the mouth fully with a jaw thrust, deepen anesthesia, or choose another size. Leaving the same device and squeezing harder is the error to avoid.

Overinflating the cuff or pushing blindly deeper does not correct a folded tip or a malposition. Insertion attempts remain limited to three.

Key principle: *When a supraglottic airway fails, change something and reinsert rather than squeezing harder.*

129. Correct answer D. Equipment faults can mimic a failed airway, so check the circuit and valve as well as the patient

A circuit disconnected at the elbow or a pressure-limiting valve left closed has been the cause of more than one reported failed airway. When mask ventilation fails, the equipment must be checked as well as the patient, and a disconnected circuit is a direct cause of a flat trace. Delaying correction until after a surgical airway accepts needless risk. A flat capnogram does not exclude a circuit fault, since it is what a disconnection produces. The same circuit serves both a mask and a supraglottic device, so the fault affects both.

Key principle: *Check the circuit and valve when ventilation fails, because a simple fault can mimic a failed airway.*

130. Correct answer D. Pass a bronchoscope loaded with a tube or exchange catheter through the device, oxygen continuing

Once ventilating, a supraglottic device can also serve as a conduit, so a bronchoscope loaded with a tube or an exchange catheter can be passed through it while oxygen continues to flow. This belongs to a team with the bronchoscope in the room and the skill to use it. Removing the device that keeps the patient alive for the sake of a conduit attempt with no fallback is the mistake to avoid. Blind passage beside the device is unreliable. A scalpel airway is for the cannot oxygenate emergency, which this patient is not in.

Key principle: *A ventilating supraglottic airway can be a conduit for bronchoscopic intubation, but never remove it without a fallback.*

131. Correct answer B. The stiffness is not muscle tone, so look for glottic obstruction, a leak, or a stiff chest

The nerve stimulator sorts the causes of a failed mask seal. Four strong twitches suggest that laryngospasm or chest wall rigidity may respond to relaxation. With no twitches the patient is fully paralyzed, so the stiffness is not muscle tone and the cause lies elsewhere, such as edema or a foreign body at the glottis, a leak, an equipment fault, or a chest made stiff by bronchospasm or aspiration. Treating a nonparalytic cause with more relaxant is a common and avoidable delay. Assuming inadequate depth ignores what the stimulator has already shown.

Key principle: *With no twitches present, further relaxant cannot help, so search for a non-muscular cause.*

132. Correct answer A. Draw up sugammadex and place it within reach before induction begins

Sugammadex must be in the room, drawn up, and within reach before induction begins, not collected after the saturation has fallen. A drug that takes a minute to find is a drug that was not available. Succinylcholine is not a safety net, because recovery of ventilation is slower and less dependable than assumed, and sugammadex does not reverse it. Relying on spontaneous recovery returns to the reasoning that the chapter rejects. The dose used for immediate reversal of rocuronium is 16 mg/kg.

Key principle: *Have sugammadex drawn up and within reach before induction, not fetched after desaturation.*

133. Correct answer E. It restores the patient's own breathing but does not clear laryngeal edema or opioid depression

Reversal is not rescue. Restoring neuromuscular function returns the patient's own breathing, but it does not remove the edema produced by several rounds of laryngoscopy or the ventilatory depression from an opioid or a hypnotic. A patient who wakes into an obstructed airway has moved from one emergency to another. Sugammadex acts only at the neuromuscular junction. Even when the pharyngeal muscles regain tone, swelling can still obstruct the airway.

Key principle: *Reversal returns neuromuscular function only; it does not treat edema, opioids, or obstruction.*

134. Correct answer C. A train-of-four ratio of at least 0.9

When the decision is to wake the patient, the block must be gone and not merely lessened, and a train-of-four ratio of at least 0.9 recorded at the stimulator is the standard. A patient with residual weakness has pharyngeal muscles that cannot hold the airway open. Four visible twitches show only that some response is present and do not exclude a

fading ratio. A subjective judgment of strength or a tetanic response is not the accepted standard. Residual block makes the patient more vulnerable to obstruction on emergence.

Key principle: *Before waking after a failed airway, document a train-of-four ratio of at least 0.9.*

135. Correct answer E. Proceed with the device after agreeing a spoken fallback with a senior colleague and the surgeon

For urgent surgery the balance shifts, and an operation in which delay threatens life may justify proceeding with a second-generation supraglottic airway or intubating through it. The decision belongs to the anesthesia team with a senior colleague, in conversation with the surgeon, and proceeding is a plan only when the fallback has been spoken aloud. Waking is the default for elective surgery but is not the only acceptable option. Proceeding without a fallback leaves the team without an exit if the device fails. Another laryngoscopy returns to a failed step in swollen tissue.

Key principle: *Proceeding with a rescue device in urgent surgery is acceptable only with a spoken fallback and senior input.*

136. Correct answer A. Her own safety, and whether waking her is more dangerous than continuing

Obstetric guidance places the safety of the mother first, so the question after a failed intubation is whether her own condition makes waking her more dangerous than continuing. Concern for the fetus alone does not usually outweigh the risk of an airway that is not secure. The obstetrician's timing preference is relevant but is not the primary determinant. Treating delay as the only outcome that matters ignores the risk of an unsecured maternal airway, which can end in harm to both patients.

Key principle: *After failed intubation in obstetrics, maternal safety decides between waking and proceeding.*

137. Correct answer D. Return to the supraglottic device or mask ventilation, not repeat laryngoscopy

Waking the patient is itself a procedure that includes reversal of residual blockade, a head-up position, suction of the pharynx, and a plan for obstruction on emergence. That plan is to return to the supraglottic device or the mask, not to repeat laryngoscopy. Another laryngoscopy in swollen tissue returns to a failed step with a falling saturation. Paralyzing again for a repeat attempt discards the purpose of waking. Removing the device that provides oxygenation leaves the obstruction untreated.

Key principle: *If the airway obstructs on emergence, go back to the supraglottic device or mask, not to the blade.*

138. Correct answer C. Light sedation, thorough topical anesthesia of the airway, and capnography confirmation before induction

An awake technique is a different anesthetic and not a delayed version of the first. The patient is told what will happen, an antisialagogue is given early, topical anesthesia is applied over many minutes, and sedation is kept light enough for the patient to breathe and cooperate. The tube is confirmed in the trachea by capnography before anesthesia is induced, and a surgical airway kit is within reach. Deep sedation removes the protection of spontaneous breathing. Inducing first defeats the technique, and a rushed topical application gives poor conditions.

Key principle: *An awake technique relies on light sedation, thorough topical anesthesia, and tube confirmation before induction.*

139. Correct answer E. Failed intubation, supraglottic airway, and mask ventilation, with capnography as the judge

Cannot intubate, cannot oxygenate is declared on the evidence of failure, not on the evidence of a low number. Intubation has failed, the supraglottic airway has failed after its permitted attempts, and the final mask attempt has failed even with two hands and airways, with a flat trace indicating no ventilation. Waiting for a particular saturation hands the decision to the monitor. The oximeter is useful but is not the declaring criterion, and no surgeon is needed to authorize the act. Late surgical airways usually follow a belief that one more attempt might work.

Key principle: *Declare cannot intubate, cannot oxygenate on documented failure of every route, not on a number.*

140. Correct answer C. Judge it by the saturation in the next minute, since an unneeded early scalpel costs only a scar

Late surgical airways usually trace to the belief that one more attempt from above might work. The measure of the decision is what the saturation does in the next minute, not what the clinician hopes. An early scalpel that proves unnecessary costs a neck scar, whereas a late scalpel that proves necessary costs a brain. Continued attempts from above are a distraction once the patient cannot be oxygenated. Waiting for a surgeon or for a threshold delays the act in the way the chapter warns against.

Key principle: *An unnecessary early scalpel costs a scar; a late necessary one can cost a brain.*

Group 8. Front-of-Neck Access

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141. Correct answer A. The lower half of the membrane, because the cricothyroid arteries run along the upper border

The membrane spans the gap between the thyroid and cricoid cartilages and is only about 9 millimeters high and 30 wide. The cricothyroid arteries run along its upper border, which is one reason the incision belongs in the lower half. The vocal folds lie about a centimeter above, so a high cut also risks the cords. Cutting through the thyroid cartilage is the error of mistaking cartilage for membrane. The first tracheal ring lies just below the cricoid, so the lower half also leads toward the airway.

Key principle: *Place the incision in the lower half of the cricothyroid membrane to avoid the arteries along its upper border.*

142. Correct answer B. Mistaking the thyroid cartilage for the cricoid and cutting too high

In many women the thyroid cartilage has no visible prominence, the membrane is narrower, and the overlying skin is thicker than expected. Palpation misses the membrane in a large share of them, and the common error is to mistake the thyroid cartilage for the cricoid and to cut too high. Ultrasound marking resolves most of that error. The other errors are not the typical pattern described. A woman with a predicted difficult airway should be marked as routinely as any other patient.

Key principle: *In women, palpation often mistakes the thyroid cartilage for the cricoid and leads to a cut that is too high.*

143. Correct answer D. When it drops into a soft depression, with the firm cricoid ring directly below

With the larynx stabilized between thumb and middle finger, the index finger travels down the midline over the thyroid notch and cartilage until it drops into a soft depression. That depression is the membrane, and the firm ring directly below it is the cricoid. The sequence of soft depression then firm ring is the reverse of the first distractor. The sternal notch is far below the membrane. Palpation should stay in the midline and be confirmed by rolling the finger across, because lateral structures can feel like the larynx in a swollen neck.

Key principle: *The membrane is the soft depression below the thyroid cartilage, with the firm cricoid ring directly beneath it.*

144. Correct answer C. A dark triangle with a bright anterior rim

In the transverse view, the thyroid cartilage appears as a dark triangle with a bright anterior rim. Sliding the probe toward the feet brings into view a bright horizontal line, the air and mucosa interface, and then the cricoid cartilage, which appears as a dark arch. The membrane lies in the short stretch between the two. The string of pearls is a longitudinal finding, not a transverse one. TACA stands for thyroid cartilage, airway, cricoid cartilage, airway.

Key principle: *In the transverse plane the thyroid cartilage is a dark triangle, then the air line, then the cricoid arch.*

145. Correct answer A. Mark the membrane with ultrasound before induction, while awake, since it is a planning tool

Ultrasound is a planning tool and not a rescue tool. In an unfolding emergency there is no time to bring in a machine, warm the gel, and learn the views, and its value lies in the minutes before induction when the patient is awake and able to lie still. The skin mark survives into the emergency and saves the search. Palpation identifies the membrane correctly in only a minority of obese patients, so skipping the scan is a gamble. Delaying the scan until after induction loses the benefit of an awake, cooperative patient.

Key principle: *Use ultrasound before induction to mark the membrane; do not rely on it during the emergency.*

146. Correct answer D. Stand at the dominant-side shoulder and open the collar while in-line stabilization is held

Position matters as much as landmarks. The operator should stand beside the patient's shoulder on the side of the dominant hand, since a clinician at the head with a low bed works with a flexed arm and a bent wrist, and the cut suffers. In a cervical collar the anterior section is opened while an assistant provides manual in-line stabilization. Neck extension over a shoulder roll brings the larynx forward and widens the membrane when it is permitted. Removing all stabilization is unsafe in trauma, and a lower bed worsens the ergonomics.

Key principle: *Position the operator beside the shoulder, and open the anterior collar with manual in-line stabilization.*

147. Correct answer B. Proceed in the midline within the membrane, ready with pressure if the vessel bleeds

A midline approach within the membrane avoids most of the vessels that sit near the field, including the pyramidal lobe, a high isthmus, a high innominate artery, and anterior jugular veins. Color Doppler can identify a crossing vessel, and a mark placed with that information is safer. In an emergency a vessel is not a reason to stop. It is a reason to be ready with pressure. Moving laterally leaves the midline where the structures are safest, and a cannula is more likely to fail.

Key principle: *A crossing vessel is a reason to be ready with pressure, not a reason to stop.*

148. Correct answer E. Exchange for a smaller cuffed tube and, if needed, widen the incision with a gentle turn of the blade

A small tube is preferable to a larger tube that will not pass the membrane. A tube that will not pass can be exchanged for a smaller cuffed tube, and the incision can be widened by a gentle turn of the blade. Force risks injury to the cartilage and trachea. Switching to a cannula and jet ventilator after a failed scalpel attempt gains little unless the trachea has clearly been entered. Persisting with a tube that will not fit repeats the failed step.

Key principle: *If the tube will not pass, use a smaller cuffed tube or widen the incision, not more force.*

149. Correct answer B. Continue 100 percent oxygen by every route still open and keep the neuromuscular block deep

The emergency is declared before the first cut, oxygen at 100 percent continues by every route still open, including nasal insufflation, and the block is kept deep so that the chest and larynx are relaxed. A last attempt at face mask ventilation can run while the kit is opened, and none of this delays the procedure because it happens in parallel. Stopping oxygen causes harm. Letting the block fade makes the larynx and chest tighter. Further laryngoscopy is a distraction once the patient cannot be oxygenated from above.

Key principle: *Keep oxygen flowing by every open route and keep the block deep while preparing to cut.*

150. Correct answer A. The tube was advanced too deeply and entered the right mainstem bronchus

The trachea below the cricoid is short, and the tube should be advanced only until the cuff has just passed through the membrane. A tube pushed deeply will enter a bronchus, most often the right, giving unilateral breath sounds. Pretracheal placement gives no waveform and no rising saturation. Cuff herniation and an obstructing bougie would not produce one-sided breath sounds with a brief rise in saturation. Capnography remains the only reliable check of placement.

Key principle: *Advance the tube only until the cuff has just passed the membrane, or it will enter a bronchus.*

151. Correct answer D. Continue, guiding the bougie by feel, because oxygen matters more than blood at this point

Bleeding is the complication seen first, but in most cases it is minor and the airway is more urgent. If blood obscures the field, continue, because the route of the bougie is guided by feel and oxygen matters more than blood. Pressure around the tube after it is placed usually controls oozing, and a significant vessel injury is managed by a surgeon once oxygenation is secure. The worst response is to stop the procedure because of blood.

Key principle: *Do not stop a front-of-neck procedure because of bleeding; secure oxygenation first.*

152. Correct answer E. Start a second approach at once: a longer incision, another operator, or a smaller tube

The goal in practiced hands is oxygen through a tube in the trachea within about a minute of the incision. If the first attempt fails, a second approach follows at once, such as a longer vertical incision, a different operator, or a smaller tube. Switching to a cannula and a jet ventilator after a failed scalpel attempt gains little unless the trachea has clearly been entered. A cannula through a bleeding track is no more likely to find the airway than the tube was. Ultrasound is a planning tool, not a rescue tool.

Key principle: *After a failed first cut, start a second approach at once rather than switching to a cannula.*

153. Correct answer C. The narrow lumen resists flow too much for a bag, so a high-pressure source is needed

A narrow-bore cannula has an internal diameter under about 2 millimeters. The resistance of such a lumen is far too high for a bag or ventilator to push gas through, so a high-pressure oxygen source is required, and this requirement is the root of nearly every problem the technique causes. A wide-bore kit with a lumen of about 4 millimeters or more can accept a conventional circuit. The problem is resistance, not length or connector type. Total upper airway obstruction makes jet ventilation more dangerous, not easier.

Key principle: *A narrow-bore cannula needs a high-pressure source because its lumen resists gas flow.*

154. Correct answer D. Inspiration for about one second and expiration for at least four seconds, at the lowest pressure giving chest rise

Oxygen is delivered through a narrow cannula by a manual jet ventilator at a regulated pressure, with an inspiratory period of about one second and exhalation lasting at least four seconds. The pressure is set at the lowest value that produces visible chest rise, because higher pressures cause barotrauma. A short exhalation allows gas to be retained, and continuous flow removes the chance to exhale. The cannula must also be fixed with one hand while the other controls the valve. Every delivery needs attention to whether the chest falls between breaths.

Key principle: *Jet ventilation needs about 1 second in, at least 4 seconds out, at the lowest effective pressure.*

155. Correct answer E. The cannula lies in the pretracheal tissue, so the oxygen is inflating the neck

A cannula placed through the membrane may sit in the pretracheal tissue rather than in the lumen, and the first jet of oxygen then inflates the neck, giving subcutaneous emphysema without chest movement. Audits of airway complications in the United Kingdom reported that emergency cannula techniques performed by anesthesiologists failed in a large proportion of attempts. Esophageal or bronchial placement would not cause a swollen neck with crepitus. A narrow cannula has no cuff to herniate. Cannulas also kink with repositioning and slip out of short tracks.

Key principle: *Neck swelling with the first jet means the cannula is in the pretracheal tissue.*

156. Correct answer A. A lumen of 4 mm or more takes a conventional circuit, but the many steps invite error

Wide-bore cannula kits advance a needle through the membrane while aspirating, and air bubbles confirm entry. A guidewire is passed, the track is dilated, and a cannula with an internal diameter of about 4 millimeters or more is advanced over the wire. Because the lumen is wide, a conventional circuit can be attached with no special pressure source. The kit has more steps than the scalpel technique, and each offers a chance for error such as a kinked wire, a cannula pushed through the posterior wall, or a dilator that will not pass.

Key principle: *Wide-bore kits permit a conventional circuit but have more steps, and thus more chances for error, than the scalpel.*

157. Correct answer B. It is an exhalation adjunct with limited evidence, and it does not replace the scalpel

Expiratory assist devices attempt to address the exhalation problem by drawing gas out actively, so exhalation does not depend on an open upper airway. Case reports and simulation work are encouraging, and some centers keep the device available. The evidence is still limited, the device is not universally stocked, and the clinician using it needs training that most departments do not provide. It is an adjunct and not yet a replacement for the scalpel.

Key principle: *Expiratory assist devices are promising adjuncts with limited evidence, not replacements for the scalpel.*

158. Correct answer B. The adolescent is managed as an adult, and the 4-year-old with a cannula technique

Pediatric guidance advises a cannula technique in children under about 8 years, because the membrane is tiny, the larynx is higher and softer, and a scalpel incision risks cartilage damage while a tube of useful size will not pass. Adolescents are managed like adults. Between those ages the decision rests on the operator's judgment about the size of the membrane. The pediatric technique is oxygen insufflation through a large intravenous cannula with a plan for definitive access, and the surgeon or otolaryngologist should be summoned early.

Key principle: *Under about 8 years use a cannula technique; adolescents are managed like adults.*

159. Correct answer C. Arrhythmia, bradycardia, or arrest, with early checks of temperature, glucose, and blood pressure

Hypoxic injury is the quiet complication. A patient who has spent minutes below a saturation of 70 percent may have cardiac or neurologic damage by the time oxygen flows, so the team should expect arrhythmia, bradycardia, and in some cases cardiac arrest, and be prepared to start chest compressions and treat the rhythm. Temperature, glucose, and blood pressure should be checked early, and the neurologic examination guides later decisions. Assuming that only bleeding matters ignores the hypoxic insult. Hoarseness is a late complication.

Key principle: *Expect arrhythmia or arrest after prolonged hypoxia, even once oxygen is flowing.*

160. Correct answer D. Decision time, attempts at each step, technique, tube size, capnography, complications, and anatomy

Documentation should be exact because the next anesthesiologist will rely on it. It should include the time of the decision, the number of attempts at each step, the technique chosen, the size and type of tube, the depth of insertion, the capnography confirmation, and any complications, and should describe the anatomy as found, including whether the membrane was palpable. The patient must also be told what happened and given an airway alert letter, and the event reported to the safety system. A few lines now save a future patient from a repeated failed attempt.

Key principle: *Document the decision time, attempts, technique, tube, confirmation, complications, and anatomy found.*

Group 9. Laryngospasm, Bronchospasm, and Aspiration

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161. Correct answer B. A mucosal reflex via the superior laryngeal nerve that adducts the cords, strongest at light planes

Laryngospasm is a protective reflex in which the superior laryngeal nerve carries stimulation from the airway mucosa to the brainstem, and the reply is sustained adduction of the vocal folds. The reflex is strongest at light planes of anesthesia, when the cough reflex is suppressed but the closure reflex remains intact. Triggers include secretions, an oral airway inserted too light, and irritant agents, and children with a recent respiratory infection or asthma are at highest risk. Recurrent laryngeal injury, lower airway smooth muscle contraction, and tongue fall do not produce a protective reflex closure.

Key principle: *Laryngospasm is a mucosal reflex closure of the cords that is strongest at light planes of anesthesia.*

162. Correct answer C. The cords were not fully shut, and the obstruction was easier to relieve than it first appeared

Other causes of obstruction such as tongue fall, mucus, a kinked circuit, or an obstructed filter can mimic the picture. The early test is simple: if a gentle jaw thrust with positive pressure opens the airway, the cords were not fully shut and the problem is easier than it appeared. Complete laryngospasm would not respond to a thrust and positive pressure alone. Bronchospasm produces wheeze, high pressures, and a prolonged expiratory phase rather than silent inspiratory crowing. A foreign body or epiglottitis would need different treatment altogether.

Key principle: *If a jaw thrust with positive pressure opens the airway, the cords were not fully shut.*

163. Correct answer A. A small bolus of propofol to deepen anesthesia and relax the cords

A small bolus of propofol deepens anesthesia and relaxes the cords in many patients, and it is the first drug to give when intravenous access exists. If the patient remains obstructed and the saturation is falling, succinylcholine follows without delay. Atropine should be ready because bradycardia follows hypoxia, but it does not open the cords. An inhaled beta-agonist treats bronchospasm, not cord closure. Naloxone does not relax the cords and could worsen laryngeal reflexes.

Key principle: *Give a small propofol bolus first for laryngospasm when intravenous access exists.*

164. Correct answer A. A reduced dose of 0.1 to 0.5 mg/kg may suffice, but in hypoxia give enough to be certain

If the patient remains obstructed and the saturation is falling, succinylcholine is given without delay. A dose below the usual intubating dose, 0.1 to 0.5 mg/kg or about 10 mg in many adults, often relaxes the cords, but in a hypoxic patient the safe course is to give enough to be certain, and the full dose of 1 to 1.5 mg/kg is available if needed. Waiting for propofol spends oxygen. Restricting the dose to a fixed small amount risks failure in a hypoxic patient. Atropine should be ready for bradycardia.

Key principle: *Succinylcholine often works at a reduced dose in laryngospasm, but in a hypoxic patient give enough to be certain.*

165. Correct answer C. Do not intubate until a relaxant is given, since attempts through shut cords provoke more reflex

Two errors cause harm in laryngospasm. The first is to ventilate hard against closed cords, which inflates the stomach and invites regurgitation. The second is to attempt intubation in a patient who is not paralyzed, because the cords are shut and the attempt provokes more reflex. When a muscle relaxant is given, the cords open, ventilation usually becomes easy, and intubation is rarely needed. The tracheal tube should still be kept ready. A capnogram waveform would mean ventilation is occurring, which argues against complete spasm.

Key principle: *Do not attempt intubation in an unparalyzed patient in laryngospasm, and do not ventilate hard against closed cords.*

166. Correct answer E. Extubate fully awake with intact reflexes, or deep if the airway is easy, but not in between

Laryngospasm clusters at induction and at emergence. At emergence it follows extubation in the stage between unconscious and awake, when the airway reflexes have returned but the patient cannot yet protect the airway or control breathing. The patient should be extubated fully awake or deeply anesthetized, in a patient with an easy airway, with the pharynx suctioned beforehand and the room kept quiet. An elective operation in a child with a recent upper respiratory infection is better postponed. Stimulating the child provokes the reflex.

Key principle: *Avoid the transition zone: extubate fully awake or deep, never in between.*

167. Correct answer B. Auscultate both sides and withdraw the tube slightly, since a bronchial tube can mimic bronchospasm

Not everything that wheezes is asthma. Mechanical causes include a tube that is kinked, bitten, or blocked, and a tube that has slipped into a bronchus, which gives unilateral breath sounds. Each cause has a different cure and the wrong cure costs minutes. Initial steps include ventilating by hand with 100 percent oxygen, passing a suction catheter, and

auscultating both sides to rule out bronchial intubation. Epinephrine and magnesium are for confirmed severe bronchospasm. Desflurane is irritant and should be avoided in bronchospasm.

Key principle: *Rule out a bronchial or kinked tube before treating wheeze as bronchospasm.*

168. Correct answer B. Propofol or ketamine for induction, and sevoflurane as the volatile agent

Propofol attenuates airway reflexes more than thiopental or etomidate, so it suits reactive airways, and ketamine dilates the bronchi. Volatile agents are bronchodilators at adequate depth, and sevoflurane is preferred because it dilates the bronchi, whereas desflurane is irritant and should be avoided. Morphine, atracurium, mivacurium, and meperidine can release histamine. Neostigmine raises secretions and bronchial tone unless an anticholinergic is paired with it, and even beta-blocker eye drops can precipitate bronchospasm.

Key principle: *In reactive airways, favor propofol or ketamine and sevoflurane, and avoid desflurane and histamine-releasing drugs.*

169. Correct answer E. Repeat albuterol as needed, add ipratropium for a second mechanism, and watch rate and potassium

Inhaled beta-agonists are the mainstay of treatment, delivered by metered-dose inhaler through an adapter in the circuit or by nebulizer into the inspiratory limb. Delivery to the lower airway is limited when flow is poor, so repeated administration is normal. Ipratropium adds a second mechanism, and the combination is common in severe cases. These drugs may raise the heart rate and lower potassium, effects that should be watched. Corticosteroids take hours to act and complement, rather than replace, bronchodilators.

Key principle: *Repeat inhaled albuterol as needed, add ipratropium in severe cases, and watch heart rate and potassium.*

170. Correct answer E. Magnesium sulfate over 20 minutes, and an epinephrine infusion titrated to effect

Refractory bronchospasm opens other options. Magnesium sulfate infused over about 20 minutes relaxes smooth muscle and helps in refractory cases, and an epinephrine infusion can be titrated to effect, with higher volatile concentrations continuing to help. Aminophylline is rarely used now because of its toxicity and modest benefit. Neostigmine increases secretions and bronchial tone, and a beta-blocker, even a topical one, can precipitate bronchospasm. Extracorporeal support has been used in life-threatening cases in centers that offer it.

Key principle: *Magnesium and an epinephrine infusion are options for refractory bronchospasm; avoid aminophylline as a first choice.*

171. Correct answer C. Delay elective surgery and optimize with bronchodilators and sometimes a short steroid course

Assessment is where prevention starts. Patients with active wheeze should have elective surgery delayed and their disease optimized with bronchodilators and sometimes a short course of steroids. Patients on regular inhalers continue them through the morning of surgery. In reactive patients, a laryngeal mask is preferred to a tube where possible, anesthesia is deepened before airway instrumentation, and an inhaled beta-agonist may be given beforehand. Nonselective beta-blockers should be avoided because they can precipitate bronchospasm.

Key principle: *Delay elective surgery in active wheeze, and continue regular inhalers through the morning of surgery.*

172. Correct answer A. Clear liquids until two hours and a light meal until six hours, so the meal is not yet allowed

Current guidance allows clear liquids until two hours before elective anesthesia, breast milk until four hours, infant formula and a light meal until six hours, and fried or fatty food only after eight hours. In this case the light meal was four hours ago and the clear liquid was 90 minutes ago, so both fall short of the intervals. These rules apply to healthy patients having elective procedures and do not guarantee an empty stomach where emptying is delayed. The fast is one piece of information among several.

Key principle: *Fasting intervals: clear liquids 2 hours, breast milk 4, formula and a light meal 6, fatty food 8.*

173. Correct answer D. Solid gastric contents, so manage the airway as for a full stomach

Gastric ultrasound estimates the stomach contents in minutes. An empty antrum appears collapsed with thick walls, clear fluid appears as a dark distended antrum, and solid contents produce a mixed echogenic pattern with a frosted glass look. Solid contents mean the stomach is not empty, so delaying anesthesia or treating the airway as for a full stomach is appropriate. A collapsed antrum would suggest an empty stomach. Clear fluid would look dark and distended, not mixed and echogenic.

Key principle: *A mixed, frosted glass antrum means solid contents: manage the airway as for a full stomach.*

174. Correct answer C. Sodium citrate, which acts within minutes, before every cesarean general anesthetic

Nonparticulate antacids such as sodium citrate act within minutes and are the standard before cesarean delivery, with 30 mL of a 0.3 molar solution orally. H2 blockers and proton pump inhibitors reduce acid production over hours, and metoclopramide speeds emptying. None prevents regurgitation, and aspiration of neutralized fluid still causes injury, though less of it. Prophylaxis is reserved for pregnant patients, emergencies, and those with recognized risk. It is not recommended routinely in healthy patients having elective surgery.

Key principle: *Give sodium citrate before every general anesthetic for cesarean delivery; routine prophylaxis is not for healthy elective patients.*

175. Correct answer E. Head-up positioning lengthens apnea time and cuts regurgitation; cricoid pressure has limited evidence

Rapid sequence induction shortens the time to airway protection: the patient is preoxygenated, the induction agent and a rapid-acting blocker are given together, and the tube is placed without mask ventilation. Succinylcholine and high-dose rocuronium both serve, depending on contraindications and the availability of sugammadex. Cricoid pressure is used by many teams though the evidence is limited, and it can worsen the laryngeal view. Head-up positioning increases safe apnea time and reduces regurgitation. If regurgitation occurs, the head is lowered and the pharynx suctioned.

Key principle: *Use head-up positioning in rapid sequence induction; cricoid pressure has limited evidence and may worsen the view.*

176. Correct answer C. Suspected solid particles or lobar collapse, with no saline lavage since it spreads the material

Bronchoscopy is indicated when solid particles are suspected or lobar collapse occurs. Saline lavage is not recommended because it spreads the material. Other management includes suction of the trachea through the tube before positive-pressure ventilation, 100 percent oxygen, PEEP, and lung-protective ventilation. Prophylactic antibiotics and corticosteroids are not recommended for chemical pneumonitis. Wheeze alone or fever is not a specific indication.

Key principle: *Bronchoscopy is for suspected solid particles or lobar collapse; do not lavage with saline.*

177. Correct answer A. She is unlikely to develop complications, and proceeding with the urgent surgery is reasonable

If the aspiration is small, the patient is stable on room air or modest oxygen, and the operation is urgent, proceeding is reasonable. Elective surgery is usually postponed after a significant event. Patients who remain free of respiratory symptoms for two hours are unlikely to develop complications, whereas those with hypoxemia, wheeze, or radiographic change need monitored care and often intensive care. Acute respiratory distress syndrome can develop over 24 to 48 hours in such patients. Antibiotics are not routine for chemical pneumonitis.

Key principle: *Two hours free of respiratory symptoms after a small aspiration predicts a low risk of complications.*

178. Correct answer B. Forceful inspiration against the obstruction created large negative intrathoracic pressure that injured capillaries

Negative pressure pulmonary edema is a noncardiogenic edema that follows forceful inspiratory effort against an obstructed upper airway, and the typical patient is young, male, muscular, and athletic. Biting a tube or laryngeal mask is a recognized cause of acute obstruction. A large negative intrathoracic pressure raises venous return and the

transmural gradient across the capillaries, which fail mechanically. Aspiration favors the right lower lobe after vomiting, cardiogenic edema comes with cardiomegaly and raised natriuretic peptide, and fluid overload does not follow the sequence of obstruction then edema within minutes.

Key principle: *Obstruction followed within minutes by pink frothy sputum and a normal heart size suggests negative pressure pulmonary edema.*

179. Correct answer D. Type II negative pressure pulmonary edema after relief of chronic obstruction, which can appear late

Type II edema follows the sudden relief of a chronic obstruction. A patient breathing against a chronic obstruction builds intrinsic positive pressure that holds the lung open, and when the obstruction is removed that pressure falls suddenly, so edema can appear after a delay of minutes to hours. The timing is easy to miss because the surgery is already over. Type I follows acute obstruction, usually within minutes. Cardiogenic edema and aspiration have different features, and treatment of the type II form is the same as type I.

Key principle: *Type II edema follows relief of a chronic obstruction and may appear minutes to hours later.*

180. Correct answer D. Diuretics are usually unnecessary, as the problem is leakiness; give oxygen and positive pressure

Treatment is supportive: oxygen, continuous positive airway pressure or PEEP, reintubation if the patient fails to oxygenate or tires, and lung-protective ventilation. Diuretics are not routinely needed because the problem is leakiness and not volume overload, and they may lower the blood pressure in a patient whose volume status is normal. They are reserved for clear volume overload. Steroids have no proven role. Most patients recover within 12 to 48 hours, and many are extubated within a day.

Key principle: *Treat negative pressure pulmonary edema with positive pressure and support; diuretics only for true volume overload.*

Group 10. Airway Obstruction, Fire, and Extubation Failure

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181. Correct answer B. High risk, because an oxidizer and an ignition source will both be near the surgical site

Surgery above the xiphoid process with an open source of oxygen is the classic high-risk configuration, because oxygen-enriched air collects under the drapes and electrosurgery is the most common ignition source. The assessment should be made at the time-out so that everyone in the room knows the case is high risk. The absence of a tracheal tube does not lower the risk, since most fires in this setting occur outside the airway. Waiting to see a spark defeats the purpose of prevention.

Key principle: *An oxidizer plus an ignition source near the field makes a case high risk, and the team must know it at the time-out.*

182. Correct answer D. A breach of the cuff becomes visible as blue fluid and gives early warning of damage

Filling the cuff with saline colored by methylene blue is a small step that makes a breach obvious, so the team is warned before the fire develops. The cuff should also be protected with moistened pledgets and the gauze kept wet. The saline does not make the tube fireproof, and the dye does not measure pressure or prevent burning. The purpose is early detection, which is added to the other precautions.

Key principle: *Colored saline in the cuff makes a laser breach visible early.*

183. Correct answer B. Confirm that the tube is rated for the wavelength of the laser in use and agree the settings

Laser-resistant tubes are specific to the wavelength of the laser, so a tube designed for a carbon dioxide laser may not protect against another. A checklist that matches the tube, the laser, and the settings before the first pulse closes a common gap. Metallic tubes can reflect the beam onto adjacent tissue, and silicone tubes burn in a different pattern

from polyvinyl chloride rather than resisting the laser. Wet gauze around an ordinary tube is not equivalent protection.

Key principle: *Match the tube, the laser wavelength, and the settings before the first pulse.*

184. Correct answer A. Use a carbon dioxide extinguisher on the burning drapes and activate the fire alarm

When a fire that is not in the airway does not go out at once, a carbon dioxide extinguisher is used because it leaves no harmful residue on the patient, and the alarm is activated. If the fire grows, the patient is evacuated, the door is closed, and the gas supply to the room is shut off. Restoring oxygen feeds the fire. Dry drapes are fuel and not a means of extinguishing, and delay in raising the alarm allows the fire to spread.

Key principle: *A fire that persists after saline needs a carbon dioxide extinguisher and the alarm.*

185. Correct answer D. Intensive care unit, because airway edema may progress for hours and ventilation may be prolonged

After an airway fire the patient goes to the intensive care unit, because airway edema can progress over hours and a burned airway may need prolonged ventilation or a tracheostomy. Mild findings at the first bronchoscopy do not predict how the swelling will evolve. A recovery room stay or ward bed does not provide the monitoring and airway support that may be needed. Resuming surgery would ignore the injury.

Key principle: *Airway fire patients are monitored in intensive care because edema progresses for hours.*

186. Correct answer A. Hyperinflation of the lung beyond the obstruction from trapped air

Aspiration is most common in children under three, and a quiet interval after choking is typical. A foreign body can act as a ball valve, letting air in during inspiration but not out during expiration, so the lung beyond it appears hyperinflated on an expiratory film. Missed objects may instead cause recurrent infection or a collapsed lobe, which show volume loss. A symmetric pattern or a subglottic narrowing does not point to a unilateral ball valve.

Key principle: *A ball valve foreign body shows unilateral hyperinflation on an expiratory film.*

187. Correct answer E. It preserves airway tone and avoids pushing the object deeper with positive pressure

Spontaneous ventilation keeps the muscle tone of the airway and avoids pushing the object distally or worsening a ball valve effect. Controlled ventilation gives immobility and reduces coughing, but it delivers positive pressure. The evidence does not favor one approach, and the choice depends on the location and size of the object, the state of the lung, and the habits of the team. Deep anesthesia with topical lidocaine on the cords is used in either approach to prevent laryngospasm.

Key principle: *Spontaneous breathing avoids positive pressure that can push a foreign body deeper.*

188. Correct answer E. Remove the object under direct vision with Magill forceps, then ventilate with oxygen

Complete obstruction in an unconscious victim is managed with laryngoscopy and removal of the object under vision with Magill forceps. Back blows and abdominal thrusts are for the conscious adult. Front-of-neck access is the next step only if the object cannot be removed and oxygenation fails. A blind finger sweep risks pushing the object deeper. The same obstruction can arise from dental fragments, throat packs, or gauze left in the pharynx.

Key principle: *In an unconscious victim, use laryngoscopy and Magill forceps; reserve front-of-neck access for failure.*

189. Correct answer C. Lost negative pleural pressure, lower lung volume, and relaxed airway tone let the mass compress the airways

In the awake patient, the negative pleural pressure of spontaneous breathing holds the airways open against the mass. Anesthesia reduces lung volume, abolishes that pressure, and relaxes the muscles that support the airway, so the mass can compress the trachea, the bronchi, or the great vessels. Compression of the right ventricular outflow tract and pulmonary artery can cause abrupt cardiovascular collapse. The other explanations do not account for the loss of the supporting forces.

Key principle: *Anesthesia removes the pressures and muscle tone that keep airways open around a mediastinal mass.*

190. Correct answer C. Keep spontaneous ventilation, avoid relaxants, and place femoral bypass access under local anesthesia

When general anesthesia cannot be avoided in a patient with a symptomatic mediastinal mass, the principles are to preserve spontaneous ventilation, avoid relaxants, position the patient to relieve compression, and arrange extracorporeal support in the room with femoral access obtained under local anesthesia. A rigid bronchoscope must be at hand. Waiting until collapse has occurred leaves no time to set up support. A relaxant removes the last margin of safety, and the supine head-down position increases compression.

Key principle: *If general anesthesia is unavoidable for a mediastinal mass, secure bypass access first and keep the patient breathing.*

191. Correct answer E. Awake fiberoptic intubation under topical anesthesia, with the neck kit open and backup options agreed

The principle for a tumor that narrows the airway is to keep the patient breathing until the airway is secured. Awake fiberoptic intubation or inhalational induction with spontaneous ventilation are accepted routes, and neuromuscular blockers are avoided until the airway has been proven. The plan should include a second and a third option, with the otolaryngologist present and the front-of-neck kit open. Relaxants given before the airway is proven, and heavy sedation, can remove the last margin of safety.

Key principle: *With a narrowed airway from tumor, keep the patient breathing and avoid relaxants until the airway is proven.*

192. Correct answer C. Awake fiberoptic intubation with preparation for a surgical airway, without waiting for more drugs

Angioedema from angiotensin-converting enzyme inhibitors responds poorly to epinephrine and steroids and often needs awake fiberoptic intubation or a surgical airway. Icatibant has not been shown to help this form, and airway management should not be delayed for it. Induction with a relaxant removes spontaneous breathing in a swollen airway that may be impossible to ventilate. Observation on the ward is unsafe once stridor has appeared.

Key principle: *ACE inhibitor angioedema with stridor needs an early secured airway, awake if possible, not more drugs.*

193. Correct answer A. Keep her calm and move to the operating room for induction by experienced hands with surgical backup

A child with fever and drooling who may have epiglottitis needs a calm environment and induction in the operating room by experienced hands. Agitation, examination of the throat, or a trip to radiology can precipitate complete obstruction. Sedation to prevent crying may depress ventilation. Nebulized epinephrine and ward observation delay definitive airway care in a child who is at risk of sudden obstruction.

Key principle: *Suspected epiglottitis: do not upset the child; induce in the operating room with experienced staff.*

194. Correct answer E. A fasted patient with reactive airways and an easy airway after hernia repair

Deep extubation avoids coughing in patients with reactive airways or fragile surgical repairs, but it is unsafe in those with obstructive sleep apnea, a full stomach, or a difficult airway. Wired jaws restrict airway access and make rescue harder if obstruction follows. The fasted patient with an easy airway and reactive airways is the one for whom the benefit may outweigh the risk, with the usual preparation for reintubation still in place.

Key principle: *Deep extubation is unsafe with sleep apnea, a full stomach, or a difficult airway.*

195. Correct answer B. Leave the tube in place, transfer to intensive care, and consider steroids before reassessing extubation

After extensive neck or facial surgery the airway may be swollen, and the safest decision is often to leave the tube in and arrange intensive care. The cuff leak test has limited predictive value, but an absent leak with visible swelling is a strong warning. Steroids act over hours, not minutes. The Bailey maneuver is unsuitable for a swollen airway, and an exchange catheter is a rescue aid for reintubation and does not make extubation of an edematous airway safe.

Key principle: *A swollen airway after prolonged surgery is a reason to leave the tube in and use intensive care.*

196. Correct answer B. Extubate awake and upright, apply CPAP or high-flow nasal oxygen, and monitor with capnography

Patients with obesity and obstructive sleep apnea are extubated awake and semi-upright, with the block fully reversed and long-acting opioids kept to a minimum. Continuous positive airway pressure or high-flow nasal oxygen immediately after extubation reduces obstruction and hypoxemia, and capnography plus a monitored bed for the first night protect against events after the patient leaves the operating room. Deep extubation is unsafe in sleep apnea. A train-of-four count of four does not show adequate recovery of the block.

Key principle: *In sleep apnea, extubate awake and upright, support with CPAP or high-flow oxygen, and monitor.*

197. Correct answer D. Withdraw the tube slightly and rotate it counterclockwise a quarter turn, then advance gently

When a tube loaded over a catheter catches on the arytenoids, withdrawing it slightly and rotating it counterclockwise a quarter turn turns the bevel the right way, and it should then be advanced without force. A smaller tube is used in this setting, not a larger one. Forcing the tube can injure the larynx, and removing the catheter discards the guide. Jet ventilation through the catheter has caused pneumothorax and death, especially when the upper airway is obstructed.

Key principle: *If the tube catches over a catheter, withdraw and rotate counterclockwise; do not force.*

198. Correct answer B. A subglottic or tracheal lesion

Inspiratory stridor suggests obstruction above or at the glottis, expiratory stridor suggests an intrathoracic lesion, and biphasic stridor suggests a subglottic or tracheal lesion. Whatever the site, the first task is to decide whether the patient is tiring, since a silent, exhausted patient is at more risk than a loud one. Distal bronchospasm produces wheeze rather than stridor.

Key principle: *Biphasic stridor points to the subglottis or trachea.*

199. Correct answer E. Bilateral vocal cord paralysis, with the cords at the midline

After thyroid surgery, bilateral cord paralysis presents with stridor at once on extubation, and the voice may be normal because the cords lie close together at the midline. Reintubation is often required, followed by otolaryngology review, steroids, and a decision about tracheostomy. Unilateral palsy causes hoarseness and aspiration but rarely stridor. A flat soft neck argues against hematoma, and the measured ratio argues against residual block.

Key principle: *Immediate stridor with a normal voice after thyroid surgery suggests bilateral cord palsy.*

200. Correct answer D. Preoxygenate head-up with high-flow nasal oxygen; best operator, video laryngoscope, bougie, small tube; no relaxant

Reintubation after extubation failure is harder than the first intubation, with a hypoxic patient and swollen or bloody tissues. Oxygenation is optimized first with a tight mask and high-flow nasal oxygen in a head-up position, the most experienced operator takes the first attempt, and equipment is chosen for a difficult airway. A relaxant should not be given to a patient whose airway cannot be mask ventilated. The airway has changed since the first intubation, so repeating the same plan is not appropriate.

Key principle: *Do not paralyze a patient who cannot be mask ventilated; prepare, oxygenate, and use the best operator and equipment.*

Group 11. Malignant Hyperthermia: Genetics and Pathophysiology

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201. Correct answer D. Propofol and rocuronium for a rapid sequence induction, with sugammadex available

Succinylcholine and all potent volatile agents are the established triggers, so a full stomach in a susceptible patient is managed with a rapid sequence induction that uses a nondepolarizing relaxant such as rocuronium. Propofol is safe, and removing the vaporizers does not make succinylcholine safe. Sevoflurane is a trigger and a mask induction does not protect the airway. Nitrous oxide is safe, but it does not change the danger of succinylcholine.

Key principle: *In a susceptible patient with a full stomach, avoid succinylcholine and use a nondepolarizing relaxant.*

202. Correct answer C. Spinal or epidural anesthesia, which is safe in susceptible patients and avoids all triggers

A susceptible woman can receive neuraxial analgesia and anesthesia without risk, and cesarean delivery under spinal or epidural anesthesia is usual. Local anesthetics of both the amide and ester type are safe. If general anesthesia is needed, a total intravenous technique without succinylcholine is used. Succinylcholine and volatile agents are triggers regardless of fetal considerations, and prophylactic dantrolene is not recommended.

Key principle: *Neuraxial anesthesia is safe in malignant hyperthermia susceptibility.*

203. Correct answer A. Mortality exceeded 70 percent before dantrolene and is now under 5 percent with early treatment

Before dantrolene became available, mortality exceeded 70 percent. With early recognition and treatment it is now under 5 percent. Cooling treats a consequence of the hypermetabolism and does not stop its cause, and the other figures do not match the historical or current data.

Key principle: *Dantrolene cut mortality from over 70 percent to under 5 percent.*

204. Correct answer D. Many carriers never declare the disease, because penetrance and expressivity vary

Inheritance is autosomal dominant with variable penetrance and variable expressivity. A person with a pathogenic variant may never react, may react to one anesthetic and not another, or may have a mild abortive episode, which is why the genetic prevalence is much higher than the clinical incidence. The disease is not recessive and is not limited to men, although reactions are more frequent in young men and children. Susceptible patients do not all react at the first exposure.

Key principle: *The gap between carriers and clinical cases reflects variable penetrance.*

205. Correct answer A. Cav1.1 in the T tubule is mechanically linked to the ryanodine receptor, which opens and releases calcium

In skeletal muscle the dihydropyridine receptor, also called Cav1.1, senses the voltage change in the transverse tubule and is mechanically linked to the ryanodine receptor type 1 in the sarcoplasmic reticulum. When the sensor moves the receptor opens and calcium floods the cytoplasm, binds troponin C, and starts contraction. The ATPase pump returns calcium to the store, which allows relaxation. In malignant hyperthermia a trigger lowers the threshold at which this release channel opens.

Key principle: *Skeletal muscle calcium release depends on a mechanical link between Cav1.1 and the ryanodine receptor.*

206. Correct answer B. Carry a letter and an alert card, and tell every clinician, including ambulance teams who may give succinylcholine

Practical measures after a diagnosis include a written letter explaining the diagnosis and the safe technique, a medical alert bracelet or card, and an entry in the medical record that every service will see. Dentists, surgeons, and emergency teams should be told, because emergency medical services sometimes give succinylcholine. Prophylactic dantrolene is not recommended, and susceptibility is lifelong and not limited to long anesthetics.

Key principle: *After a diagnosis, give a letter and alert card and make sure every clinician, including emergency teams, knows.*

207. Correct answer A. Sustained contraction depletes ATP, the membrane fails, and potassium, myoglobin, and creatine kinase leak out

In the hypermetabolic muscle, ATP is consumed faster than it can be made. When ATP runs out the cell membrane fails and potassium, myoglobin, and creatine kinase pour into the circulation. Hyperkalemia then causes arrhythmia, and myoglobin damages the kidney. The kidney is a victim and not the origin, and the acid-base disturbance in the reaction is acidosis rather than alkalosis.

Key principle: *Energy failure in muscle releases potassium and myoglobin into the blood.*

208. Correct answer C. It is delayed after the acute episode and uses fresh muscle tested within hours at a specialized center

The contracture test is performed at a few specialized centers on fresh living muscle that is tested within hours of biopsy. It is limited to patients above a minimum age or weight and is normally postponed for a period after an acute episode, as advised by the testing center. Frozen muscle cannot be used. Genetic testing does not detect every susceptible person, because many families carry variants that are not yet identified, and the test is done in a bath, not by exposing the patient.

Key principle: *Contracture testing needs fresh muscle at a specialized center and is delayed after an acute episode.*

209. Correct answer E. The excess calcium comes from the sarcoplasmic reticulum, not through the membrane channels these drugs block

Dantrolene is the specific treatment because the sarcoplasmic reticulum is the source of the calcium, and not the cell membrane. Calcium channel blockers act on membrane channels and do not control the intracellular store. They do not bind the ryanodine receptor, and the problem is not the speed of their onset. The cardiac isoform of the receptor is normal in this disease, so the heart is a victim of the metabolic derangement and not its origin.

Key principle: *The calcium in malignant hyperthermia comes from the sarcoplasmic reticulum, so membrane channel blockers do not treat it.*

210. Correct answer D. Tissues keep releasing volatile agent, and calcium is already loose in the muscle

The reaction does not end at the moment the agent is stopped, because tissue stores of the volatile continue to be released and calcium is already loose in the myoplasm. That lag is the reason treatment must continue after the trigger is removed. Heat is a consequence, not an independent driver of the calcium release. Dantrolene stabilizes the receptor but does not wash out the agent; washout depends on high-flow fresh gas and hyperventilation.

Key principle: *The reaction outlasts the trigger, so treatment must continue after the agent is stopped.*

211. Correct answer A. Plan a trigger-free anesthetic while this is clarified, because one positive answer is enough

The preoperative interview is the first defense. A single positive answer, such as an unexplained fever in a relative during surgery or dark urine after exercise, is enough to plan a trigger-free anesthetic while the question is clarified. Prior uneventful anesthetics do not exclude susceptibility, and many patients who react have previously tolerated triggers. A normal creatine kinase does not exclude the disorder, and prophylactic dantrolene is not recommended.

Key principle: *One positive answer in the history justifies a trigger-free anesthetic.*

212. Correct answer E. A trigger-free technique with total intravenous anesthesia and no volatile agent or succinylcholine

The STAC3 disorder, first described in Native American families, includes cleft palate and scoliosis and carries a high risk of malignant hyperthermia. These patients should receive a safe anesthetic, which means avoiding all potent volatile agents and succinylcholine and using total intravenous or regional techniques. Volatile agents are triggers whatever their potency, and nitrous oxide is simply safe and does not offset a volatile agent. The risk is not confined to adults.

Key principle: *STAC3 myopathy carries a high risk, so use a trigger-free technique.*

213. Correct answer B. Succinylcholine can cause sustained contracture in myotonia that mimics malignant hyperthermia but is distinct

Myotonic disorders pose a related problem. Succinylcholine can cause a sustained contracture that makes ventilation and intubation impossible, and volatile agents may provoke myotonic reactions in some patients. These events are not malignant hyperthermia, though they can look like it, and a clinician who cannot distinguish them quickly is best advised to treat the dangerous possibility first. Avoiding succinylcholine is the prevention.

Key principle: *Myotonic contracture after succinylcholine can mimic malignant hyperthermia but is a separate process.*

214. Correct answer C. Take a careful anesthetic history, discuss genetic testing, and plan a trigger-free anesthetic

Exertional heat stroke and exertional rhabdomyolysis have been linked to RYR1 variants in a proportion of patients, and some prove to be susceptible to malignant hyperthermia on testing. A person with unexplained severe heat illness during exercise deserves a careful anesthetic history and, in many centers, a discussion of genetic testing. A trigger-free anesthetic is the prudent plan meanwhile. A normal creatine kinase does not exclude susceptibility, and prophylactic dantrolene is not recommended.

Key principle: *Unexplained exertional heat stroke or rhabdomyolysis suggests possible RYR1 susceptibility.*

215. Correct answer B. Prophylactic dantrolene is not recommended; use the trigger-free technique and keep dantrolene available

Prophylactic dantrolene adds side effects and is unnecessary if triggers are avoided. The correct approach is a trigger-free technique with a prepared machine, with dantrolene available in the building in case of an unexpected reaction. The other options propose prophylaxis on the basis of duration, route, or technique, none of which is recommended.

Key principle: *Avoid triggers rather than giving prophylactic dantrolene, but keep dantrolene available.*

216. Correct answer D. Residual agent leaches from vaporizers, absorbent, and rubber circuit parts, so remove them, change the circuit, and flush

Volatile agent in vaporizers, absorbent, and rubber components of the breathing circuit can leach into the gas stream after the machine appears to be empty, and small concentrations may be enough for the most sensitive patients. Preparation includes removing or disabling vaporizers, replacing the circuit and reservoir bag, flushing the machine according to the manufacturer's guidance, or using activated charcoal filters. The risk is greater with older machines. The other statements do not reflect these mechanisms.

Key principle: *Prepare the machine by removing vaporizers, changing the circuit, and flushing or using charcoal filters.*

217. Correct answer C. Ask about weakness, hypotonia, and family history, and go trigger-free only if the answers are uncertain

Children with strabismus or ptosis were once thought to carry a high risk, and a few have an underlying myopathy, but most do not. A reasonable approach is to ask about weakness, hypotonia, and family history in every child with a muscle or eye condition, and to use a trigger-free technique when the answers are uncertain. A careful preoperative evaluation is worth more than a list of associated conditions. Prophylactic dantrolene is not recommended.

Key principle: *Screen children with eye or muscle conditions by history, and use trigger-free anesthesia if uncertain.*

218. Correct answer B. Treat as a possible abortive reaction, check creatine kinase, and arrange testing and counseling

Malignant hyperthermia covers a spectrum from the fulminant reaction to an abortive form in which only a few features appear. The abortive form is more common, harder to recognize, and may resolve once the trigger is stopped, but it still warrants follow-up. Creatine kinase checks for muscle injury, and the patient should be referred for testing and counseling. Rigidity is not required for an abortive episode, and exposing the patient again to a volatile agent would be unsafe.

Key principle: *A resolved abortive episode still needs follow-up, testing, and counseling.*

219. Correct answer D. It scores rigidity, muscle breakdown, acidosis, and temperature after the event, ranking likelihood 1 to 6

The scale scores rigidity, muscle breakdown, respiratory acidosis, temperature rise, cardiac involvement, and other indicators such as family history. The raw score converts to a rank from 1, meaning the diagnosis is almost never, to 6, meaning almost certain. It was designed for retrospective use and research and supports the decision to test the patient and family. It does not replace clinical judgment at the bedside and does not replace testing.

Key principle: *The Larach scale ranks likelihood from 1 to 6 after the event and does not guide bedside treatment.*

220. Correct answer E. Refer for contracture testing, and manage her and relatives as susceptible until then

Genetic testing examines only variants that are known, and many families carry variants not yet identified, so a negative result does not exclude susceptibility. When the test is negative and suspicion is high, the contracture test is the next step. Until then the patient is treated as susceptible, and relatives are managed as susceptible unless they have a negative contracture test. Single variant testing of relatives is possible only if a causative variant has been found.

Key principle: *A negative genetic test does not exclude susceptibility; go on to contracture testing.*

Group 12. Treating a Malignant Hyperthermia Crisis

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221. Correct answer E. Suspect absorbed insufflation gas, increase minute ventilation, and watch for tachycardia, rigidity, or fever

A rising end-tidal carbon dioxide needs a systematic check: ventilation, increased production from fever, sepsis, thyroid storm, or shivering, and anything absorbed, as with laparoscopic insufflation. Malignant hyperthermia enters the list when production is raised and no other source is present, particularly after a volatile agent or succinylcholine. Here the timing with insufflation and the absence of other features point to absorbed gas, with continued vigilance for the signs of a reaction. The plan should change at once if tachycardia, rigidity, or other features appear.

Key principle: *Check for absorbed insufflation gas and other causes of raised carbon dioxide before diagnosing malignant hyperthermia.*

222. Correct answer C. Hyperventilate at two to three times usual minute ventilation with high-flow 100 percent oxygen

After the trigger is stopped, a high fresh gas flow of 100 percent oxygen washes the volatile agent out of the circuit and the patient. Hyperventilation to two or three times the usual minute ventilation removes the excess carbon dioxide and lowers the anesthetic concentration in the lungs. A low flow would allow rebreathing of agent, and keeping the usual ventilation would leave carbon dioxide accumulating. Oxygen at 100 percent is preferred over a nitrous oxide mixture in this setting.

Key principle: *Wash out the volatile with high-flow 100 percent oxygen and two to three times the usual minute ventilation.*

223. Correct answer A. A second team member, such as a nurse or colleague, so the leader stays at the head of the bed

Calling the hotline helps in two ways: the expert confirms the plan and advises on dosing and complications, and the act of calling gives a second person a defined task, which keeps the leader at the head of the bed. The call can be made by a nurse or a second anesthesiologist, and the record of the call becomes part of the documentation. Delaying the call until after the crisis wastes the advice, and the leader should not abandon the bedside to make it.

Key principle: *Assign the hotline call to a second person so the leader can keep directing care.*

224. Correct answer A. Propofol and an opioid infusion, because a paralyzed hypermetabolic patient may be aware

The anesthetic is converted to a non-triggering technique. Total intravenous anesthesia with propofol and an opioid maintains unconsciousness while the volatile agent is washed out, and a benzodiazepine can be added. Nondepolarizing relaxants are safe, but they do not treat the rigidity of malignant hyperthermia because the defect is distal to the neuromuscular junction. The patient must remain adequately anesthetized, since paralysis can hide awareness.

Key principle: *Convert to total intravenous anesthesia and keep the patient deeply anesthetized.*

225. Correct answer D. Ask the surgeon to stop or finish quickly, and continue with total intravenous anesthesia during treatment

The patient cannot be stabilized while the operation continues, so the surgeon is asked to stop or finish quickly. If bleeding or another life-threatening problem requires completion, the surgeon proceeds under total intravenous anesthesia while the anesthesia team continues treatment. The surgeon can also help with cooling by irrigating open body cavities with cold saline. Dantrolene and ventilation are the priorities, and the operation is secondary. Desflurane is a trigger.

Key principle: *In a crisis, the surgery is secondary; stop or finish quickly under intravenous anesthesia.*

226. Correct answer B. Transfer to a hospital with an intensive care unit, whatever the apparent response

A patient who has had a suspected reaction at an outpatient center must go to a hospital, whatever the apparent response. The metabolic derangement may take hours to correct, muscle injury is still evolving, and recrudescence is a recognized hazard that needs monitoring and continued dantrolene in an intensive care unit. Observation at the center, early discharge, or discharge on oral dantrolene all leave a recurrence unattended. Such sites need dantrolene in stock and a plan to transfer the patient.

Key principle: *After a reaction at an outpatient site, transfer to a hospital with intensive care.*

227. Correct answer C. Regular insulin 0.1 units/kg (2 units) with 25 percent dextrose 2 mL/kg

For children, regular insulin is given at 0.1 units/kg with 25 percent dextrose at 2 mL/kg, which is 0.5 g/kg of glucose. For a 20-kg child that is 2 units of insulin with 40 mL. The adult dose would be excessive and risks severe hypoglycemia, and insulin without enough dextrose has the same risk. Blood glucose should be checked hourly and potassium every 15 to 30 minutes.

Key principle: *Pediatric hyperkalemia dosing is weight-based: insulin 0.1 units/kg with 2 mL/kg of 25 percent dextrose.*

228. Correct answer B. 3.5 mL, which is 175 mg

The initial dose is 2.5 mg/kg, so a 70-kg patient needs 175 mg. At a concentration of 50 mg/mL this is 3.5 mL, so a single vial supplies the dose in a small volume. The other volumes correspond to errors in calculation. The small volume is the main practical advantage of the nanocrystalline formulation over conventional vials, which need large volumes of sterile water.

Key principle: *Calculate 2.5 mg/kg and divide by 50 mg/mL for the nanocrystalline formulation.*

229. Correct answer D. Use a secure large or central vein, watch the site, and give it by a dedicated route without mixing

The dantrolene solution is strongly alkaline, and tissue necrosis can follow leakage into the subcutaneous space. A secure large vein or a central line is preferred, the site should be watched closely, and the drug should not be mixed with other medications. It should be given as a rapid bolus through a free-flowing line. Saline or dextrose causes the drug to precipitate, so sterile water is used. The nanocrystalline formulation has the same pH problem.

Key principle: *Give dantrolene through a secure large vein, with a dedicated line and no mixing.*

230. Correct answer C. About 36 vials or the equivalent, reachable from any anesthetizing location within 10 minutes

A hospital that provides anesthesia should hold enough dantrolene to treat a large patient for the first hours, with a common recommendation of 36 conventional vials or an equivalent amount of the concentrated product. It should be stored where it can reach any anesthetizing location within about 10 minutes. Expiry dates should be checked on a schedule, and stock replaced after any use. Offices and ambulatory sites need the same supply or a clear transfer agreement.

Key principle: *Stock about 36 conventional vials, or the equivalent, within about 10 minutes of every anesthetizing site.*

231. Correct answer C. 1 mg/kg intravenously every 4 to 6 hours for at least 24 hours

After the first response the usual practice is 1 mg/kg intravenously every 4 to 6 hours, which is about 80 mg for this patient, or a continuous infusion, for at least 24 hours. The half-life is roughly 5 to 12 hours depending on the

formulation, so the reaction can recur as the level falls. Once-daily dosing leaves long gaps, an oral route is not preferred in the first hours, and 10 mg/kg is the cumulative ceiling for reconsidering the diagnosis and not a single dose.

Key principle: *Maintenance dantrolene is 1 mg/kg every 4 to 6 hours for at least 24 hours.*

232. Correct answer A. Use actual body weight and round up, because under-dosing leaves the reaction untreated

Dantrolene is dosed on actual body weight and not ideal weight. In an obese patient this means a large dose and a long preparation, but under-dosing leaves the reaction untreated, so doses should be rounded up and not down. When the weight is unknown it should be estimated, since an approximate dose given early is better than an accurate dose given late. Delay in dantrolene is the strongest predictor of complications.

Key principle: *Dose dantrolene on actual body weight and round up.*

233. Correct answer A. Dantrolene can cause weakness lasting hours, so support ventilation until he is clearly strong

Muscle weakness after dantrolene can last for hours and may impair airway reflexes and breathing, so ventilation should be supported until the patient is clearly strong. Dantrolene also enhances the effect of nondepolarizing neuromuscular blockers. Its action is on the sarcoplasmic reticulum and not at the neuromuscular junction, so neostigmine is not the remedy. Phlebitis, dizziness, nausea, and drowsiness are other common effects. The liver injury seen with chronic oral use is not relevant to a short emergency course.

Key principle: *Dantrolene causes hours of muscle weakness, so support ventilation until strength is clear.*

234. Correct answer E. Give dantrolene to the mother, because an untreated reaction is the greater risk to the fetus

Pregnancy is not a reason to withhold dantrolene. The drug crosses the placenta, but the effect of a short course on the fetus is small compared with the danger of an untreated reaction, and the mother is treated first. After the crisis, dantrolene passes into breast milk, and breastfeeding is generally deferred while the drug is cleared. Cooling alone does not stop the hypermetabolism, and calcium channel blockers should be avoided with dantrolene.

Key principle: *Pregnancy does not justify withholding dantrolene.*

235. Correct answer D. The esophageal reading, because a core site reflects the trend most rapidly

The core temperature is measured by an esophageal, bladder, or rectal probe, because skin and axillary readings underestimate the core, especially in a shocked patient. The esophageal probe reflects the trend most rapidly, whereas rectal and bladder probes lag behind, which is relevant when deciding when to stop cooling at about 38 degrees. A trend over several minutes is more informative than a single value.

Key principle: *Use a core temperature site, ideally esophageal, to guide cooling.*

236. Correct answer B. Repeat potassium every 15 to 30 minutes at first, and glucose hourly while insulin is active

Potassium levels change quickly. As dantrolene takes effect and the membranes recover, potassium returns into cells, and overcorrection with insulin and dextrose can cause hypokalemia or hypoglycemia. Potassium and glucose should be measured frequently, every 15 to 30 minutes at first, with glucose hourly while insulin is active, and treatment adjusted to the results. Creatine kinase peaks much later and cannot guide potassium therapy.

Key principle: *Recheck potassium every 15 to 30 minutes and glucose hourly after insulin therapy.*

237. Correct answer C. Keep ventilation high to clear bicarbonate-derived carbon dioxide, and repeat the blood gas

Acidosis in malignant hyperthermia has a respiratory part from carbon dioxide and a metabolic part from lactate and muscle breakdown. Bicarbonate is used when the pH is below about 7.2 and the base deficit is large, but it produces carbon dioxide, so ventilation must be increased to clear it. A repeated gas measures the response and prevents overtreatment with alkali. Reducing ventilation would worsen the respiratory acidosis.

Key principle: *Bicarbonate generates carbon dioxide, so keep ventilation high and recheck the gas.*

238. Correct answer E. Titrate dantrolene to effect, and use beta-blockers with caution because they can mask recurrence

Sinus tachycardia is a sign of hypermetabolism and settles as dantrolene works, so treating the cause comes before rate control. Beta-blockers are used with caution because they can mask signs of recurrence and may worsen the circulation. Calcium channel blockers are avoided because of the interaction with dantrolene. Stopping dantrolene removes the only specific treatment, and a vagal maneuver does not address the metabolic cause.

Key principle: *Treat the cause of sinus tachycardia with dantrolene; beta-blockers may mask recurrence.*

239. Correct answer C. Check platelets, PT, fibrinogen, and D-dimer, transfuse by results, and request surgical review of the thighs

Hyperthermia and muscle damage can trigger disseminated intravascular coagulation, which appears as bleeding from lines and wound sites and abnormal clotting tests. A platelet count, prothrombin time, fibrinogen, and D-dimer guide transfusion. Swollen, tense limbs suggest compartment syndrome, which needs surgical review. Heparin would worsen bleeding, and delaying the work-up to see the creatine kinase peak risks missed complications.

Key principle: *Check coagulation after a severe reaction and examine the limbs for compartment syndrome.*

240. Correct answer E. Treat as recurrence: repeat dantrolene, cool, correct metabolic abnormalities, and reassess

Registry data suggest that roughly one in five patients has a second episode, usually within the first 16 hours, with the signs of the original reaction. The response is the same: repeat dantrolene, cool, correct the metabolic abnormalities, and re-evaluate. Dantrolene should remain at the bedside until the observation period is complete. Antipyretics are ineffective because the fever is not centrally mediated, and a volatile agent would trigger the reaction again.

Key principle: *Recurrence is common in the first day; treat it as a new crisis.*

Group 13. Hypermetabolic Mimics

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241. Correct answer B. Cyproheptadine given through the nasogastric tube

This is moderate serotonin syndrome, which is treated by stopping serotonergic drugs, giving benzodiazepines, and adding cyproheptadine, a serotonin antagonist. Cyproheptadine has no parenteral form, so it must go by mouth or nasogastric tube. Bromocriptine is a dopamine agonist that can worsen serotonin toxicity and is reserved for neuroleptic malignant syndrome. Antipyretics fail because the fever comes from muscle activity and not from a raised hypothalamic set point. Propranolol can mask tachycardia and cause hypotension without treating the cause.

Key principle: *Cyproheptadine is the serotonin antagonist for moderate serotonin syndrome and is given only enterally.*

242. Correct answer E. Blockade or loss of dopaminergic signaling in the hypothalamus and basal ganglia

Neuroleptic malignant syndrome is a state of central dopamine deficiency. The hypothalamus governs heat dissipation and the basal ganglia govern movement, so loss of dopamine signaling there produces failure of heat loss, rigidity, and autonomic instability. Excess serotonergic stimulation causes serotonin syndrome, which is marked by clonus and hyperreflexia. Uncontrolled sarcoplasmic calcium release describes malignant hyperthermia, and catecholamine excess describes pheochromocytoma crisis. A raised set point would respond to antipyretics, which is not the mechanism here.

Key principle: *Neuroleptic malignant syndrome is dopamine deficiency in the hypothalamus and basal ganglia.*

243. Correct answer A. Onset over days after a dopamine antagonist, before anesthesia

Tempo is the first clue: neuroleptic malignant syndrome develops over one to three days after a new drug or dose change, whereas malignant hyperthermia appears within minutes or hours of a volatile agent or succinylcholine. Here

the illness predates any trigger. Rigidity, fever, autonomic lability, a high creatine kinase, and lactic acidosis can all occur in both conditions, so none of them separates the two. The history and timeline do the discriminating work.

Key principle: *Onset over days before anesthesia points to neuroleptic malignant syndrome, not malignant hyperthermia.*

244. Correct answer A. Ondansetron

In patients with Parkinson disease, central dopamine antagonists can worsen the dopamine deficiency and precipitate rigidity and a neuroleptic malignant picture. Metoclopramide, droperidol, haloperidol, and promethazine are all avoided for this reason. Ondansetron is a serotonin antagonist without dopamine blocking effect and is usually acceptable. The levodopa schedule should also be continued, by nasogastric tube if the patient cannot swallow.

Key principle: *Use ondansetron, not dopamine antagonists, for nausea in Parkinson disease.*

245. Correct answer C. Morphine

Linezolid inhibits monoamine oxidase, so serotonergic agents add real risk. Meperidine, tramadol, methadone, and dextromethorphan all contribute serotonergic activity in the perioperative period. Morphine and hydromorphone are considered safer opioid options in this setting. The medication review before surgery costs only a few minutes and prevents most of these events.

Key principle: *In a patient on a monoamine oxidase inhibitor or linezolid, choose morphine or hydromorphone over meperidine, tramadol, or methadone.*

246. Correct answer A. Ephedrine

Indirect-acting vasopressors such as ephedrine release stored catecholamines, and in the presence of monoamine oxidase inhibition this can produce a severe hypertensive reaction. Direct-acting vasopressors such as phenylephrine and norepinephrine act on receptors without relying on stored transmitter and are the safer choices. Morphine and hydromorphone are also considered safer opioid options than meperidine in these patients. The key step is recognizing the interaction before the drug is drawn up.

Key principle: *Avoid indirect-acting vasopressors such as ephedrine in patients on monoamine oxidase inhibitors.*

247. Correct answer E. Give calcium, then insulin with glucose, and increase ventilation

Rhabdomyolysis releases potassium and myoglobin, and the heart is at risk from the potassium while the kidneys are at risk from the myoglobin. Hyperkalemia with electrocardiographic change is treated early with calcium, insulin with glucose, and ventilation, along with brisk fluids to protect the kidneys. Dialysis is reserved for refractory hyperkalemia or renal failure and is not the first step. Fluids alone are too slow for ECG changes, a beta-blocker does not treat hyperkalemia, and dantrolene alone does not address the immediate threat.

Key principle: *Treat rhabdomyolytic hyperkalemia early with calcium, insulin with glucose, and ventilation; reserve dialysis for refractory cases.*

248. Correct answer D. It reduces conversion of thyroxine to triiodothyronine

Beta-blockade controls the adrenergic effects of thyroid storm, and propranolol additionally reduces conversion of thyroxine to the more active triiodothyronine. Blocking stored hormone release is the role of iodine, and stopping synthesis is the role of the thionamides. Displacement from binding proteins raises free hormone and is the reason aspirin is avoided. Binding hormone in the gut describes cholestyramine, reserved for refractory cases.

Key principle: *Propranolol blocks adrenergic effects and also reduces peripheral T4 to T3 conversion.*

249. Correct answer D. It reduces hormone conversion and covers adrenal insufficiency

Glucocorticoids reduce peripheral conversion of thyroxine and cover the relative adrenal insufficiency that can accompany severe stress, which matters in a hypotensive patient. Synthesis is blocked by thionamides, and gut binding is the role of cholestyramine. Displacement of bound hormone would raise free levels and make the storm worse. The fever of storm is not from a raised set point, which is why acetaminophen and cooling are used and not corticosteroids for antipyresis.

Key principle: *Steroids in thyroid storm reduce conversion and cover relative adrenal insufficiency.*

250. Correct answer C. Flaccid muscles with a history of a stopped antithyroid drug

Rigidity is absent in thyroid storm and the onset is usually later than the first hour, whereas the history of a stopped thyroid drug raises the probability of storm directly. A rising end-tidal carbon dioxide, tachycardia, acidosis, and rising temperature all occur in malignant hyperthermia and do not discriminate. The history and muscle examination do most of the work, because thyroid function tests take too long to guide the first hour.

Key principle: *Flaccidity plus a thyroid history separates thyroid storm from malignant hyperthermia.*

251. Correct answer E. Alpha blockade for one to two weeks, then a beta-blocker, with salt and fluids

Preparation changes outcomes more than any intraoperative drug. Alpha blockade with phenoxybenzamine or doxazosin for one to two weeks lowers blood pressure and restores the contracted plasma volume, and liberal salt and fluids support this. A beta-blocker is added only after alpha blockade is established, because beta blockade alone leaves alpha stimulation unopposed. Fluid restriction works against restoring volume, and relying on intraoperative vasodilators alone forgoes the main benefit.

Key principle: *Start alpha blockade first, add a beta-blocker later, and encourage salt and fluids before pheochromocytoma surgery.*

252. Correct answer B. Give fluid and norepinephrine, and check the glucose level

After the vein is ligated the catecholamine drive disappears and leaves a vasodilated, volume-depleted patient, so volume and norepinephrine are the priorities, along with close attention to glucose. Phentolamine and esmolol would lower the pressure further. Fluid restriction deepens the volume deficit. Glucagon stimulates catecholamine release from these tumors and is a recognized provoking drug, and it has no role here.

Key principle: *After tumor vein ligation, treat hypotension with volume and norepinephrine and watch the glucose.*

253. Correct answer B. Stop catecholamine pressors, start alpha blockade, cool, and support organs

A multisystem pheochromocytoma crisis can look exactly like sepsis or malignant hyperthermia, and catecholamine vasopressors given in error may worsen it. The paroxysmal history and the stress cardiomyopathy are the clues. Treatment combines alpha blockade, cooling, organ support, and early tumor removal when stable. A beta-blocker before alpha blockade is dangerous, and dantrolene, iodine, and thionamides do not address the catecholamine surge.

Key principle: *Multisystem pheochromocytoma crisis mimics sepsis; avoid catecholamine pressors and use alpha blockade.*

254. Correct answer D. Treat the shivering, warm gently, and follow the temperature trend

Shivering on emergence raises oxygen consumption and carbon dioxide production by a large factor and can reproduce a hypermetabolic picture. A cool patient with a rising carbon dioxide is far more likely to be shivering or overshooting during rewarming than to have malignant hyperthermia, although inspired carbon dioxide and tone should still be checked, as was done here. Treatment is gentle warming and control of the shivering. The other options treat diseases that nothing in the picture supports.

Key principle: *A cool patient with rising carbon dioxide on emergence is more likely shivering than hypermetabolic.*

255. Correct answer E. Sepsis or catecholamine excess

A pure metabolic acidosis with a high lactate points toward sepsis or catecholamine excess. Malignant hyperthermia and other muscle breakdown states produce a mixed respiratory and metabolic acidosis with high potassium and a markedly raised creatine kinase. Neuroleptic malignant and serotonin syndromes also involve muscle activity and often a high creatine kinase. Rebreathing from an equipment fault would raise carbon dioxide and cause a respiratory component to the acidosis.

Key principle: *Pure metabolic acidosis with high lactate favors sepsis or catecholamine excess over muscle breakdown.*

256. Correct answer B. Volatile agent over desiccated absorbent, generating heat and carbon monoxide

Desiccated absorbent is a recognized hazard, because volatile agents flowing over dry granules can generate heat and carbon monoxide, and the canister may be hot. Malignant hyperthermia does not heat the canister, and exhausted absorbent or valve faults produce rebreathing of carbon dioxide and not a hot canister. Absorbent color is also an

unreliable guide, since exhausted granules can revert to their original color after a rest. The remedy is mechanical: change the absorbent and check the circuit.

Key principle: *Dry absorbent with a volatile agent can overheat and make carbon monoxide.*

257. Correct answer C. Venous carbon dioxide embolism or tension pneumothorax

A sudden collapse with a fall in end-tidal values suggests venous gas embolism or tension pneumothorax, and the insufflation should be stopped and the cause treated. Steady peritoneal absorption of carbon dioxide produces a gradual rise in end-tidal values that responds to increased ventilation. Extraperitoneal gas gives a sudden or disproportionate rise with subcutaneous emphysema. Malignant hyperthermia and rebreathing both raise end-tidal carbon dioxide, which is the opposite of this finding.

Key principle: *A sudden fall in end-tidal carbon dioxide with collapse during insufflation suggests gas embolism or pneumothorax.*

258. Correct answer D. A probe in the lower esophagus

Core sites such as the esophagus, nasopharynx, bladder, and rectum follow the true temperature, while skin and axillary sites lag or mislead, and a probe lying against a warming device can give a spurious fever. A fever without a rise in carbon dioxide production also points away from malignant hyperthermia, so confirming the reading prevents an unnecessary crisis response. The esophageal probe is the practical core site during general anesthesia.

Key principle: *Confirm fever with a core temperature site before accepting a skin or axillary reading.*

259. Correct answer A. Stop the transfusion, return the bag to the blood bank, and treat shock

Fever with hypotension, hemoglobinuria, and surgical-field oozing suggests an acute hemolytic transfusion reaction. The transfusion is stopped, the bag, tubing, and patient samples are returned for checking, the clerical record is reviewed, and the shock and coagulopathy are treated. A febrile nonhemolytic reaction alone would be benign, but this picture is not. Continuing the unit, even slowly, risks further harm, and dantrolene treats a different disease.

Key principle: *Fever with hypotension and hemoglobinuria during transfusion means stop the unit and check for hemolysis.*

260. Correct answer E. Stop the volatile, hyperventilate, give dantrolene, and look for mimics in parallel

Malignant hyperthermia is a late rise in temperature after earlier signs of increased metabolism, so waiting for fever is a mistake. When it cannot be excluded in the first minutes, the team stops triggering agents, hyperventilates with oxygen, gives dantrolene, and continues the search for mimics at the same time. The harm from unnecessary dantrolene is small and the harm from withholding it is large. Thyroid tests take too long, and completing a workup first delays the specific drug.

Key principle: *If malignant hyperthermia cannot be excluded, treat it and search for mimics in parallel.*

Group 14. Local Anesthetic Pharmacology and the Mechanism of Toxicity

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261. Correct answer C. Higher affinity for open or inactivated channels than for resting channels

Local anesthetics bind a site within the pore of the voltage-gated sodium channel and have higher affinity for channels that are open or inactivated. A tissue firing rapidly spends more time in those states, so block accumulates. Binding from the extracellular side is incorrect because the charged form acts from inside the cell. Potassium channel blockade is a separate contribution to arrhythmia, and tissue diffusion is not the mechanism of use dependence.

Key principle: *Use-dependent block arises because affinity is higher for open and inactivated channels.*

262. Correct answer A. The uncharged form crosses the membrane, and the charged form binds inside

Local anesthetics are weak bases that work from inside the cell. The uncharged form crosses the nerve membrane, re-equilibrates inside, and the charged form then binds within the pore of the channel. The charged form does not cross

the membrane readily, and the binding site is intracellular. Acidosis, not alkalosis, increases the charged fraction and traps drug within neurons.

Key principle: *The uncharged base enters the cell and the charged form binds the channel from inside.*

263. Correct answer E. Inhibitory cortical neurons are blocked before excitatory neurons

At rising concentrations, inhibitory neurons in the cortex are blocked before excitatory ones, so the first effect is disinhibition, with numbness, tinnitus, restlessness, and then twitching and seizure. At higher concentrations both pathways are blocked and excitation gives way to coma and apnea. The order is not due to agonist action or to cardiac blockade coming first. The sequence can be hidden by sedation, which raises the seizure threshold.

Key principle: *Early neurologic excitation reflects blockade of inhibitory neurons first.*

264. Correct answer D. Hypercarbia and acidosis raise brain delivery and trap the charged drug

A seizing patient who is not breathing becomes hypercarbic and acidotic within a minute. Hypercarbia increases cerebral blood flow and delivers more drug to the brain, and acidosis increases the charged fraction inside neurons and traps it there, so both lower the seizure threshold and make the heart more vulnerable. Acidosis does not release drug from neurons, and alkalosis is not the issue. This is why airway, oxygenation, and ventilation are first priorities.

Key principle: *Hypercarbia and acidosis lower the seizure threshold, so ventilate early.*

265. Correct answer D. A widening QRS complex with a slowing sinus rate

Early cardiac effects of sodium channel blockade include PR prolongation, widening of the QRS complex, sinus slowing, and ST segment changes, and these can precede symptoms and the fall in pressure. A wide complex with a slow rate is a warning of channel blockade and not a benign finding. A shortened PR interval, a faster rate, and isolated atrial beats do not fit. A pre-existing bundle branch pattern is not a new change.

Key principle: *A widening QRS with slowing after a large block signals channel blockade and demands action.*

266. Correct answer A. Low concentrations constrict vessels and higher concentrations dilate them

Vascular effects are biphasic: low concentrations constrict smooth muscle, which is why some agents prolong their own action, while higher concentrations cause vasodilation. Ropivacaine and levobupivacaine are more vasoconstrictive than bupivacaine at clinical concentrations. In toxicity, vasodilation joins myocardial depression to produce profound hypotension that responds poorly to fluid and ordinary doses of vasopressor. Epinephrine is not required for the vascular effects.

Key principle: *Local anesthetic vascular effects are biphasic, with vasodilation at toxic concentrations.*

267. Correct answer B. Ropivacaine, an amino-amide

Amino-esters are hydrolyzed in plasma by pseudocholinesterase and produce para-aminobenzoic acid, which is responsible for the allergic reactions once common with esters. Procaine, tetracaine, and benzocaine share that family and metabolite. Amino-amides, such as ropivacaine, are metabolized in the liver, and true allergy to an amide is rare. Many reported amide reactions are caused by preservatives or the vasoconstrictor additive, which should still be considered.

Key principle: *Esters release para-aminobenzoic acid and are the family linked to true allergy; amides rarely cause it.*

268. Correct answer C. Degree of protein binding

Protein binding determines how long the drug remains in the tissue and therefore the duration of block. Lipid solubility determines potency and the ease of entering nerve membranes, and the dissociation constant determines how much drug is uncharged at physiologic pH and therefore the speed of onset. Pseudocholinesterase hydrolyzes esters and is not a determinant of amide duration. Bupivacaine is both highly lipid soluble and highly protein bound, which makes it potent, long acting, and more dangerous in the circulation.

Key principle: *Dissociation constant sets onset, lipid solubility sets potency, and protein binding sets duration.*

269. Correct answer A. Slow absorption from tumescent infiltration delays the peak

Large doses of lidocaine given by tumescent infiltration or topically to the airway can produce toxic concentrations hours later, because absorption is slow and the peak is delayed. An intravascular injection would have caused symptoms within a minute or two. Lidocaine is an amide metabolized by the liver, so ester hydrolysis and para-aminobenzoic acid are not involved. Methemoglobinemia is associated with prilocaine and benzocaine and presents with cyanosis and a fixed saturation.

Key principle: *Tumescent and topical lidocaine can reach toxic levels hours after the procedure.*

270. Correct answer C. Chloroprocaine

Chloroprocaine is an amino-ester hydrolyzed by plasma pseudocholinesterase within about a minute in adults, so systemic reactions are less likely and shorter, and accumulation is unlikely. The other agents are amides cleared by the liver with elimination half-lives of hours. Bupivacaine in particular is the more dangerous drug once it reaches the circulation. Even so, chloroprocaine is not safe to inject intravascularly on purpose.

Key principle: *Chloroprocaine's rapid plasma hydrolysis gives it a favorable systemic margin.*

271. Correct answer D. The syringe holds 200 mg, which exceeds the limit of 150 mg

A 0.5 percent solution contains 5 mg in each milliliter, so 40 mL holds 200 mg. The commonly cited limit for plain bupivacaine is 2.5 mg/kg, which for 60 kg is 150 mg, and it must not exceed 175 mg. The figure of 225 mg applies only to bupivacaine with epinephrine and is also capped by weight at 3 mg/kg. Confusing percent with milligrams is a classic error, so the number should be said aloud before injection.

Key principle: *A 0.5 percent solution is 5 mg per mL; calculate the milligrams before injecting.*

272. Correct answer A. Toxicity is additive and the combined fractions of 1.1 exceed the safe total

When different local anesthetics or different operators contribute, toxicity is treated as additive. Each dose is expressed as a fraction of its own maximum and the sum must stay below one. Here 0.6 plus 0.5 gives 1.1, which exceeds the safe total. Separate sites do not remove the systemic effect, and relative potency does not make the arithmetic independent. A single named person should keep the tally.

Key principle: *Add each local anesthetic as a fraction of its maximum and keep the sum below one.*

273. Correct answer E. Stop the infusion before the block, or count its dose in the total

Systemic lidocaine infusions are safe at the low therapeutic range, but risk rises when a regional block with the same or an additive agent is added, and when hepatic dysfunction or conduction disease is present. Stopping the infusion before a large block, or including it in the total dose, removes the hazard. Increasing the rate or delaying review adds exposure. Lidocaine and bupivacaine toxicity are additive, so ignoring the infusion is an error.

Key principle: *Count or stop a running lidocaine infusion before adding a large regional block.*

274. Correct answer B. It reduces peak levels by slowing absorption, less so for long-acting agents

Epinephrine added to a short-acting agent such as lidocaine reduces the peak plasma concentration by slowing absorption, but it has less effect with long-acting agents. It does not remove the risk of toxicity, because an intravascular injection still acts immediately and a depot can still peak later. It constricts vessels rather than dilating them. Dose limits for bupivacaine with epinephrine are only modestly higher than for plain solutions.

Key principle: *Epinephrine slows absorption of short-acting agents much more than long-acting ones.*

275. Correct answer C. Anesthesia masks early signs, and infants bind less drug and clear it slower

Blocks in children are usually done under general anesthesia, so early brain symptoms are absent and the first sign is commonly an arrhythmia, an ECG change, or hypotension. Infants have lower binding protein levels, immature hepatic enzymes, and slower clearance, so free concentrations are higher. Caudal blocks are absorbed rapidly, not poorly. Seizures can occur in children, but masking by anesthesia explains their absence here.

Key principle: *In children under general anesthesia, toxicity often shows first as an ECG change.*

276. Correct answer A. Bupivacaine is contraindicated since tourniquet failure releases drug systemically

In intravenous regional anesthesia the tourniquet is the safety device, and the drug enters the systemic circulation when the cuff is released or if it leaks or fails. Only short-acting, low-toxicity agents are appropriate, and bupivacaine is contraindicated. The cuff should stay inflated for a minimum time, and deflation should be staged. Failure of the tourniquet is the main preventable hazard.

Key principle: *Never use bupivacaine for intravenous regional anesthesia; tourniquet failure is the main hazard.*

277. Correct answer D. Inhibition of CYP1A2 by fluvoxamine slowing ropivacaine clearance

Fluvoxamine inhibits CYP1A2, which metabolizes ropivacaine and lidocaine, so clearance falls and drug accumulates during a continuous infusion. Ropivacaine is an amide cleared by hepatic enzymes, so pseudocholinesterase is irrelevant. Enzyme induction would lower, not raise, levels. A medication list should be read with the local anesthetic in mind, and a new neurologic symptom during an infusion should be treated as toxicity until proven otherwise.

Key principle: *Check for CYP inhibitors, since fluvoxamine slows ropivacaine and lidocaine clearance.*

278. Correct answer B. A high hepatic extraction ratio with flow-dependent clearance

Lidocaine has a high hepatic extraction ratio, so its clearance depends on liver blood flow, which falls in heart failure, with beta-blockers, and under volatile anesthesia. Bupivacaine and ropivacaine are cleared at a rate that depends more on enzyme activity than on blood flow. Lidocaine is an amide and not an ester, so cholinesterase is not the main route. Its active metabolites are cleared by the kidney and can accumulate with long infusions.

Key principle: *High extraction means lidocaine clearance tracks liver blood flow.*

279. Correct answer B. A postoperative rise in alpha-1 acid glycoprotein lowers the free fraction

Alpha-1 acid glycoprotein binds basic drugs with high affinity and rises after surgery and trauma as an acute-phase protein, so the free fraction falls over the first postoperative days. This is one reason prolonged infusions of ropivacaine and bupivacaine are tolerated for a time. The tolerance is temporary and does not replace vigilance. Pseudocholinesterase acts on esters, not amides, and binding is lower, not higher, in neonates, pregnancy, and liver disease.

Key principle: *Acute-phase alpha-1 acid glycoprotein lowers free drug early after surgery.*

280. Correct answer C. Total weight overestimates the safe dose, so lean weight is more prudent

Maximum doses are expressed in milligrams per kilogram and ignore the site and the patient's condition. A dose calculated on total body weight in an obese patient overestimates what the patient can safely absorb, because lean mass and vascular beds do not scale with fat. Dosing on ideal or lean weight is more prudent, and the lowest effective volume and concentration is always the better choice. Obesity does not protect against toxicity.

Key principle: *Dose local anesthetics on lean or ideal weight in obese patients.*

Group 15. Recognizing and Treating Local Anesthetic Systemic Toxicity

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281. Correct answer B. Beta-blockers can blunt the marker, so a negative result is unreliable

An intravascular marker dose of epinephrine causes a brief rise in heart rate and blood pressure, but it can be missed in patients taking beta-blockers, in older adults, in laboring women, and under general anesthesia. A negative response therefore does not exclude an intravenous catheter, and doses should be given in increments with continued observation. Conversely, a fast, brief tachycardia immediately after injection should be treated as a warning and not as anxiety.

Key principle: *A negative epinephrine marker response does not exclude intravascular injection.*

282. Correct answer A. Treat local anesthetic toxicity as the leading cause until excluded

Sedation and vasovagal reactions can cause bradycardia and hypotension after a block, but they usually respond to simple measures. The rule is that if the patient deteriorates in the hour after a block and the cause is not clear, local

anesthetic toxicity is the leading diagnosis until excluded. A plasma level returns too late and treatment never waits for it. Anaphylaxis and ischemia stay on the differential but should not displace the toxicity plan.

Key principle: *Unexplained deterioration within an hour of a block is toxicity until proven otherwise.*

283. Correct answer D. Urticaria with bronchospasm and facial angioedema

Anaphylaxis brings urticaria, bronchospasm, and angioedema, which are not features of local anesthetic toxicity. Perioral numbness, metallic taste, seizure, widening QRS, PR prolongation, and ventricular ectopy are signs of sodium channel and central nervous system effects of the drug. The differential must be held deliberately, but the cutaneous and airway findings point away from toxicity.

Key principle: *Urticaria, bronchospasm, and angioedema point to anaphylaxis, not local anesthetic toxicity.*

284. Correct answer E. The diagnosis is clinical, and treatment never waits for the level

A plasma drug level takes far too long to return, and the level that causes harm varies from patient to patient. The diagnosis is made at the bedside from timing, exposure, and the combination of symptoms and electrocardiographic findings. Samples can be sent afterward for documentation. No threshold is needed to start lipid emulsion in the setting of cardiovascular involvement.

Key principle: *Diagnose local anesthetic toxicity at the bedside and do not wait for drug levels.*

285. Correct answer C. Early signs of toxicity that can precede overt bradycardia

Cardiac monitoring changes can be subtle in the first minutes: a rise in T wave amplitude, a prolonged PR interval, and slight widening of the QRS complex can precede overt bradycardia, and a brief episode of ventricular ectopy may be the only warning in a sedated patient. A clinician watching only the numerical heart rate and saturation can miss all of this, which is why the waveform should be watched during and after a large block. A high spinal would follow neuraxial injection and has its own pattern.

Key principle: *Watch the ECG waveform, not just the numbers, after a large block.*

286. Correct answer A. Bag-mask ventilation with an oral airway and 100 percent oxygen

A seizing patient cannot be intubated until the seizure is controlled or the muscles are relaxed, and a patient who is hypoxic and acidotic is easy to lose. Bag-mask ventilation with an oral airway usually suffices at first, and oxygenation and ventilation treat the toxicity itself because hypoxia and acidosis worsen both the brain and heart effects. Nasal cannula is inadequate for an apneic patient, and a surgical airway is not an initial step. Intubation is planned with a drug combination that does not worsen cardiovascular depression.

Key principle: *Start with bag-mask ventilation and 100 percent oxygen; intubate in a controlled way.*

287. Correct answer D. Give a small dose of succinylcholine, with the airway secured

Seizures should be stopped promptly because muscle activity generates acid and consumes oxygen. If they persist despite benzodiazepines, a small dose of a neuromuscular blocker such as succinylcholine stops the muscle activity so that acidosis and hypoxemia can be limited, provided the airway is secured. Large doses of propofol are best avoided with any cardiovascular instability. Lidocaine would compound sodium channel blockade, and waiting allows continuing acidosis.

Key principle: *For refractory seizures, paralyze briefly with the airway secured to limit acidosis.*

288. Correct answer E. Sodium channel block causes conduction block atropine cannot reverse

Hypotension and bradycardia call for the usual supportive treatment, but the heart and vasculature respond poorly. Atropine may be tried, though it often fails in the face of conduction block, because the problem lies in sodium channel blockade within the conducting tissue and not in vagal tone. Epinephrine in small doses is the vasopressor of choice. Lipid emulsion does not inactivate atropine, and the failure is not caused by muscarinic blockade.

Key principle: *Atropine often fails in toxicity because the bradycardia is conduction block.*

289. Correct answer E. Stored where blocks are done, with large syringes, tubing, and a dose card

A lipid rescue kit should hold 20 percent emulsion in the volume needed for an adult, large syringes, tubing, and a card with the bolus and infusion doses written in plain figures. It should be stored where blocks are performed and not in a distant pharmacy, and it should be checked on a schedule. Simulation exercises expose problems such as who draws up the emulsion and who calls the bypass center before they matter.

Key principle: *Keep the lipid kit, complete with a dose card, where blocks are done.*

290. Correct answer B. Use a larger vein or pressure bag because the emulsion is viscous

Because the emulsion is viscous, it flows slowly through a small catheter, and a large vein or a pressure bag may be needed. These details are best learned before they are needed. Diluting is not part of the regimen, propofol is not a substitute for lipid emulsion because its fat content is only ten percent, and lowering the dose delays effective treatment. Waiting wastes the early window.

Key principle: *Lipid emulsion is viscous; use a large vein or pressure bag.*

291. Correct answer E. It extracts lipophilic drug from tissue and gives the heart fatty acids

Several mechanisms probably contribute. The lipid sink explanation holds that the expanded lipid phase in plasma extracts lipophilic local anesthetic from tissues and lowers concentration at the target organs. A second contribution is metabolic, supplying fatty acid substrate to a myocardium whose fatty acid transport has been blocked. Other proposed effects include a positive inotropic action and protection during low flow. Emulsion does not occupy the sodium channel or speed metabolism, and displacing drug from binding proteins would raise free drug.

Key principle: *Lipid acts as a sink and as fatty acid fuel for the starved myocardium.*

292. Correct answer B. Lipemia distorts many assays, so draw samples before lipid if it causes no delay

Lipemic blood interferes with many laboratory assays, producing spurious values for hemoglobin, electrolytes, and creatinine, so samples should be drawn before lipid whenever this does not delay treatment. Treatment never waits for samples, but a pre-lipid sample is valuable. The results are artifactual and not true changes. Samples after the event remain useful for documentation and follow-up of complications.

Key principle: *Draw blood before lipid if it does not delay treatment, because lipemia distorts assays.*

293. Correct answer D. Lipid can clog the oxygenator of an extracorporeal circuit

Lipid can clog the oxygenator of an extracorporeal circuit, which is a practical point to raise with the perfusion team if bypass is under consideration. This does not mean lipid should be withheld, and it does not remove the need for extracorporeal support when the circulation does not recover. The call to the bypass center should be made early, because transfer takes time. Heparin and bypass are not contraindicated by lipid.

Key principle: *Warn the perfusion team that lipid can clog an oxygenator.*

294. Correct answer A. Plausible mechanism, animal data, recoveries after failed care, and low harm

The evidence is weak by the standards of clinical trials, and publication bias affects reported successes. The case for treatment rests on a plausible mechanism, a consistent signal in animals, many reports of recovery after failed standard care, and a low rate of serious harm. In a disease with high mortality and no good alternative, most authorities judge that the balance favors giving it early. The treatment is not harmless, since pancreatitis and fat overload have been reported.

Key principle: *Lipid is recommended on mechanism, animal data, and case reports despite no human randomized trials.*

295. Correct answer C. Observe closely with emulsion ready, and treat any hemodynamic change

In isolated, brief central nervous system symptoms that settle with supportive care and a benzodiazepine, close observation may be enough, with the emulsion drawn up and ready. If the seizure is prolonged or recurrent, or if any cardiovascular change, a widening QRS complex, or an arrhythmia develops, lipid should be given without delay once the airway is secured. Waiting for cardiac arrest is a failure to apply available treatment, and discharging from monitoring ignores a possible delayed rise in the level.

Key principle: *Brief resolved central nervous system signs allow observation with lipid ready; any cardiovascular change triggers it.*

296. Correct answer A. Stop the infusion and treat as toxicity until proven otherwise

A continuous infusion should be stopped at the first sign of toxicity, and any new neurologic complaint during an infusion is treated as toxicity until proven otherwise. The catheter should not be reused for injection until the cause is understood. Increasing the infusion or injecting through the catheter adds drug, and attributing symptoms to opioids risks missing the diagnosis. Catheter migration into a vein is a recognized cause.

Key principle: *Stop a catheter infusion at the first sign of toxicity.*

297. Correct answer C. Aspirate and repeat a test dose because the catheter may have moved

Catheter migration into a vein or the subarachnoid space is a recognized cause of toxicity, so aspiration and test dosing are repeated at every top-up in labor and when a catheter has been moved. Doses are given in increments with observation. Epidural catheters in laboring women are more likely to sit in an engorged epidural vein, and the pregnant heart is more sensitive to bupivacaine. Early symptoms such as perioral tingling should not be dismissed as anxiety.

Key principle: *Repeat aspiration and test dosing at each top-up and after any catheter movement.*

298. Correct answer A. Electrolytes, gas, lactate, creatine kinase, troponin, and echo if needed

After a significant event, electrolytes, a blood gas, lactate, creatine kinase, and troponin should be checked, with lipase measured if abdominal pain develops after lipid therapy. An echocardiogram is reasonable if ventricular function was compromised, and the neurologic examination should be repeated because a prolonged seizure or hypoperfusion may cause injury. Specialist input from cardiology or neurology is sought for any persistent abnormality. A drug level does not replace this assessment.

Key principle: *After recovery, check metabolic, cardiac, and neurologic status, with echocardiography if function was compromised.*

299. Correct answer E. She may have a block once the cause is known, using lower, slower doses

A patient who has had toxicity is not barred from regional anesthesia in the future, but the cause of the event should be established first. If the cause was an intravascular injection or an excessive dose, the technique can be repeated with lower doses, slower injection, ultrasound guidance, and more intensive monitoring. If an underlying mitochondrial or cardiac disorder is suspected, specialist advice is warranted. The patient should carry a record of the event and share it with each new team.

Key principle: *Toxicity does not bar future regional anesthesia once the cause is addressed.*

300. Correct answer D. Report to a registry and review dose, technique, and response without blame

Reporting the event is part of the duty, since voluntary registries collect cases and supply much of the evidence used to refine treatment. Institutions should treat the event as a patient safety incident and conduct a structured review of how the dose was calculated, how the injection was performed, how monitoring was arranged, and how the response unfolded. The aim is learning and process change and not blame. Disclosure to the patient and family is also part of care.

Key principle: *Report and review toxicity events for learning, not blame.*

Group 16. Preventing Toxicity and Managing Block and Neuraxial Complications

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301. Correct answer C. It is a useful habit but cannot exclude a vessel, so inject incrementally

Aspiration is a necessary habit, but a negative result does not exclude intravascular placement. Collapsed veins, small vessels, and multiorifice catheters all give false reassurance. For this reason aspiration is paired with incremental

injection, with pauses and questions to the patient. Ultrasound reduces risk but does not make a negative aspirate conclusive. A positive aspirate remains a clear warning, so the habit has value for catheters as well as needles.

Key principle: *Negative aspiration never clears an injection; pair it with incremental dosing.*

302. Correct answer D. Stop injecting, call for help, and prepare to treat local anesthetic toxicity

A metallic taste, perioral tingling, or lightheadedness during injection is an early symptom of intravascular drug or rising plasma concentration. The injection stops at once, help is called, and the team prepares for toxicity with the lipid emulsion kit and resuscitation equipment. Monitors often change only after symptoms, so stable readings do not justify continuing. Slower completion or smaller increments still deliver more drug into a possible intravascular site, and further sedation removes the patient's ability to report worsening.

Key principle: *Any early toxicity symptom during injection means stop, call for help, and prepare.*

303. Correct answer E. The patient cannot report early symptoms, so a monitor change may be the first sign

An awake or lightly sedated patient reports a metallic taste, tingling, or lightheadedness long before a monitor changes. Deep sedation removes this early warning and also raises the seizure threshold, so the first indication of trouble may be hemodynamic or electrocardiographic. The trade should be made consciously, with tighter monitoring and smaller increments if deeper sedation is needed. The other options describe effects that sedation does not have.

Key principle: *Deep sedation silences the earliest warning of local anesthetic toxicity.*

304. Correct answer B. The pressure may signal intrafascicular placement, so stop injecting and reposition the needle

An opening pressure above about 15 psi is a commonly used warning of possible intrafascicular placement, and the injection should stop. Lower pressures do not exclude intraneural injection, and the absence of pain does not confirm a safe position, especially if the patient is sedated. High pressure is a sign of resistance to flow, not of a vascular tip, and an epinephrine marker has no role in assessing it. Reposition the needle and recheck the pressure before injecting again.

Key principle: *High opening pressure is a stop signal even when the patient has no pain.*

305. Correct answer E. Small-bore connectors for neuraxial use that cannot join intravenous tubing

Connectors designed for neuraxial and regional use cannot be joined to intravenous tubing, which removes the possibility of the wrong-route error by design. Reminders and expertise depend on vigilance, which fails under pressure. An epinephrine marker tests for intravascular injection of local anesthetic, not for a misconnected line. Labeling syringes and reading them aloud add further protection, but engineering controls are the most reliable.

Key principle: *Design out wrong-route errors with connectors that cannot be misconnected.*

306. Correct answer A. Cephalad spread of the hyperbaric solution favored by head-down positioning

Hyperbaric solutions respond to gravity, so head-down positioning soon after injection encourages ascent of the block. Pregnant patients also need smaller intrathecal doses, so spread to the cervical region is more likely. The picture of arm numbness, speech difficulty, and hypotension reflects a rising block. Intravascular injection would cause immediate neurologic excitation, subdural block is typically delayed and patchy, and allergy would show skin and airway signs.

Key principle: *Head-down position after a hyperbaric spinal can send the block too high.*

307. Correct answer C. It is mainly from medullary hypoperfusion and drug effect, not diaphragm palsy

When a block reaches the brainstem the patient loses consciousness and stops breathing, and the apnea is attributed more to medullary hypoperfusion and drug effect than to paralysis of the diaphragm. The practical point is that the patient cannot be relied upon to keep breathing, so ventilation is assisted and the airway secured early. Local anesthetic does not destroy the phrenic nerve, and pulmonary edema is not the mechanism. Intercostal weakness contributes to difficulty breathing but does not explain loss of drive.

Key principle: *Apnea in total spinal is a brainstem problem, so ventilate and secure the airway.*

308. Correct answer D. Control the airway, ventilate, and support circulation until the effect fades

Retrobulbar and peribulbar blocks can reach the brainstem through the optic nerve sheath. The presentation is sudden loss of consciousness or apnea minutes after the block, and the management is airway, ventilation, circulation, and time. Lipid emulsion treats systemic toxicity, and antihistamines and steroids do not treat this picture. Imaging before resuscitation delays the support that the patient needs. Recovery is usually complete if hypoxia and prolonged hypotension are avoided.

Key principle: *Brainstem anesthesia after an eye block is supportive: airway, ventilation, circulation.*

309. Correct answer B. Deliver the fetus emergently to improve the effectiveness of compressions

The maternal circulation is the fetal circulation, and resuscitation of the mother is the first priority. If the mother does not respond within about four to five minutes, emergency delivery is performed to improve the effectiveness of compressions and the chance of survival of both. The neonatal team is called early whenever maternal collapse occurs. Waiting for the block to recede or for fetal documentation wastes time, and lipid emulsion does not treat a high spinal.

Key principle: *Maternal arrest without response by about five minutes calls for emergency delivery.*

310. Correct answer A. Beyond about 8 to 12 hours after the onset of the deficit

Recovery is often good if compression is relieved within a few hours of the onset of deficit, and the deficit is often permanent if relief is delayed beyond about 8 to 12 hours. The narrow window explains why imaging and neurosurgical consultation proceed together and why every step that causes delay is questioned. Longer intervals describe the time over which an abscess may evolve, not the window for a hematoma.

Key principle: *Hematoma decompression must happen within hours; beyond 8 to 12 hours deficits are often permanent.*

311. Correct answer B. Preserved motor function allows new weakness from a hematoma to be noticed

Infusions should use dilute solutions that preserve motor function so that new weakness is noticed, and patients with neuraxial catheters should have scheduled motor and sensory checks. A dense block can mask the early signs of a hematoma and delay the response. Dilution does not alter the bleeding risk, does not prevent migration, and does not change the need to time catheter removal to the anticoagulant. Any new deficit prompts the infusion to be stopped so an examination can be done.

Key principle: *Use dilute neuraxial infusions so that new weakness is not hidden.*

312. Correct answer C. Preserved motor function lets new weakness from a hematoma be noticed

Guidelines from the American Society of Regional Anesthesia and Pain Medicine specify, drug by drug, the interval after a dose before a neuraxial procedure, the interval after a procedure before the next dose, and when a catheter can be removed. Intervals depend on the agent, the dose, renal function, and the presence of a catheter, so the guideline is consulted and not recalled. Using another drug's interval or a single interval ignores these differences. Postoperative checks detect a hematoma but do not prevent it.

Key principle: *Look up the anticoagulant interval for the specific drug; do not rely on memory.*

313. Correct answer B. Consult the current society guideline for that agent and document the plan

Staphylococcus aureus from the patient's skin is the most common organism in epidural abscess. Empiric treatment covers methicillin-resistant staphylococci and gram-negative organisms and is adjusted when culture results are known. Surgical drainage is added for patients with neurologic deficit or a large collection. Oropharyngeal streptococci are associated with meningitis after neuraxial block, not with typical catheter abscess, and fungal or single-agent coverage is too narrow.

Key principle: *Epidural abscess is usually Staphylococcus aureus; start broad coverage and image.*

314. Correct answer C. Staphylococcus aureus, covering resistant staphylococci and gram-negative rods

Meningitis after neuraxial block is usually caused by streptococci from the operator's oropharynx and is prevented by wearing a mask. Alcohol-based chlorhexidine skin preparation allowed to dry, sterile gloves, and a sterile field further reduce infection. Needle gauge and injection speed do not affect contamination, and routine antibiotic prophylaxis is not part of the preventive plan described.

Key principle: *A mask worn by the operator prevents meningitis from oropharyngeal streptococci.*

315. Correct answer D. The operator wearing a face mask during the procedure

Anterior spinal artery syndrome produces painless motor loss with preserved vibration sense, because the dorsal columns are supplied separately. Abscess brings back pain and fever, and transient neurologic symptoms bring pain without a deficit. Cauda equina syndrome has followed continuous spinal anesthesia with concentrated lidocaine, not an epidural infusion. Even so, any new or unexplained deficit after a neuraxial block is treated as hematoma or abscess until imaging says otherwise.

Key principle: *Painless motor loss with preserved vibration sense suggests anterior spinal artery syndrome.*

316. Correct answer D. Anterior spinal artery syndrome with cord ischemia

Transient neurologic symptoms follow some spinal anesthetics with back and buttock pain but no neurologic deficit. The absence of fever, weakness, sensory loss, and sphincter involvement argues against abscess, hematoma, and arterial infarction. These patients still need a careful examination, and any progressive or unexplained deficit prompts imaging. The condition is not an infection or compressive lesion.

Key principle: *Back and buttock pain without deficit after spinal anesthesia is transient neurologic symptoms.*

317. Correct answer A. Transient neurologic symptoms after spinal anesthesia

Most postoperative neurologic deficits arise from surgery, tourniquet, or positioning, so the first question is whether something treatable exists, such as a tight cast or dressing, compartment syndrome, or a surgical cause. After a neuraxial block a hematoma or abscess would be added to that list. A careful examination, documentation, and early neurologic referral follow. Electromyography is most informative several weeks after injury. Most deficits improve over weeks to months, and patients should be told this honestly while the evaluation continues.

Key principle: *Exclude treatable causes first; electromyography is informative weeks later.*

318. Correct answer E. Seek treatable causes such as a tight dressing, document, and refer early

A small pneumothorax can enlarge under positive pressure and become a tension pneumothorax, and nitrous oxide expands closed air spaces. Tension pneumothorax is treated with immediate needle decompression followed by a chest drain, and treatment does not wait for imaging. Increasing pressure would worsen the tension, and bronchodilators do not treat it. Nitrous oxide should be avoided in a patient who has had a block near the pleura.

Key principle: *Tension pneumothorax under positive pressure needs immediate decompression.*

319. Correct answer A. Immediate needle decompression, then a chest drain, without waiting for imaging

A dense block can mask compartment syndrome, and severe escalating pain unresponsive to analgesia is the usual warning. Breakthrough pain, tense swelling, or pain on passive stretch must prompt urgent assessment and not an additional bolus. In high-risk patients the anesthesiologist and surgeon should agree on a plan with dilute solutions that leave some sensation and examine the limb often. Compression would worsen the pressure, and a denser block would further hide the process.

Key principle: *Breakthrough pain under a block is compartment syndrome until proven otherwise.*

320. Correct answer B. Arrange urgent assessment for compartment syndrome, not another bolus

Horner syndrome, with ptosis, miosis, and facial flushing, results from spread to the sympathetic chain and is a recognized effect of blocks around the neck. In a stable patient the response is to recognize it, explain it, and monitor until it resolves. Intravascular injection, total spinal, and pneumothorax would present with seizures, loss of consciousness, or respiratory and hemodynamic compromise, none of which are present here.

Key principle: *Horner syndrome after a neck block is benign spread to the sympathetic chain.*

Group 17. Perioperative Hypersensitivity: Mechanisms

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321. Correct answer C. The term is retired; classify reactions by mechanism, and treatment is the same

The term anaphylactoid was used for reactions that resembled anaphylaxis but did not involve IgE, and it has been retired because the clinical picture and the treatment are the same. Allergic reactions are those mediated by the immune system through IgE or other pathways, and nonallergic reactions arise when a drug activates mast cells directly. All pathways converge on the same cells and mediators, so the same urgent treatment applies. The mechanism matters later, when deciding which drugs are safe in the future.

Key principle: *Retire the word anaphylactoid; classify by mechanism and treat the same way.*

322. Correct answer E. Antigen-specific IgE must first be made and bound to mast cells and basophils

On first exposure the immune system makes antigen-specific IgE, which binds the high-affinity receptor on mast cells and basophils, leaving the patient sensitized but without symptoms. On re-exposure the antigen links two or more IgE molecules on the cell surface, and the cross-linking triggers degranulation within seconds to minutes. Mast cells are already mature in the tissues, and histamine is preformed. Complement is not required for the IgE pathway.

Key principle: *IgE-mediated reactions need prior sensitization, so they follow a second or later exposure.*

323. Correct answer C. Dose-dependent activation of MRGPRX2 on mast cells without prior sensitization

MRGPRX2 on connective tissue mast cells binds a range of cationic molecules, including neuromuscular blockers with a tetrahydroisoquinoline structure such as atracurium and cisatracurium. Activation produces degranulation without prior sensitization, depends on dose and rate, and may explain reactions with negative skin tests. An IgE mechanism would require earlier sensitization, and IgG immune complexes are described mainly in animal models. Bradykinin accumulation produces angioedema without this rapid mast cell picture.

Key principle: *MRGPRX2 activation can cause first-exposure reactions with negative skin tests.*

324. Correct answer B. Complement activation producing C3a and C5a, which trigger degranulation

Some drugs activate complement directly, producing the small peptides C3a and C5a, which bind receptors on mast cells and drive degranulation. Contrast media, liposomal products, and some plasma expanders have been linked to this complement activation-related pseudoallergy. A first-exposure reaction with no specific IgE argues against sensitization. Opioid histamine release and bradykinin accumulation do not fit a liposomal product at first infusion.

Key principle: *Complement activation can cause a first-exposure reaction without IgE.*

325. Correct answer A. Stem cell factor acting on its KIT receptor on the cell surface

Mast cells mature in the tissues under the influence of stem cell factor acting on the KIT receptor. Immunoglobulin E binding sensitizes existing cells but does not drive their maturation, and C5a and histamine are activating or effector signals. Clonal mast cell disease is usually driven by a mutation in the KIT gene, which is why it increases the number of mast cells.

Key principle: *Mast cells mature through stem cell factor and KIT.*

326. Correct answer D. Repeat in one to two hours and add a later baseline; normal does not exclude it

Tryptase rises within about half an hour of onset, peaks in the first one to two hours, and then falls with a half-life of a couple of hours. Samples are taken as soon as treatment allows, again an hour or two later, and a baseline is obtained after recovery, since the rise above baseline is what supports mast cell activation. A normal tryptase does not exclude anaphylaxis, especially in reactions driven by basophils. A single late sample would miss the peak.

Key principle: *Sample tryptase early, repeat in one to two hours, and obtain a baseline later.*

327. Correct answer E. Platelet-activating factor

Platelet-activating factor is formed from membrane lipids after activation and is a potent cause of hypotension and bronchospasm, thought to be a major contributor to severe reactions. Histamine, tryptase, chymase, carboxypeptidase,

heparin, and tumor necrosis factor are preformed and stored in granules, ready for release within seconds. Leukotrienes and prostaglandin D2 are also made after activation but are formed from arachidonic acid.

Key principle: *Platelet-activating factor is newly made and drives severe shock and bronchospasm.*

328. Correct answer A. Mast cell mediators in the heart causing coronary vasospasm

The heart contains its own mast cells, and their mediators can cause coronary vasospasm, arrhythmia, and depressed contractility, a pattern sometimes called Kounis syndrome. This explains ischemic changes in a patient with normal coronary anatomy. Bronchospasm may add to hypoxemia but does not cause ST elevation by itself, and potassium release and drug-induced plaque erosion are not described mechanisms.

Key principle: *Allergic coronary vasospasm (Kounis syndrome) can cause ischemia in anaphylaxis.*

329. Correct answer C. Hypotension, tachycardia, and a rise in airway pressure

Anesthesia alters the presentation: drugs reach the circulation at once, drapes hide the skin, positive-pressure ventilation conceals respiratory distress, and anesthetic agents blunt compensation. The dominant early signs are hypotension, tachycardia, and increased airway pressure, while rash and swelling come late or not at all. An unconscious patient cannot report itching or doom. Fever is not part of the immediate picture.

Key principle: *Under anesthesia, expect circulatory and airway pressure changes before skin signs.*

330. Correct answer D. Clonal mast cell disease, often from a KIT gene mutation

Clonal mast cell disease increases the number of mast cells, is usually driven by a mutation in the KIT gene, and is associated with severe and recurrent anaphylaxis. A raised baseline tryptase should prompt this consideration, and these patients may need premedication on specialist advice, avoidance of mast cell triggers, and a rescue plan. IgA deficiency is a risk for transfusion reactions, G6PD deficiency matters for methylene blue, and pholcodine sensitization relates to neuromuscular blockers.

Key principle: *A raised baseline tryptase should prompt consideration of mastocytosis.*

331. Correct answer B. It acts on alpha and beta receptors and limits mediator release

Alpha stimulation constricts vessels and reduces capillary leak, beta-1 stimulation increases the force and rate of the heart, and beta-2 stimulation relaxes bronchial smooth muscle and reduces further mediator release from mast cells. No other single drug does all of these things, so it remains first-line and delay is the most important modifiable factor in outcome. Antihistamines and steroids act more slowly and on a narrower range of effects. Epinephrine does not block H1 receptors, remove released mediators, or prevent IgE binding.

Key principle: *Epinephrine opposes the whole cellular and vascular picture; adjuncts do not.*

332. Correct answer B. Ongoing release of lipid mediators and cytokines can cause relapse hours later

Some patients improve and then relapse hours later in a biphasic pattern, and a few have a protracted course lasting many hours despite treatment. These patterns reflect continuing release of lipid mediators and cytokines and persistence of the trigger in the body. A normal blood pressure at first recovery therefore does not mean the reaction is finished, and observation in a monitored setting is part of management. The other options are not described mechanisms.

Key principle: *Do not treat a reaction as finished when blood pressure first normalizes.*

333. Correct answer E. Intradermal tests at high concentration can be falsely positive from histamine release

Skin prick and intradermal tests, specific IgE assays for a few agents, and the basophil activation test all contribute, and each has limits. Intradermal tests with neuromuscular blockers at high concentration can give false positives from nonspecific histamine release, and tests may be negative in non-IgE reactions. Interpretation by an allergist who knows the concentrations, timing, and history separates true allergy from pseudoallergic response. Specific IgE assays exist only for a few agents such as latex and chlorhexidine.

Key principle: *Testing must be interpreted by an allergist; high-concentration tests can mislead.*

334. Correct answer B. Sugammadex can cause anaphylaxis within minutes, even after prior exposure

Sugammadex, a cyclodextrin, has been associated with anaphylaxis, usually within minutes of injection and sometimes in patients who have received it before. A patient who deteriorates after reversal may be reacting to sugammadex and not to residual relaxant or surgical stimulus. Neostigmine and glycopyrrolate have also been implicated, so reversal agents and adjuncts are both on the list of suspects. The incidence is low, and the drug remains valuable.

Key principle: *Consider sugammadex as the trigger when a patient deteriorates after reversal.*

335. Correct answer A. Previous long-term exposure to NPH insulin

Protamine reactions are more frequent in patients exposed to neutral protamine Hagedorn insulin, in men who have had a vasectomy, and perhaps in those with fish allergy. Egg allergy, shellfish allergy, and penicillin labels are not recognized risk factors for protamine reactions. A reaction after heparin reversal should therefore prompt consideration of protamine.

Key principle: *Prior NPH insulin or vasectomy raises the risk of protamine reactions.*

336. Correct answer E. Testing to find safe alternatives, since cross-reactivity is common but not universal

Cross-reactivity among neuromuscular blockers is common because shared ammonium groups are the antigens, roughly 40 percent in the UK audit as a minimum estimate, but it does not follow a simple pattern by chemical class. Formal testing finds safe alternatives, whereas blanket avoidance deprives the patient of drugs likely to be safe. Premedication does not prevent anaphylaxis, and repeating the culprit is unsafe.

Key principle: *After a relaxant reaction, test formally; do not avoid all agents or assume safety by class.*

337. Correct answer C. Clarify the history, recognize histamine release, and use a synthetic opioid

Morphine and codeine are histamine releasers and cause flushing, itching, and a fall in blood pressure that do not involve IgE. True IgE-mediated opioid allergy is rare and tends to involve the natural phenanthrene opioids, while synthetic opioids share little structure. A detailed account of what happened, at what dose and speed, is more useful than the label. Codeine is also a histamine releaser and a natural opioid, and premedication does not make a rapid bolus safe.

Key principle: *Most opioid reactions are pseudoallergic; clarify the history and consider a synthetic opioid.*

338. Correct answer D. No proof they prevent severe reactions, and they may mask early signs

Corticosteroids and antihistamines have been used before known exposures, but there is no proof that they prevent severe immediate reactions, and they may blunt early signs and delay recognition. They have a role in preventing reactions to some radiographic contrast agents and in managing mastocytosis on specialist advice. For most perioperative risk, the strategy is avoidance of the suspected trigger, readiness for rescue, and attentive observation at exposure. Neither drug replaces epinephrine.

Key principle: *Premedication does not prevent perioperative anaphylaxis.*

339. Correct answer D. Shellfish allergy does not predict reaction to contrast or povidone-iodine

Shellfish allergy does not predict reactions to iodinated contrast or povidone-iodine antiseptic beyond the general increase in risk in atopic patients. Treating it as a contraindication removes safe options without benefit. Similar myths concern egg and soy with propofol and sulfonamide antibiotics with nonantibiotic sulfonamide drugs. Premedication is not required on this basis.

Key principle: *Shellfish allergy is not a reason to withhold contrast or povidone-iodine.*

340. Correct answer E. Arrange supervised testing or an oral amoxicillin challenge to delabel her

More than nine in ten patients with a penicillin label are not allergic when tested properly, and the label leads to broader or less effective prophylaxis with higher infection rates. Patients with a low-risk history can often be tested or challenged with oral amoxicillin under supervision, and most are cleared. Untested exposure at induction is unsafe and confuses any reaction, and blanket avoidance of all beta-lactams ignores low cross-reactivity. Even without delabeling, a detailed record allows a safe alternative such as cefazolin.

Key principle: *Delabel low-risk penicillin allergy with supervised testing or challenge.*

Group 18. Recognizing and Treating Anaphylaxis Under Anesthesia

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341. Correct answer A. Bronchospasm, possibly early anaphylaxis, so search for the trigger

A rising peak airway pressure, a slow expiratory phase, and a capnogram that slopes upward like a shark fin point toward the small airways. Bronchospasm can be the first or only sign of anaphylaxis, although isolated bronchospasm with a stable pressure is more often a primary airway problem, so the trigger is sought and treatment readied. The other conditions would give an absent or abnormal baseline waveform, or a different pattern of compliance change.

Key principle: *A shark-fin capnogram with high airway pressure means small-airway bronchospasm.*

342. Correct answer E. Falling output with airway narrowing, as in anaphylaxis, so treat promptly

A falling end-tidal carbon dioxide with a rising airway pressure is an unusual combination that should make the clinician stop and think. It reflects falling cardiac output together with airway narrowing, which suits anaphylaxis, and the plethysmographic waveform often shrinks first as peripheral vessels dilate. Deepening the anesthetic would add vasodilation, and a circuit disconnection would lower pressure, not raise it. Reading the monitors together shows the pattern sooner than any single number.

Key principle: *Falling end-tidal carbon dioxide with rising airway pressure suggests anaphylaxis.*

343. Correct answer A. Late bradycardia signals imminent arrest, so escalate and prepare compressions

Some patients become bradycardic late in a severe reaction, a paradoxical response that signals imminent arrest and must never be mistaken for reassurance. The low pressure and poor waveform confirm that the circulation is failing. The response is to escalate epinephrine and fluid, call for help, and prepare to begin compressions. Reducing epinephrine or relying on atropine alone ignores the underlying shock.

Key principle: *Bradycardia with hypotension late in anaphylaxis signals arrest, not recovery.*

344. Correct answer B. Skin signs are unreliable, and some severe reactions never produce a rash

Skin signs are the sign everyone expects and the least dependable. The chest and arms may be under drapes, the flush may not appear until the circulation recovers, and a minority of severe reactions never produce a rash. Finding urticaria confirms the diagnosis, but finding clear skin does not exclude it. Tryptase is not available at the bedside and does not guide immediate treatment, and clear skin does not separate allergic from nonallergic mechanisms.

Key principle: *Clear skin does not exclude anaphylaxis.*

345. Correct answer D. 0.5 mg intramuscularly into the anterolateral thigh

Without intravenous access, epinephrine is given intramuscularly into the anterolateral thigh, where absorption is faster than from the arm or the subcutaneous tissue. The adult dose is 0.5 mg of the 1 mg/mL preparation, repeated every 5 minutes if needed. Fifty micrograms is the intravenous titrated bolus dose and is too small intramuscularly. The 1 mg/mL concentrate must never be given intravenously by mistake.

Key principle: *No intravenous access: epinephrine 0.5 mg intramuscularly into the anterolateral thigh.*

346. Correct answer D. Use the 100 mcg/mL syringe, giving 50 mcg intravenously every 1 to 2 minutes

Titrated intravenous boluses of 50 mcg of the 100 mcg/mL preparation, repeated every 1 to 2 minutes until the pressure responds, are safer than a single large dose because the clinician sees the effect between doses. The 1 mg/mL concentrate is for intramuscular use only and must never be given intravenously by mistake. A 10-minute interval is too slow in shock, and 1 mg is the arrest dose. A prefilled, labeled syringe in the anaphylaxis kit prevents dilution errors.

Key principle: *Intravenous titration uses 50 mcg of the 100 mcg/mL preparation every 1 to 2 minutes.*

347. Correct answer C. 20 mcg

The weight-based intravenous bolus for children is 1 mcg/kg, repeated as needed, which gives 20 mcg for a 20-kg child. Two micrograms is a tenfold underdose, and 200 micrograms is a tenfold overdose. A dose of 500 micrograms is the intramuscular maximum of 0.5 mg and is not an intravenous bolus. A diluted syringe prepared in advance limits calculation and dilution errors.

Key principle: *Pediatric intravenous epinephrine bolus is 1 mcg/kg.*

348. Correct answer C. Start an epinephrine infusion titrated to blood pressure

Three failed boluses is a reasonable signal to begin an epinephrine infusion, in the range of 0.05 to 0.1 mcg/kg/min, titrated to blood pressure with an arterial line, and to call a second anesthesiologist. Antihistamines and steroids act too slowly to reverse shock. Phenylephrine offers no inotropic support, and stopping epinephrine removes the beta effects the patient needs. Repeated boluses alone do not provide a sustained effect.

Key principle: *After three failed boluses, start an epinephrine infusion.*

349. Correct answer B. Wait until stable and give it slowly, since it does not treat shock

H1 blockers relieve itching and urticaria but do not treat shock or bronchospasm, and rapid intravenous injection of some of them can lower blood pressure further. Chlorpheniramine, if used, is given slowly after stabilization, and a non-sedating oral antihistamine is preferred later. Injectable promethazine should be avoided. Drawing up adjuncts must not delay a further dose of epinephrine.

Key principle: *Antihistamines are slow adjuncts after stabilization, never a substitute for epinephrine.*

350. Correct answer C. Lengthen the expiratory time, for example by lowering the rate

In bronchospasm a long expiratory time is allowed to avoid breath stacking, which can itself cause hypotension by raising intrathoracic pressure and reducing venous return. Raising the rate or the tidal volume shortens expiration or adds air that cannot leave. High positive end-expiratory pressure adds to air trapping. Epinephrine remains the first treatment, followed by inhaled albuterol, a volatile agent, and ketamine or magnesium for resistant cases.

Key principle: *In bronchospasm, prolong expiration to prevent breath stacking.*

351. Correct answer B. Magnesium sulfate 2 g intravenously over 20 minutes

For bronchospasm, epinephrine comes first, then inhaled albuterol, with an inhaled volatile agent considered for its bronchodilating effect and ketamine or magnesium for resistant cases. Magnesium is given as 2 g intravenously over 20 minutes in an adult. Corticosteroids act too slowly to matter in the first hours, and antihistamines do not treat bronchospasm. A beta-blocker would oppose epinephrine and worsen bronchoconstriction.

Key principle: *Resistant bronchospasm in anaphylaxis: add magnesium or ketamine after epinephrine and albuterol.*

352. Correct answer D. Current treatment with a serotonergic antidepressant

Methylene blue is reserved for truly refractory patients. It can precipitate serotonin syndrome in patients taking serotonergic drugs, causes hemolysis in glucose-6-phosphate dehydrogenase deficiency, and transiently distorts pulse oximeter readings. The suggested dose range is 1 to 2 mg/kg intravenously given slowly, and use is off-label. Penicillin labels, statins, and mild asthma are not relevant cautions.

Key principle: *Check for serotonergic drugs and G6PD deficiency before methylene blue.*

353. Correct answer E. Norepinephrine, since phenylephrine lacks inotropy and may cause bradycardia

Norepinephrine provides strong alpha constriction with some beta support and is the usual companion to epinephrine. Pure alpha agonists such as phenylephrine can restore pressure but offer no inotropic support and may provoke reflex bradycardia in a patient who is already struggling. Vasopressin acts through V1 receptors that remain responsive when adrenergic receptors are downregulated, so it is a useful addition. Phenylephrine does not trigger mast cells.

Key principle: *Choose norepinephrine or vasopressin over phenylephrine as the second vasopressor.*

354. Correct answer E. Give titrated epinephrine now, since it is low risk if the diagnosis differs

When hypotension and bronchospasm appear together shortly after a drug, anaphylaxis is treated first, because epinephrine is unlikely to harm a patient who turns out to have something else. The working assumption in any unexplained collapse is that a drug, disinfectant, dye, or equipment item is the cause until another explanation earns its place. Tryptase is a later confirmatory test, steroids and antihistamines cannot reverse shock, and imaging delays the one effective treatment.

Key principle: *Treat first when hypotension and bronchospasm follow an exposure.*

355. Correct answer D. Notify the extracorporeal team early while resuscitation continues

Extracorporeal membrane oxygenation has rescued patients with refractory shock or arrest from anaphylaxis, and any center that offers it should be contacted early. The window for a good result is short, and early notification costs little because the team can prepare while resuscitation continues. Anaphylactic arrests are often reversible, and prolonged efforts have succeeded. A patient with this degree of shock cannot go to a ward.

Key principle: *Call the extracorporeal team early in refractory shock.*

356. Correct answer C. Check ECG and troponin, repeat echocardiography, and monitor for pulmonary edema

Transient myocardial dysfunction follows a significant fraction of severe reactions, so an electrocardiogram and troponin help, and a repeat echocardiogram within a day shows whether the ventricle has recovered. Non-cardiogenic pulmonary edema can develop in the hours after resuscitation, so lung-protective ventilation and monitoring continue. Infusions are weaned one at a time, watching for rebound vasodilation as epinephrine clears. Large fluid volumes are given because of capillary leak, so prophylactic diuresis is not indicated.

Key principle: *After severe anaphylaxis, check for myocardial dysfunction and watch for pulmonary edema.*

357. Correct answer A. Grade II, a multisystem reaction that is not life threatening

Grade I is limited to skin and mucous membranes. Grade II involves measurable multisystem effects that are not life threatening, such as moderate hypotension or tachycardia. Grade III is life threatening with shock, severe bronchospasm, or serious arrhythmia, and grade IV is arrest. Grade II and above call for epinephrine, an organized resuscitation, and a decision about the operation. The scale describes the worst point of the event.

Key principle: *Grade II means measurable multisystem effects that are not life threatening.*

358. Correct answer D. Need for repeated epinephrine doses or an infusion in the episode

A biphasic reaction is a second episode hours after the first, without further exposure. It cannot be predicted reliably from the severity of the first event, though patients who needed repeated epinephrine doses or an infusion are at higher risk. Any patient with a perioperative reaction should be observed in a monitored environment for several hours at least, and those with severe or refractory reactions for a day or more. A patient on an epinephrine infusion should not go to a general ward.

Key principle: *Higher-need first episodes carry higher biphasic risk, so observe them longer.*

359. Correct answer C. Say a reaction occurred and will be investigated, naming no culprit

Communication begins as soon as the patient can understand: a reaction occurred, it has been treated, tests will follow, and the team will arrange them. Early guesses about the culprit are often wrong, and a patient wrongly labeled allergic to a drug can lose access to a useful agent for life. A written note with the date and drugs given helps the next clinician. Withholding information delays referral, and a blanket label would deny safe anesthetics. Samples and syringes are preserved, and referral is arranged before discharge.

Key principle: *Tell the patient what happened, but do not name a single culprit before testing.*

360. Correct answer B. Treat as a nonpregnant adult, adding left uterine displacement, without delay

In a pregnant patient the mother comes first, because her survival is the best treatment for the fetus. The uterus is displaced to the left, oxygen and epinephrine are given as in a nonpregnant adult, and fetal monitoring should not

delay maternal treatment. If the patient arrests, standard resuscitation applies, and the obstetric team prepares for early delivery. Withholding epinephrine or delaying treatment for fetal reasons risks both lives.

Key principle: *In pregnancy, treat the mother first, displace the uterus, and do not delay epinephrine.*

Group 19. After the Event: Diagnosis and Future Anesthetics

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361. Correct answer A. Tryptase stays raised for hours, whereas histamine returns to baseline within about 30 minutes

Tryptase is a protease stored in mast cell granules that is stable in blood for hours, so a sample drawn after stabilization still reflects the reaction. Histamine peaks within minutes and is gone in half an hour or less, which makes it practical only when an immediate sample is possible. Tryptase does not identify the trigger, and it rises within about thirty minutes and peaks at one to two hours, not instantly. Histamine release is not limited to IgE-mediated events.

Key principle: *Tryptase is the practical marker because it persists for hours, while histamine disappears within minutes.*

362. Correct answer E. Labeling every tube with draw time and onset time and telling the laboratory it is a timed series

The value of a tryptase series depends on the clock times of the reaction onset and of each draw, and the laboratory result is meaningless without them. Staff who see several tubes from one patient may assume duplication and discard some, so a note explaining that it is a timed series is needed. Pooling samples destroys the kinetic information, and delaying the tubes defeats the purpose of capturing the peak. Late reactions after emergence are sampled on the same principle as intraoperative ones.

Key principle: *Label every tube with draw time and onset time, and tell the laboratory it is a timed series.*

363. Correct answer A. A normal tryptase does not exclude it, as some reactions occur without a measurable rise

Reactions driven mainly by basophils, milder reactions, and some non-IgE mechanisms can occur without a measurable tryptase rise, and a poorly timed sample can miss the peak. The clinical picture and response to epinephrine carry weight, and the result must be read beside the history and the timeline. A normal value neither proves an IgE mechanism nor names a culprit. Histamine drawn a day later would be uninformative because it clears within minutes.

Key principle: *A normal tryptase never excludes anaphylaxis.*

364. Correct answer A. Explain that tryptase shows mast cell activation but cannot name the trigger

A raised tryptase tells the clinician only that mast cell mediators were released. It does not identify the trigger and does not distinguish IgE-mediated reactions from direct mast cell activation. The allergy field should state the suspected substances and the certainty of suspicion, and the allergy clinic updates it after testing. Naming a culprit before testing risks a false label that follows the patient for life, while leaving the field blank loses useful information.

Key principle: *Tryptase confirms mast cell activation, not the identity of the trigger.*

365. Correct answer C. Direct mast cell activation through MRGPRX2 without prior sensitization

Several neuromuscular blockers, some antibiotics, and some opioids activate mast cells through the MRGPRX2 receptor without any prior sensitization or IgE antibody. This explains reactions on first exposure and reactions with raised tryptase but unremarkable skin tests. The treatment is the same, but the implications for future anesthetics differ, since the patient may tolerate other drugs of the class at standard doses. Complement-mediated and T cell mechanisms would not account for a tryptase rise.

Key principle: *MRGPRX2 activation explains first-exposure reactions with raised tryptase and negative IgE-based tests.*

366. Correct answer B. Hereditary alpha tryptasemia caused by extra copies of the alpha tryptase gene

A smaller elevation of baseline tryptase, with a family pattern, points to hereditary alpha tryptasemia, a common inherited condition due to extra copies of the alpha tryptase gene and associated with more severe reactions. Values persistently above about twenty raise the possibility of systemic mastocytosis, and no single raised value confirms it. The baseline was drawn well after recovery, so residual activation is unlikely, and her kidney function is normal. These patients merit allergy or hematologic evaluation.

Key principle: *A modest persistent baseline elevation with a family history suggests hereditary alpha tryptasemia.*

367. Correct answer E. Draw the first sample now; a sample taken during resuscitation is still valuable

The first sample should not be delayed for the sake of tidiness, and a sample drawn during a resuscitation is better than a perfect sample taken an hour later. Even a sample drawn during cardiac arrest is valuable, because the result is read against the patient's own baseline drawn later. The site should be away from an infusion that may have contained the suspected drug. The clock time of the draw and of the first sign should be recorded.

Key principle: *Do not delay the first tryptase; draw it during resuscitation and record the clock time.*

368. Correct answer C. Personally refer her to a drug allergy clinic, sending the record and tryptase results

Every patient with a suspected perioperative allergic reaction should be referred to a clinic with drug allergy experience, and audits show many are never referred or are anesthetized again with the probable culprit. The anesthesiologist who managed the event should make the referral personally and send the record and tryptase results, since the surgical team may not recognize the need. Waiting leaves the label unresolved, and blanket avoidance of anesthesia is neither necessary nor safe.

Key principle: *The anesthesiologist who managed the reaction should personally make the allergy referral.*

369. Correct answer B. Substances missing from the record may go untested, leaving the culprit unidentified

Test panels should cover every substance the patient was exposed to in the hour or so before the reaction, including disinfectants, dyes, and latex. The allergy service relies on the anesthetic record to decide what to test, and a drug that is missing from the record cannot be tested. Documenting doses, times, and products used for preparation, catheters, and gloves during the event is worth more than a reconstruction afterward. Tryptase does not identify the trigger.

Key principle: *An incomplete anesthetic record produces an incomplete test panel.*

370. Correct answer A. Opioids and some relaxants cause wheals through direct histamine release above validated doses

Skin testing uses a prick test first, then intradermal testing at concentrations validated to avoid nonspecific irritation, with positive and negative controls read after fifteen to twenty minutes. Concentrations above the validated level produce false positives, especially with opioids, atracurium, and mivacurium, which cause wheals by direct histamine release. The wheal here does not prove IgE-mediated allergy. The sequence and reading time described in the distractors are incorrect.

Key principle: *Use validated concentrations; irritant doses of opioids and some relaxants give false-positive wheals.*

371. Correct answer B. A systemic reaction to the test would be harder to treat on a beta-blocker

Skin tests carry a small risk of a systemic reaction, particularly with the drug that probably caused the event. Testing is therefore deferred when a beta-blocker cannot be stopped, and in pregnancy, because treating a test reaction would be harder. It is antihistamines, not beta-blockers, that suppress the wheal and must be stopped before testing. Beta-blockers do not alter mast cell stores or create false positives.

Key principle: *Defer skin testing when a systemic reaction would be harder to treat, as with a beta-blocker.*

372. Correct answer D. The negative result does not clear it, as sensitivity is incomplete for several drugs

Specific IgE assays are useful for latex, chlorhexidine, penicillins, and a few other agents, and especially when skin testing is unsafe. Their sensitivity is incomplete for several drugs, including succinylcholine and morphine, so a negative result does not clear the drug. Cross-reactivity among neuromuscular blockers is common, so a negative assay for one agent cannot clear the class. Repeating the assay daily has no rationale.

Key principle: *A negative specific IgE does not clear succinylcholine or morphine.*

373. Correct answer E. A basophil activation test performed at a specialized center

The basophil activation test measures the response of the patient's own basophils to a drug in the laboratory and is helpful when skin tests are equivocal, particularly for neuromuscular blockers and sugammadex, although only specialized centers offer it. Provocation testing is not used to confirm the drug that probably caused a severe reaction. Higher intradermal concentrations produce false positives. Plasma histamine and complement fragments have a minor role and do not resolve this question.

Key principle: *Use basophil activation testing when skin tests for relaxants or sugammadex are equivocal.*

374. Correct answer A. Use it for patient A to confirm tolerance, not to confirm the culprit in patient B

Drug provocation involves giving graded doses under observation to confirm tolerance. It is routine for local anesthetics and for many antibiotics when skin tests are negative and the drug is needed. It is not used to confirm the drug that probably caused a severe reaction. For that situation the allergy service relies on skin tests and, where necessary, basophil activation.

Key principle: *Provocation confirms tolerance of needed drugs; it is not used to prove a probable culprit.*

375. Correct answer D. Ask the allergy service for a letter classifying each agent and listing alternatives

A letter that says only that the patient is allergic to anesthesia is nearly useless and can do harm by prompting the next team to avoid a technique the patient could safely have. The anesthesiologist should expect a specific letter that names each substance tested, classifies it as confirmed, probable, possible, or excluded, and lists the alternatives found negative, and should return to the allergy service if it is not. Premedication does not prevent anaphylaxis, and disregarding the note is unsafe.

Key principle: *Request a specific allergy letter that classifies every agent tested.*

376. Correct answer D. Propofol may be used, since egg and soy allergy are not contraindications to it

Some labels do not need to restrict care at all, and egg and soy allergy are not contraindications to propofol. Recognizing this during preoperative evaluation prevents unnecessary changes to the plan and keeps a useful drug available. Premedication with a steroid and an antihistamine does not prevent anaphylaxis in any case. Neither testing nor a challenge is required before use.

Key principle: *Egg and soy allergy do not contraindicate propofol.*

377. Correct answer B. Shellfish allergy does not predict reactions to contrast or povidone-iodine

Iodine is not the allergen in shellfish allergy, so the label does not predict reactions to iodinated contrast or to povidone-iodine. Making this distinction during preoperative evaluation prevents unnecessary changes and keeps useful agents available. Premedication would not be required and would not prevent a true reaction. Both distractors that split the two products rest on the same false premise about iodine.

Key principle: *Shellfish allergy is not a reason to avoid contrast or povidone-iodine.*

378. Correct answer C. Infuse it slowly, over at least an hour, to avoid histamine release

Vancomycin is often used when a beta-lactam must be avoided, and it must be given slowly to avoid histamine release. Whenever possible the allergy service should be asked which beta-lactams are safe, since many patients tolerate drugs with dissimilar side chains and are better served by an effective drug. Steroid and antihistamine premedication does not prevent anaphylaxis and can mask early signs. Rapid injection invites the very reaction the precaution is meant to avoid.

Key principle: *Give vancomycin slowly to avoid histamine release.*

379. Correct answer E. Avoid relaxants, using a supraglottic airway with propofol and a short-acting opioid

The simplest safeguard for a relaxant allergy is a technique that limits exposure, such as regional anesthesia, a supraglottic airway with spontaneous ventilation, or intubation with propofol and a short-acting opioid. Cross-

reactivity within the class is common because of shared ammonium groups, which is why alternatives are tested and not assumed. Sugammadex has itself caused reactions, and premedication does not prevent anaphylaxis.

Key principle: *Limit exposure by avoiding relaxants when possible; never assume an untested relaxant is safe.*

380. Correct answer C. Give an honest account, notify risk management, preserve drugs and tubing, and sample tryptase

When an event ends in death, the family is owed an honest, compassionate account from a senior clinician, with a plain statement of what is known and a promise to share what is learned. Risk management should be notified early, the institution's rules on reporting to the medical examiner or coroner followed, and drugs, syringes, tubing, and the record preserved. A post-mortem tryptase sample from a peripheral vein can help the pathologist. Naming a culprit before testing risks a false conclusion, and silence is read as concealment.

Key principle: *After a fatal reaction, disclose honestly, notify risk management, preserve evidence, and sample tryptase.*

Group 20. Hemorrhagic Shock Physiology

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381. Correct answer A. A shock index, heart rate over systolic pressure, above 0.9 and rising

Shock index is a bedside tool that performs better than either heart rate or systolic pressure alone. A value above 0.9, and especially one at or above 1, predicts the need for transfusion and higher mortality in trauma and obstetric hemorrhage, and here it is about 1.1. Pregnancy conceals early loss, and an unchanged early hemoglobin gives no comfort because whole blood is being lost. A value near 0.6 is normal and would not be a reason for concern.

Key principle: *A shock index above 0.9 predicts transfusion even when blood pressure looks acceptable.*

382. Correct answer E. Pressure holds until a quarter to a third of volume is lost, so tachycardia and poor perfusion come first

A healthy child maintains a normal blood pressure until a quarter to a third of the blood volume has been lost, relying on a rapid heart rate and vasoconstriction. Tachycardia, poor capillary refill, cool extremities, and altered behavior are therefore the early signs, and hypotension is a very late one. Waiting for hypotension means waiting for decompensation. This child should be treated as having significant hemorrhage until proven otherwise.

Key principle: *In children, hypotension is a very late sign of hemorrhage.*

383. Correct answer B. The pressure is low relative to his baseline, so look for bleeding and consider volume

Baseline blood pressure matters, and a pressure of 100 in a chronic hypertensive patient is already low. Older patients have stiff ventricles and blunted catecholamine responses and may not mount a tachycardia, so a heart rate of 78 does not reassure. Organ dysfunction from the preceding hypertension and atherosclerosis follows quickly when perfusion falls. Hemoglobin is also unreliable early in hemorrhage and should not set the timing of action.

Key principle: *Judge pressure against the patient's baseline; 100 mm Hg is low in a chronic hypertensive patient.*

384. Correct answer E. Oxygen extraction rose from about 25 toward 70 percent until the reserve ran out

At rest the tissues extract only about a quarter of the oxygen delivered, so delivery can fall considerably before consumption is affected. As delivery drops, extraction rises toward sixty or seventy percent and mixed venous saturation falls. Below a critical threshold consumption becomes delivery dependent, anaerobic metabolism begins, and lactate rises, so a patient can look well and then deteriorate within minutes. Early hemoglobin does not fall in acute hemorrhage, and compensation protects the brain and heart at the expense of other organs.

Key principle: *Rising oxygen extraction hides hemorrhage until the reserve is exhausted, and then collapse is abrupt.*

385. Correct answer D. Protein C activation and glycocalyx loss cause anticoagulation and hyperfibrinolysis

Roughly a quarter to a third of severely injured patients arrive with abnormal clotting before any fluid has been given. Tissue injury combined with shock activates protein C and degrades the endothelial glycocalyx, producing anticoagulation and hyperfibrinolysis, called acute traumatic coagulopathy. Dilution, citrate, and hypothermia may

add to the problem later but cannot explain a defect that exists on arrival. A clinician who thinks only of dilution will wait for a problem that has already begun.

Key principle: *Acute traumatic coagulopathy exists before dilution, driven by protein C activation and glycocalyx loss.*

386. Correct answer B. Platelet adhesion is impaired and clotting enzymes slow below about 35 degrees

Below about 35 degrees Celsius platelet adhesion is impaired and the enzymes of the clotting cascade slow progressively, so coagulation fails more severely as the temperature approaches 32. Hypothermia also depresses the myocardium, provokes arrhythmias, and shifts the oxygen dissociation curve to the left, which impairs unloading. Anesthetized patients cannot shiver, so the clinician must do the work of thermoregulation on their behalf.

Key principle: *Hypothermia below about 35 degrees Celsius impairs platelets and slows the clotting cascade.*

387. Correct answer D. Restore perfusion and stop the bleeding; bicarbonate does not correct the coagulopathy

Acidosis arises from anaerobic metabolism and from the chloride load of large volumes of saline, and below a pH of about 7.2 the activity of coagulation enzyme complexes drops sharply. Treating the cause, by restoring perfusion and controlling bleeding, is the only reliable remedy. Sodium bicarbonate has not been shown to correct the coagulopathy and is reserved for extreme acidosis with refractory shock. Saline would add a hyperchloremic acid load.

Key principle: *Treat acidosis by restoring perfusion and stopping bleeding; bicarbonate does not fix coagulopathy.*

388. Correct answer B. The damage control thresholds are met, so shorten the operation and pack

The anesthesiologist should tell the surgeon plainly when the patient crosses the thresholds for damage control, which are usually a temperature below 34 degrees Celsius, a pH below 7.2, and a coagulopathy that is clinically evident. Here all three are met. No amount of resuscitation outpaces an open vessel, so source control by packing or a shortened operation is the priority. Calcium replacement is part of the resuscitation but does not change the decision.

Key principle: *Temperature below 34, pH below 7.2, and clinical coagulopathy signal the need for damage control.*

389. Correct answer E. Measure ionized calcium on the blood gas and replace calcium

Ionized calcium falls when citrate in stored blood binds it, and shock, hypothermia, and liver hypoperfusion aggravate this. Low calcium weakens the heart, relaxes vessels, and impairs clot formation, producing a pressure that does not respond to volume or vasopressors, a prolonged QT interval, and poor clotting. The blood gas analyzer measures the ionized fraction directly and quickly, so treatment need not wait for a serum total. Repeated vasopressor boluses treat the effect and not the cause.

Key principle: *Refractory hypotension, long QT, and oozing during massive transfusion suggest low ionized calcium.*

390. Correct answer D. Give tranexamic acid now, since hyperfibrinolysis responds to it and benefit falls with time

Hyperfibrinolysis, in which clots are destroyed almost as fast as they form, is associated with high mortality and responds to antifibrinolytic drugs. This is why tranexamic acid is given early and not held until laboratory values return, and viscoelastic tests show the pattern as rapid clot breakdown. The benefit falls as time passes, and the drug should not be started more than three hours after injury. Platelets and cryoprecipitate do not treat the fibrinolysis itself.

Key principle: *Give tranexamic acid early in hemorrhage, within three hours of injury, without waiting for laboratory results.*

391. Correct answer E. At induction, with forced-air warming, warmed fluids, and a core temperature probe

Warming should begin before it seems necessary in any operation where major bleeding is likely, starting the forced-air warmer at induction, warming the room, and using an in-line device that brings every fluid and blood product to body temperature. A reliable core probe shows whether warming is working. Anesthetized patients cannot shiver and so cannot defend themselves, which makes waiting for a sign of cold unreliable. Anesthetic drugs, exposure of the abdomen, and cold products all drain heat from the start.

Key principle: *Start warming at induction when major bleeding is likely; anesthetized patients cannot shiver.*

392. Correct answer A. He is likely to respond to volume, since variation above about 13 percent predicts it

In a patient on controlled ventilation with a regular rhythm and tidal volumes of at least 8 mL/kg, respiratory variation in the arterial pressure or plethysmograph of more than about 13 percent predicts a response to volume. The method fails with spontaneous breathing, arrhythmias, low tidal volumes, an open chest, or right heart failure, none of which applies here. Treating the number as meaningless or as fluid overload would withhold the resuscitation the patient needs.

Key principle: *Respiratory variation above about 13 percent predicts volume responsiveness when its conditions are met.*

393. Correct answer A. Weigh the sponges, subtract the irrigation, and remember estimates tend to be low

Visible losses are easiest to quantify by weighing the sponges and subtracting irrigation fluid from the canister, and estimates of blood loss are usually too low, so a candid estimate should be sought from the surgeon. The hemoglobin is unreliable in early hemorrhage, and urine output reflects perfusion rather than the volume lost. Concealed losses in the retroperitoneum, pelvis, thigh, chest, gut, or uterus can hold liters with little external sign.

Key principle: *Quantify visible loss by weighing sponges and subtracting irrigation; estimates are usually too low.*

394. Correct answer C. Ongoing hypoperfusion, so look for bleeding and reassess resuscitation

Arterial lactate and base deficit reflect tissue hypoperfusion and correlate with the volume of loss, the need for transfusion, and mortality. A rising lactate in a patient who appears stable is an early warning, whereas a slow fall after resuscitation indicates recovery. Point-of-care results are available within minutes, and gases should be sent every fifteen to thirty minutes during active bleeding. Attributing the trend to anesthetic depth repeats the classic mistake.

Key principle: *A rising lactate in a patient who looks stable is an early warning of continued bleeding.*

395. Correct answer E. Hypovolemia, although vasodilatory states can look similar

An underfilled, hyperdynamic left ventricle with walls nearly touching at end systole and a collapsing inferior vena cava supports hypovolemia. These findings can also appear in vasodilatory states, so they are interpreted with the clinical picture. Echocardiography also helps exclude mimics such as tamponade and a failing right ventricle, which would show an effusion or a dilated right heart. A poorly contracting left ventricle would not be hyperdynamic.

Key principle: *An underfilled, hyperdynamic left ventricle with a collapsing vena cava supports hypovolemia.*

396. Correct answer A. Vasopressors can mask a volume deficit, so bleeding control and volume are needed

Fluid and blood buy time, but they cannot substitute for controlling the bleeding. Vasopressors can maintain perfusion pressure while volume is restored, but used alone they hide a volume deficit and increase organ ischemia, as the rising lactate and falling urine output show here. Norepinephrine is acceptable as a bridge after control of bleeding or while volume is restored, and vasopressin is reserved for refractory hypotension. A target pressure does not guarantee tissue perfusion.

Key principle: *Vasopressors are a bridge; they cannot replace volume or source control.*

397. Correct answer B. Have blood ready, draw up a vasopressor, and give a reduced dose of a stable agent slowly

Induction removes the sympathetic drive that has been sustaining the pressure, and positive pressure ventilation reduces venous return, so the pressure falls. Prepare before the first drug with blood in the room, a vasopressor drawn up, and large-bore access, then reduce the dose of an agent that preserves hemodynamics, such as ketamine at a reduced dose, and give it slowly after good preoxygenation. Ask the surgeon to be ready so the time at the lowest pressure is short. Waiting for a normal pressure is not possible while the bleeding continues.

Key principle: *In hypovolemic induction, prepare blood and pressors first and reduce the dose of a hemodynamically stable agent.*

398. Correct answer C. Consider a femoral arterial line, as radial pressure may underestimate central pressure

In profound shock with high-dose vasopressors, the radial pressure can read lower than the central pressure, and a femoral line may give a truer value. Escalating vasopressors against a falsely low reading risks excessive vasoconstriction and organ ischemia. Central venous pressure is a poor predictor of volume status and should be used only as a trend. The arterial line remains valuable for beat-to-beat pressure, sampling, and dynamic indices, but placement should not delay the start of resuscitation.

Key principle: *In profound shock on vasopressors, radial pressure can underestimate central pressure; consider a femoral line.*

399. Correct answer A. Remove air from bags and tubing, and never leave a pressure bag on an empty container

Pressure bags on crystalloid and blood bags speed flow through standard sets and are appropriate for hemorrhage. Every pressurized system carries the risk of air embolism, so air must be removed from bags and tubing, and a pressure bag must never be left unattended on an empty container. Blood warmers should be used for any substantial volume because cold blood worsens the hypothermia that drives coagulopathy. Limiting pressure bags to crystalloid would defeat their purpose.

Key principle: *Purge air and never leave a pressure bag on an empty container to avoid air embolism.*

400. Correct answer E. Central venous pressure predicts volume status poorly and should not guide this decision

Central venous pressure is a poor predictor of volume status and should be used as a trend, if at all. In active hemorrhage with hypotension, transfusion should follow the physiology and the clinical trajectory and not a single filling pressure. Start blood when bleeding is brisk and the circulation is unstable. Dynamic indices, lactate, base deficit, and ultrasound give better information than a static pressure.

Key principle: *Do not use a single central venous pressure to decide whether a bleeding patient needs blood.*

Group 21. Massive Transfusion and Balanced Resuscitation

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401. Correct answer B. That definition is retrospective, so activate on predicted need from mechanism and physiology

The traditional definition of ten or more units in twenty-four hours can be met only in retrospect, and the other definitions also describe what has already happened. The practical question is whether the patient is likely to need a large volume, and it must be answered early. Late activation costs lives, whereas an unnecessary activation costs only some wasted preparation. Laboratory thresholds lag behind the bleeding.

Key principle: *Activate the protocol on prediction of need, not on a retrospective definition.*

402. Correct answer D. The Assessment of Blood Consumption score is 2, which predicts massive transfusion

The score counts penetrating mechanism, a positive focused ultrasound of the abdomen, a systolic pressure of 90 mm Hg or lower, and a heart rate of 120 or higher. This patient has a positive ultrasound and a systolic pressure of 88, for a score of 2, since the mechanism is blunt and the heart rate is below 120. Two or more predict a need for massive transfusion with good sensitivity, and activation should not wait for laboratory data.

Key principle: *An ABC score of 2 or more (penetrating, positive ultrasound, systolic 90 or lower, heart rate 120 or higher) predicts massive transfusion.*

403. Correct answer C. Six red cells, six plasma, and one apheresis platelet unit, with later coolers sent until canceled

Good protocols make activation a single act, and the first cooler commonly holds six units of red cells, six of plasma, and one apheresis platelet unit, ready within minutes. Later coolers follow automatically until the protocol is canceled,

so the team never has to ask again while their hands are full. Sending red cells alone, or waiting for laboratory results, reproduces the unbalanced resuscitation the protocol is meant to prevent.

Key principle: *One call sends a fixed 1:1:1 cooler, and further coolers follow until the protocol is canceled.*

404. Correct answer B. Deactivate the protocol, return unused products, and move to goal-directed therapy

Deactivation is as important as activation. When bleeding is surgically controlled, the circulation is stable, and laboratory values are within target, the blood bank should stop the cycle of coolers and receive unused products promptly. Continued transfusion beyond need exposes the patient to circulatory overload, acute lung injury, thrombosis, and infection, and wastes a scarce resource. The team then switches from fixed ratios to goal-directed replacement, which is not the same as stopping all products permanently.

Key principle: *Deactivate the protocol once bleeding is controlled and values are at target.*

405. Correct answer C. It is low for pregnancy, where fibrinogen normally rises, and signals consumption

In normal pregnancy the fibrinogen concentration rises, so a value that would be acceptable in a nonpregnant patient signals early consumption in obstetric hemorrhage. Fibrinogen falls early in obstetric bleeding, and the target is above 200 mg/dL. Waiting for spontaneous correction would allow the defect to worsen. This is a reason to prioritize early replacement of fibrinogen, which is the scaffold of the clot, and not a reflection of sampling error.

Key principle: *In obstetric hemorrhage a fibrinogen near 180 mg/dL is already low and signals consumption.*

406. Correct answer A. Return or discard them, since the limit outside a validated cooler is often 30 minutes

Blood products leave the blood bank in a validated cooler, and if they are removed and left on the counter, the red cells must be used or returned within a defined window, often thirty minutes, or discarded. Visual inspection or label checking does not restore storage conditions, and blood must never be warmed in a microwave. A cooler that sits in the corridor for an hour is a wasted cooler. Practice and a checklist reduce such failures.

Key principle: *Red cells outside a validated cooler beyond the defined window must be returned or discarded.*

407. Correct answer C. Survivor bias, since longer survivors were more likely to receive more plasma

The early studies from Iraq and Afghanistan suggested that near-equal plasma and red cell numbers improved survival, but patients who survived long enough to receive more plasma were more likely to have a high ratio on paper. The multicenter observational study used methods that accounted for time and found that higher early ratios of plasma and platelets to red cells were associated with lower mortality in the first hours. The distractors describe flaws that were not the issue.

Key principle: *Survivor bias weakened the early ratio studies; later work accounted for time.*

408. Correct answer D. One apheresis unit equals roughly six pooled units, so the cooler approximates a 1:1:1 ratio

A one-to-one-to-one ratio means one unit of plasma and one platelet dose for each unit of red cells. Since one apheresis platelet unit is roughly equal to six units pooled from whole blood, a cooler with six red cell units, six plasma units, and one apheresis unit approximates the ratio. The aim is to give the patient something close to whole blood in composition, and the task is to see that plasma and platelets are delivered at the same pace as the red cells.

Key principle: *One apheresis platelet unit is about six pooled units, so 6:6:1 approximates 1:1:1.*

409. Correct answer C. A hematocrit of about 29 percent, a platelet count of about 80,000, and factor activity of about 65 percent

Reconstituted whole blood is a weaker product than fresh whole blood, because each component is diluted by the storage solution and the anticoagulant. Modeling one unit of each component in the usual proportions gives a hematocrit of about 29 percent, a platelet count of about 80,000, and clotting factor activity of about 65 percent of normal. This explains why patients receiving balanced products still develop low fibrinogen, low calcium, and hypothermia, and why those problems must be sought and treated.

Key principle: *Reconstituted 1:1:1 blood is dilute, so check fibrinogen, calcium, and temperature.*

410. Correct answer E. Give platelets, since above 100,000 is advised in brain injury or massive bleeding

A platelet count above 50,000 is the usual target in bleeding patients, but above 100,000 is recommended in patients with traumatic brain injury or ongoing massive bleeding. This patient has both, so additional platelets are indicated. Cryoprecipitate supplies fibrinogen and does not raise the platelet count. Lower targets would expose the patient to further intracranial and surgical bleeding.

Key principle: *Aim for platelets above 100,000 in traumatic brain injury or ongoing massive bleeding.*

411. Correct answer A. Do not start it now, since benefit falls with time and it may harm beyond three hours

Tranexamic acid belongs in the first minutes and not in the second hour, since the benefit falls as time passes and the drug may be harmful if started more than three hours after the injury. The regimen is given within three hours of injury or onset of bleeding. Doubling the dose or tying it to fibrinogen replacement does not address the timing problem. Hemostatic resuscitation relies on early antifibrinolytic therapy and limited crystalloid.

Key principle: *Start tranexamic acid within three hours of injury; later use may be harmful.*

412. Correct answer B. The platelet count, which is below the target of 50,000 per microliter

Targets in European guidance include a fibrinogen of 150 to 200 mg/dL or higher, a platelet count above 50,000, a prothrombin time ratio below 1.5, and a hemoglobin of 7 to 9 grams per deciliter. Only the platelet count of 42,000 falls outside these targets. The fibrinogen is above the range, the prothrombin time ratio is below 1.5, and the hemoglobin is within the accepted range. Goal-directed therapy corrects the specific defect rather than repeating fixed packs.

Key principle: *After the initial packs, replace the specific defect: platelets above 50,000, fibrinogen 150 to 200 or higher, prothrombin time ratio below 1.5.*

413. Correct answer D. It is a reasonable alternative, since a multicenter trial found no difference in outcome

Proponents point to small volumes, no thawing delay, and lower donor exposure, but the evidence comes largely from observational and single-center studies, and clotting factors other than those in the concentrates are not replaced. A multicenter trial of viscoelastic-guided therapy against conventional laboratory-guided therapy in trauma found no difference in its primary outcome. The approach is therefore a reasonable alternative and not a clear improvement, and combining a ratio-based start with goal-directed continuation is the usual practice.

Key principle: *Viscoelastic-guided factor concentrates are a reasonable alternative, not a proven improvement.*

414. Correct answer B. Calcium gluconate peripherally, in larger volume because it has less elemental calcium

Calcium chloride delivers about three times the elemental calcium of the same volume of calcium gluconate, but it is irritating and extravasation causes tissue necrosis, so it needs a secure large-bore or central line. The gluconate salt is safer through a peripheral vein, at a larger dose such as 3 g. Delay is harmful because low calcium impairs contractility, vascular tone, and clotting. Under anesthesia hypocalcemia shows as a failing circulation, not as symptoms.

Key principle: *Use calcium gluconate peripherally and calcium chloride through secure central access.*

415. Correct answer E. Infuse the calcium through a separate port, or flush the tubing between the two

Calcium infused into the same tubing as citrated blood can cause clotting in the line, so a separate port should be used or the tubing flushed between. Other salts of calcium behave similarly, and slowing the red cells does not remove the interaction. Calcium requirements vary widely, so the level and not a fixed dose should be followed, with each dose and level documented. Overshoot into hypercalcemia has its own cardiac effects.

Key principle: *Give calcium through a separate port or flush the tubing; do not mix it with citrated blood.*

416. Correct answer D. Use fresher or washed red cells, give calcium early, and infuse slowly peripherally

Children and neonates are more vulnerable to the metabolic effects of transfusion because a single unit represents a large fraction of blood volume, hepatic metabolism of citrate is immature, and a bolus of potassium from an old unit

can cause arrest. Fresher, sometimes washed, red cells are used, calcium is given early when plasma or large volumes are given, and infusion is slow through a peripheral line when possible. The margin for error is smaller than in adults.

Key principle: *In neonates use fresher or washed red cells, early calcium, and slow peripheral infusion.*

417. Correct answer E. Metabolism of citrate from transfused blood to bicarbonate

Acid-base disturbance changes during a massive transfusion. Early on the patient is acidotic from lactate and the acidity of stored blood, but later the citrate in the transfused blood is metabolized to bicarbonate and a metabolic alkalosis can develop. Lactate and chloride excess cause acidosis, not alkalosis, and potassium leak does not explain this picture. Bicarbonate should not be relied on except in extreme acidosis, and large volumes of chloride-rich saline should be avoided.

Key principle: *Late alkalosis after massive transfusion reflects citrate metabolized to bicarbonate.*

418. Correct answer E. The low titer reduces the risk of hemolysis when the recipient is not group O

Low-titer group O whole blood contains red cells, plasma, and platelets in one unit at roughly the proportions that balanced resuscitation tries to reconstruct, and it is stored cold. The plasma of a group O donor carries anti-A and anti-B antibodies, so a low titer reduces the risk of hemolysis when the recipient is not group O. It does not change the citrate content, so calcium and fibrinogen still need attention. Rh sensitization is a separate issue handled by choice of Rh-negative units.

Key principle: *Low anti-A and anti-B titers make group O whole blood safer for non-group-O recipients.*

419. Correct answer D. Pause salvage or use a separate suction device while these agents are in use

Topical hemostatic agents based on microfibrillar collagen and thrombin can be aspirated and reinfused with disastrous effect, so they must be kept out of the suction. Hypotonic irrigation, antibiotic solutions, hydrogen peroxide, and alcohol can also cause hemolysis or toxicity. When such agents are in use, salvage should be paused or a separate suction used, and the anesthesiologist should state this before the case starts. Washing and filtering are not reliable defenses against these materials.

Key principle: *Keep topical hemostatic agents out of cell salvage suction.*

420. Correct answer B. A low magnesium level from citrate binding and intracellular loss

Magnesium falls with citrate binding and with the loss of intracellular stores, and a low value worsens arrhythmias and the response to calcium. Glucose can also shift in either direction, so the blood gas analyzer that measures glucose and ionized magnesium should be read at each check. Any value outside the normal range should be corrected, and the electrolyte problems should be treated together and not in sequence. The other distractors do not describe the recognized pattern.

Key principle: *Check glucose and magnesium during massive transfusion; low magnesium worsens arrhythmias and calcium response.*

Group 22. Coagulopathy and Hemostatic Therapy

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421. Correct answer A. They reflect only the early plasma phase of clotting, not platelets or fibrinolysis

The prothrombin time and activated partial thromboplastin time measure the first few seconds of clot formation in plasma, so they cannot show platelet function, clot strength, or lysis. Viscoelastic testing uses whole blood and follows the clot through formation, maturation, and dissolution, and the results are available in minutes. Conventional tests run at 37 degrees and so miss the effect of hypothermia, rather than correcting it. They do not measure final clot strength, and they do not measure fibrinogen, which needs its own assay.

Key principle: *Conventional coagulation tests are slow and blind to platelet function, clot strength, and lysis.*

422. Correct answer D. Withhold prophylactic plasma, since the INR predicts bleeding poorly in cirrhosis

In cirrhosis the fall in procoagulant factors is balanced by a fall in anticoagulant proteins, so clot formation is often normal and the INR does not predict bleeding well. Prophylactic plasma for a raised INR is usually unhelpful and can cause circulatory overload. The claim that the INR reliably predicts bleeding ignores this balance. Plasma has a large volume, which is why overload is a real risk. Vitamin K treats warfarin effect, and nothing here suggests warfarin.

Key principle: *Do not treat the INR in cirrhosis with prophylactic plasma; viscoelastic testing shows the true clotting state.*

423. Correct answer A. Slow the transfusion, give a loop diuretic, and support oxygenation

Dyspnea with hypertension, raised venous pressure, a high natriuretic peptide, and a positive fluid balance in an elderly patient with heart and kidney disease receiving rapid plasma is circulatory overload, which responds to a diuretic and a slower infusion. Anaphylaxis would show bronchospasm, urticaria, and hypotension. Bacterial contamination would produce fever, rigors, and shock. Hemolysis would produce hypotension, dark urine, and oozing, and this patient is hypertensive.

Key principle: *Circulatory overload shows hypertension and a high natriuretic peptide and responds to a diuretic.*

424. Correct answer C. Stop the infusion, culture patient and product, give antibiotics, and notify the blood bank

Fever, rigors, and shock during or soon after a platelet infusion suggest bacterial contamination, which is a particular risk with platelets because they are stored at room temperature. The product must be stopped, both patient and product cultured, broad-spectrum antibiotics started, and the blood bank told so that products from the same donor can be quarantined. An antihistamine and antipyretic would suit a mild febrile or allergic reaction. Overload does not cause fever and shock. Anaphylaxis would show a rash or bronchospasm, and the transfusion should not be continued.

Key principle: *Fever and shock after platelets: suspect contamination, stop, culture, treat, and notify the blood bank.*

425. Correct answer B. Washed cellular products or units from IgA-deficient donors if available

Severe allergic reactions in IgA-deficient recipients with anti-IgA antibodies behave like anaphylaxis, and future transfusions should use washed products or units from IgA-deficient donors. Leukoreduction lowers febrile reactions but does not remove plasma proteins. Irradiation prevents graft-versus-host disease and does not alter plasma antibodies. Premedication is unreliable against a reaction of this severity.

Key principle: *IgA-deficient recipients with anti-IgA antibodies need washed or IgA-deficient donor products.*

426. Correct answer D. Antagonism of inhibitory receptors in the brain by the drug itself

At high doses, particularly in cardiac surgery, tranexamic acid causes seizures by antagonizing inhibitory receptors in the brain, and the risk rises with age, renal impairment, and bypass. Because the drug is cleared by the kidney, renal failure raises the level. Thromboembolic events have not clearly increased in trials. Hyponatremia is a feature of desmopressin, and the sodium here is normal. Heparin rebound would cause bleeding, not a seizure with a normal electrolyte profile.

Key principle: *High-dose tranexamic acid causes seizures; reduce the dose in renal failure.*

427. Correct answer E. Withhold it, since benefit is lost and harm is possible beyond three hours

The benefit of tranexamic acid in trauma is time dependent, with the greatest effect in the first hour and a possible increase in harm when it is started more than three hours after injury. At four hours it should not be given routinely. Starting late does not increase benefit. Giving only the infusion does not solve the time problem. A lytic pattern on testing would be a separate indication that needs to be weighed by the team, and the chapter does not support late routine use.

Key principle: *Tranexamic acid should be given within three hours of injury; later use may harm.*

428. Correct answer A. Fibrinogen concentrate, since it mixes in minutes and cryoprecipitate is noninferior

Fibrinogen concentrate is a standardized, pasteurized product that can be reconstituted in minutes and gives a predictable rise. A randomized cardiac surgery trial found cryoprecipitate noninferior for blood component use, so the choice rests on availability, speed, and cost, and speed favors concentrate here. Plasma contains too little fibrinogen and adds volume. The idea that concentrate is inferior contradicts the trial. Platelets do not correct a fibrinogen deficit.

Key principle: *Fibrinogen concentrate and cryoprecipitate are both acceptable; choose by speed and availability.*

429. Correct answer C. Replace fibrinogen, platelets, and factors as needed and withhold an antifibrinolytic

This is consumptive coagulopathy from sepsis, and after treating the cause the aim is to replace fibrinogen, platelets, and factors to a safe level. Antifibrinolytics are used only when viscoelastic testing shows a lytic pattern, because blocking fibrinolysis in a thrombotic state can cause harm, and no lysis is present. Recombinant factor VII is a last-line agent and adds arterial thrombotic risk. Vitamin K does not address consumption, and observation alone leaves a bleeding patient behind.

Key principle: *In DIC, replace what is consumed and give antifibrinolytics only for a proven lytic pattern.*

430. Correct answer D. Give tranexamic acid and bring fibrinogen to target before other factor products

After the source is controlled and temperature, calcium, and pH are corrected, the sequence is an antifibrinolytic, then fibrinogen replacement to the target, then platelets and prothrombin complex concentrate or plasma according to the clotting time. Concentrate given before fibrinogen is corrected invites thrombosis and may not work. Recombinant factor VII is reserved for failure of everything else. Plasma gives a small amount of fibrinogen in a large volume and does not replace the deficit.

Key principle: *Sequence: control the source, correct physiology, antifibrinolytic, fibrinogen, then platelets and factors.*

431. Correct answer C. An increase in arterial thrombotic events such as stroke

Recombinant activated factor VII can stop refractory bleeding but increases arterial thrombotic events, and it works only after temperature, pH, calcium, fibrinogen, and platelets have been corrected, which has been done here. Hyponatremia is a risk of desmopressin. Pulmonary vasoconstriction is a feature of protamine reactions. Anti-IgA antibodies matter for plasma-containing products in IgA-deficient recipients, not for this agent.

Key principle: *Recombinant factor VII is a last-line agent with a risk of arterial thrombosis.*

432. Correct answer B. Stop all heparin, including flushes, and start a non-heparin agent

A falling platelet count five to ten days after heparin exposure with new thrombosis indicates heparin-induced thrombocytopenia. All heparin, including flushes and coated catheters, should be stopped and replaced with a non-heparin anticoagulant. Lowering the dose does not remove the trigger, and platelets may feed thrombosis. Low-molecular-weight heparin can also provoke the syndrome. Protamine neutralizes heparin but the antibody persists, so resuming heparin on recovery repeats the exposure.

Key principle: *A falling platelet count with thrombosis after heparin means stop all heparin and use a non-heparin agent.*

433. Correct answer D. Desmopressin, watching for hyponatremia and tachyphylaxis

Desmopressin increases von Willebrand factor and factor VIII release and improves platelet function in uremia, and dialysis also helps. Its adverse effects are hyponatremia and tachyphylaxis. Protamine neutralizes heparin and has no role in uremic platelet dysfunction. Fibrinogen is normal here, so replacing it does not help. Idarucizumab is the antidote for dabigatran only.

Key principle: *Uremic platelet dysfunction responds to desmopressin and dialysis.*

434. Correct answer B. Protamine at 1 milligram per milligram of enoxaparin, accepting only partial reversal

Protamine neutralizes only about 60 percent of the anti-factor Xa activity of low-molecular-weight heparin, so for enoxaparin given within eight hours the dose is 1 milligram per milligram, with a smaller second dose if bleeding

continues. Expecting complete reversal ignores this limit. Vitamin K acts on vitamin K dependent factors and takes hours. Idarucizumab is specific for dabigatran. Fondaparinux, in contrast, has no antidote.

Key principle: *Protamine only partly reverses low-molecular-weight heparin.*

435. Correct answer E. Four-factor prothrombin complex concentrate according to the local protocol

Bleeding on apixaban, rivaroxaban, or edoxaban is treated with four-factor prothrombin complex concentrate in many guidelines, and this is the practical choice when andexanet alfa is unavailable, as it has been in the United States since December 2025 after the Food and Drug Administration concluded that its risks outweighed its benefits. Idarucizumab binds only dabigatran. Protamine neutralizes heparin. Plasma is bulky and does not reliably reverse these drugs.

Key principle: *Xa inhibitor bleeding: four-factor concentrate, or andexanet alfa where licensed.*

436. Correct answer C. Do not transfuse routinely, since a randomized trial showed harm in this setting

A randomized trial in intracerebral hemorrhage showed harm from platelet transfusion in patients on antiplatelet drugs, so it should not be given routinely there. The fact that aspirin acts for the life of the platelet is the reason the idea is tempting, but the trial result takes precedence in this setting. Protamine reverses heparin. Prothrombin complex concentrate replaces vitamin K dependent factors and does not restore platelet function.

Key principle: *Do not give platelets routinely for intracerebral hemorrhage on antiplatelet drugs.*

437. Correct answer C. Stop the infusion, give cryoprecipitate, and give an antifibrinolytic

Alteplase has a short half-life but its depletion of fibrinogen and plasminogen lasts a day or more, so the effect outlasts the drug. The infusion should be stopped, fibrinogen replaced with cryoprecipitate, an antifibrinolytic given, and platelets or plasma used as indicated. Waiting leaves a bleeding patient untreated. Protamine acts on heparin. Vitamin K dependent factors are not the problem with a thrombolytic.

Key principle: *Thrombolytic effects outlast the half-life; replace fibrinogen and give an antifibrinolytic.*

438. Correct answer E. Reversal does not shorten the interval, so general anesthesia is often safer

Guidelines set intervals between the last anticoagulant dose and needle placement that depend on the drug and renal function, and after reversal the intervals do not shorten because the drug can return as the antidote clears. Spinal hematoma has devastating consequences, so general anesthesia is often safer when the coagulation state is uncertain. The INR is not a reliable measure of Xa inhibitor effect, and a catheter adds exposure rather than safety.

Key principle: *Reversal of an anticoagulant does not shorten the interval before neuraxial needle placement.*

439. Correct answer A. Do not transfuse, since a restrictive threshold of 7 to 8 applies in stable patients

Randomized trials in critically ill adults and in cardiac and orthopedic surgery support a hemoglobin threshold of 7 to 8 grams per deciliter in patients who are not actively bleeding, with higher thresholds for active coronary ischemia. This patient is stable, asymptomatic, and above the threshold. The lag of hemoglobin behind loss applies to active hemorrhage, not here. A threshold of 10 exposes patients to transfusion risks without a benefit, and an assumed future fall is not an indication.

Key principle: *In stable patients use a restrictive hemoglobin threshold of 7 to 8 grams per deciliter.*

440. Correct answer E. Run a platelet mapping assay, since standard assays miss P2Y12 inhibitor effects

Standard viscoelastic assays are insensitive to aspirin and to P2Y12 inhibitors unless a platelet mapping assay is used, so a normal tracing does not exclude drug-induced platelet dysfunction. Accepting the tracing as normal would miss the cause. Repeating the same assay does not make it sensitive to these drugs, although a poorly handled sample can mislead. An anti-factor Xa level measures heparin or Xa inhibitors, not antiplatelet effect.

Key principle: *Standard viscoelastic tests miss aspirin and P2Y12 effects; use platelet mapping.*

Group 23. Obstetric and Trauma Hemorrhage

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441. Correct answer A. Early hemorrhage; a shock index above one is a warning, so call for help and treat now

Pregnancy expands blood volume by 40 to 50 percent, so a mother can lose a large fraction of her circulating volume before the pressure falls, and the heart rate rises late. The shock index of about 1.1 here is above one and is an early warning, and the patient has signs of hypovolemia with ongoing loss. Waiting for hypotension means acting too late. Quantitative measurement corrects the usual underestimate rather than overestimating. Treating a quiet, flat patient as merely anxious is the classic error.

Key principle: *Pregnancy hides hemorrhage; act on the shock index and the quantified loss before the pressure falls.*

442. Correct answer E. Examine the lower genital tract for lacerations

A firm uterus with an intact placenta and persistent bright bleeding after an instrumented delivery points to trauma, one of the four Ts, such as a cervical or vaginal laceration. Naming all four causes early avoids treating atony while a laceration bleeds. Carboprost and further oxytocin treat tone, which is not the problem. A balloon tamponades the uterine cavity and would not control a laceration of the lower tract.

Key principle: *A firm uterus with ongoing bright bleeding: look for trauma.*

443. Correct answer A. Give a small dose slowly, then an infusion, since a bolus can cause hypotension

Oxytocin is the first-line uterotonic and is given as a slow injection followed by an infusion. A rapid bolus causes vasodilation, hypotension, and tachycardia, and in a hypovolemic mother the fall can be dangerous. Delaying the uterotonic leaves the uterus bleeding. Methylergonovine should not be given as an intravenous bolus, and as a vasoconstrictor it is contraindicated in hypertensive disease.

Key principle: *Give oxytocin slowly; a rapid bolus causes hypotension in a hypovolemic mother.*

444. Correct answer D. Methylergonovine given intramuscularly

Methylergonovine is a vasoconstrictor and is contraindicated in hypertension and preeclampsia. Carboprost can cause bronchospasm and is avoided in asthma, which this patient does not have. Misoprostol has no major contraindications. Tranexamic acid is part of the standard response to postpartum hemorrhage and should be given early. Choosing the second-line drug depends on comorbidity.

Key principle: *Avoid methylergonovine in hypertension and preeclampsia.*

445. Correct answer C. Place an intrauterine balloon while the obstetrician prepares surgical options

When drugs fail, mechanical measures follow, and an intrauterine balloon can tamponade a bleeding cavity quickly while surgical control such as compression sutures, ligation, or hysterectomy is prepared. Embolization is suited to stable patients and takes time, so it is not appropriate in an unstable one. Repeating drugs that have failed wastes time. Observation with massage alone leaves a hypotensive patient bleeding, and delay in escalation is a recognized cause of death.

Key principle: *When drugs fail, escalate promptly: balloon, sutures, ligation, embolization in stable patients, or hysterectomy.*

446. Correct answer B. Convert early to general anesthesia, with a reduced-dose ketamine or etomidate induction

A mother under regional anesthesia can continue it while stable, but conversion to general anesthesia should be made early when bleeding is brisk, she is unstable, or coagulopathy raises concern about further neuraxial manipulation. Induction doses are reduced, and ketamine or etomidate is a common choice. A full-dose epidural bolus risks profound sympathetic block and hypotension. Continuing the epidural alone accepts a failing block in an unstable patient. A new spinal adds neuraxial manipulation in a patient developing coagulopathy.

Key principle: *Convert to general anesthesia early in a bleeding, unstable, or coagulopathic obstetric patient.*

447. Correct answer E. Cell salvage is acceptable in obstetrics with a leukocyte-depletion filter

Cell salvage is acceptable in obstetrics with a leukocyte-depletion filter and is increasingly used, particularly where major blood loss is anticipated, as in accreta. The statement that it is unacceptable reflects an older concern that the filter addresses. It is not restricted to patients who refuse blood, and it should be set up before incision rather than delayed until the hemoglobin is low.

Key principle: *Cell salvage with a leukocyte-depletion filter is acceptable in obstetrics.*

448. Correct answer D. Stabilize her and arrange transfer to a center with a multidisciplinary team

Planned management in a center with blood bank depth, interventional radiology, and intensive care dramatically reduces morbidity compared with unexpected management, and a facility without these resources is hazardous. If a patient presents unexpectedly with bleeding, she should be stabilized and transferred where possible. Attempts to remove the placenta can cause torrential bleeding. Waiting for labor or delaying to term risks bleeding in an unprepared setting.

Key principle: *An unexpected bleeding accreta in a poorly equipped hospital: stabilize and transfer.*

449. Correct answer D. Ask the surgeon to pause and pack the field while volume is caught up

The pace of the operation determines the blood loss, and pausing to catch up on volume is a legitimate request when the patient is behind. Product replacement should be early and not wait for hypotension. Attempts to remove the placenta can cause torrential bleeding. A vasopressor treats a pressure and not a bleed. Deferring products until the dissection ends worsens the deficit.

Key principle: *Pausing the surgery to catch up on volume is a legitimate request.*

450. Correct answer C. Avoid neuraxial blockade while coagulopathic and replace fibrinogen early

Abruption is a leading cause of obstetric coagulopathy, with early consumption of fibrinogen, which is the first factor to fall and predicts severity. Coagulation should be measured before neuraxial blockade, and regional anesthesia avoided when the patient is already coagulopathic, so fibrinogen is replaced early with cryoprecipitate or concentrate. An acceptable platelet count does not offset a low fibrinogen. Plasma alone does not correct fibrinogen well. A smaller needle does not remove the risk of spinal hematoma.

Key principle: *In abruption, check coagulation first and avoid neuraxial blockade when coagulopathic.*

451. Correct answer A. Uterine rupture

Uterine rupture occurs in patients with prior uterine surgery or obstructed labor and presents with pain, loss of fetal heart rate, and shock. Placenta previa produces bleeding that is typically painless. Amniotic fluid embolism causes collapse with coagulopathy, but the prior uterine surgery and pain with loss of the fetal heart tracing favor rupture. Inversion produces sudden shock after delivery. In each case the approach is to call for help, resuscitate, and treat the cause.

Key principle: *Prior uterine surgery with pain, lost fetal heart rate, and shock suggests uterine rupture.*

452. Correct answer C. No mortality difference, but better hemostasis and fewer early exsanguination deaths with 1 to 1 to 1

Mortality at 24 hours and 30 days did not differ significantly between the groups, but the 1 to 1 to 1 group had more patients achieving hemostasis and fewer deaths from exsanguination in the first day. For this reason most centers use balanced ratios from the first minutes. The trial did not show a mortality benefit at 30 days, did not show harm, and did not favor the lower plasma and platelet ratio.

Key principle: *Balanced ratios improved early hemostasis and reduced early exsanguination deaths.*

453. Correct answer B. Add large-bore access above the diaphragm and use a warmed rapid infuser

In torso injuries, large-bore intravenous access above the diaphragm is preferred because fluid infused below the diaphragm can escape through an injured vessel. A warmed rapid infuser delivers volume quickly without cooling the patient. More lines below the diaphragm share the same flaw. A tibial intraosseous line is also below the injury and has lower flow rates.

Key principle: *Torso trauma: use large-bore access above the diaphragm.*

454. Correct answer B. Advocate temporary closure and transfer to the intensive care unit for resuscitation

Damage control surgery controls bleeding by packing, ligation, shunting, or temporary closure, accepts contamination, and leaves definitive repair for a later operation. The first operation should be completed in about an hour, with transfer to the intensive care unit for warming and resuscitation. Staying longer in a cold, acidotic, coagulopathic patient invites death on the table. Normalizing the temperature in the operating room prolongs the same danger.

Key principle: *In damage control surgery, stop early and finish the resuscitation in the intensive care unit.*

455. Correct answer D. It is reserved for penetrating injury with recent loss of cardiac activity

Resuscitative thoracotomy is reserved for penetrating injury with recent loss of cardiac activity, where it permits cross-clamping of the aorta, relief of tamponade, and internal cardiac massage. It is not indicated for blunt trauma with a prolonged absence of pulse. It does not replace blood or balanced resuscitation, and it is not a treatment for stable patients. The decision belongs to the trauma team, with the anesthesiologist managing volume and drugs.

Key principle: *Resuscitative thoracotomy is for penetrating injury with recent loss of cardiac activity.*

456. Correct answer B. Use a vasopressor only as a short bridge during induction while blood is given

Observational studies associate early vasopressor use in trauma with higher mortality, partly because of sicker patients and partly because vasoconstriction in a hypovolemic patient worsens tissue perfusion. A brief infusion during induction or while blood is being prepared is a reasonable bridge, with the plan to replace it with volume. A vasopressor treats a pressure and not a bleed. Withholding both blood and pressors leaves a patient in shock untreated.

Key principle: *A vasopressor treats a pressure, not a bleed; use it only as a short bridge to blood.*

457. Correct answer C. Start a vasopressor, since neurogenic shock requires more than volume

Neurogenic shock produces hypotension with bradycardia and requires vasopressors and not volume alone. The cord is perfused by the mean arterial pressure, and prolonged hypotension worsens the neurologic injury, so permissive hypotension is not appropriate. The mean pressure target is set by local protocol and the spine team, traditionally 85 to 90 mm Hg. Large crystalloid volumes do not correct the loss of sympathetic tone, and there is no evidence of bleeding to justify massive transfusion.

Key principle: *Neurogenic shock needs vasopressors, and spinal cord injury excludes permissive hypotension.*

458. Correct answer A. Probable occult shock; the heart rate is unreliable, so use lactate and base deficit

Older adults on beta-blockers cannot mount a tachycardia, so the usual signs of shock fall away and a normal heart rate is misleading. Lactate, base deficit, and trends in pulse pressure and the arterial waveform are more reliable, and here they show hypoperfusion in a patient with a pelvic fracture. Treating the numbers on the screen as reassuring ignores this. Attributing the picture to the beta-blocker alone withholds needed treatment, and nothing suggests sepsis.

Key principle: *In a patient on a beta-blocker, trust lactate and base deficit more than the heart rate.*

459. Correct answer C. Give blood products and adjust the target upward, since prolonged hypoperfusion is unsafe

Permissive hypotension is acceptable for an hour or two when the operating room is minutes away. When transport takes hours, the duration of hypoperfusion makes prolonged hypotension unsafe, and prehospital teams in such settings give blood products and adjust targets upward. Maintaining a low target for three hours accumulates an oxygen debt. Crystalloid to a high pressure dilutes clotting factors and may dislodge clot. Withholding all infusion ignores ongoing loss.

Key principle: *Prolonged transport makes prolonged permissive hypotension unsafe.*

460. Correct answer E. Restore normal perfusion now, since the bleeding is controlled and hypotension is harming him

Permissive hypotension is a bridge to surgical control and is typically limited to an hour or two. Once bleeding is controlled, the target changes to restoration of normal perfusion, and a patient left hypotensive after the surgeon has finished is being harmed, as the rising lactate suggests. The anesthesiologist should say so. Continuing the strategy as a precaution accumulates an oxygen debt. Deferring the decision to a later team leaves hypoperfusion untreated now.

Key principle: *When bleeding is controlled, permissive hypotension ends and normal perfusion is restored.*

Group 24. Intraoperative Cardiac Arrest: Causes and Response

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461. Correct answer D. Wipe away the pooled solution, move oxygen off the chest, clear everyone, and shock

The operating room is oxygen-rich, and drapes and alcohol-based solutions are flammable, so a discharge can ignite a fire if solution has pooled under the pads or oxygen flows across the chest. The remedy is to wipe away pooled solution, keep the circuit away from the shock path, and make sure everyone is clear, which takes seconds and should not delay the shock. Waiting for evaporation delays treatment of a shockable rhythm. Fire risk is not limited to cautery. Drugs do not replace the shock.

Key principle: *Clear pooled prep solution and oxygen in seconds, then shock without delay.*

462. Correct answer E. Place a backboard, change compressors, and coach two inches deep at 100 to 120 per minute

The target rate is 100 to 120 per minute and the depth at least two inches but not more than two and four-tenths in an adult, with full recoil. The mattress absorbs force, so a backboard helps, and compressors should change every two minutes because fatigue degrades quality after about a minute. A low end-tidal value suggests inadequate output. Slowing the rate lowers cardiac output, excessive depth risks injury, and keeping a tired compressor degrades quality more than a short pause.

Key principle: *Use a firm surface, 100 to 120 per minute, adequate depth, and rotate compressors every two minutes.*

463. Correct answer B. One breath about every six seconds with no pause in compressions

After the airway is secured, ventilation is one breath about every six seconds without pausing compressions. Excess ventilation raises intrathoracic pressure and reduces venous return, which lowers coronary perfusion. The compression-to-breath cycle with pauses applies before the airway is secured. Faster rates worsen the problem. Withholding ventilation completely leaves the patient hypoxic.

Key principle: *After intubation, ventilate at about one breath every six seconds without pausing compressions.*

464. Correct answer A. As soon as it can be prepared, then every three to five minutes

For non-shockable rhythms such as asystole and pulseless electrical activity, epinephrine should be given as early as possible, since the benefit of early dosing is greater than in shockable rhythms, and it is repeated every three to five minutes. Delaying for rhythm checks or cycles loses that benefit. Waiting for the cause to be corrected defeats the purpose, since the drug supports perfusion while the search goes on.

Key principle: *In non-shockable arrest give epinephrine early and repeat every three to five minutes.*

465. Correct answer C. Lipid emulsion with reduced epinephrine doses

Seizure followed by cardiovascular collapse after a regional block is local anesthetic systemic toxicity, which calls for lipid emulsion and smaller epinephrine doses along with standard resuscitation. Dantrolene treats malignant hyperthermia. Calcium and bicarbonate are not recommended routinely and do not treat this toxicity. Standard-dose epinephrine alone leaves the toxin untreated and the cause is the priority in the operating room.

Key principle: *Local anesthetic toxicity arrest: lipid emulsion and smaller epinephrine doses.*

466. Correct answer B. Stop the volatile agent; give dantrolene with active cooling

Rising carbon dioxide, rigidity, and a climbing temperature indicate malignant hyperthermia, which ends in arrest if missed, and the cause-directed treatment is to stop the triggering agent and give dantrolene with cooling alongside standard resuscitation. Lipid emulsion is for local anesthetic toxicity. Thrombolysis is for massive pulmonary embolism. Calcium and insulin treat hyperkalemia, although hyperkalemia can accompany the syndrome.

Key principle: *Malignant hyperthermia arrest: dantrolene and cooling.*

467. Correct answer D. Epinephrine with rapid intravenous volume

Bronchospasm and cardiovascular collapse within minutes of a neuromuscular blocker and an antibiotic indicate anaphylaxis, which calls for epinephrine and volume. Lipid emulsion treats local anesthetic toxicity, and no local anesthetic was given. Dantrolene treats malignant hyperthermia, which would show rising carbon dioxide and temperature. Needle decompression treats tension pneumothorax, in which breath sounds would be absent on one side.

Key principle: *Anaphylaxis arrest: epinephrine and volume.*

468. Correct answer A. Pause briefly to check the rhythm, the arterial waveform, and a pulse

A sudden jump in end-tidal carbon dioxide during compressions can be the first sign of return of circulation, especially with a pulsatile arterial waveform, and should prompt a pause to check. The return of an arterial waveform, a rise in carbon dioxide, and a palpable pulse are better evidence than the rhythm on the screen. Ignoring the sign delays recognition. A low output would show a low value, not a high one. Compressing faster does not help if circulation has returned.

Key principle: *A sudden rise in end-tidal carbon dioxide during compressions may signal return of circulation; pause and check.*

469. Correct answer B. Give small epinephrine boluses and treat the cause, without waiting for a flat line

A patient with a pulse but a failing circulation can sometimes be rescued with small, repeated boluses of epinephrine, such as 10 to 100 micrograms, and treatment of the cause before compressions are needed. Full arrest doses are avoided while a pulse is present. Compressions begin if the pressure is unrecordable and the capnography trace has collapsed, which has not happened here. Waiting for the monitor to flatten is the recognized error in peri-arrest care.

Key principle: *In peri-arrest, give small epinephrine boluses and treat the cause; do not wait for a flat line.*

470. Correct answer C. Move the pads about a hand's breadth from the generator, and do not delay the shock

When a patient has a pacemaker or implanted defibrillator, the pads should be kept about a hand's breadth away from the generator, and the shock should not be delayed to find a better site. Placing pads over the generator risks device damage and poor current delivery. Waiting for a representative delays treatment of a lethal rhythm. The device has already failed to terminate the rhythm, so withholding an external shock is unsafe.

Key principle: *Keep pads a hand's breadth from an implanted device, and do not delay the shock.*

471. Correct answer E. Deliver one shock, then resume compressions at once for a full two minutes

The guidelines have moved toward a one-shock strategy because the pause for a rhythm check after each of three stacked shocks costs more than it returns. After the shock, compressions resume immediately for a full two-minute cycle without stopping to check the rhythm. A long pulse check delays perfusion, and checking early shortens the cycle. Pauses should stay under ten seconds.

Key principle: *Give one shock, then resume compressions at once for two minutes.*

472. Correct answer A. Pseudo-pulseless electrical activity, which needs the cause treated as well

When ultrasound shows organized cardiac motion with no palpable pulse, the state is sometimes called pseudo-pulseless electrical activity, and it responds to treatment of the underlying cause rather than to compressions alone. A damped line does not explain the absent pulse and weak motion. Defibrillation is not indicated for an organized rhythm that is not ventricular tachycardia or fibrillation. Cardiac motion on echo excludes asystole.

Key principle: *Organized cardiac motion without a pulse: treat the cause, not just the rhythm.*

473. Correct answer D. Call the extracorporeal support team now, since the decision must come early

Extracorporeal support is a rescue for selected patients when conventional resuscitation is failing, and the team needs time to cannulate, so the decision to call must be made early, often within 10 to 15 minutes. Candidates are young patients with a presumed reversible cause such as drug toxicity, pulmonary embolism, or hypothermia and no contraindication to anticoagulation. Waiting 45 minutes forfeits the opportunity. Age over 60 is not a requirement. Drug toxicity is itself a reason to call.

Key principle: *Call early for extracorporeal support in suitable patients with a reversible cause.*

474. Correct answer D. Give a small dose of an amnestic agent once perfusion is stable

A patient who has regained circulation but is paralyzed and no longer receiving anesthetic may be aware, so a small dose of an amnestic agent is appropriate once perfusion is stable. Drugs that were stopped wash out slowly and in a low-flow state accumulate in the tissues, so the patient may be profoundly vasodilated, and a full dose of an anesthetic or a full concentration of the volatile agent risks hypotension. Assuming unconsciousness after an arrest is not safe in a paralyzed patient.

Key principle: *After return of circulation, give a small amnestic dose to a paralyzed patient once perfusion is stable.*

475. Correct answer A. Vasopressin, since it works when acidosis blunts catecholamine response

Vasopressin no longer has a place as a routine arrest drug, but it serves in the peri-arrest and post-resuscitation period when vasoplegia complicates the picture, because it acts when acidosis and catecholamine exhaustion have blunted the response to epinephrine. It can be used with norepinephrine and volume to keep cerebral and coronary perfusion adequate. Arrest doses of epinephrine are not for a patient with a pulse. Routine bicarbonate and magnesium are not recommended without a specific indication.

Key principle: *Vasopressin helps post-resuscitation vasoplegia when catecholamine response is blunted.*

476. Correct answer A. Remove the insufflation, release retraction, and control bleeding

Surgeons have specific duties in an arrest: stop the operation or make the patient safe, control bleeding with packing or pressure, release retraction, and remove insufflation. Insufflation can impede venous return and the surgeon's knowledge of the field may name the cause. Continuing the dissection or raising pressure endangers the patient. The surgeon is a member of the team, not a spectator, and should be told the expected role in the first minute.

Key principle: *The surgeon helps: release insufflation and retraction, control bleeding, and share what happened in the field.*

477. Correct answer B. Pause at the next check to restate the working diagnosis and read the causes aloud

Fixation, in which the team repeats the same therapy while the cause lies elsewhere, is the hardest error to recognize, and a scheduled pause at each cycle to restate the working diagnosis helps. A reader who reads the list of reversible causes aloud at every rhythm check supports this without overriding the leader. Giving epinephrine more often does not address a cause. Deferring the search wastes the time in which the cause is treatable. The leader should decide with input from the team, not hand the diagnosis to one member.

Key principle: *Break fixation with a scheduled pause to restate the diagnosis and read the causes aloud.*

478. Correct answer C. Continue, since a reversible cause is being treated and the rhythm is shockable

Perioperative arrest often has a reversible cause, a healthy underlying brain, and a team that can treat it, so resuscitation may justifiably continue longer than in other settings and there is no fixed duration. Effort should continue while a reversible cause is being treated and some sign of potential recovery exists, such as a shockable rhythm. No single finding, including a low end-tidal value, should decide the question alone.

Key principle: *There is no fixed duration; continue while a reversible cause is being treated and signs of recovery persist.*

479. Correct answer E. Complete an abbreviated operation, a decision made jointly and stated aloud

The decision about the operation comes after the heart restarts. If the arrest was caused by the surgery itself, such as massive bleeding, the operation may need to be completed in an abbreviated fashion. The decision belongs to the surgeon and anesthesiologist together, is based on the stability of the patient, and should be stated aloud so the whole team understands what comes next. Stopping with the source untreated invites another arrest, and a unilateral decision excludes the surgeon who knows the field.

Key principle: *After return of circulation, decide jointly about the operation and state the plan aloud.*

480. Correct answer C. Speak soon, plainly, and in person with the surgeon, without speculating or assigning blame

Communication with the family is the joint responsibility of the anesthesiologist and surgeon, and should be soon, plain, and in person, describing what happened, what was done, what is known and not known, and what happens next. Speculation about causes before the review is complete and assigning blame should be avoided. Prompt, honest disclosure is ethically required and protects the relationship. Silence until the review ends, a premature definitive cause, or a written statement alone each fall short.

Key principle: *Disclose promptly, honestly, and in person; do not speculate or assign blame.*

Group 25. Arrest in Special Positions and Populations

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481. Correct answer B. One pad over the right scapula and the other at the left mid-axillary line

In a prone patient who cannot be turned promptly, an anterior-posterior arrangement (one pad over the right scapula and the other at the left mid-axillary line, or on the anterior chest if reachable) or a bi-axillary arrangement is acceptable. The first shock should not be delayed for a perfect arrangement when any workable position exists. Pads over the lumbar spine or buttocks lie below the heart, so little current crosses the myocardium. Pads placed over the occiput are remote from the heart. Overlapping pads shunt current across the skin and do not traverse the chest.

Key principle: *Do not delay the first shock in a prone arrest; use an anterior-posterior or bi-axillary pad arrangement.*

482. Correct answer D. The surgeon releases the pins, one named person protects the head, and another removes the frame

Release from fixation is the step rehearsed least, and the surgeon is the person who can release pins and clamps fastest. Roles for protecting the head and neck and for removing the frame should be named before the operation so that the turn takes about a minute. Assigning roles during the arrest wastes the minutes that matter. Finishing the closure first delays resuscitation, because the surgical site is not the priority when the heart has stopped. One clinician working alone cannot release the pins and manage the airway at the same time.

Key principle: *Name who releases the pins, who protects the head, and who removes the frame before the operation begins.*

483. Correct answer A. Concealed retroperitoneal hemorrhage from injury to a great vessel

Great vessel injury during lumbar discectomy produces hemorrhage into the retroperitoneum with little evidence at the wound, and it fits the steady fall in pressure, the rising heart rate, and the low end-tidal carbon dioxide of a low-flow state. Venous air entry occurs when the surgical site is above the heart, not below it. A kinked or displaced tube would be shown by the capnography trace and the ventilation. Abdominal compression from the frame would have appeared at positioning and not after an hour of stability. The capnography trace and arterial line help separate airway from circulatory causes.

Key principle: *A dry wound with falling pressure after lumbar surgery suggests concealed retroperitoneal bleeding.*

484. Correct answer E. The surgeon begins internal cardiac massage through the open chest at once, before any other step

With the chest already open, the surgeon can begin internal massage immediately, which is more effective than external compressions in most cases. Rolling the patient or closing the chest first delays circulation and makes the compressions less effective. External compressions in the lateral position are ineffective, which is why a closed-chest lateral arrest calls for a supine position. Waiting for the table to be flattened leaves the heart without flow. Internal defibrillation can follow using paddles on the heart at low energy.

Key principle: *In a lateral arrest with an open chest, the surgeon starts internal massage immediately.*

485. Correct answer A. Turn her supine, begin compressions, and return to two-lung ventilation with 100 percent oxygen

Compressions are ineffective in the lateral position, so the table is flattened, supports are removed, and the patient is rolled supine. One-lung ventilation is abandoned in favor of two-lung ventilation with pure oxygen, and the tube position is checked, with a fiberoptic examination if resistance or loss of capnography persists. Continuing one-lung ventilation leaves a large shunt. Delaying oxygen to check the tube accepts avoidable hypoxemia. Compressions over the dependent chest wall do not generate adequate flow, and a vasopressor first delays circulation.

Key principle: *In a closed-chest lateral arrest, turn supine and switch to two-lung ventilation with pure oxygen.*

486. Correct answer C. Lower him supine or head-down, flood the field with saline, and start compressions

In a sitting or beach chair arrest the patient is lowered to the supine or head-down position at once, the field is flooded with saline if air is suspected, and compressions begin. The delay lies in unlocking the chair or head holder, so the release should be practiced before the first incision. Compressions in the sitting position are ineffective. Drugs and fluid first postpone the interventions that matter. Aspiration of a line may help, but it must not delay compressions and repositioning.

Key principle: *Sitting arrest: lower the patient, flood the field with saline, and begin compressions.*

487. Correct answer D. Give amiodarone and use small-dose epinephrine or none, on the surgeon's advice

After cardiac surgery the usual full-dose epinephrine bolus is not recommended, because it can cause severe hypertension and tachyarrhythmia when circulation returns and may disrupt suture lines. A single antiarrhythmic, with amiodarone preferred, is given while the chest is prepared, and epinephrine is given in small doses or not at all on the advice of the surgeon or intensivist. Calcium and magnesium are for specific indications, not routine use. Withholding every drug is not required. Atropine has no role in ventricular fibrillation.

Key principle: *After cardiac surgery, use amiodarone and avoid routine full-dose epinephrine; mechanical fixes come first.*

488. Correct answer B. Proceed to emergency reopening of the chest without waiting for imaging

A drain that suddenly stops with falling pressure and rising filling pressures suggests tamponade from a clotted drain. Ultrasound can be unreliable in the early postoperative chest, so the decision to reopen should not wait for imaging. Insisting on confirmation, transfer to a scanner, or prolonged compressions all consume the minutes that decide survival. Gentle clearing of the tubes may be tried per local practice, but it must not replace preparation for reopening in a peri-arrest patient.

Key principle: *Suspected tamponade after cardiac surgery: reopen the chest without waiting for imaging.*

489. Correct answer E. Connect them to the external pacemaker and pace at a high rate at maximum output

In asystole or severe bradycardia after cardiac surgery, the wires are connected to an external pacemaker and the heart is paced at a high rate at maximum output, in ventricular or dual-chamber mode as appropriate. Turning the pacemaker off is the step for pulseless electrical activity, where pacing may mask fibrillation. Low output may fail to capture. Deferring pacing for epinephrine wastes the advantage of wires already in place. Transcutaneous pads are a backup if the wires fail.

Key principle: *Pace through the wires in asystole after cardiac surgery; switch the pacemaker off in pulseless activity.*

490. Correct answer C. 20 joules

Internal defibrillation applies paddles directly to the heart, so the energy needed is far lower than for external shocks, and the first shock starts at 20 joules. External-range settings of 100 to 360 joules risk myocardial injury when delivered directly to the epicardium. Energy can be increased if the first shock fails. The setting is chosen by the person holding the paddles in agreement with the anesthesiologist, who checks that the device is not in synchronized mode.

Key principle: *Internal defibrillation starts at 20 joules, far below the external setting.*

491. Correct answer B. Place new venous access above the diaphragm and give the drug through it

In late pregnancy the enlarged uterus compresses the vena cava, so drugs given below the diaphragm may reach the heart slowly. Access should be placed above the diaphragm. A femoral line shares the same problem. The endotracheal route is unreliable and not preferred when venous access can be obtained. Waiting for delivery withholds therapy during the period when it is needed, and hysterotomy is a separate decision with its own timing.

Key principle: *In late pregnancy, use venous access above the diaphragm for resuscitation drugs.*

492. Correct answer A. Intubate with a smaller tube using a video laryngoscope and ventilate at 10 breaths per minute

Pregnant women have a higher incidence of failed intubation and tolerate apnea poorly, so the most experienced operator should intubate with a smaller tube and a video laryngoscope, with a supraglottic device as backup. Ventilation is with 100 percent oxygen at 10 breaths per minute, because hyperventilation raises intrathoracic pressure and impairs venous return even more in the gravid patient. Prolonged mask ventilation increases the risk of aspiration and hypoxemia. A larger tube and a faster rate both worsen the situation.

Key principle: *In maternal arrest, the most experienced operator secures the airway; ventilate at 10 breaths per minute.*

493. Correct answer E. Stop the magnesium infusion and give 1 g of calcium chloride intravenously

Magnesium toxicity in a patient with reduced renal function causes loss of reflexes, respiratory depression, and cardiac conduction disturbances. The infusion is stopped and intravenous calcium is the antidote, either 1 g of calcium chloride or 3 g of calcium gluconate. Halving the infusion leaves toxicity in place. Naloxone treats opioid effects and lipid emulsion treats local anesthetic toxicity, neither of which is suggested here. Dantrolene would worsen weakness.

Key principle: *Magnesium toxicity: stop the infusion and give intravenous calcium.*

494. Correct answer C. Support the airway, give magnesium sulfate, and control the blood pressure

A seizure in late pregnancy with hypertension is eclampsia until shown otherwise, and magnesium sulfate is the treatment. The seizing patient can become hypoxic and arrest if the airway is not protected, so airway support, oxygen, and blood pressure control are first measures. Lipid emulsion has no role when no local anesthetic has been given. Delivery may be needed later but does not replace stabilization. Imaging comes after the patient is stable, and delaying airway support risks hypoxic arrest.

Key principle: *Seizure in late pregnancy with hypertension: protect the airway and give magnesium sulfate.*

495. Correct answer E. Neither displacement nor hysterotomy is indicated, because the fundus is below the umbilicus

Left uterine displacement is used when the fundus reaches or exceeds the level of the umbilicus, which corresponds to about 20 weeks of gestation, and resuscitative hysterotomy is considered under the same condition. At 14 weeks the uterus does not compress the great vessels enough to matter, and delivery would not relieve aortocaval compression or offer a viable infant. Table tilt reduces compression effectiveness. The timing of four minutes applies only when the uterus is at or above the umbilicus.

Key principle: *Displacement and hysterotomy apply only when the fundus is at or above the umbilicus.*

496. Correct answer E. Give 100 percent oxygen, hyperventilate, and give a pulmonary vasodilator

Pulmonary hypertensive crises respond to oxygen, hyperventilation, and pulmonary vasodilators, but only if the diagnosis is made. Hypoxemia, hypercapnia, and acidosis raise pulmonary vascular resistance, so allowing

hypercapnia or lowering oxygen worsens the crisis. A fluid bolus does not address the right-sided afterload. Compressions are reserved for loss of perfusion that persists despite correction of oxygenation. A beta blocker would reduce an already compromised cardiac output.

Key principle: *Pulmonary hypertensive crisis: oxygen, hyperventilation, and a pulmonary vasodilator.*

497. Correct answer A. 1.2 mL

The pediatric arrest dose of epinephrine is 0.01 mg/kg, which is 0.1 mL/kg of the 0.1 mg/mL solution, repeated every 3 to 5 minutes. For 12 kg this is 0.12 mg, or 1.2 mL. Tenfold errors, such as 12 mL or 0.12 mL, are the commonest and most dangerous, particularly when decimal points are misplaced. Half and double doses are also wrong. A length-based tape or a preprinted chart reduces arithmetic errors under stress.

Key principle: *Pediatric epinephrine is 0.01 mg/kg, or 0.1 mL/kg of the 0.1 mg/mL solution.*

498. Correct answer B. 0.1 mg, because a minimum single dose applies

Atropine for pediatric bradycardia from a vagal cause is 0.02 mg/kg intravenously, with a minimum single dose of 0.1 mg. Oxygenation and ventilation come first, and they have already been addressed here. Giving less than the minimum may produce paradoxical slowing. Larger doses add toxicity without benefit, and an adult dose would be a tenfold or greater overdose. If the rate stays below sixty with poor perfusion, compressions and epinephrine follow.

Key principle: *Pediatric atropine is 0.02 mg/kg with a minimum single dose of 0.1 mg.*

499. Correct answer D. 15 compressions to 2 breaths

With two rescuers and no advanced airway, the ratio for a child is 15 to 2, whereas a single rescuer uses 30 to 2. The ratio of 3 to 1 is for newborn resuscitation. Continuous compressions with a breath every 2 to 3 seconds apply only after an advanced airway is in place. Ventilation matters more in children than in adults because hypoxia is the commonest cause of arrest, so effective breaths carry the same priority as compressions.

Key principle: *Two rescuers, no advanced airway, child: 15 to 2.*

500. Correct answer B. Use a radiant warmer and plastic wrap, and check the glucose early in the resuscitation

Newborns lose heat rapidly and are prone to hypoglycemia, so a radiant warmer and plastic wrap are used for preterm infants and the glucose is checked early and low values treated. Towels on an open table and a warm room alone do not prevent heat loss in a small infant. A wet blanket evaporates heat. Skin-to-skin placement under drapes is impractical while the mother is anesthetized and the infant needs resuscitation. Giving dextrose before measuring risks both delay and error.

Key principle: *For preterm infants use a radiant warmer and plastic wrap, and check glucose early.*

Group 26. Peri-Arrest Arrhythmias, Pacing, and Cardioversion

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501. Correct answer D. Apply pacing pads and prepare to pace, because the block may progress without warning

Fixed PR intervals with sudden dropped beats and a wide QRS describe Mobitz type II block, which arises below the node and can progress to complete block without warning. Pacing capability should be ready even though she is stable now. Mobitz type I, with progressive PR prolongation, is the benign nodal form. Atropine acts on the node and is unreliable in infranodal block. Esmolol and diltiazem would slow conduction further and could precipitate complete block.

Key principle: *Mobitz type II block needs pacing readiness because it can progress to complete block suddenly.*

502. Correct answer E. Glucagon, high-dose insulin with dextrose, and calcium

Bradycardia from beta-blockers and calcium channel blockers is typically refractory to atropine. Treatment includes glucagon, high-dose insulin, and calcium in addition to pacing. Exceeding the atropine maximum adds toxicity

without reliable benefit. Antibody fragments treat digoxin toxicity and bicarbonate treats sodium channel blockade, while naloxone and flumazenil address sedative effects that do not explain this picture.

Key principle: *Beta-blocker bradycardia refractory to atropine: glucagon, high-dose insulin, calcium, and pacing.*

503. Correct answer B. Treat the bradycardia promptly with atropine and have epinephrine ready

Sugammadex has been associated, rarely, with marked bradycardia and even asystole within minutes of administration. The rate should be watched, atropine and epinephrine kept immediately at hand, and the slowing treated promptly rather than attributed to emergence. Neostigmine itself causes bradycardia through muscarinic stimulation and would worsen it. A second dose of sugammadex does not treat the cause. Adenosine slows nodal conduction and would deepen the bradycardia.

Key principle: *Bradycardia after sugammadex needs prompt treatment; do not attribute it to emergence.*

504. Correct answer D. Confirm oxygenation, ventilate with 100 percent oxygen, and give atropine

A second dose of succinylcholine in a child, particularly without an anticholinergic, can cause profound bradycardia, and bradycardia in a child is hypoxia until proven otherwise. Oxygenation and ventilation are confirmed first and atropine is given for the vagal effect. Observation is unsafe when the pressure is low, because infants depend on rate to maintain output. A shock is for tachyarrhythmia, an adult infusion rate is inappropriate, and neostigmine would add muscarinic bradycardia.

Key principle: *A second dose of succinylcholine in a child can cause profound bradycardia; ventilate and give atropine.*

505. Correct answer D. Continue the beta-blocker, avoid QT-prolonging antiemetics, keep electrolytes in range, and have magnesium ready

A patient with long QT syndrome needs a plan to avoid triggers: avoid QT-prolonging antiemetics such as ondansetron and droperidol, keep potassium and magnesium in range, minimize sympathetic surges, continue beta-blockers, and have magnesium and a defibrillator ready. Stopping the beta-blocker removes protection. A smooth emergence matters, so light anesthesia with abrupt waking is a poor choice. Amiodarone prolongs the QT interval and is not the rescue drug for torsades. Atropine would raise the sympathetic tone and the rate unpredictably.

Key principle: *Long QT: continue beta-blockers, avoid QT-prolonging drugs, correct electrolytes, and keep magnesium ready.*

506. Correct answer C. Synchronized cardioversion starting at 200 joules biphasic, with the pads already placed

Atrial fibrillation with hypotension and altered mentation is unstable, and the treatment is synchronized cardioversion, with the 2025 starting energy for atrial fibrillation or flutter at 200 joules biphasic. A very low energy is likely to fail. An unsynchronized shock risks inducing ventricular fibrillation. Esmolol is for rate control in a stable patient and could worsen the hypotension. Adenosine does not treat atrial fibrillation.

Key principle: *Unstable atrial fibrillation: synchronized cardioversion at 200 joules biphasic.*

507. Correct answer A. Immediate unsynchronized defibrillation

Pulseless polymorphic ventricular tachycardia is treated as cardiac arrest with an immediate unsynchronized shock, because the device often cannot synchronize to a changing waveform. Synchronized cardioversion is for rhythms with a pulse and may not deliver at all in this setting. Magnesium and amiodarone are not substitutes for a shock in a pulseless patient. Adenosine as a diagnostic test wastes time and is unsafe in an irregular wide-complex rhythm.

Key principle: *Pulseless polymorphic tachycardia: shock at once, unsynchronized.*

508. Correct answer A. Give a reduced dose, because dipyridamole potentiates the effect of adenosine

Adenosine is much more potent in patients taking dipyridamole or carbamazepine, and in transplanted hearts, so a reduced dose is necessary. Higher doses would be needed with methylxanthines such as caffeine or theophylline, not with dipyridamole, which blocks adenosine reuptake. Giving the usual dose risks prolonged asystole. Verapamil is a second-line option. A stable patient does not need electricity, and adenosine is not contraindicated with a reduced dose.

Key principle: *Reduce the adenosine dose with dipyridamole, carbamazepine, or a transplanted heart.*

509. Correct answer C. Recognize flutter and control the ventricular rate with esmolol or diltiazem

Adenosine is not a treatment for atrial fibrillation or flutter. It transiently blocks the node and unmasks the flutter waves, which makes the diagnosis. In a stable patient the rate is then controlled with esmolol, diltiazem, or amiodarone, and synchronized cardioversion is used if the patient becomes unstable. Repeating adenosine adds risk without benefit. Atropine would increase the ventricular rate, calcium has no role, and an unsynchronized shock is wrong in a patient with a pulse.

Key principle: *Adenosine unmasks flutter but does not treat it; control the rate if stable.*

510. Correct answer D. Control the rate and plan anticoagulation before any attempt at cardioversion

Fibrillation lasting more than forty-eight hours carries a risk of thromboembolism with cardioversion, so in a stable patient the rate is controlled with esmolol, diltiazem, or amiodarone and anticoagulation is discussed with the surgeon and cardiologist. Immediate cardioversion is reserved for instability. Adenosine does not treat atrial fibrillation. Chemical conversion carries the same embolic risk as electrical conversion. A rate of 130 in a postoperative patient deserves control and a search for triggers.

Key principle: *Atrial fibrillation lasting over 48 hours: rate control and anticoagulation planning before cardioversion.*

511. Correct answer E. Restore blood pressure and hemoglobin, slow the heart rate, and give oxygen

Most perioperative ischemia is a supply-demand imbalance, from tachycardia and pain on the demand side and hypotension, anemia, and hypoxemia on the supply side. The treatment is to correct the cause: restore the pressure, correct the anemia, slow the rate, and oxygenate. Nitroglycerin can drop the pressure further in a hypovolemic patient. A beta-blocker bolus worsens hypotension. Anticoagulation in an actively bleeding surgical patient is weighed with the surgeon and not given reflexively.

Key principle: *Perioperative ischemia is usually supply-demand imbalance; treat pressure, anemia, rate, and oxygenation.*

512. Correct answer B. Have pacing ready, because passage may produce complete heart block

In a patient with left bundle branch block, damage to the right bundle during catheter passage can produce complete heart block, so a pacing capability should be ready before the catheter is floated. Lidocaine and procainamide do not prevent block and procainamide can slow conduction. Floating without the balloon increases the chance of trauma and ectopy. Pads attached to a device that is switched off cannot pace in time.

Key principle: *Have pacing ready before floating a pulmonary artery catheter in left bundle branch block.*

513. Correct answer A. Interference is unlikely at this distance, but the pulse should be monitored by plethysmography

Monopolar electrosurgery below the umbilicus carries a low risk of interference because the current path is far from the device, whereas surgery above the umbilicus raises the risk. Bipolar cautery and ultrasonic scalpels do not generate the same interference. Even so, the electrocardiogram can be distorted by cautery, so a plethysmograph or arterial line confirms that a pulse is present. A recent interrogation is reassuring. Programming is not routinely required for low-risk surgery.

Key principle: *Risk of interference is low below the umbilicus; confirm the pulse with plethysmography.*

514. Correct answer B. Ask the device team to program asynchronous pacing and suspend tachycardia therapies

A magnet over a defibrillator usually suspends tachycardia detection and therapy but does not change the pacing function, so a pacemaker-dependent patient with a defibrillator may need reconfiguration. The device team can set an asynchronous mode and turn off therapies, and external pads stay in place until therapies are restored. Cautery near the neck is a high-risk site. Devices are not shielded, and removing pads leaves the patient unprotected.

Key principle: *A magnet on a defibrillator does not make pacing asynchronous; reconfigure if the patient is dependent.*

515. Correct answer D. At least 8 cm from the generator, preferably anterior-posterior

Pads should be placed at least about 8 cm from the generator, preferably in an anterior-posterior arrangement, and never over the generator. A shock should not be delayed to find an ideal position or to interrogate the device, which is checked afterward because a shock may change its settings. Reduced energy risks failure to defibrillate. Chest compressions are safe. Treat the arrest first and the device later.

Key principle: *Place pads at least 8 cm from the generator and never delay a shock for the device.*

516. Correct answer C. Confirm therapies are restored and interrogate the device before he leaves monitoring

Normal settings must be restored after surgery, and the device should be checked if interference was significant, if the device was reconfigured, if cardioversion or defibrillation was used, or if surgery was close to it. Interrogation should occur before the patient leaves a monitored setting. Therapies do not return automatically after programmer suspension. Waiting for symptoms is unsafe because a lethal arrhythmia may be the first sign. A patient with the device left off is exposed to arrhythmia, and the loss may not be discovered until much later.

Key principle: *Restore and check the device before the patient leaves monitoring.*

517. Correct answer C. Check and replace magnesium, because low magnesium can prevent potassium correction

Hypokalemia is often accompanied by hypomagnesemia, which can prevent correction of potassium until magnesium is replaced. Magnesium is also a safe empiric therapy for ventricular ectopy and torsades. Insulin with dextrose and bicarbonate lower potassium further by shifting it into cells. Calcium does not raise potassium. Amiodarone prolongs the QT interval and does not address the cause, and treating the rhythm without the number risks repeated failure.

Key principle: *Refractory hypokalemia: replace magnesium too.*

518. Correct answer A. Calcium chloride via a large vein, titrated to the ionized calcium

Hypocalcemia is common after massive transfusion and prolongs the QT interval and depresses contractility. Ionized calcium, not total calcium, is the relevant measurement, and treatment is intravenous calcium chloride or gluconate through a large vein titrated to the ionized value. Magnesium alone does not correct the deficit. Bicarbonate can lower ionized calcium further. Atropine does not address a low ionized calcium and the problem here is not bradycardia.

Key principle: *After massive transfusion, treat low ionized calcium with intravenous calcium.*

519. Correct answer E. Digoxin-specific antibody fragments, with potassium and magnesium correction

Digoxin toxicity can produce nearly every arrhythmia, including bidirectional ventricular tachycardia, and hypokalemia and hypomagnesemia enhance it. Treatment includes correction of electrolytes, magnesium, and digoxin-specific antibody fragments for life-threatening rhythms. Cardioversion in digoxin toxicity can precipitate refractory ventricular fibrillation and does not remove the cause. Amiodarone, procainamide, and verapamil do not reverse toxicity and may worsen conduction or hypotension.

Key principle: *Life-threatening digoxin toxicity needs antibody fragments and correction of electrolytes.*

520. Correct answer B. Rewarm actively and continue resuscitation, because shocks may fail until he is warm

Below about 30 degrees Celsius the heart is irritable and ventricular fibrillation may occur with minimal stimulation, and defibrillation attempts may fail until the patient has been rewarmed. The temperature must be corrected as part of the treatment, and resuscitation continues during rewarming. Repeated high-energy shocks without rewarming are unlikely to succeed. Stopping is premature in a hypothermic patient. Drugs are often ineffective in the cold, and pacing does not treat ventricular fibrillation.

Key principle: *In severe hypothermia, rewarm; defibrillation may fail until the patient is warm.*

Group 27. Post-Arrest Care and Mechanical Support

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521. Correct answer E. Decide together and explicitly, favoring damage control closure if he cannot tolerate more surgery

After return of circulation the team must decide whether to stop, abbreviate, or complete the procedure, based on whether the cause was found and corrected, whether the patient can tolerate more surgery, and the risk of leaving the operation unfinished. In an emergency laparotomy the answer is often a damage control closure. A shared decision made out loud protects the patient from pressure to finish simply because the incision is open. Unilateral decisions by either party, or waiting for improvement while operating, risk further injury.

Key principle: *After return of circulation, decide together whether to stop, abbreviate, or finish the operation.*

522. Correct answer D. Blood gas with potassium, glucose, lactate, and hemoglobin, a twelve-lead electrocardiogram, an echocardiogram, and a drug review

The reversible causes are hypoxia, hypovolemia, acidosis, potassium disturbance, hypothermia, tension pneumothorax, tamponade, toxins, and thrombosis. A blood gas, an electrocardiogram, a bedside echocardiogram, and a review of the drugs given in the preceding ten minutes cover most of them within minutes. The cause shapes the vasopressor choice and candidacy for extracorporeal support, so delaying the search is unwise. Head imaging and electroencephalography address the brain, not the cause. Catheterization is not required for every arrest, since many are not coronary.

Key principle: *Search for the cause in the first hour with a gas, an electrocardiogram, an echocardiogram, and a drug review.*

523. Correct answer E. Aim for normothermia and prevent fever, because hypothermia would worsen clotting

Current guidance asks for active temperature control in every comatose survivor, with a target anywhere from mild hypothermia to normothermia. A patient who is bleeding, coagulopathic, or hemodynamically fragile after a surgical arrest is a poor candidate for deliberate cooling, because hypothermia worsens clotting, provokes arrhythmia, and raises infection risk. Normothermia with fever prevention is safer. Cooling toward 32 or 33 degrees would deepen the coagulopathy. Waiting for fever to reach 38.5 degrees allows the injury that fever causes.

Key principle: *A bleeding or coagulopathic survivor should be kept normothermic, with fever prevented.*

524. Correct answer D. Maintain temperature control for at least 36 hours, rewarm slowly, and prevent fever

Current guidance asks that temperature control be maintained for at least 36 hours and that fever be prevented for 36 to 72 hours. Rewarming should be slow and controlled, because rebound hyperthermia is common and harmful. Stopping at 12 hours or letting the patient drift to a fever removes the protection against the second wave of injury. Stopping temperature monitoring is unsafe. A week of cooling is not supported and adds infection and coagulation risks.

Key principle: *Hold temperature control for at least 36 hours and rewarm slowly.*

525. Correct answer E. It hides seizures and clouds the examination, so doses and times must be recorded

Shivering raises oxygen consumption and defeats cooling, so sedation and, when needed, neuromuscular blockade are part of the plan. The cost is that blockade hides seizures and weakens the neurologic examination, so the sedatives and blockers given, with their times and durations, must be recorded to decide when an examination can be trusted. Blockade does not raise intracranial pressure in general, does not prevent cooling, and does not block temperature probes. Cortical evoked potentials are recorded centrally and are not abolished by a peripheral blocker in the same way.

Key principle: *Neuromuscular blockade hides seizures and clouds the examination; record every dose.*

526. Correct answer A. Bilaterally absent cortical responses on somatosensory evoked potentials

Bilaterally absent cortical somatosensory evoked potential responses are among the most specific findings for a poor outcome, and a prediction is credible only when two or more tests agree and no confounder explains them. A reactive continuous background argues for a better outlook. An early examination under sedation is unreliable. Neuron-specific enolase is interpreted at 48 and 72 hours, not at 6 hours. Myoclonus within the first 72 hours is a warning sign and not a stand-alone predictor.

Key principle: *Bilaterally absent cortical evoked potentials are highly specific, and prognosis needs concordant tests.*

527. Correct answer C. Obtain electroencephalography to separate myoclonus from seizure, and treat seizures promptly

Seizures are treated promptly, but prophylactic anticonvulsants are not recommended. Myoclonus is distinguished from seizure by electroencephalography, because the two are managed differently and carry different prognostic weight. Aggressive suppression of every epileptiform discharge is not required. Myoclonic status within 72 hours is a warning sign for prognosis, so ignoring it is wrong. Multiple drugs given at once cause sedation that clouds the later examination.

Key principle: *Treat seizures promptly, give no prophylaxis, and use electroencephalography to separate myoclonus from seizure.*

528. Correct answer A. Insulin infusion to keep the glucose between 70 and 180 mg/dL, with frequent checks

Glucose is managed to avoid both hypoglycemia below 70 mg/dL and hyperglycemia above 180 mg/dL, because both injure the brain and tighter control has caused harm. An insulin infusion with checks every 1 to 2 hours at first achieves this. Tight control to 80 to 110 risks hypoglycemia. Leaving a glucose of 260 untreated or waiting for 300 allows hyperglycemic injury. Deliberate hyperglycemia is not neuroprotective.

Key principle: *Keep glucose between 70 and 180 mg/dL after arrest.*

529. Correct answer D. A tidal volume of 6 to 8 mL/kg of predicted body weight, with low plateau pressures

Many post-arrest patients have aspirated, are volume overloaded, or have contused lungs, and the ventilator strategy for acute respiratory distress syndrome applies. Tidal volumes are 6 to 8 mL/kg of predicted body weight, not actual weight, with low plateau pressures, and the PaCO2 target is 35 to 45 mm Hg confirmed on a blood gas. Larger volumes cause lung injury. Permissive hypercapnia raises intracranial pressure through vasodilation. An unmeasured tidal volume cannot be protective.

Key principle: *Use 6 to 8 mL/kg of predicted body weight after arrest.*

530. Correct answer A. Support the pressure with a vasopressor, give fluid sparingly, and keep positive end-expiratory pressure low

A distended right ventricle depends on adequate coronary perfusion pressure, so systemic pressure is maintained with a vasopressor and fluid is given sparingly. Hypoxemia, hypercapnia, and acidosis raise pulmonary vascular resistance and should be corrected, high positive end-expiratory pressure should be avoided, and an inhaled pulmonary vasodilator can lower right-sided afterload without lowering the systemic pressure. Large boluses overdistend the ventricle. An intravenous vasodilator lowers systemic pressure and coronary perfusion. Hypercapnia raises pulmonary vascular resistance.

Key principle: *In right ventricular failure after arrest, support the pressure, limit fluid, and avoid high positive end-expiratory pressure.*

531. Correct answer B. Look for low cardiac output, ongoing bleeding, or an undrained source of sepsis

Lactate is high immediately after resuscitation and should fall as flow is restored, so the trend matters more than a single value. A rising lactate despite a mean pressure of 65 or more calls for a closer look at cardiac output, ongoing bleeding, and an undrained source of sepsis. The number to chase is perfusion, and pressure is only one route to it. Bicarbonate treats the number and not the cause, and stopping the vasopressor risks hypotension.

Key principle: *A rising lactate despite adequate pressure means perfusion is still inadequate; find out why.*

532. Correct answer E. Short-acting agents such as propofol and remifentanil, with a train-of-four before assessment

Short-acting agents such as propofol and remifentanil clear quickly and allow a meaningful neurologic assessment within hours of stopping the infusion. Long-acting benzodiazepines and accumulating opioid metabolites can cloud the picture for days, especially in kidney or liver impairment. Neuromuscular blockers should be used sparingly and for the shortest time, and a train-of-four should confirm recovery before assessment. Recording the last dose of each drug on the chart lets the examiner know what is still circulating.

Key principle: *Choose short-acting sedatives and confirm blocker recovery before examining.*

533. Correct answer B. A young patient with a witnessed shockable arrest from massive pulmonary embolus and 10 minutes of low flow

The best candidates have a witnessed arrest, ideally one witnessed by the anesthesiologist, with a shockable rhythm or a reversible cause such as pulmonary embolism, coronary occlusion, hypothermia, or drug toxicity. They are young and well enough to benefit and have a short low-flow time, since outcomes deteriorate steeply as compressions continue. Unwitnessed arrests with long compressions, irreversible causes, and end-stage disease are poor candidates. Extracorporeal support treats nothing by itself, so a reversible problem is essential.

Key principle: *Select extracorporeal resuscitation for witnessed arrests with a reversible cause and short low-flow time.*

534. Correct answer C. Benefit is limited to selected patients treated quickly by experienced teams; large out-of-hospital trials were not positive

One single-center randomized trial in refractory ventricular fibrillation reported higher survival with an invasive pathway that included extracorporeal support, whereas the larger Prague and INCEPTION trials in out-of-hospital arrest did not show a significant benefit. Observational in-hospital data are more favorable, especially from centers that deploy teams quickly. The fair summary is that benefit lies in selected patients treated fast by experienced teams, and indiscriminate use is neither effective nor a sound use of a scarce resource.

Key principle: *Extracorporeal resuscitation helps selected patients treated fast; the large trials were not clearly positive.*

535. Correct answer A. The circuit carries most of the output, so a low end-tidal value is expected

End-tidal carbon dioxide may read low once the circuit carries most of the output, because little blood passes through the native lungs. A falling value after flow is established does not by itself signal trouble, and ventilation can be confirmed by other means such as breath sounds and arterial gas. Reintubation, stopping flow, or replacing the oxygenator on this finding alone would be unnecessary and could be harmful. Good pressure and lower limb saturation suggest the circuit is working.

Key principle: *Low end-tidal carbon dioxide after full flow starts is expected, not a sign of tube failure.*

536. Correct answer E. Individualize heparin, possibly lower or delayed, and check platelets, fibrinogen, and hemoglobin

Heparin is given at cannulation and then adjusted by activated clotting time or other protocols, and a freshly operated patient may need a lower target or a delay. Platelets, fibrinogen, and hemoglobin are checked often because the circuit consumes them, and bleeding from anticoagulation, cannulation sites, and consumptive coagulopathy is a leading complication. A standard bolus ignores the surgical wounds. Permanent omission of anticoagulation risks circuit thrombosis. Daily checks are too infrequent, and fixed dosing ignores the changing clotting status.

Key principle: *Individualize anticoagulation after surgery and monitor platelets, fibrinogen, and hemoglobin often.*

537. Correct answer B. Arrest time, suspected cause, drugs given, cooling plan, and sedatives that cloud the examination

A clear handover gives the intensive care team the time of the arrest, the suspected cause, the drugs given, the cooling plan, and the sedatives that will cloud the examination. Handing over only the rhythm and the last pressure sends the receiving team in without what it needs. Transport also needs a transport ventilator, full monitoring, a defibrillator, resuscitation drugs, and an infusion pump with working batteries, with pressure and oxygen targets checked before moving and again on arrival.

Key principle: *Hand over the arrest time, cause, drugs, cooling plan, and sedatives that cloud the examination.*

538. Correct answer D. Add a dated addendum explaining the correction, leaving the original entry unaltered

The anesthesia record should be factual and contemporaneous, and an entry must never be altered after the fact. If a correction is needed, a dated addendum is added. Overwriting or deleting entries damages the integrity of the record and the credibility of the clinicians. Leaving an error uncorrected leaves the receiving team and the review with wrong data. The surgeon should not edit the anesthesia record, which belongs to the anesthesia team.

Key principle: *Never alter an entry; correct with a dated addendum.*

539. Correct answer C. Notify the organ procurement organization per policy, and let trained specialists raise it

When the evidence points to a poor outcome, the discussion shifts to goals of care guided by what the patient would have wanted. Organ donation is a further consideration, and the local organ procurement organization is notified according to policy. Trained specialists, and not the clinical team, then raise the subject with the family, which separates the prognostic conversation from the request. Raising donation early or by the treating team blurs those roles. Delaying notification until after withdrawal can lose the opportunity.

Key principle: *Notify the organ procurement organization per policy; trained specialists, not the treating team, raise donation.*

540. Correct answer A. Treat the cause; immediate angiography has shown no benefit without ST elevation

ST elevation or a new bundle branch block in a patient whose arrest may have had a coronary cause calls for emergency angiography, whereas trials of immediate angiography in patients without ST elevation showed no benefit. Many intraoperative arrests are not coronary in origin, and a freshly operated patient may not tolerate the anticoagulation intervention requires. Here the cause is hemorrhage and should be treated. Thrombolysis or heparin in a bleeding patient is hazardous. The cause search must not wait for neurologic recovery.

Key principle: *Without ST elevation, immediate angiography has not shown benefit; treat the identified cause.*

Group 28. Embolic Crises

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541. Correct answer D. It is more soluble in blood than nitrogen, so it enters the bubble faster than nitrogen leaves it

Nitrous oxide is far more soluble in blood than nitrogen, so it enters a gas bubble faster than nitrogen can diffuse out, and the bubble grows. The same reasoning applies to pneumocephalus and other closed gas spaces. High-concentration oxygen has the opposite effect because it lowers the nitrogen content of the bubble and encourages resorption. The other options misstate the solubility relationship or invent effects that nitrous oxide does not have.

Key principle: *Nitrous oxide expands gas bubbles; stop it at once and use 100 percent oxygen.*

542. Correct answer C. Induction with ketamine or etomidate, with a vasopressor infusion immediately available

In right ventricular failure, a loss of systemic pressure reduces right coronary perfusion and can start a downward spiral, and intubation with positive pressure raises pulmonary vascular resistance further. Ketamine or etomidate preserve hemodynamics better than propofol, and a vasopressor must be ready. Large fluid loads overfill the right ventricle and worsen septal bowing. Volatile induction and heavy sedation both risk hypotension or hypercapnia and are not safer.

Key principle: *Induce a failing right ventricle with ketamine or etomidate and a vasopressor at hand.*

543. Correct answer C. Norepinephrine, which raises systemic pressure and right coronary flow while sparing pulmonary resistance

Acute right ventricular failure from an embolus is supported with a vasopressor that raises systemic pressure and so restores right coronary perfusion while sparing pulmonary vascular resistance. Norepinephrine, and vasopressin as an alternative, fit this goal. The other options either lack a reliable systemic pressure effect or are not preferred in this setting.

Key principle: *Support the failing right ventricle with norepinephrine or vasopressin.*

544. Correct answer D. Reduce tidal volume, PEEP, and inspiratory time while keeping oxygenation and normocapnia

High tidal volumes, high positive end-expiratory pressure, and long inspiratory times raise intrathoracic pressure, reduce venous return, and increase right ventricular afterload. The goal is good oxygenation and normocapnia with the lowest pressures that achieve them. Permissive hypercapnia is not appropriate because acidosis raises pulmonary vascular resistance. Larger breaths or more pressure worsen the right ventricle without fixing dead space.

Key principle: *In right ventricular failure ventilate gently and avoid hypercapnia and acidosis.*

545. Correct answer B. Place the patient head down for removal and apply an occlusive dressing afterward

Air enters a central venous tract when venous pressure falls below atmospheric pressure, as in a sitting patient who is hypovolemic or taking deep breaths. Head-down positioning raises venous pressure at the site, and an occlusive dressing seals the tract afterward. Deep inspiration and an upright position widen the gradient and favor air entry. A dry gauze does not seal the tract.

Key principle: *Remove central lines head down and seal the tract with an occlusive dressing.*

546. Correct answer C. The rate of entry, because a rapid bolus can form an air lock in the right ventricular outflow tract

The rate of entry matters more than the total volume. A large bolus delivered quickly, on the order of a few milliliters per kilogram, forms an air lock in the right ventricular outflow tract and stops the circulation, while slow entry of a similar volume may be tolerated. Pressurized infusers, cell salvage systems, and large-bore lines can pump air directly into the vein, so de-airing every line matters.

Key principle: *With venous air, the speed of entry matters more than the total volume.*

547. Correct answer B. It is a clinical diagnosis of exclusion, because no test is reliable at the bedside

Amniotic fluid embolism is diagnosed clinically, after other causes such as hemorrhage, pulmonary embolism, anaphylaxis, and eclampsia are considered. No bedside test is reliable. D-dimer is nonspecific, and capnography and echocardiography show only the right heart effects shared with other emboli. Most affected women have no identifiable risk factor, so the absence of one does not help.

Key principle: *Amniotic fluid embolism is a clinical diagnosis of exclusion, so think of it early.*

548. Correct answer E. Left uterine displacement, with fetal delivery within about five minutes if arrest persists

After about twenty weeks, manual left uterine displacement relieves aortocaval compression during compressions, and delivery of the fetus within about five minutes improves the chance of maternal return of circulation. The obstetric team is called at the first sign of collapse because the decision to deliver takes time to execute. A full lateral position makes effective compressions impractical, and waiting for tests or a rhythm change only delays treatment.

Key principle: *In a pregnant arrest, displace the uterus and deliver within about five minutes.*

549. Correct answer B. Continue CPR for 60 to 90 minutes after fibrinolysis to allow the drug to act

In arrest from presumed embolism, fibrinolysis takes time to lyse the obstructing clot, so compressions are continued for 60 to 90 minutes afterward, weighing the surgical bleeding risk. A short resuscitation would end before the drug can act. Venoarterial extracorporeal support is the bridge when conventional measures fail and the team is available.

Key principle: *After fibrinolysis for embolic arrest, keep doing CPR for 60 to 90 minutes.*

550. Correct answer C. Alert the extracorporeal team early for venoarterial support to bypass the obstructed pulmonary circulation

Venoarterial support bypasses the obstructed pulmonary circulation, unloads the right ventricle, and keeps the patient alive while fibrinolysis acts or an embolectomy is arranged. Cannulation takes time and the window to protect the brain is short, so the team is called early. Venovenous support does not bypass the right ventricle, and balloon pumps and left-sided devices do not address the obstruction.

Key principle: *Call the extracorporeal team early in refractory embolic arrest.*

551. Correct answer D. A sudden change from the known baseline remains significant despite the abnormal gradient

Patients with chronic obstructive disease or obesity may have a wide baseline gradient, so the change from baseline matters more than the absolute number. A sudden change in a patient with an abnormal baseline is still significant. Recording a baseline early, with a paired arterial sample in high-risk cases, makes later interpretation reliable. Hyperventilation lowers the value gradually, not abruptly.

Key principle: *Interpret the end-tidal change against the patient's own baseline gradient.*

552. Correct answer A. Sodium bicarbonate

Sodium bicarbonate is buffered to carbonic acid and then to carbon dioxide, which transiently raises the end-tidal value and can mask a falling trend. This is a recognized pitfall in interpreting capnography during a crisis. The other agents do not generate carbon dioxide. Interpretation of the trend should account for recent drug administration.

Key principle: *Bicarbonate transiently raises end-tidal carbon dioxide and can mislead.*

553. Correct answer E. It is among the earliest signs that compressions or reperfusion therapy are working

During resuscitation, a rising end-tidal carbon dioxide reflects restoration of pulmonary blood flow, and it is among the earliest signs that compressions or reperfusion are working. Recording the value at regular intervals lets the team see whether interventions are helping. It is not an artifact or a sign of deterioration in this setting.

Key principle: *A rising end-tidal value during arrest suggests that flow is returning.*

554. Correct answer B. Neuraxial anesthesia lowers venous thrombosis after hip and knee surgery

Neuraxial anesthesia reduces venous thrombosis after hip and knee surgery, probably through improved lower limb blood flow. It does not replace early mobilization, compression, and timed anticoagulation. The anesthesiologist should know when the last anticoagulant dose was given and when the next is planned, so that the risk of spinal hematoma can be balanced against the risk of thrombosis.

Key principle: *Neuraxial anesthesia lowers postoperative venous thrombosis but does not replace prophylaxis.*

555. Correct answer C. D-dimer is nearly always elevated after surgery, so clinical suspicion should drive imaging

After an operation the D-dimer is nearly always elevated, so it has little discriminating value, and emergency department scoring systems are less helpful in postoperative patients. Embolism joins the differential of atelectasis, aspiration, residual neuromuscular blockade, opioid effect, and pulmonary edema when the oxygenation problem is out of proportion to the other findings. Clinical suspicion should drive imaging.

Key principle: *After surgery, do not rely on D-dimer; let clinical suspicion guide imaging.*

556. Correct answer E. Confusion and hypoxemia on day two with a petechial rash on the chest and axillae

Fat embolism syndrome appears usually within one to three days after long-bone fracture or orthopedic surgery, with hypoxemia, confusion, and a petechial rash over the chest, axillae, and conjunctivae. Treatment is supportive, and no drug has been proven to prevent or reverse it. Wheezing with urticaria suggests anaphylaxis, and an isolated swollen calf points toward venous thrombosis rather than the syndrome.

Key principle: *Think fat embolism for hypoxemia, confusion, and petechiae one to three days after fracture or nailing.*

557. Correct answer E. Epinephrine in small intravenous boluses

For collapse during bone cement implantation syndrome, communicate at once, support with oxygen and vasopressors, treat arrhythmia, and use small epinephrine boluses, which are often the most effective agent. Fluid alone is insufficient when the right ventricle is failing, and steroids have no established role in reversing the reaction. Nitrous oxide is harmful because it expands gas bubbles in the circulation.

Key principle: *Use small epinephrine boluses for cement-related collapse.*

558. Correct answer A. Widen the search to hemorrhage, tamponade, or anaphylaxis, as the findings argue against embolism

A dilated, hypokinetic right ventricle, a bowed septum, an underfilled left ventricle, and right heart clot make embolism likely enough to treat. The absence of those findings in a hypotensive patient argues for another cause, such as hemorrhage, tamponade, or anaphylaxis, and the search should widen at once. Fibrinolysis in a hypovolemic or bleeding patient could be harmful.

Key principle: *Normal right heart findings in shock argue against embolism; widen the differential.*

559. Correct answer E. Announce the finding at once so the surgical and cardiac teams can adjust plans

A mobile clot in transit in the right atrium, or across a patent foramen ovale, carries a high risk of further embolism and paradoxical stroke and is a medical and surgical emergency. Options include surgical embolectomy, thrombolysis, and catheter-based removal. The anesthesiologist should announce it. Flushing the clot onward would cause the embolus the team wants to avoid.

Key principle: *Announce thrombus in transit; it is an emergency even when the patient is stable.*

560. Correct answer C. The McConnell sign, supporting acute pulmonary embolism

The McConnell sign, hypokinesis of the right ventricular free wall with sparing of the apex, is an additional clue to acute pulmonary embolism, alongside a dilated right ventricle, septal bowing, tricuspid regurgitation, and a small hyperdynamic left ventricle. It does not describe tamponade or left-sided disease. No single sign is specific, but the combination points to an obstructive process.

Key principle: *Free wall hypokinesis with a normal apex is the McConnell sign of pulmonary embolism.*

Group 29. Obstructive and Pressure Crises

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561. Correct answer D. Flat neck veins occur in hypovolemia and tracheal deviation is late, so the diagnosis remains likely

Diagnosis of tension pneumothorax is clinical. Distended neck veins support it but are absent in hypovolemic patients, and tracheal deviation appears late. Rising airway pressure, falling saturation, hypotension, and unilateral hyperresonance with diminished breath sounds are enough to act. Waiting for a radiograph in an unstable patient is an error.

Key principle: *Do not wait for neck veins, tracheal shift, or a radiograph to treat tension pneumothorax.*

562. Correct answer A. Finger thoracostomy in the fifth intercostal space anterior to the midaxillary line, then a chest tube

In a ventilated patient the preferred method is finger thoracostomy, in which a small incision in the fifth intercostal space just in front of the midaxillary line and blunt dissection open the pleura, followed by a chest tube. This converts a tension pneumothorax into a simple open one. Needle decompression is an acceptable alternative when the operator is not confident. Waiting is unsafe, because each positive pressure breath adds more air.

Key principle: *In a ventilated patient with tension pneumothorax, finger thoracostomy is preferred when the operator can perform it.*

563. Correct answer A. Use a long, large-bore catheter, because failure is most common in muscular and obese patients

Needle decompression requires a large catheter long enough to cross the chest wall, and its failure rate is highest in muscular and obese patients, so a longer catheter is used in those patients. The fifth intercostal space anterior to the midaxillary line or the second intercostal space in the midclavicular line are accepted sites. Delay for imaging is inappropriate in an unstable patient.

Key principle: *Needle decompression fails when the catheter is too short for a thick chest wall.*

564. Correct answer D. A mainstem intubation can produce absent sliding on the nonventilated side

Absent lung sliding can result from mainstem intubation, pleural adhesions, or apnea, so the finding is read in the clinical context. Cephalad movement of the carina during Trendelenburg positioning can push the tube into a mainstem bronchus. A lung point, the clinical findings, and a response to decompression increase the likelihood of pneumothorax. Checking tube depth and listening to both lungs are fast steps that eliminate this branch.

Key principle: *Absent sliding can come from a mainstem tube, so read it with the clinical picture.*

565. Correct answer E. Decompress both sides of the chest during resuscitation as a reversible cause

If the patient arrests, bilateral decompression is performed during resuscitation, because tension pneumothorax is one of the reversible causes of arrest. After a procedure, the possibility of a second pneumothorax on the other side is considered. Epinephrine does not replace decompression. Delays for imaging reduce the chance of return of circulation.

Key principle: *In arrest from suspected tension pneumothorax, decompress both sides during compressions.*

566. Correct answer D. A pneumothorax in the ventilated lung is an emergency because no other lung is available

During one-lung ventilation, a pneumothorax in the ventilated lung is an emergency, since no other lung is available for gas exchange. Treatment is decompression, usually with the surgeon's help and with the nonventilated lung used to restore two-lung ventilation as soon as feasible. Increasing tidal volume adds pressure and worsens the tension. The collapse of the surgical lung does not protect the dependent lung.

Key principle: *Pneumothorax in the lone ventilated lung during one-lung ventilation is an emergency.*

567. Correct answer B. Pulsus paradoxus is unreliable during positive pressure ventilation, when the pattern may reverse

Pulsus paradoxus, an inspiratory fall in systolic pressure of more than ten millimeters of mercury, is unreliable during positive pressure ventilation, and the pattern may reverse. Beck's triad is present in only a minority of patients and is hard to detect under drapes. Under anesthesia, a rising central venous pressure, narrowing pulse pressure, low-voltage complexes with electrical alternans, and equalized diastolic pressures are more helpful. Echocardiography is the diagnostic tool of choice.

Key principle: *Do not rely on pulsus paradoxus or Beck's triad to diagnose tamponade under anesthesia.*

568. Correct answer A. Perform transesophageal echocardiography to look for a localized clot

After cardiac surgery, tamponade is often localized, with clot compressing one chamber while the rest of the pericardium looks empty. Chest tubes that stop draining suddenly after brisk drainage suggest that clot is retaining blood. A negative transthoracic study should not be accepted when the picture is suggestive, and transesophageal echocardiography is the investigation of choice. Drugs that slow the heart or reduce tone are avoided.

Key principle: *After cardiac surgery, a negative transthoracic study does not exclude localized tamponade.*

569. Correct answer A. Give fluid and an inotropic agent, with a vasopressor, without delaying drainage

A fluid bolus raises filling pressure above the pericardial pressure and transiently improves output. Epinephrine or dobutamine maintains heart rate and contractility, and phenylephrine or norepinephrine supports coronary perfusion. Drugs that slow the heart or reduce tone are avoided, and positive end-expiratory pressure is kept low. These measures must not delay drainage, which is the treatment.

Key principle: *Temporize tamponade with volume, chronotropy, and pressure support, but drain without delay.*

570. Correct answer C. Clot cannot be aspirated, so surgical drainage by subxiphoid window or sternotomy is needed

Needle pericardiocentesis under echocardiographic guidance relieves tamponade from medical effusions and some traumatic ones, but blood that has clotted cannot be aspirated. In trauma or after cardiac surgery the treatment is surgical, by subxiphoid window or sternotomy. Removing even a small volume can restore the circulation because the pericardial pressure-volume curve is steep. A drain is left in place and the patient is monitored for reaccumulation.

Key principle: *Clotted pericardial blood needs surgical drainage, not needle aspiration.*

571. Correct answer C. Induce with the surgeon scrubbed, blood and vasopressors running, and no delay to incision

Penetrating cardiac injury behaves like a different disease from slow effusions. Bleeding into a tight pericardium can produce shock within minutes, and the treatment is surgical release and repair. Anesthesia is induced with the surgeon scrubbed and ready, blood and vasopressors running, and minimal delay between induction and incision. Drugs that reduce tone or heart rate can precipitate arrest.

Key principle: *For penetrating cardiac injury, induce only when the surgeon is ready to cut.*

572. Correct answer D. Major vessel injury with concealed retroperitoneal hemorrhage

Unexpected hypotension at trocar insertion should raise the possibility of major vessel injury with hemorrhage into the retroperitoneum, which can be well concealed. The pressure of insufflation can tamponade a bleeding vessel, so that bleeding appears only on desufflation. The correct response is rapid conversion to an open procedure, large-bore access, blood in the room, and clear communication with the surgeon.

Key principle: *Hypotension at trocar placement means vascular injury until proven otherwise.*

573. Correct answer B. Intra-abdominal hypertension, a sustained pressure of 12 mm Hg or higher without organ dysfunction

Normal intra-abdominal pressure is about 5 to 7 mm Hg. Intra-abdominal hypertension is a sustained pressure of 12 mm Hg or higher, and the syndrome requires a sustained pressure above 20 mm Hg with new organ dysfunction. This patient has hypertension without the syndrome, so treatment is conservative: sedation, neuromuscular blockade, gastric and rectal decompression, and fluid optimization, with escalation if the pressure rises or organs fail.

Key principle: *Compartment syndrome needs pressure above 20 mm Hg plus new organ dysfunction.*

574. Correct answer A. Instill a small volume of saline and read at end expiration, zeroed at the midaxillary line

The bladder, filled with a small volume of saline, transmits intra-abdominal pressure to a transducer zeroed at the midaxillary line, and the reading is taken at end expiration in the supine patient. Central venous pressure is distorted by transmitted intrathoracic pressure and is not a substitute. Larger volumes or other positions give inaccurate values.

Key principle: *Measure bladder pressure with a small saline volume, at end expiration, zeroed at the midaxillary line.*

575. Correct answer E. Give calcium chloride intravenously and prepare bicarbonate and volume

Decompression produces a dramatic improvement in blood pressure, but reperfusion injury with hyperkalemia, acidosis, and arrhythmia may follow. The anesthesiologist should prepare calcium, bicarbonate, and volume in advance. Calcium chloride stabilizes the myocardial membrane and is preferably given centrally. Closing the abdomen would restore the pressure and defeat the purpose of the operation.

Key principle: *Expect reperfusion hyperkalemia and acidosis after abdominal decompression.*

576. Correct answer C. Cephalad movement of the carina has pushed the tube into the right mainstem bronchus

Pneumoperitoneum pushes the diaphragm cephalad, reduces functional residual capacity and compliance, and raises peak airway pressure. Trendelenburg positioning moves the carina cephalad and can push a tube into a mainstem

bronchus, usually the right, with absent breath sounds on the other side. Checking the tube depth and listening to both lungs is a fast step. An unexplained rise in pressure also raises the possibility of subcutaneous emphysema or pneumothorax.

Key principle: *Head-down laparoscopy can advance the tube into a mainstem bronchus.*

577. Correct answer E. Use low insufflation pressure and a slow rate, with an arterial line and preloading

Patients with severe valvular disease, heart failure, pulmonary hypertension, or recent infarction tolerate the rise in afterload and fall in preload poorly. The anesthesiologist may request lower insufflation pressure, slower insufflation, and less extreme positioning, and may place an arterial line to watch the effect beat to beat. Volume is given in anticipation. Where the cardiac risk is high enough, an open procedure may be safer.

Key principle: *With limited cardiac reserve, use gentler pneumoperitoneum and prepare the circulation first.*

578. Correct answer D. An end-expiratory hold gives total PEEP but underestimates it in active breathers

An end-expiratory hold closes the circuit at end expiration for a few seconds, and the equilibrated pressure is the total PEEP, from which the set PEEP is subtracted. It underestimates the problem in patients who are actively breathing or have severe flow limitation. A more direct measure is the volume of gas released during a prolonged apnea of 20 to 60 seconds. The number is a guide to lengthening expiration, not a target to chase.

Key principle: *The end-expiratory hold can underestimate auto-PEEP in active breathers or severe flow limitation.*

579. Correct answer B. Lower the rate and tidal volume, shorten inspiratory time, and accept hypercapnia

After the emergency, the strategy lengthens expiration by lowering the rate and tidal volume and shortening inspiratory time, accepting hypercapnia guided by an arterial gas. Ratios of inspiration to expiration of one to four or more are typical. Inverse ratios and pauses worsen gas trapping. External PEEP below the intrinsic level may help some patients with flow limitation, but it is harmful in others.

Key principle: *In auto-PEEP, lengthen expiration and tolerate hypercapnia.*

580. Correct answer A. Cardiac tamponade or a pulmonary embolus

Low peak pressure with hypotension and a high venous pressure suggests tamponade or an embolus. High peak pressure with hypotension suggests tension pneumothorax, auto-PEEP, or bronchospasm, and a tight abdomen suggests pneumoperitoneum or compartment syndrome. Running through these four obstructive causes with a disconnection, an ultrasound probe, a bladder pressure, and a look at the field rarely misses the cause.

Key principle: *Peak pressure helps separate causes of obstructed circulation: high points to the chest, low to the pericardium or lung circuit.*

Group 30. Hemodynamic Collapse and Metabolic Emergencies

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581. Correct answer B. Ventricular failure, a pump problem where an inotrope is reasonable once reversible causes are excluded

Low pressure with a low cardiac index, high filling pressures, cool extremities, and poor venous saturation indicate ventricular failure, in contrast to vasoplegia with normal or high output. The first questions are whether a reversible cause such as ischemia, arrhythmia, or obstruction is present. Restoring rhythm and correcting ischemia often matter more than the choice of inotrope. A transesophageal echocardiogram helps sort which treatment is paying.

Key principle: *Low output with high filling pressures means a pump problem, not pipes or preload.*

582. Correct answer C. Methylene blue risks hemolysis in glucose-6-phosphate dehydrogenase deficiency, so another agent is better

Methylene blue inhibits guanylate cyclase and nitric oxide synthase and raises vascular resistance in vasoplegia, but it causes hemolysis in glucose-6-phosphate dehydrogenase deficiency, can transiently lower pulse oximeter readings,

and may increase pulmonary vascular resistance. It also precipitates serotonin syndrome with serotonergic drugs. Hydroxocobalamin, which binds nitric oxide, is another rescue option.

Key principle: *Check glucose-6-phosphate dehydrogenase status and serotonergic drugs before methylene blue.*

583. Correct answer B. Milrinone is a long-acting vasodilator, so a vasopressor commonly accompanies it

Milrinone raises contractility without using the beta receptor, which helps in beta-blocked patients, but it is a vasodilator with a long half-life, so a vasopressor commonly accompanies it. The dose is lowered in renal impairment. Dobutamine raises output with little change in pressure and may lower it through vasodilation. Epinephrine raises pressure and output but also lactate and arrhythmia risk.

Key principle: *Milrinone works without beta receptors but vasodilates and lasts long.*

584. Correct answer E. Pulmonary vasoconstriction from protamine; stop it and support the right ventricle

The most dangerous protamine reaction is catastrophic pulmonary vasoconstriction, with acute right ventricular failure, rising pulmonary artery pressure, and falling systemic pressure. Slow administration, ready epinephrine, and a ready inhaled vasodilator reduce the harm. If it occurs, stop the infusion, consider returning to bypass, and support the right ventricle. Rapid injection causes histamine release, and an immune reaction produces anaphylaxis, which are different patterns.

Key principle: *A protamine reaction with high pulmonary pressure is right ventricular failure; stop and support.*

585. Correct answer C. Lighten the anesthetic and give phenylephrine or norepinephrine as needed

Anesthetic-related vasodilation from volatile agents, propofol, opioids, and neuraxial blockade is far more common than true vasoplegia. Reducing the dose of the offending agent and giving phenylephrine or norepinephrine restores the pressure within minutes. A pressure that responds quickly to a modest dose of vasopressor is anesthetic effect, and one that does not respond to escalating doses is something else. Reaching for rescue drugs skips the simpler step.

Key principle: *Lighten the anesthetic first; a quick vasopressor response means anesthetic effect.*

586. Correct answer B. The Cushing reflex from raised intracranial pressure

A slow heart with hypertension and irregular breathing in a patient with a head injury is the Cushing reflex, a response to raised intracranial pressure. Treating the rate alone would fix a number and miss the diagnosis. The clinician asks why the heart is slow, checking oxygenation, the airway, and the cause, before reaching for atropine.

Key principle: *Bradycardia with hypertension and irregular breathing after head injury is the Cushing reflex.*

587. Correct answer A. The Bezold-Jarisch reflex, triggered by mechanoreceptors in an underfilled left ventricle

The Bezold-Jarisch reflex, in which mechanoreceptors in an underfilled left ventricle trigger paradoxical bradycardia and vasodilation, accounts for sudden bradycardia and loss of consciousness after spinal anesthesia, particularly in hypovolemic or upright patients. Blockade of the upper thoracic cardiac accelerator fibers adds to it by letting vagal tone dominate. Young, healthy men are the group most often affected in reviews of arrest during spinal anesthesia.

Key principle: *The Bezold-Jarisch reflex explains sudden bradycardia after a spinal in an underfilled patient.*

588. Correct answer D. Doses below 0.5 mg may paradoxically slow the rate, so a full dose is given

An antimuscarinic such as atropine or glycopyrrolate is given in a full dose, because small doses, below 0.5 mg for atropine, may paradoxically slow the rate. Dosing is 0.5 to 1 mg repeated every 3 to 5 minutes to a maximum of 3 mg. Glycopyrrolate has a slower onset. If these fail, small boluses of epinephrine are used, followed by pacing.

Key principle: *Do not underdose atropine; small doses can slow the heart.*

589. Correct answer B. Glucagon, with calcium and high-dose insulin with dextrose

Severe beta-blocker toxicity responds to glucagon, which acts independently of the beta receptor, and to calcium and high-dose insulin with glucose. Nausea is common with glucagon, so an antiemetic is often needed. A call to the poison center, early extracorporeal support, and attention to potassium and magnesium may be lifesaving. Antibody fragments are specific for digoxin, and lipid emulsion is reserved for refractory calcium channel blocker poisoning.

Key principle: *Beta-blocker poisoning: glucagon, calcium, and high-dose insulin.*

590. Correct answer C. Peaked T waves, prolonged PR, loss of P waves, widened QRS, sine wave, then fibrillation or asystole

The electrocardiogram in hyperkalemia progresses from peaked T waves to a prolonged PR interval, loss of P waves, widened QRS complex, a sine wave pattern, and finally ventricular fibrillation or asystole. Low potassium produces flattened T waves and U waves. Low voltage with alternans suggests tamponade. A value from a blood gas analyzer can be read in a minute and should be sought early in any unexplained arrhythmia.

Key principle: *Hyperkalemia progresses from peaked T waves to widened QRS to sine wave and arrest.*

591. Correct answer D. Calcium acts in minutes for half an hour; insulin with dextrose acts in 15 to 30 minutes

Treatment follows a hierarchy. Intravenous calcium stabilizes the myocardium within minutes and lasts about half an hour, so it is repeated if the electrocardiogram remains abnormal. Insulin and dextrose then shift potassium into cells within fifteen to thirty minutes, with an inhaled beta-agonist and bicarbonate in acidosis. Dialysis removes potassium definitively. Calcium does not lower the potassium.

Key principle: *Give calcium first for membrane stability, then shift potassium with insulin and dextrose.*

592. Correct answer A. Rocuronium, with sugammadex available, to avoid a massive potassium rise

In patients with upregulated extrajunctional acetylcholine receptors, including burns beyond the first day, denervation, prolonged immobilization, crush injury, severe infection, and neuromuscular disease, succinylcholine can cause a massive and fatal rise in potassium. Rocuronium with sugammadex offers an alternative for rapid sequence induction. Reducing the dose of succinylcholine does not remove the risk.

Key principle: *Avoid succinylcholine in burns beyond day one; use rocuronium with sugammadex.*

593. Correct answer D. Calcium, bicarbonate, and insulin with dextrose are given, but inhaled beta-agonists are not

In cardiac arrest from hyperkalemia, calcium and bicarbonate, with insulin and dextrose, are commonly given with compressions, although the guideline rating of their effect on survival is not well established. Inhaled beta-agonists are not recommended in arrest. Resuscitation continues longer than usual because the cause is reversible. Calcium stabilizes the myocardial membrane and does not worsen hyperkalemia.

Key principle: *In hyperkalemic arrest give calcium and bicarbonate, skip inhaled beta-agonists, and persist.*

594. Correct answer E. Euglycemic ketoacidosis from the sodium-glucose cotransporter 2 inhibitor

Euglycemic ketoacidosis is a risk in patients taking sodium-glucose cotransporter 2 inhibitors, because glucose can be near normal despite severe acidosis. These drugs are stopped several days before elective surgery. The response is insulin and fluid, with dextrose as needed. Chloride-rich saline produces a normal gap acidosis, and respiratory acidosis would not explain ketones or a high gap.

Key principle: *Normal glucose does not exclude ketoacidosis in patients on sodium-glucose cotransporter 2 inhibitors.*

595. Correct answer A. Bicarbonate generates carbon dioxide, so ventilation must be increased and a repeat gas obtained

Bicarbonate is used with restraint. It generates carbon dioxide that must be exhaled, so it fails in patients who cannot raise their minute ventilation, and it can worsen intracellular acidosis. It is considered for severe acidemia with hemodynamic compromise, for hyperkalemia with acidosis, and for specific toxicities such as sodium channel blocker poisoning. A modest dose is given, ventilation is increased, and a blood gas repeated. Treating the cause remains the principal task.

Key principle: *Bicarbonate makes carbon dioxide; give a modest dose and increase ventilation.*

596. Correct answer D. Hypocalcemia from citrate binding ionized calcium, impairing contractility and coagulation

Citrate in blood products binds ionized calcium, and the liver may not clear it in shock, so hypocalcemia develops and impairs contractility and coagulation. Giving calcium with large-volume transfusion, measuring ionized calcium and potassium frequently, using warmed products, and choosing fresher or washed units for infants and renal failure keep these problems manageable. Stored cells also leak potassium.

Key principle: *Give calcium and check ionized calcium frequently during massive transfusion.*

597. Correct answer C. Inadequate ventilatory compensation that can be corrected with ventilator changes

Winter's formula predicts the expected carbon dioxide for a given bicarbonate, and a measured value above the prediction suggests inadequate ventilatory compensation. In an anesthetized patient this is a ventilator setting, so minute ventilation can be increased. The anion gap, calculated from sodium, chloride, and bicarbonate with correction for low albumin, separates lactate, ketones, toxins, and renal failure from chloride excess.

Key principle: *A carbon dioxide above the Winter prediction means the ventilator is under-compensating.*

598. Correct answer B. Give hydrocortisone now, drawing blood first only if the dose is not delayed

Diagnosis of adrenal crisis in the operating room is clinical, and the treatment is given without waiting for confirmation. Blood for cortisol and corticotropin can be drawn before the first dose, but the dose should not be delayed. A random cortisol that is low or even in the normal range during shock supports the diagnosis, while a high value argues against it. Dexamethasone does not interfere with the cortisol assay and can be used if testing is pending.

Key principle: *In suspected adrenal crisis, treat first and test second.*

599. Correct answer E. Sudden loss of catecholamine excess with vasodilation and possible adrenal insufficiency

After resection of a pheochromocytoma, the sudden loss of catecholamine excess produces profound vasodilation and hypotension, and relative adrenal insufficiency may coexist. In patients having adrenal or pituitary surgery, the surgical and endocrine teams plan hydrocortisone replacement in advance, and the anesthesiologist gives it at the appropriate point and watches for hypotension that signals a shortfall.

Key principle: *Expect hypotension after pheochromocytoma resection and plan steroid coverage.*

600. Correct answer E. Octreotide, because sulfonylurea overdose causes recurrent hypoglycemia

The effect of a single dextrose bolus is brief, especially after long-acting agents, so glucose is rechecked after fifteen minutes. Sulfonylurea overdose causes recurrent hypoglycemia that responds to octreotide. Glucagon is an alternative when access is not available but is less effective in patients who are malnourished or have liver disease. Severe hypoglycemia is a neurologic emergency.

Key principle: *Recurrent hypoglycemia after a sulfonylurea responds to octreotide.*